Letian Huang (Ed.)

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Machine Learning and Soft Computing

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Letian Huang
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Machine Learning and Soft Computing

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Preface

The 9th International Conference on Machine Learning and Soft Computing (ICMLSC), held from January 24–26, 2025, in Tokyo, Japan, was a resounding success. Organized by Chuo University, Japan, co-organized by Keiai University, Japan, and supported by the University of Münster, Germany, and Concordia University, Canada, the conference provided a vibrant platform for scholars and practitioners to exchange ideas in the dynamic fields of Machine Learning and Soft Computing.

The ICMLSC Conference aimed to foster discussions on new issues, grapple with complex problems, and uncover advanced solutions, thus shaping emerging trends in these areas. The conference received a total of 121 full-text submissions. Through a rigorous and meticulous review process, 52 papers were selected for inclusion in these two proceedings volumes, resulting in an acceptance rate of 42%. Additionally, there were 37 abstract submissions, out of which 12 were accepted for presentation at the conference, with an acceptance rate of 32%. These abstract-based presentations, however, are not part of these proceedings.

The review process was carefully organized. A diverse panel of experts from the international academic and industrial communities, serving as members of the Program Committee, was responsible for evaluating the submissions. Each paper was assigned to at least three reviewers who thoroughly examined the quality of research, novelty of ideas, and potential impact. The reviewers provided detailed feedback, and the Program Committee engaged in extensive discussions to ensure that only the most meritorious papers were selected.

The papers in this volume cover a wide range of topics, including novel machine learning algorithms, applications of soft computing in various domains such as healthcare, finance, and robotics, and the intersection of machine learning and soft computing techniques for more efficient problem-solving. They represent cutting-edge research in the field, and we believe they will serve as valuable resources for researchers, students, and industry professionals alike.

We would like to express our sincere gratitude to all the authors who contributed their work to the conference. Their research efforts were the driving force behind the success of this event. We are also deeply indebted to the Program Committee members and the Organizing Committee members for their tireless dedication in soliciting, reviewing, and organizing the conference program.

We hope that readers will find these proceedings both informative and inspiring, and that they will contribute to further advancements in the exciting fields of Machine Learning and Soft Computing.

January 2025 Letian Huang

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Contents – Part I

Multimodal Data Analysis and Model Optimization

Utilizing Demographic Data and Insurance Claims History to Develop Machine Learning for Assessing Cardiovascular Disease Risk 3
Napat Uraisomsurat, Tanet Sriamorn, and Paisit Khanarsa

Comprehensive Framework for Artificial Intelligence and Big Data Integration in Highe Education 16
Hasan Kadhem

Data Envelopment Analysis on Response Surface Method for Efficient Parameterization of Evolutionary Algorithms in Industrial Applications 32
Atiwat Nanphang and Pongchanun Luangpaiboon

Feature Niching Based Differential Evolution for Feature Selection on High-Dimensional Data 44
Biyu Yin, Mingwei Wang, Maolin Chen, and Wei Liu

Evaluating the Performance of Open-Source LLMs in Local RAG Systems: A Practical Study on Low-Carbon Data Applications 55
Yu-Jie Chang, Tai-Yi Yu, and Chung-Hao Chang

Data Selection for Close-Domain Data in Medical Continual Pretraining: A Case Study on Data Selection via Importance Resampling (DSIR) 67
Chatiyar Ardchon, Can Udomcharoenchaikit, Nonthakit Chaiwong, and Paisit Khanarsa

Improving Domain-Specific Data Question Answering with Deep and Cross-Lingual Transfer Learning 80
Mohammed Alsuhaibani and Mirza Omer Beg

Leveraging Co-Occurrence Graphs and NLP for Enhanced Job-Skill Matching 94
Peera Sittiros, Nirach Romyen, Tanatorn Tanantong, and Krittakom Srijiranon

Risk Estimation with Active Labeling 103
Alessandro Magnani, Esteban Arcaute, and Shie Mannor

Basic Theories of Machine Learning and Emerging Application Technologies

Using Machine Learning Techniques to Discriminate Good and Poor Sleepers in Virtual Reality Environment 123
En-Chen Chen and Tsai-Yen Li

San Jose Urban Forest – An Open-Source Tree Canopy Surveying and Assessment Tool 145
Martin Alvarez Lopez, Manan Choksi, Sai Yaaminie Ganda, Vaibhavi Hiteshkumar Savani, and Bernardo Flores

Towards a Machine Learning-Based Approach to Predicting Stock Price Volatility and Its Associated Risk in Egypt 157
John El Gallab, Noha Abdelbary, Mustafa Elfar, Fady Abouelghit, Basma Rady, and Roba Bairakdar

Lift and Shift of Model Code Using Machine Learning Microservices with Generative AI Mapping Layer in Enterprise SaaS Applications 171
Venkata Duvvuri

Machine Learning-Driven Extended Creativity for Reshaping Traditional Artistic Pieces 192
Chutisant Kerdvibulvech

Towards AI/ML-Powered Hybrid Project Management Strategy for the Healthcare Sector 202
Manisha Khadgi and Fareed Ud Din

Enhancing Textual Deception Detection: A Fused Handcrafted Feature Approach with Machine Learning Models 216
Omar S. Fahmy, Kamel A. Elhaddad, Khaled M. Badran, and Mohamed K. Elhadad

Machine Learning Models for Validating the Self-declaration Conformity Assessment: Risk Evaluation 231
A. M. Sadek, Mohammed S. AlRashidi, Yousef M. AlMutiri, AbdulRahman M. AlQahtani, and Turki S. AlQahtani

Enterprise Credit Rating Framework Based on Risk Contagion Graph Neural Network 243
Chun Cheng and Cuicui Luo

Intelligent Recommendation System Design and Privacy Security

Emotionally Unbiased Reinforcement Learning for Equilibrium-Seeking in Conflict-Driven Multi-agent Systems 257
Raj Kashikar

Diversified Conversational Recommendation System 267
Jia-Chi Hsu, Szu-Hao Huang, Chiao-Ting Chen, and Cheng-Jhang Wu

Rule Extraction with Reject Option 278
Cecilia Sönströd and Ulf Johansson

Poisoning Attacks Against Security-Aware Federated Recommendation System 301
Chi Lee, Szu-Hao Huang, and Chiao-Ting Chen

Applying Artificial Intelligence in Taiwan's Fin-Tech: Exploring Usage Intention of Robo-Advisor Service 314
Chieh-Peng Lin, Shin-Ru Lin, and Chou-Kang Chiu

Detecting Social Bots Using Neural Networks with Social, Word Embedding, and Temporal Features 322
I-Hsien Ting, Kazunori Minetaki, and Mei-Yun Hsu

Exploring Information Presentation and Sentiment Experience in Generative AI-Controlled Interfaces: A Case Study on Electric Bicycle Interaction 334
Chung-Shun Feng, Tsu-Wu Hu, and Yu-Che Huang

Advanced Generative AI: A Multi-modal Approach Through Conversational Chatbot and Intelligent Question Answering for Enhanced Customer Experience and Assistance in Air Travel Domain 344
Vipin Mishra, Venkata Duvvuri, and Renuka Chintalapati

Author Index 363

Contents – Part II

Deep Learning Models and High-Performance Computing

Reinforcement Learning for ETF Rebalancing Under Index Composition Changes 3
Mei-Hua Wu, Pin-Chieh Ho, Chih-Chung Chang, Szu-Hao Huang, and Chiao-Ting Chen

Predicting Potential Soil and Groundwater Contamination Risks from Gas Stations Using a Deep Learning Technique 15
I.-Cheng Chang, Shen-De Chen, Yu-Jie Chang, Huey-Long Chen, and Tai-Yi Yu

Retinex-Infused Deep Learning Network for Image Dehazing 25
Ali Murtaza, Uswah Khairuddin, Ahmad A. M. Faudzi, and Kazuhiko Hamamoto

Development of a Vision-Based Navigation and Landing System Based on Deep Learning and Image Segmentation Methods with Runway Information for the Landing of a Fixed-Wing UAV 37
Wen-Hao Liao and Ying-Chih Lai

A Deep Learning Approach to Parking Space Detection: Achieving High Accuracy and Scalability with FastAPI and Alexnet 48
Waraporn Viyanon

Forest Fire and Smoke Detection Using Image-Based Deep Learning 58
Napatsorn Tantasanee and Thitirat Siriborvornratanakul

Federated Learning for Privacy-Preserving: Current Status and Future Directions 71
Ankit Chouhan and JingTao Yao

Data-Driven Complex System Modeling and Intelligent Optimization Algorithms

Geographical Zoning for Electric Vehicle Charging Station Planning by Using Hierarchical Clustering 87
Nutjaree Ludda and Pornthipa Ongkunaruk

An Efficient Fine-Tuning Strategy of Large Language Models for Unit Test Code Generation via Multi-Task Prompt Learning 99
Gang Chen, Xinyi Liu, Bin Chen, Zeke Li, Haiwei Fan, Peng Cao, and Guangqi Wen

Solving a Home Healthcare Routing and Scheduling Problem with Real-World Features 111
Roberto Montemanni

DNA Sequence Design Based on Dual-Sorting Constrained Multi-objective Evolutionary Algorithm 120
YuXin Mei, Kai Zhang, and XianHang Luo

A Novel Constrained Multi-objective Evolutionary Algorithm for Solving VRPSPDTW 134
ChengXi Liang, Kai Zhangr, Ni Wu, and Ling Zhang

Credit Card Fraud Detection with Deep Multilayer Perceptrons 146
Sadia Rahman and JingTao Yao

Evaluating Calibration Techniques for Reliable Predictions 159
Aicha Maalej, Ulf Johansson, and Tuwe Lofstrom

Artificial Bee Colony Optimization for EEG Channel Selection in Subject Independent Motor Imagery BCI 176
Sorawit Therdkiattikoon and Suthathip Maneewongvatana

Evaluating the Impact of Multi-sensory Virtual Reality in Prostate Cancer Health Education: Enhancing Learners' Learning Interest, Emotional Participation and Knowledge Level 186
Hao Fang, Xingyu Chen, Seng Yue Wong, Kenny S. L. Cheah, Peilin Li, and Huaijun Sun

Image Analysis and Processing Methods Based on AI

Self-attention Network with Dynamic Perception for Crowd Counting 201
Xiao Hu, Yiqiu Huang, Qinxia Hu, and Zhiwei Guo

Biometric Authentication Using Unnaturally Bended Palm Shape 211
Genta Koike and Hiroshi Dozono

A VLC Mapping-Based Lossless Data Hiding Method for JPEG Images with Stable File Size Increment 221
Chi-Cheng Yu and Chen-Yi Lin

Investigation of Frame Differences as Motion Cues for Video Object Segmentation 228
Sota Kawamura, Hirotada Honda, Shugo Nakamura, and Takashi Sano

A Hybrid Reversible Data Hiding Approach for JPEG Images Based on DCT Coefficients 236
Ping-Lin Liu and Chen-Yi Lin

Ablation Studies of KRAS in Colon Cancer: Predictive Power and Diagnostic Accuracy 243
Venkata Duvvuri, Murali R. Kuracha, Yuchiao Shaw, Zhenzhen Ren, Sushma Konduri, and Renuka Chintalapati

Hydrotherapy Hip Abduction Angle Range Classification via Hybrid Joint Vector YOLO NAS Model (JV-YOLO NAS) 256
Aly Khaled Dawood, Amr Emad Mekki, Uswah Khairuddin, and Noureldin S. Eissa

Automated Intelligent Image-Based Tomato Sorting Prototype 271
Mohammed Abu Mallouh, Rafah Al-Gabr, Fatima Ahmad, and Basel Jouda

Detailed Lower Body Gait Analysis for Hydrotherapy Exercise Using Modified YOLOv7 Pose 281
Amr Emad Mekki, Aly Khaled Dawood, Uswah Khairuddin, and Noureldin S. Eissa

Analysis of Feature Subspace and Explainability of CNNs for Classification of High Resolution Remote Sensing Images 295
Krishna Gupta and Manisha Parlewar

Author Index 305

Multimodal Data Analysis and Model Optimization

Utilizing Demographic Data and Insurance Claims History to Develop Machine Learning for Assessing Cardiovascular Disease Risk

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Abstract. Cardiovascular diseases are challenging to treat and can lead to severe complications or death. Insurance companies face difficulties predicting future risks of cardiovascular disease, complicating health insurance renewals and preventive care for at-risk individuals. Therefore, early identification of those at risk is essential for effective insurance management. This paper applies tree-based machine learning models to predict future cardiovascular disease risks using demographics, insurance claims, air quality, and illness summary features. Four tree-based models were implemented: Decision Tree, Random Forest, Extreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LightGBM). The performance of the models using only age and gender was compared to those incorporating all available features. XGBoost demonstrated the best performance, achieving sensitivity ranging from 0.7 to 0.758 and specificity ranging from 0.496 to 0.772.

Keywords: Critical illness · Health insurance · Machine learning · Tree base model

1 Introduction

The associations between air pollution and its harmful effects on nearly all organs of the body are undeniable. Studies have shown that air pollution exacerbates all major cardiovascular diseases (CVDs) [1]. In 2019, air pollution was responsible for 22% of ischemic heart disease (IHD) deaths and 15% of stroke deaths worldwide. This widespread exposure and its impact on cardiovascular health pose a significant challenge to global health [2]. The prevalence of cardiovascular disease continues to rise and remains difficult to treat, often requiring prolonged treatment durations and potentially leading to complications or even death. The factors contributing to an individual's susceptibility to critical illness is diverse and challenging to predict. COVID-19 has been a significant contributor, with the increase in heart disease from 2020 to 2021 accounting for over 4% of the recent decline in life expectancy [3]. Therefore, there is an urgent need for effective risk assessment and preventive healthcare.

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In recent years, there has been a notable increase in the number of individuals purchasing insurance, reflecting a growing awareness of health-related issues [4].

Insurance companies accumulate vast amounts of claims history data daily. However, this data has not been utilized to its full potential [4]. Cardiovascular disease typically takes years to develop, progressing gradually from early risk factors to more severe conditions. Often, these risk factors first manifest as underlying or less severe illnesses. In the insurance industry, existing health insurance claims provide valuable insights into customers' health histories and lifestyles [4, 5]. These records provide insurers with the ability to assess individual risks more accurately, supporting better future risk estimation and targeted preventive care for high-risk customers [6].

Due to the inadequacies of current illness risk estimator, the objective of this paper is to leverage insurance data to predict critical illness risk of cardiovascular diseases, particularly within group health insurance data. Insurance data, which contains a wealth of information about policyholders' demographics, claims histories, and treatment histories, has the potential to enhance predictive analytics in this domain. Analyzing claim patterns and gaining insights into critical illness can provide insurance companies with valuable tools that can be utilized in various impactful ways.

2 Related Work

Cardiovascular diseases (CVDs) have consistently been the leading cause of death globally, significantly contributing to both the loss of health and increased healthcare costs. It is crucial to assess the risk of cardiovascular disease and stroke, as this helps individuals identify potential risks early and take proactive steps to avoid contributing factors [7]. In the healthcare domain, illnesses are classified according to the International Classification of Diseases, Tenth Revision (ICD-10) [8]. Many factors are related to cardiovascular disease, including unhealthy nutrition, physical inactivity, dyslipidemia, hyperglycemia, high blood pressure, overweight and obesity, demographics, thrombosis, and physiological factors [9].

In 2022, Jan Carlo T. Arroyo and Allemar Jhone P. Delima [10] developed a cardiovascular disease prediction model using 12 demographic and blood test results. The GA-ANN performed the best in terms of accuracy. The effects of changes in neurons, layers, activation functions, and optimizers were analyzed. In 2023, Krishna Mridha et al. [11] developed a web application for automated stroke prediction using a binary classification model, addressing a 1:20 data imbalance with SMOTE oversampling and focusing on features like demographics, BMI, work type, glucose levels, and illness history. Six machine learning models, including Logistic Regression, SVM, Random Forest, Naïve Bayes, KNN, and XGBoost, were tested, emphasizing results with and without data leakage. Also in 2023, Ryan Marcus, Jeremy M. Lupague, et al. [12] utilized 2021 BRFSS data from the WHO to integrate a machine learning model for comprehensive health risk assessment, selecting the model with the highest AUC score to differentiate cardiovascular disease from overall health, with Logistic Regression outperforming others and highlighting gender and diabetes as key features. In 2024, Utkarsh Madaksira [13] applied machine learning to predict cardiovascular disease risk, transforming related illnesses such as cancer, arthritis, and depression, along with lifestyle factors like exercise, smoking history, and alcohol consumption, into predictive features. Among five

models tested, including Logistic Regression, Random Forest, K-Nearest Neighbors, Naïve Bayes, and Decision Tree, Random Forest demonstrated the best performance.

Most research studies utilize data on food consumption behavior, lifestyle patterns, blood test results, exercise history, smoking, and alcohol consumption. These are often difficult for insurance companies to obtain. Therefore, an interesting gap in the literature is identified. Although insurance companies may not have access to detailed behavioral data of their customers, such data is likely reflected in the form of disease claims, particularly those related to cardiovascular disease. Claim patterns and the complex relationships between diseases could be analyzed using tree models, as they are effective at capturing intricate relationships. This approach allows for the visualization of these relationships in a tree structure, providing deeper insights into the connections between illnesses [14]. To explore this further, the study experiments with four machine learning models: Decision Tree, Random Forest, Extreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LightGBM), comparing their performance to identify the most effective approach for analyzing healthcare data.

3 Methodology

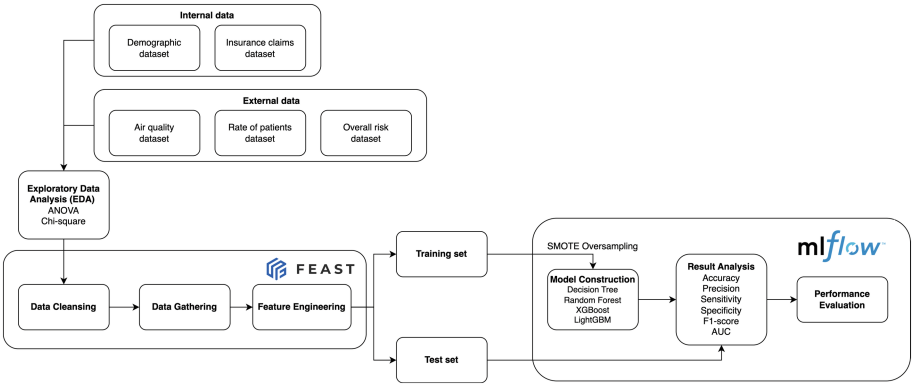

Fig. 1. System overview of machine learning process for predicting cardiovascular disease risk.

The system overview of machine learning process for predicting cardiovascular disease risk presented in this paper is divided into two primary components, as illustrated in Fig. 1. The first section examines the findings from exploratory data analysis (EDA) conducted after collecting data from the demographic dataset, the insurance claims dataset, the air quality dataset, and patient rate statistics. Subsequently, the Feast Feature Store is utilized for feature engineering, data collection, and data cleansing as part of the machine learning lifecycle process. The data is then split into two sets: 80% for training and 20% for testing, enabling the tracking of model performance during experiments with MLFlow in the second section. The baseline model's performance is compared to that of a tuned model for each model type, following the same workflow.

3.1 Dataset and Preprocessing

Both internal and external data are collected for the baseline model and classification model, with a total of 594 features used in the experiment. As seen in Table 1, internal data includes information on demographic features and health insurance claim features (quarterly data) from the Sunday Insurance data, during 2015 to 2024. Table 1 presents a transformation of historical claim records into a summary of claim information on a quarterly basis. The claim types include OPD and IPD, while disease types encompass 138 categories, such as respiratory system, digestive system, diabetes, obesity and infection. The "total agreed amount" represents the total sum of health insurance claims

Table 1. Internal data features for demographic and health insurance claims

Feature type	Feature name	Datatype	Definition
Demographic	gender	string	Male or Female
	age	int	The insured's age (years)
	industry major group	string	Type of work
Insurance claim	count_ <claim type>	int	Count number of claims in each claim type (OPD, IPD)
	count_ <disease type>	int	Count number of claims in each disease type (138 disease types, such as abnormal blood exam, breast disease, obesity, circulatory system, respiratory system, digestive system, diabetes, eye disease and infection.)
	sum_total_agree_amt	float	Sum of health insurance claim in each customer
	sum_ <admitted type> _days	int	Sum of days stayed in each admitted type (regular room or ICU)
	count_chronic_disease_claim_frq	int	Count number of chronic disease claims
	count_surgery_ claim_frq	int	Count number of surgery claims

for each customer. The admitted type shows the count of days a customer stayed in a regular room or ICU. Additionally, chronic disease or surgery records the number of times chronic diseases or surgeries were claimed.

Table 2 presents air quality features collected from the Pollution Control Department (Thailand). This dataset comprises daily air quality readings obtained from air quality monitoring stations. The daily air indicator values are averaged over the past year and include metrics such as particulate matter 2.5 μm (PM2.5), particulate matter 10 μm (PM10), ozone, carbon monoxide, sulfur dioxide, and the air quality index (AQI). Additionally, a summary of illness reports from the Ministry of Public Health (Thailand), collected from all health services, is used to generate patient rate features and an overall risk feature by considering disease prevalence per 100,000 people. These features are calculated based on the average of relevant ICD-10 codes for major diseases in each region and gender. The study specifically focuses on five diseases: cancer, cardiovascular disease, lung disease, liver disease, and chronic kidney failure.

For data preprocessing, a regular expression was used to match the ICD-10 pattern, which begins with an English letter followed by two digits. Categorical features, such as gender and industry major group, were processed using one-hot encoding. Internal data mapped by gender, province, and postcode was merged with external data, and to enable the model to understand new customer data from unfamiliar provinces, postcodes and provinces were converted into risk features derived from external data. The dataset used in this study is highly imbalanced, with a 1:289 ratio, comprising 1,385 samples from the cardiovascular disease group and 399,676 samples from the healthy group. To address this imbalance, the synthetic minority oversampling technique (SMOTE) was applied during resampling to enhance the model's ability to learn from the minority class.

Table 2. External data features from air quality indicators and illness statistics

Feature type	Feature name	Datatype	Definition
Air quality	avg_ <air quality indicator> _value	float	Average air quality last n year (particulate matter 2.5 μm, particulate matter 10 μm, ozone, carbon monoxide, sulfur dioxide and air quality index)
Illness statistics	<critical illness> _prevalence_rate	float	Critical illness 100,000 prevalence (cancer, cardiovascular disease, lung disease, liver disease, chronic kidney failure)
	overall_illness_risk	float	Overall illness 100,000 prevalence

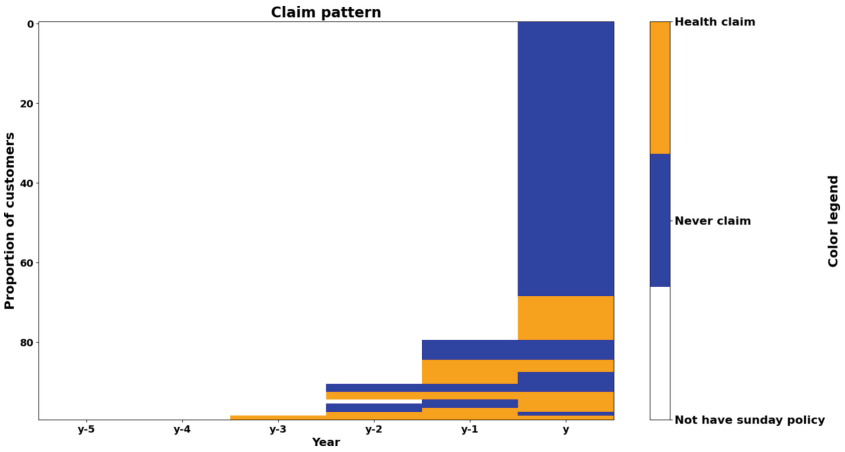

Fig. 2. Distribution of health insurance claim patterns over policy years

3.2 Exploratory Data Analysis on Claim Patterns of Cardiovascular Disease-Related Conditions

Figure 2 illustrates an analysis of historical claim data, where the x-axis represents the years preceding each customer's most recent policy, and the y-axis represents the proportion of customers. The yellow sections indicate at least one health insurance claim in a given policy year, blue represents having a policy without any claims in that year, and white indicates no policy during that year. The analysis reveals that most customers have a low claim frequency, which highlights the importance of including external data to improve the analysis. Moreover, most customers hold policies for only a single year. To address these challenges, only data from each customer's most recent policy will be used to keep the number of features manageable and avoid missing values.

3.3 Experimental Design

In the insurance industry, gaining insights from data is as important as the prediction model's results [4]. Therefore, tree-based models are well-suited for identifying significant claim patterns associated with diseases and capturing complex relationships. Four tree-based models were used in the experiments: Decision Tree, Random Forest, Extreme Gradient Boosting (XGBoost) and Light Gradient Boosting Machine (LightGBM).

A Decision Tree predicts outcomes by splitting data into branches based on feature values, but it is prone to overfitting. Random Forest mitigates overfitting by combining multiple Decision Trees and averaging their predictions to improve performance [15]. XGBoost builds trees sequentially, correcting errors from previous trees to enhance performance through gradient boosting [16]. Similarly, LightGBM, another gradient boosting model, is optimized for speed and efficiency by using a leaf-wise splitting strategy rather than level-wise, making it particularly effective for large datasets [17]. In the baseline setup, all four models used default hyperparameters, and no adjustments

were made to the binary classification threshold. Additionally, 3-fold cross-validation was applied to evaluate performance and reduce biased errors.

Most insurance companies now calculate premiums based on the level of illness risk, with age and gender being among the most significant factors [18]. The baseline model estimated the risk of critical diseases based solely on age and gender. For the baseline model, default hyperparameters were used along with the default threshold for binary classification. As shown in Table 1 and 2, demographic features, insurance claim features, air quality features, and illness report summaries were selected based on visual exploration of their relationships and statistical measures of relationship strength with the target variable. To eliminate feature redundancy, multicollinearity analysis was applied to examine correlations between independent features and to exclude highly correlated ones. A summary of the selected feature groups, determined using ANOVA and Chi-square tests, is presented in Table 3.

3.4 Evaluation Metrics

This paper utilized fundamental metrics for evaluating the model, including F1-score, accuracy, precision, sensitivity, specificity, and the area under the ROC curve (AUC). These metrics are calculated using the following formulas:

$$Accuracy = \frac{True\ Positives + True\ Negatives}{True\ Positives + False\ Positices + True\ Negatives + False\ Negatives} \tag{1}$$

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives} \tag{2}$$

$$Sensitivity = \frac{True\ Positives}{True\ Positives + False\ Negatives} \tag{3}$$

$$Specificity = \frac{True\ Negatives}{True\ Negatives + False\ Positives} \tag{4}$$

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{5}$$

In the healthcare and health insurance domain, the focus is primarily on sensitivity and specificity to evaluate how well a model can identify patients and healthy customers [19]. This paper aims to balance the model's performance between sensitivity and specificity to effectively distinguish between the two categories. Insurance companies provide preventative healthcare to high-risk consumers to lower their risk and reduce the severity of claims. However, they aim to avoid allocating resources to provide preventive treatment for low-risk clients. The model was built using binary classification, but insurance companies need probability distributions to assess individual risk levels. Given the ratio of healthy customers to those with cardiovascular disease, the industry prefers a right-skewed probability distribution for healthy customers, indicating low risk. In contrast, probabilities for high-risk cardiovascular patients should exhibit a left-skewed distribution.

4 Result and Discussion

The results of this paper are divided into three sections: Sect. 1 covers the key insights from feature selection, Sect. 2 presents the model performance reports, and Sect. 3 discusses the key insights from feature importance based on tree-based models.

4.1 Key Insights from Feature Selection

Table 3. Summary of selected features for cardiovascular risk prediction

Data source	Feature type	Selected feature
Internal	Demographic	gender, age, industry_major_group
	Insurance claim	normal room admitted day, ICU room admitted day, total agree, OPD claim, IPD claim, surgery claim, chronic disease claim
	Related illness	Cerebrovascular, Diabetes, Hypertension, Obesity, Circulatory symptom, Metabolic disorder
External	Air quality	PM2.5, PM10, Ozone, Carbon monoxide, Nitrogen dioxide, Sulfur dioxide, Air quality index (AQI)
	Illness statistics	Cancer rate, cardiovascular disease rate, Liver disease rate, Chronic kidney failure rate

ANOVA and Chi-square tests were used to select 177 features, as shown in Table 3. Key insights were derived from the factors and claim patterns associated with cardiovascular disease. These factors include prolonged exposure to poor air quality, which increases the risk of respiratory ailments, cardiovascular disorders, and other chronic conditions. External data, such as illness report summaries, emerged as crucial features, highlighting that individuals living in areas with a high prevalence of cardiovascular diseases are significantly more likely to develop similar conditions. Additionally, many patients had a history of surgery within 4 to 12 months before their first cardiovascular disease claim, with a substantial number being admitted to the ICU or regular hospital rooms during recovery. Moreover, Table 3 indicates that Internal data indicates that customers often exhibit conditions such as diabetes, hypertension, obesity, chronic kidney disease (CKD), or metabolic disorders 4 to 12 months before their first cardiovascular disease claim.

4.2 Model Performance

Table 4 presents the hyperparameter configurations for each model. The hyperparameters were selected, and the experiments were tracked using MLFlow. Each model type achieves optimal performance when its hyperparameters are properly tuned. From Table 5, it is evident that the accuracy, specificity, and AUC of the all-feature models are significantly better than the baseline performances. Specifically, accuracy increases

from a range of 0.428 - 0.541 to 0.729 - 0.772, specificity improves from 0.426 - 0.541 to 0.729 - 0.772, and AUC rises from 0.5 - 0.58 to 0.781 - 0.857. In contrast, the precision and F1-score of the all-feature models show only slight improvements. These enhancements highlight the substantial impact of incorporating additional features.

Table 4. Hyperparameter configurations for models in cardiovascular risk prediction

Model Type	Hyperparameters
Decision Tree	Max depth(2), Min sample split(12), Min samples leaf(12), Criterion(entropy)
Random Forest	Max depth(4), N estimators(20), Min samples split(20), Min samples leaf(20), Criterion(gini)
XGBoost	Max depth(5), Eta(0.08), N estimators(20), Max delta step(4), Min child weight(8)
LightGBM	Max depth(4), Learning rate(0.01), N estimators(80), Min child weight(12), Colsample bytree(0.8), Drop rate(0.25)

Table 5. Baseline performance of tree-based models for CVD risk prediction

Baseline performance				
	Decision Tree	Random Forest	XGBoost	LightGBM
Accuracy	0.428	0.428	0.496	0.541
Sensitivity	0.785	0.785	0.700	0.663
Specificity	0.426	0.426	0.496	0.541
Precision	0.005	0.005	0.005	0.005
F1-score	0.010	0.010	0.010	0.010
AUC	0.500	0.500	0.580	0.543

Moreover, XGBoost stands out as the best-performing model across all metrics. In medical models, the primary focus is on sensitivity and specificity to evaluate the model's ability to accurately identify both sick and healthy individuals. For all-feature performances of Decision Tree and Random Forest, there is a trade-off between specificity and sensitivity. As sensitivity increases (0.754 for both Decision Tree and Random Forest), specificity tends to decrease (0.729 for Decision Tree and 0.726 for Random Forest). In contrast, both XGBoost and LightGBM show improvements in both sensitivity and specificity, with sensitivity values of 0.758 for both models and specificity values of 0.772 for XGBoost and 0.747 for LightGBM. This indicates that XGBoost and LightGBM not only achieve slightly higher sensitivity than Decision Tree and Random Forest but also maintain or improve specificity, suggesting a more balanced performance in identifying both sick and healthy individuals. Although the sensitivity and specificity

Table 6. All feature performance of tree-based models for CVD risk prediction

All feature performance				
	Decision Tree	Random Forest	XGBoost	LightGBM
Accuracy	0.729	0.726	0.772	0.747
Sensitivity	0.754	0.754	0.758	0.758
Specificity	0.729	0.726	0.772	0.747
Precision	0.010	0.010	0.012	0.011
F1-score	0.019	0.019	0.023	0.021
AUC	0.783	0.781	0.857	0.843

of XGBoost and LightGBM are comparable, XGBoost outperforms LightGBM in other metrics.

The distribution of XGBoost prediction probabilities was analyzed, as the goal of this study is to estimate risk levels. In the insurance industry, most customers are healthy, while a smaller portion comprises individuals with cardiovascular conditions. Consequently, the predicted probability distribution for the healthy group is expected to be right-skewed, as the model should predict low risk levels for them. Conversely, the distribution for customers with cardiovascular diseases is expected to be left-skewed, as the model should identify them as having higher risk levels. Figure 3 illustrates the predicted probabilities from the baseline model. In this figure, the distribution of cardiovascular disease customers is right-skewed, with the model predicting most health risk levels between 0.1 and 0.3. These predictions do not align with the expected probabilities, which should be higher for these customers. In contrast, Fig. 4, which incorporates all features, shows a right-skewed distribution for healthy customers, indicating that most of them have a low risk of cardiovascular disease.

Additionally, Fig. 4 demonstrates that customers with cardiovascular disease exhibit a high-risk probability, reflected in the left-skewed distribution. The behavior of the all-feature model aligns well with the characteristics of insurance data.

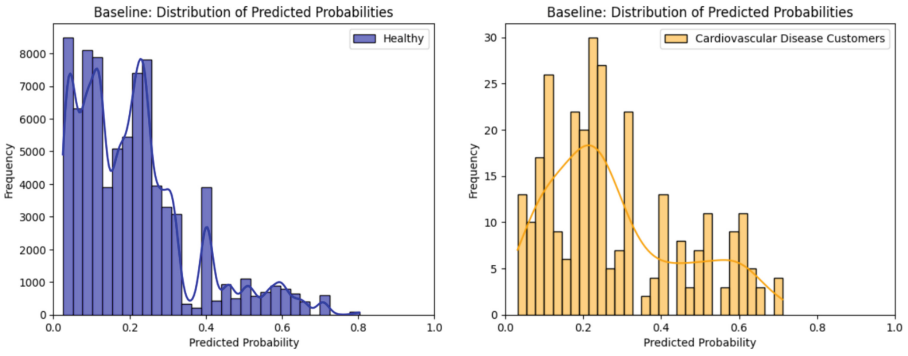

Fig. 3. Baseline model healthy customers and CVD customers, respectively

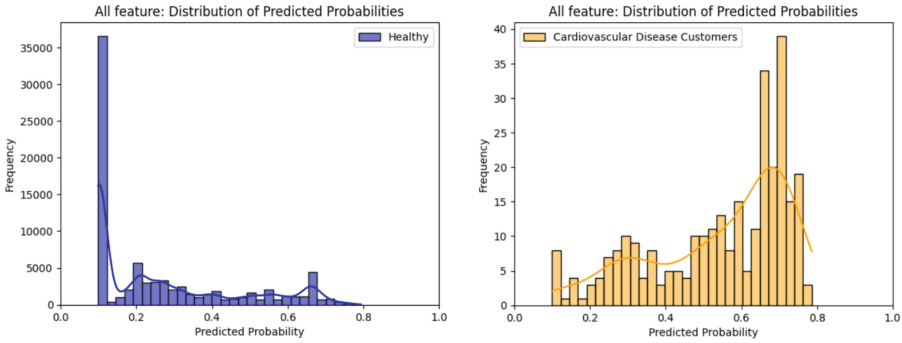

Fig. 4. All feature model healthy customers and CVD customers, respectively

4.3 Key Insights from Feature Importance Based on XGBoost

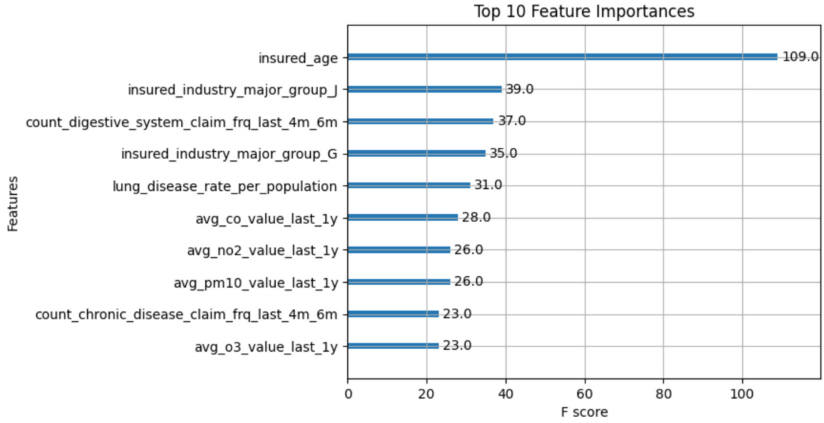

Fig. 5. Top 10 selected features related to cardiovascular disease from XGBoost

After selecting the best model, the XGBoost feature importance tool was employed to analyze the significance of various features. As shown in Fig. 5, the top 10 features included age, industry major groups (Information and Communication; Wholesale and Retail Trade; Repair of Motor Vehicles and Motorcycles), claim history (digestive system and chronic diseases), air quality indicators (Particulate Matter with a diameter of 10 μm, Carbon Monoxide, Nitrogen Dioxide, and Ozone), and lung disease rates per population. These features were summarized into key insights based on the feature importance extracted from the model. These findings are valuable for identifying cardiovascular-related features, especially in machine learning feature engineering.

5 Conclusion

This paper demonstrates the advantage of using claim histories, air quality, and patient statistics in predicting cardiovascular disease risk. The accuracy, precision, F1-score, AUC score, and specificity of each model type show better performance compared to the baseline. Decision Tree and Random Forest exhibit lower sensitivity due to a trade-off with specificity. However, XGBoost and LightGBM, using gradient boosting to combine and refine trees, outperform the baseline in sensitivity and specificity. The probabilities generated by XGBoost provide appropriate values for cardiovascular disease risk level analysis. Future developments could leverage claim history data to analyze customers' past cardiovascular-related illnesses. Improvements could also be achieved by integrating data from the Jolly application, supported by Sunday Technologies Co., Ltd., a claims platform that provides access to policy details and life-supporting features. Key metrics, such as calories burned, steps taken, sleep hours, and heart rate, could be incorporated as features to enhance the model's predictive capabilities in real-world scenarios.

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Comprehensive Framework for Artificial Intelligence and Big Data Integration in Highe Education

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Abstract. Artificial Intelligence and big data technologies are the rising tides of strategic decision-making in higher education. This research introduces a comprehensive framework that leverages advanced AI technologies to provide intelligent, data-driven insights across institutional leadership levels. The framework identifies multiple sophisticated AI technologies such as deep learning, machine learning, natural language processing, and generative AI to analyze complex institutional data emanating from diverse sources. By synthesizing information from academic records, performance metrics, institutional repositories, and external data sources, the system generates actionable insights that support evidence-based decision-making. We proposed specialized prototypes dashboards for three key institutional roles: University President, Provost, and Department Chair, each designed to address unique strategic and operational challenges. The research highlights the transformative potential of AI in providing comprehensive, predictive intelligence that enables proactive institutional management.

Keywords: big data · deep learning · generative AI · higher education · decision system

1 Introduction

The existing higher education environment is shaped by complex challenges on multiple critical dimensions. On one hand, financial pressures have become more pronounced, as institutions strive to cope with restrictions on funding, the influence of globalization, changing market dynamics, and rising operational costs [9]. On the other hand, regulatory bodies are increasing their demands on institutions, for which strict quality assurance processes and the maintenance of accreditations are required [4]. Rapid technological advancement has brought along opportunities and challenges since the new e-learning platforms, digital tools, and information sources require significant institutional change and investment [1,2,10]. Additionally, institutions face other challenges: navigating complex research funding landscapes, fostering collaborations, and respond to the increasing global demand for education.

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In this complex landscape, institutions of higher education generate large amounts of data coming from diverse sources. These include, but are not limited to, administrative systems, governance frameworks, academic operations, research activities and outputs, learning management systems, and institutional support services. This rapid growth of the volume and variety of data brings both opportunities and challenges to institutions of higher education. The search for actionable insights from these data sources is gaining in intensity as a way to address existing challenges and predict future trends. This skill has become indispensable for effective decision-making, strategic planning, and maintaining a competitive advantage in the global education market.

The integration of artificial intelligence with big data analytical platforms offers possible solutions to address these multifaceted issues by taking advantage of the vast amount of data available [10]. This paper explore how such framework can be effectively implemented within higher education institutions to support operational efficiency, improve decision-making processes, and eventually achieve better education outcomes.

The remainder of this paper is organized as follows: The related work section discusses existing research and applications in higher education. Next, the proposed framework is detailed, followed by use cases and a prototype implementation to demonstrate its effectiveness. Lastly, the paper concludes with key findings, contributions, and recommendations for future research.

2 Related Work

Recent years have seen tremendous impact in the areas of Artificial Intelligence and Big Data technologies on analytics and decision-making in higher education. This section presents a comprehensive review of current applications and research initiatives in this domain.

2.1 Performance Prediction Models

Performance prediction research moved from simple statistical frameworks to complex machine learning methodologies. Initial investigations used regression analysis to forecast student outcomes by analyzing past academic records [31]. The most recent work has moved forward in applying deep learning architectures that incorporate a variety of data inputs, such as interactions within learning management systems, attendance trends, and measures of engagement [3,26]. These advanced models have demonstrated increased precision in predicting educational outcomes, thus enabling proactive interventions in the educational process.

Notable applications include neural network models [24] developed for continuous performance assessment and ensemble learning approaches [12] that combine multiple predictors to achieve enhanced accuracy. Researchers have given a strong focus on the interpretability of such models, facing the critical requirement of clear and explained predictions in educational systems [3].

2.2 At-Risk Student Detection Systems

Proactive identification of at-risk students through early intervention has been a major use of educational analytics [15,19]. Recent research focuses on developing real-time monitoring systems that integrate multifactorial risk indicators. These systems monitor trends in student behavior, academic performance, and engagement indicators to identify potential academic difficulties before they manifest in negative outcomes [6,14].

More complex machine learning approaches, particularly those incorporating temporal data analysis, have shown promising results in the timely identification of students in need of extra support [28]. One of the main focus areas has been developing interpretable risk indices and actionable feedback mechanisms.

2.3 Learning Systems and Feedback Analytics

Adaptive learning systems have evolved to provide personalized feedback based on in-depth analysis of learner data [20]. More recent research examines the effectiveness of immediate feedback mechanisms and their impact on student engagement as well as learning outcomes. These systems use natural language processing methods [8,27] to evaluate student responses and provide contextually relevant feedback.

Other research has investigated the role of automated assessment mechanisms in providing immediate formative feedback to learners [7]. More recent developments include intelligent tutoring systems [17,18] that adapt to different learning patterns and provide personalized educational pathways based on continuous performance analysis.

2.4 Course Recommendation Systems

Course recommendation research has gone from simple rule-based approaches to complex collaborative filtering techniques. Furthermore, modern systems are increasingly taking into account diverse factors-academic performance, professional goals, and success trends in peers-to provide course recommendations tailored to individual learners. Recent studies have also focused on the application of deep learning models in summarizing difficult trends in course selection and academic performance [16,25,36].

Innovative methodologies encompass hybrid recommendation systems [11,35] that integrate content-based and collaborative filtering techniques, utilizing both past performance metrics and prevailing academic trends. The objective of these systems is to enhance course selection to improve academic results and ensure alignment with career pathways.

2.5 Student Skill Estimation and Competency Tracking

Skill estimation research has evolved to cater to the assessment of both technical and interpersonal skills. Modern approaches use multimodal data analysis for

tracing skill improvement during various educational activities. Advanced algorithms have been developed to map course outcomes to competencies required by the industry [5,32], thus better curricula in education and labor market needs can be aligned.

Recent research has concentrated on the creation of detailed skill taxonomies and evaluation frameworks that integrate both structured and unstructured learning experiences. These frameworks utilize advanced data mining methodologies [23,33] to discern skill deficiencies and propose specific learning enhancements.

2.6 Planning and Scheduling Optimization

Educational planning and scheduling research has moved from simple timetabling solutions to complex optimization systems. The current approaches use constraint programming and genetic algorithms [13,22] in order to optimize the allocation of resources while considering multiple stakeholder preferences. Such complex constraints handled by these systems include faculty availability, room capacity, and student preferences.

2.7 Emerging Areas and Future Directions

While the previously mentioned applications represent established areas of research, recent developments point to a broader range of potential uses for AI and Big Data in higher education. New applications in diverse areas are being researched.

Research productivity analysis and collaboration recommendation systems have shown promise in improving academic research outputs [21,29]. Institutional effectiveness monitoring through integrated data has shown promise in improving the quality of administrative decision-making [34]. Strategic enrollment management with predictive analytics [30] has been one of the important institutional planning tools.

The integration of multiple data sources and analytical approaches enables more comprehensive understanding of educational processes and outcomes. Such a comprehensive approach toward educational analytics has great potential to expand the application of artificial intelligence and Big Data beyond the traditional learning analytics, and hence, promote evidence-based decision-making in all aspects of higher education management and practice.

3 The Framework

The proposed framework represents a transformative approach in applying Artificial Intelligence and Big Data technologies to the environment of higher education. The framework, through the systematic integration of advanced computational methods, intends to revolutionize institutional decision-making, enhance personalized learning experiences, and improve operational efficiency. Compared

to traditional data management methodologies, this framework uses a holistic, multi-layered structure that focuses on data integrity, accessibility by stakeholders, and ethical guidelines.

The framework's core objective is to create a comprehensive, adaptive ecosystem that transforms raw institutional data into actionable strategic insights. The integrated data sources, bound together with big data storage and advanced AI-driven analytics, are capable of establishing more agile, data-informed educational strategies for the institution.

3.1 Multidimensional Data Sources

The framework is based on a strong and diversified data collection strategy, including internal and external sources.

Internal Sources: The framework systematically integrates a wide range of institutional data, including:

- *Academic Ecosystem Data:* Curriculum repositories, learning management systems (LMS), course development platforms, and academic performance tracking systems.
- *Administrative Infrastructure:* comprises of human resources management, financial systems, institutional governance records, and strategic planning documentation.
- *Student Lifecycle Information:* Robust student profiles, enrolment data, academic performance measures, longitudinal academic trajectories and student engagement measures.
- *Institutional Governance Documentation:* Organizational bylaws, policy frameworks, strategic planning documents, and quality assurance guidelines.
- *Qualitative Performance Metrics:* Institutional quality assurance records, stakeholder feedback mechanisms, compliance certifications, and institutional performance benchmarks
- *Community and Stakeholder Engagement Data:* Alumni network contact, institutional event participation, community outreach initiatives, and collaborative research projects.

External Sources: To provide contextual and comparative insights,the framework is designed to incorporate external data sources:

- *Digital Social Engagement Platforms:* Professional and social networks for integrated stakeholder engagement analysis.
- *Academic Research Repositories:* Indexed research databases, academic publication networks, and scholarly communication platforms
- *Comparative Institutional Intelligence:* National and international educational reports, sector-specific benchmarking data
- *Institutional Performance Indicators:* Global and regional ranking systems, comparative institutional performance metrics
- *Regulatory and Accreditation Frameworks:* National and international accreditation reports, quality assurance standards, and compliance documentation

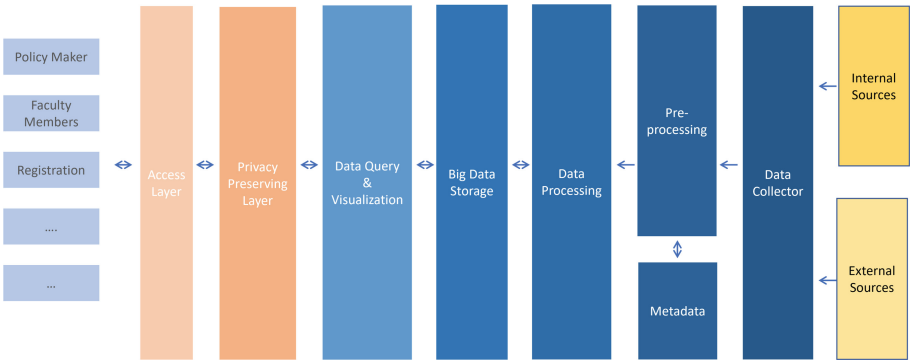

Fig. 1. The proposed framework

3.2 Processing Architecture

The framework employs a multi-layered architecture for effective management and analysis of educational data. This architecture is shown in Fig. 1.

Data Collection Layer: This layer gathers information from multiple internal and external sources. Its main features include:

- **Advanced Data Capture:** Advanced aggregation techniques for holistic data collection.
- **Data Integrity Protocols:** Mechanisms for validation of authenticity and reliability.
- **Dynamic Source Integration:** Adaptive interfaces to accommodate emerging data sources.

Pre-processing and Metadata Management: This layer refines raw data into analysis-ready formats:

- **Advanced Data Cleansing:** Algorithms to handle data quality issues, such as inconsistencies and outliers.
- **Semantic Normalization:** Standardizing data across heterogeneous sources.
- **Metadata Enrichment:** Adding contextual, structural, and provenance information.
- **Data Quality Assurance:** Continuous validation of data quality throughout the pipeline.

Data Processing Layer: The core of the framework uses AI and analytics to transform data:

- **Feature Engineering:** Sophisticated techniques for feature extraction and selection.

- **AI Model Development:** Adaptive algorithms for predictive and prescriptive analytics.
- **Multidimensional Data Transformation:** Operations on varied data types and structures.

Big Data Storage Layer: A scalable, distributed storage system designed for educational data ecosystems:

- **Distributed Storage:** Parallel processing, fault tolerance, load balancing, and horizontal scalability.
- **Hybrid Data Persistence:** The management of structured and unstructured data with unified access mechanisms.
- **Data Redundancy and Recovery:** Disaster recovery and backup protocols.

Data Query and Visualization Layer: This layer provides insights through advanced querying and visualization techniques:

- **Interactive Dashboards:** Role-based, customizable interfaces.
- **Predictive Analytics:** Visual representations of future trends.
- **Dynamic Visualizations:** Tools for real-time data exploration and filtering.

Privacy-Preserving Layer: A comprehensive security framework ensuring data protection:

- **Privacy Mechanisms:** Methods such as differential privacy, homomorphic encryption, and data anonymization.
- **Regulatory Compliance:** Adherence to GDPR, FERPA, and other standards.
- **Ethical AI Governance:** Ensuring fairness, transparency, and mitigation of biases in AI models.

Access Layer: This layer manages user access and authorization securely:

- **Granular Access Control:** Role-based permissions and real-time adjustments.
- **Stakeholder-Specific Interfaces:** Customizable access for institutional leadership, faculty, stuff, researchers, and students.
- **Secure Data Transmission:** End-to-end encryption with cryptographic protection mechanisms.

4 Prototype and Use Cases Example

This section presents a comprehensive prototype and illustrative use cases of the proposed artificial intelligence and big data framework for higher education. We demonstrate a specialized dashboards designed for the institutional leadership, with emphasis on the disruptive potential of advanced AI technologies applied to the field of institutional strategic planning and decision-making.

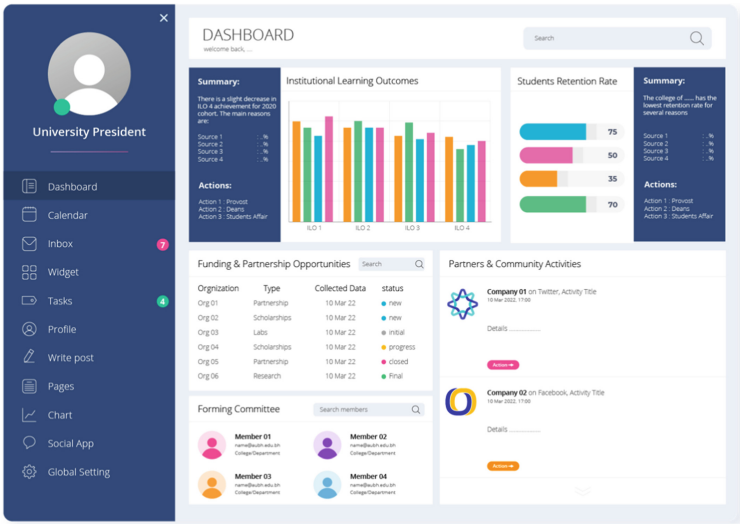

Fig. 2. President's Dashboard.

4.1 President's Dashboard

The President's dashboard represents a sophisticated intelligent system designed to provide holistic, data-driven insights into institutional performance. In higher education, strategic decision-making relies critically on comprehensive, nuanced understanding of institutional dynamics. Figure 2 illustrates an exemplary dashboard with its five critical use cases: achievements in Institutional Learning Outcomes, student retention rates, funding and partnership opportunities, partner and community activities, and intelligent committee formation support.

Analysis of the attainment of Institutional Learning Outcomes (ILOs) serves as a critical strategic tool for improving institutional efficiency and raising educational standards. By fusing data from employer surveys, internship evaluations, alumni feedback, and analytics from social media, this system builds a holistic framework for performance evaluation. Advanced deep learning algorithms expose major performance gaps while explaining complex interrelations within the institutional data. Machine learning models predict future outcomes, and generative AI composes actionable summaries and strategic recommendations, taking unrefined data to meaningful insights for better decision-making and driving continuous improvement.

Analyzing student retention rates is one of the most vital strategic tools in higher education, which has important implications for institutions undergoing financial and academic challenges. Combining data from several sources, including student information systems, academic records, and engagement measures, this system pinpoints patterns and risk factors bearing on student persistence. Advanced deep learning models analyze these variables, and machine learning methods forecast potential retention challenges to enable timely inter-

ventions. Generative AI can also support decision-making, in that it creates detailed reports with specific recommendations and resource allocation strategies to help at-risk students.

The critical challenge of institutional resource development and strategic collaboration is addressed through funding and partnership opportunities analysis. This use case has the potential to change how institutions develop their advancement strategies using AI. By synthesizing public financial data, analyzing news sources, tracking institutional sponsorships, reviewing communication histories, and monitoring potential funding landscapes, the system creates a dynamic intelligence gathering mechanism. Natural language processing technologies analyze complex textual information and bring out subtle insights on possible opportunities. Deep learning algorithms recognize intricate patterns and prospective strategic alignments, while generative AI technologies generate detailed summaries of opportunities and propose engagement strategies.

Community engagement tracking is a strategic tool that shifts institutional interactions from reactive to proactive. In recognition of the increasing importance of community relationships, this system provides real-time analysis using data from news outlets, social media, event participation, and collaborative projects. AI technologies perform multi-dimensional analyses to deliver actionable insights for the strengthening and effective management of community relationships.

Committee formation is a strategic tool aimed at optimizing team composition with respect to effective collaboration. It takes into account the profiles of faculty and staff, past committee performance, histories of collaboration, and professional strengths to suggest optimal configurations of a team. The platform uses deep learning algorithms for analyzing interpersonal dynamics and natural language processing to mine professional insights in order to develop AI-driven recommendations that enhance team effectiveness.

The dashboard's most powerful feature remains its interactive functionality, letting users drill down to more detailed information by college, department, academic program, or any custom grouping. This turns the system from a mere static reporting tool into one that is dynamic and intelligent, offering contextual analysis and actionable insights to really underpin truly data-informed strategic decision-making.

4.2 Provost's Dashboard

The Provost's dashboard is a modern, advanced strategic management tool for the monitoring of academic activities, institutional performance indicators, and strategic development. Using state-of-the-art artificial intelligence technologies, this dashboard transforms complex institutional data into actionable insights to empower evidence-based decision-making across different critical domains in academic leadership. Figure 3 presents an example of a dashboard with five different use cases that focus on the monitoring of academic activities.

College Educational Outcomes (CEOs) is the key insights into the academic performance of institutions, identifying variations across colleges and across an

Fig. 3. Provost's Dashboard.

institution, enable targeted interventions aimed at improving educational quality. This analysis draws from a variety of data sources: student records, course evaluations, learning management systems, graduate employment data, alumni surveys, and accreditation feedback. Deep learning algorithms recognize patterns in performance, while machine learning models forecast outcomes. Generative AI synthesizes summaries and recommends particular action plans for performance improvement.

The Student Attendance Widget provides a solution to the problems of student engagement and proactive intervention by monitoring attendance trends in order to identify potential risks related to academic disengagement and attrition. It combines data from multiple sources, including learning management systems, student information systems, classroom management applications, and campus access control systems, thus providing an all-inclusive view of student engagement. Using artificial intelligence technologies, this data is analyzed in which deep learning algorithms identify patterns of attendance behavior and predict academic risks. Machine learning models make individualized recommendations for intervention strategies; generative AI creates pinpointed suggestions for academic supports, faculty engagement, and retention efforts.

Tracking accreditation progress and status is an important academic governance tool, providing wide monitoring of the processes involved in institutional and programs accreditation. It measures completeness in self-evaluation reports, identifies responsible parties for compliance with accreditation standards, and clarifies document requirements. Analyzing the institutional repositories, document management systems, historical reports, and regulatory frameworks, this system creates a dynamic tool for accreditation. Natural language processing

extracts the key requirements, and identifies gaps; deep learning algorithms recommend documentation strategies. Generative AI provides contextual suggestions to strengthen accreditation applications.

Regulatory Bodies Tracking is critical in ensuring the institutions' compliance and strategic agility. The system automates the tracking of regulatory changes by identifying impacts on policies, academic programs, and operations. This feature gives a proactive approach to potentially impactful regulatory changes through bringing in data derived from government publications, regulatory websites, legal databases, and institutional policies. Natural language processing scans these sources for relevant changes, while machine-learning models assess their institutional impact. Later, generative AI makes tentative policy adjustments and pinpoints the affected areas.

Hiring Forecast and Search Committee Formation is a state-of-the-art tool in the domain of strategic human resource management. It supports workforce planning and recruitment strategies through the analysis of hiring needs, faculty performance, departmental growth, and candidate profiles. The system aggregates data from HR systems, performance evaluations, departmental plans, and external networks to produce hiring insights. The use of machine learning will forecast staffing needs, deep learning will evaluate candidate fit, and generative AI will help in shortlisting candidates and recommending search committee formations.

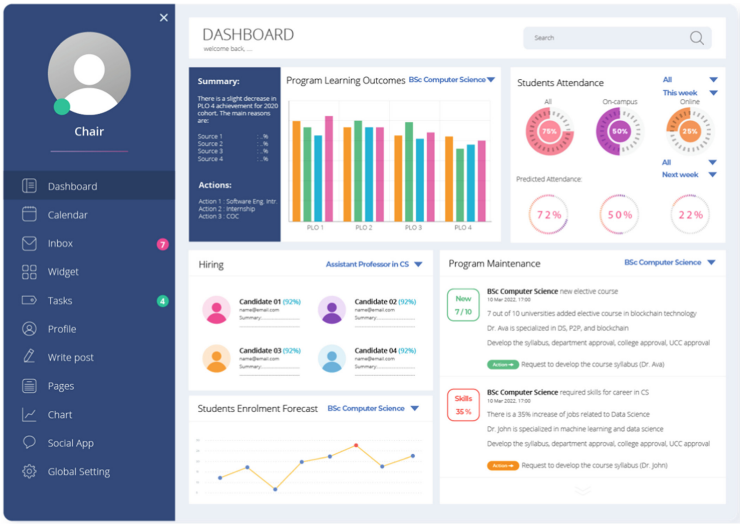

Fig. 4. Department Chair's Dashboard.

4.3 Department Chair's Dashboard

The Department Chair's dashboard emerges as a sophisticated strategic management tool designed to provide granular insights into departmental academic performance, resource allocation, and strategic development. By leveraging advanced artificial intelligence technologies, this dashboard transforms complex departmental data into actionable intelligence, supporting nuanced decision-making across critical academic management domains. Figure 4 illustrates an exemplary dashboard featuring five pivotal use cases addressing departmental academic management.

Program Outcome Assessment is an important tool in enhancing the effectiveness of the academic program. This use case allows detailing the assessment of program-level achievements that can be improved. Determine strengths and areas to work on with precision by leveraging these data sources: Academic records, Course evaluations, LMS Data, Graduate employment statistics, employer feedback, program assessments. Deep learning algorithms analyze these datasets, uncovering performance patterns; machine learning models extrapolate future outcomes. Generative AI collates all this into summaries and provide actionable recommendations for programmatic improvement and strategic growth.

The Hiring Decision Support offers an evidence-based approach to staff planning and recruitment, meaning that any decision on the hiring of a certain academic can be easily aligned with the department's goals and strategic vision. It harmonizes data coming from human resource systems, faculty assessment, strategic plans, outputs of research studies, professional networks, among others, to offer holistic insights on staffing. Machine learning algorithms calculate the need for staffing and deep learning for candidate screening. Generative AI fast-tracks the tasks of searching and recommending prospective candidates or their evaluation by the search committees.

The Program Maintenance Tool is a new application for the continual adaptation and relevance of academic programs. This state-of-the-art system provides automated, data-driven recommendations for program changes, ensuring that educational offerings remain aligned with dynamic market demands, emerging technologies, and industry standards. What makes this tool very important is its ability to bring about agile and responsive curriculum development. It does this by analyzing labor market statistics, researching technological trends, industry-specific publications, and benchmark data from comparator institutions. The system churns out comprehensive recommendations on program changes. Natural language processing technologies constantly scan dozens of data stores, including those for nascent skills demand and trending industry themes. Sophisticated deep learning algorithms score departmental programs against outside standards, while generative artificial intelligence predicts specific curriculum adjustments: course additions, deletions, or modifications. The tool extends its functionality to support implementation, tracking approval processes and suggesting appropriate faculty for new course development based on their expertise, research profiles,

and professional experience. This holistic approach ensures that program updates are not only strategically sound but also practically implementable.

Student Enrollment Forecasting is essential to strategic academic resource planning, which provides programmatic student enrollment projections and identifies the courses and faculty resources needed. Analyzing historical enrollment data, demographic trends, economic indicators, marketing data, and external market research, the system produces accurate forecasts of enrollments. Machine learning algorithms forecast enrollment trends, while deep learning reveals factors that influence program choices. Generative AI enhances reporting and generates recommendations for course offerings and faculty resource planning.

5 Conclusion and Future Work

This study presents an innovative framework that integrates artificial intelligence and big data aimed at revolutionizing strategic decision-making processes within higher education institutions. Through the incorporation of sophisticated AI technologies at various institutional tiers, the suggested framework exemplifies a holistic strategy for utilizing data-informed intelligence in the domains of academic administration and strategic planning.

The research contributes to the multi-level dashboard architecture, offering tailored strategic insights for various institutional roles, and demonstrates the potential of advanced AI technologies in educational management integrating diverse data sources for comprehensive institutional intelligence. It also presents innovative use cases to address some of the most pressing challenges in higher education strategic planning.

Continuing research is necessary to develop more complex machine-learning algorithms, improve natural language processing capabilities, and develop more complex predictive models that can accurately model the complex dynamics present in educational ecosystems. More comprehensive research is also needed to develop realistic implementation strategies that address several challenges, including reliance on high-quality and accessible data, fragmented technological infrastructure, and significant costs and resource requirements. Ethical concerns such as algorithmic bias, compliance with regulatory standards, and ensuring data privacy add to its complexity. Additionally, skill gaps among staff, institutional resistance to change, and the dynamic nature of the higher education landscape further complicate implementation. Addressing these limitations demands robust institutional commitment, ongoing research, and collaborative efforts to refine the framework and ensure its adaptability, fairness, and effectiveness in diverse educational contexts.

The proposed framework is an initial step toward a more complex, data-driven approach to higher education management. As artificial intelligence technologies continue to evolve, the possibilities of truly transformational educational strategies seem increasingly promising.

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Data Envelopment Analysis on Response Surface Method for Efficient Parameterization of Evolutionary Algorithms in Industrial Applications

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Abstract. The selection of parameters is a critical factor in the efficacy of Evolutionary Algorithms (EA) in resolving complex constrained industrial problems, such as multi-pass turning optimization. In order to identify optimal and resilient parameter configurations prior to the application of EA, this paper introduces a novel hybrid approach that integrates CCR Data Envelopment Analysis (DEA) with both Best Practice Frontier (BPF) and Worst Practice Frontier (WPF). The methodology's efficacy is evaluated on the basis of industrial models such as the single-pass, multi-pass turning, and tension/compression spring design. The results emphasize that the compromise desirability approach, which is derived from the geometric mean of response and time efficiency, substantially enhances parameter selection. Parameter D (random seed) is consistently significant, with optimal values of 0.0 in both BPF and WPF scenarios. Parameter C (population size) fluctuates between 15 (BPF) and 25 (WPF) as a result of its influence on time efficiency. The integration of DEA and RSM enhances the quality of EA solutions, with Parameter D consistently achieving optimal performance levels across all models, and Parameter C influencing time efficiency significantly. This resulted in an average production cost reduction of 12% across the tested models. Additionally, a modified Super- Efficiency BCC model was employed to further refine the efficiency evaluation of decision-making units. This hybrid approach is a valuable instrument for addressing complex industrial optimization challenges, as it ensures optimal and robust parameter selection through the combined use of BPF and WPF.

Keywords: Evolutionary Algorithm (EA) · CCR DEA · BCC DEA · Modified Super-Efficiency · ANOVA · Response and Time Efficiency

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1 Introduction

Optimization techniques are crucial in Industry 4.0, reducing operational costs and improving efficiency in manufacturing. Evolutionary Algorithms (EA) are effective for solving complex, multi-objective optimization problems, such as those in turning operations. However, their performance depends on parameter selection, with suboptimal choices leading to poor results, particularly in large-scale applications.

This paper proposes a hybrid framework combining Data Envelopment Analysis (DEA) with EA to optimize parameter selection. DEA, a non-parametric method for evaluating decision-making units, identifies optimal configurations using Best Practice Frontier (BPF) and Worst Practice Frontier (WPF). This ensures robust parameter settings, enhancing EA's effectiveness and aligning with intelligent algorithms for solving complex industrial challenges. The study focuses on multi-pass and single-pass turning operations, assessing models like Hati and Rao, Ermer and Kromodihardjo, Petropoulos, and Iwata, Oba, and Murotsu. These models, along with tension/compression spring design, provide relevant contexts for applying the DEA-RSM methodology to address constraints and minimize production costs.

A two-level factorial design refines DEA evaluations by systematically analyzing parameter impacts, enabling precise configurations. Results show substantial cost reductions, improved efficiency, and solution accuracy compared to traditional methods. Simulations with real-world data validate the framework's scalability and robustness in diverse applications. The paper reviews industrial models, details the DEA-RSM hybrid method, presents numerical results highlighting cost savings and performance gains, and concludes with future research directions to broaden its application to other industrial challenges.

2 Industrial Optimization Models

In order to optimize production processes in turning operations, a variety of models have been developed (Table 1). The Hati and Rao model (M1) is dedicated to the reduction of production costs in the multi-pass turning of mild steel workpieces with a carbide tool. It takes into account limitations such as the allowable depth of cut, input rates, and cutting velocities. In order to guarantee optimal performance under controlled conditions, additional constraints include restricting cutting force, cutting power, tool life, and temperature. Similarly, the Ermer and Kromodihardjo (M2) and Petropoulos (M3) models are dedicated to reducing the production cost of single-pass turning operations. It is intended for operations that necessitate more stringent control over the feed rate and horsepower, while still adhering to the constraints of surface quality, horsepower, and feed rate.

At the same time, the Iwata, Oba, and Murotsu (M4) model is designed to optimize the multi-pass turning of medium carbon steel workpieces using a carbide tool, with restrictions on cutting force, cutting velocities, and input rates. This model also guarantees the stability of the cutting region and restricts the amount of power consumed and surface irregularity The final problem is dedicated to the reduction of the weight of a tension/compression spring design (TCSD or M5) while taking into account constraints

such as shear stress, surge frequency, and minimal deflection. The mean coil diameter (x_2), wire diameter (x_1), and number of active coils (x_3) comprise the design variables [1].

Table 1. Mathematical models

Model	Mathematical Function	Objective or Constraints
M1	Min Cost = $n\left(3141.59V^{-1}f^{-1}d^{-4}\right) + 2.879 \times 10^{-8}V^4f^{0.75}d^{-0.025} + 10$	Objective
	$50 \leq V \leq 400$ m/min	Min and max cutting speeds (V)
	$0.30 \leq f \leq 0.75$ mm/rev	Maximal and minimal feed rates (f)
	$1.20 \leq d \leq 2.75$ mm	Range of allowable depths of cut (d)
	$F_c \leq 85$ kg; where $F_c = \left(28.10V^{0.07} - 0.525V^{0.5}\right)d \times f\left(1.59 + 0.946\frac{(1+x)}{\sqrt{(1-x)^2}+x}\right)$ and $x = \left(\frac{V}{142}\exp(2.21f)\right)^2$	Cutting force (F_c)
	$P_c \leq 2.25$ kW; where $P_c = \frac{0.746F_cV}{4500}$	Cutting power (P_c)
	$25 \leq TL \leq 45$ min; where $TL = 60\left(\frac{10^{10}}{V^5f^{1.75}d^{0.75}}\right)$	Tool life (TL)
	$T \leq 1000$; where $T = 132V^{0.4}f^{0.2}Vd^{0.105}$	Temperature (T)
	$\frac{A}{d} = n$	Limitations on the value of the depth of cut in removing 'A' in 'n' passes
M2	Min Cost $= (1.2566V^{-1}f^{-1}) + 1.77 \times 10^{-8}V^3f^{0.16} + 0.2$	Objective
	$f \leq 0.1$ in/rev	Feed rate (f)
	HP ≤ 4 hp; where $HP = 2.39V^{0.91}f^{0.78}d^{0.75}$	Horsepower (HP)
	SF ≤ 50 μ in; where $SF = 204.62 \times 10^6V^{-1.52}f^{1.004}d^{0.25}$	Surface finish (SF) and $d = 0.2$
	$0 \leq V \leq 500$	Min and max cutting speeds (V)

(*continued*)

Table 1. (*continued*)

Model	Mathematical Function	Objective or Constraints
	$0.\ 0 \leq f \leq 0.1$	Min and max feed rates (f)
M3	Min Cost $= \left(452V^{-1}f^{-1}\right) + 10^{-5}V^{2.33}f^{0.4}$	Objective
	$P_c \leq 5.5$; where $P_c = 10.6 \times V^{-1.52}f$	Horsepower (HP)
	$0.0 \leq f \leq 0.5$	Min and max feed rates (f)
	$0 \leq V \leq 500$	Min and max cutting speeds (V)
M4	Min Cost $= \sum_{i=1}^{n}\left(3927V_i^{-1}f_i^{-1}\right) + (1.95 \times 10^{-8}V_i^{2.88}f_i^{-1}\exp(5.88f_i)d_i^{-1.117} + 60$	Objective
	$0.001 \leq f \leq 5.6$ mm/rev	Min and max feed rates (f)
	$14.13 \leq V \leq 1005.3$ m/min	Min and max cutting speeds (V)
	$0 \leq d \leq A$ mm, where 'A' is the depth of material to be cut	Min and max depth of cut (d)
	$F_c \leq 170$ kg; where $F_c = 290.73V^{-0.1013}f^{0.725}d$	Max cutting force (F_c)
	$fV^2 \geq 2230.5$	Stable cutting region related to the cutting surface
	$0.356\ f^2 \leq$ Hmax; where Hmax ranges from 0.01 to 0.06 mm	Max allowed surface roughness (Hmax)
	$P_c = 7.5$ kW; where $P_c = \frac{F_c V}{4896}$	Max power consumption (P_c)
	$\sum_{i=1}^{n} d_i = A$	The sum of depths of cut of the 'n' passes used to remove the total depth 'A' of the material

(*continued*)

Table 1. (*continued*)

Model	Mathematical Function	Objective or Constraints
M5	Min Weight $= (x_3 + 2)x_2x_1^2$	Objective
	$1 - \frac{x_2^3 x_3}{71785x_1^4} \leq 0$	Constraint
	$1 - \frac{x_2^3 x_3}{71785x_1^4} \leq 0$	Constraint
	$\frac{4x_2^3 - x_1x_2}{12566(x_2x_1^3 - x_1^4)} + \frac{1}{5108x_1^2} - 1 \leq 0$	Constraint
	$\frac{x_2 + x_1}{1.5} - 1 \leq 0$	Constraint
	$0.05 \leq x_1 \leq 2.0$	Min and max level of x_1
	$0.25 \leq x_2 \leq 1.3$	Min and max level of x_2
	$2 \leq x_3 \leq 15$	Min and max level of x_3

3 Hybridization of Data Envelopment Analysis, Response Surface and Evolutionary Algorithms

A hybrid approach integrating Response Surface Methodology (RSM) with Data Envelopment Analysis (DEA) effectively optimizes meta-heuristic algorithm parameters. DEA evaluates the efficiency of parameter configurations as decision-making units (DMUs), while RSM models input-output relationships and refines configurations. This combination enhances algorithm performance, solving complex problems with improved solution quality and reduced computation time.

Evolutionary Algorithms (EA), which excel by mimicking natural selection, rely on parameters like convergence, mutation rate, population size, and random seed for optimal performance. The hybrid method complements EA by systematically refining these parameters, making both approaches invaluable for tackling challenges like waste management and machining operations.

3.1 Data Envelopment Analysis

Data Envelopment Analysis (DEA) evaluates the efficiency of decision-making units (DMUs) in converting multiple inputs into outputs without assuming a specific functional relationship [2–4]. It identifies efficient DMUs on the efficient frontier, with efficiency scores ranging from 0 to 1 based on proximity to optimal performance [5–8].

DEA includes models like CCR, which assumes constant returns to scale, and BCC, which accounts for variable returns to scale [9, 10]. Models can be input-oriented, minimizing inputs for fixed outputs, or output-oriented, maximizing outputs for fixed inputs.

For example, in healthcare, DEA assesses resource use—such as doctors and budgets—against outcomes like patient satisfaction and treatment numbers [11]. Widely used for benchmarking and strategy development, DEA highlights resource inefficiencies and guides improvements, making it a valuable tool for operational research, policy development, and performance optimization across sectors [12].

3.2 Evolutionary Algorithm (EA)

Evolutionary Algorithms (EAs) use natural selection processes like crossover, mutation, and selection to evolve a population of candidate solutions toward an optimal result. Each candidate represents a viable solution, and evolutionary processes refine them to improve outcomes, making EAs effective for complex optimization problems. Implementation involves initializing a population, defining a fitness function to evaluate solutions, selecting top performers, and applying genetic operations. Key parameters influence EA performance: Population Size balances computing load and solution diversity, Mutation Rate prevents premature convergence, Convergence Criteria dictate stopping conditions, Random Seed ensures reproducibility, and Maximum Time without improvement limits computation. Proper parameter tuning enhances performance, enabling faster and more accurate optimization solutions.

3.3 Adaptive Parameter Setting Method (APSM) Using RSM

A structured approach is implemented to evaluate the performance of various configurations in order to generate both BPF and the WPF using the APSM. This approach distinguishes between two extremes: the BPF, which emphasizes optimal performance, and the WPF, which symbolizes inefficiency. APSM offers valuable insights into the influence of various input parameters on overall efficiency by comparing these two frontiers, thereby enabling the identification of the factors that contribute to high and low performance.

The process commences with the design of experiments, which are typically conducted using a two-level factorial design [13]. This approach generates a diverse array of configurations, including both optimal and suboptimal parameter settings. The decision-making units (DMUs) are derived from these configurations and are employed for additional analysis to conduct a thorough assessment of both high-efficiency and low-efficiency scenarios. Following this, the data is normalized to account for scale differences, thereby guaranteeing that the performance measures (r_j)—response (R) and running time (T)—are comparable across various parameter settings.

The subsequent phase entails the determination of performance metrics, including execution time and response time, for numerous replicates after the DMUs have been established. The outputs for DEA are determined by calculating the mean and standard deviation of these measures in forms of the compromise desirability function (CD) of the smaller the better. EA calculates efficiency scores for each configuration, identifying the Best Practice Frontier (BPF) with the most efficient DMUs and the Worst Practice Frontier (WPF) with the least efficient [14, 15].

The Super Efficiency CCR model ranks efficient DMUs with scores above 1, providing greater differentiation by excluding the evaluated DMU from the reference set.

ANOVA is used to identify statistically significant parameters and their optimal values [16, 17]. In the final APSM stage, the desirability function, combining mean and standard deviation, determines optimal parameter settings through compromise desirability (CD). Results are compared with ANOVA and Super Efficiency DEA rankings to ensure precision. APSM manages BPF and WPF scenarios separately, offering reliable recommendations for enhancing algorithm performance in manufacturing, waste management, and other industrial applications (Fig. 1).

```
APSM_PseudoCode:
    Generate DMUs (Decision-Making Units) via Design of Experiment
    Measure Performance for Each DMU from the EA
    Normalize the Performance Data to Ensure Comparability across Configurations
    Calculate Efficiency Scores using CCR and BCC DEA for BPF and WPF
    Fit a Response Surface Model (RSM) to the Performance Metrics
    Perform Analysis of Variance ANOVA for Iterative Refinement
    Calculate Compromise Desirability (CD) via Mean and Standard Deviation
    Select the Optimal Parameter Configuration after Validation
End APSM_PseudoCode
```

Fig. 1. APSM pseudocode for parameter fine-tuning

4 Numerical Results

The numerical results section evaluates the performance of models (M1 to M5) under Best Practice Frontier (BPF) and Worst Practice Frontier (WPF) scenarios, focusing on key operational parameters identified through ANOVA. It highlights how parameters like mutation rate, population size, and convergence criteria influence efficiency and performance, with fine-tuning yielding superior outcomes across scenarios. Key parameters include Parameter A (convergence, tested at 0.0015 and 0.002), Parameter B (mutation rate, tested at 0.00007 and 0.00008), Parameter C (population size, set at 15 and 25), Parameter D (random seed, fixed at 0.0), and Parameter E (maximum time without improvement, set at 25 and 35).

These two-level evaluations provide insights into parameter impacts on model efficacy under BPF and WPF conditions. The analysis emphasizes the critical role of parameter optimization in improving algorithm performance. Subsequent subsections delve deeper into statistical insights and recommendations for further performance enhancement. Parameter D frequently exhibits statistical significance in Model M1 analysis, especially under Best Practice Frontier (BPF) and Worst Practice Frontier (WPF) scenarios for response and time efficiency. Parameter D's effect on system performance may indicate a major operational or design variable that directly affects efficiency. Parameter D's significance in all instances implies that it is crucial to performance optimization regardless of system circumstances. The moderate significance of Parameter C under

Table 2. Significant parameters for BPF and WPF-CCR DEA based on factorial analyses

Model	Efficiency	Scenario	Significant Parameters (p-value < 0.05)	Comments
M1	R	BPF	D (p-value $= 0.010$)	D is significant under BPF for M1-R.
M1	R	WPF	D (p-value $= 0.001$)	D is highly significant under WPF for M1-R.
M1	T	BPF	C (p-value $= 0.036$)	C is significant under BPF for M1-T.
M1	T	WPF	C (p-value $= 0.027$), D ($p = 0.013$)	C and D are significant under WPF for M1-T (Fig. 2).
M2	R	BPF	None (p-value > 0.05)	No significant parameters for M2-R under BPF.
M2	R	WPF	D (p-value $= 0.000$)	Parameter D is highly significant under WPF for M2-R.
M2	T	BPF	C (p-value $= 0.000$), D (p-value $= 0.000$)	C and D are highly significant under BPF for M2-T (Fig. 2).
M2	T	WPF	D (p-value $= 0.002$)	C and D are highly significant under BPF for M2-T.
M3	R	BPF	D (p-value $= 0.020$)	D is significant under BPF for M3-R.
M3	R	WPF	D (p-value $= 0.003$)	D is highly significant under WPF for M3-R.
M3	T	BPF	D (p-value $= 0.003$)	D is highly significant under BPF for M3-T.
M3	T	WPF	D (p -value $= 0.000$)	D is highly significant under WPF for M3-T.
M4	R	BPF	A (p-value $= 0.022$)	Parameter A is significant under BPF for M4-R.
M4	R	WPF	D (p-value $= 0.003$)	D is highly significant under WPF for M4-R.
M4	T	BPF	C (p-value $= 0.019$), D (p-value $= 0.045$)	C and D are significant under BPF for M4-T.
M4	T	WPF	C (p-value $= 0.000$), D (p-value $= 0.000$), E (p-value $= 0.001$)	C, D, and E are highly significant under WPF for M4-T.,
M5	R, T	BPF,WPF	None (p-value > 0.05)	No significant parameters for M5-R.

time efficiency emphasizes its relevance in boosting operational speed, indicating that tweaks to this parameter might improve Model M1's time consumption (Table 2).

Parameter D plays a crucial role in Model M2's BPF and WPF scenarios, particularly for time efficiency, as it influences machine settings or operating timings that affect productivity and cost-effectiveness. Its consistent importance highlights its role in optimizing performance, with neglect potentially leading to suboptimal results. In the BPF scenario, Parameter C impacts time efficiency, indicating its value in ideal operational conditions. Together, these findings emphasize Parameter D's resilience across scenarios and Parameter C's role in enhancing performance under specific conditions.

In Model M4, Parameter A is significant for response efficiency in BPF, while Parameters C, D, and E are crucial for time efficiency under WPF. Parameter A likely influences system inputs or environmental factors to improve reaction times in ideal settings, whereas Parameters C, D, and E ensure efficiency in stressed or suboptimal conditions. This distinction underscores the need for context-specific optimization (Table 3).

Table 3. Significant parameters and optimal levels for BPF and WPF on ANOVA results

Model	Performance Measure	Significant Parameters	BPF Proper Levels	WPF Proper Levels
M1	R	D	D = 0.0	D = 0.0
M1	T	C, D	C = 15, D = 0.0	C = 25, D = 0.0
M1	CD	None		
M2	R	None		
M2	T	C, D	C = 15, D = 0.0	D = 0.0
M2	CD	C	C = 15	D = 0.0
M3	R	D	D = 0.0	D = 0.0
M3	T	D	D = 0.0	D = 0.0
M3	CD	C	C = 25	
M4	R	A, D	A = 0.0015	D = 0.0
M4	T	C, D	C = 15, D = 0.0	C = 25, D = 0.0
M4	CD	D	D = 0.0	C = 25, D = 0.0
M5	R	None		
M5	T	None		
M5	CD	D	D = 1.0	C = 25

Significant parameters like D and C frequently emerge as critical across models, directly influencing cost-effectiveness and efficiency. Parameter D consistently impacts time efficiency, resource allocation, and process control, driving performance improvements under both optimal and suboptimal conditions. Similarly, Parameter C enhances throughput and reduces production time. These results highlight the importance of carefully tuning these parameters to maximize system performance.

ANOVA identified key parameters and their optimal values for models M1 to M5 under BPF and WPF scenarios. Parameter D is critical for M1's responses and time efficiency, with an ideal value of 0.0, while Parameter C affects time efficiency, set at 15 under BPF and 25 under WPF. M2 follows similar patterns, emphasizing Parameters C and D for time efficiency. For M3, Parameter D remains essential for both response and time efficiency at 0.0. In M4, Parameter A (0.0015) influences response efficiency under BPF, while Parameters C and D are crucial for time efficiency. No significant parameters were found for M5. Fine-tuning these parameters enhances model performance across scenarios.

Additionally, the composite desirability (CD) ANOVA analysis across models M1 to M5 showed varying degrees of significance for different parameters, with performance optimization based on the super efficiency CCR method, which extends the traditional CCR model by ranking efficient units beyond 1 to identify the Best Practice Frontier. No parameters were statistically significant in Model M1, suggesting no major factor effect on performance. In Model M2, Factor C (population size) had a p-value of 0.172, but it was not statistically significant, thus C = 15 was selected. Factor C had a p-value of 0.192 in Model M3, with C = 25 performing best. Model M4 indicated significant impacts for Factor C (p = 0.040) and Factor D (p = 0.060), with ideal values of 15 and 0. Factor D was statistically significant with a p-value of 0.023 for Model M5, with the maximum CD at D = 1 (Fig. 2). Super efficiency CCR improved ranking and selection of efficient parameter settings across different models, especially in the BPF scenario.

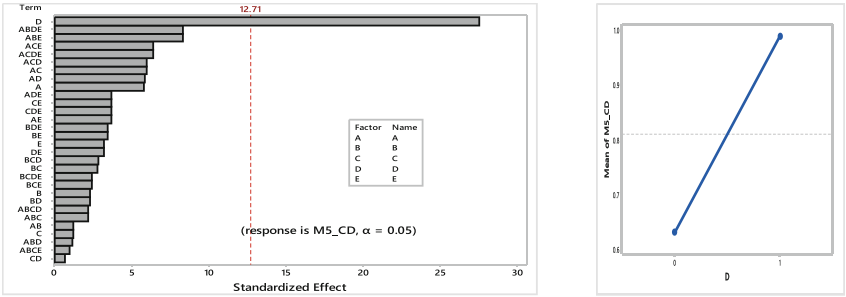

Fig. 2. The normal plot of effects for M5 with CD efficiency and its main effect

This study introduces an additional experiment that employs the BCC DEA model of the BPF to assess parameter efficacy; however, models M2 through M4 did not produce significant results. Nevertheless, the modified super-efficiency BCC model analysis for M1 identified significant parameters that influenced efficiency, including A (0.002), B (0.00007), and E (25), which consistently generated the highest efficiency score of 1.4000. The performance was further improved by the interactions between these parameters, which suggests that the optimization of algorithm efficiency is contingent upon the fine-tuning of these critical parameters. The BCC DEA model's efficiency scores were mostly 1, suggesting limited performance variance among configurations, resulting in non-significant WPF findings.

5 Conclusion and Discussion

The proposed Adaptive Parameter Setting Method (APSM) offers a reliable solution for optimizing production processes, particularly in turning operations, by generating both Best Practice Frontiers (BPF) and Worst Practice Frontiers (WPF). Numerical research highlighted Parameter D (random seed) as essential for reproducibility and stability in Evolutionary Algorithms, maintaining an ideal value of 0.0 across scenarios. Its regulation minimizes unpredictability, ensuring consistent outcomes critical for applications like multi-pass turning or spring design, reducing costs while maintaining reliability.

Parameter C (population size) enhanced time efficiency, with optimal values between 15 and 25. These findings demonstrate the APSM technique's effectiveness in addressing industrial optimization challenges. APSM effectively addresses operational constraints while maximizing efficiency when applied to industrial models such as Hati and Rao, Ermer and Kromodihardjo, Iwata, Oba, and Murotsu, as well as the tension/compression spring design. The method provides a comprehensive assessment of parameter configurations by utilizing both DEA and a two-level factorial design, enabling manufacturers to identify optimal settings for increased efficiency and cost reduction.

In the Hati and Rao model for multi-pass turning of mild steel, APSM minimizes production costs while managing constraints like cutting force, tool life, and power consumption. For single-pass turning in the Ermer and Kromodihardjo model, APSM optimizes horsepower, feed rates, and surface quality, ensuring cost-effectiveness. In the Iwata, Oba, and Murotsu model, APSM stabilizes cutting forces and velocities for medium carbon steel. For tension/compression spring design, APSM optimizes parameters within mechanical constraints, avoiding inefficiencies by analyzing best and worst practice frontiers.

Future research could enhance APSM by integrating machine learning techniques like reinforcement learning or neural networks to adjust parameters in real-time and predict optimal settings using historical data. Fuzzy logic could address uncertainties in chaotic production environments, while advanced multi-objective optimization methods, such as Pareto optimization, could help balance conflicting goals like cost and quality, providing robust solutions for complex industrial scenarios.

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Feature Niching Based Differential Evolution for Feature Selection on High-Dimensional Data

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Abstract. Feature selection (FS) is regarded as a combinatorial optimization problem that aims to search the optimal feature subset from large number of features. Evolutionary algorithm (EA) is an effective approach to solve FS problem. However, most EA-based FS methods tend to face an overwhelming number of feature combinations, which enlarges the solution space and makes it difficult to find the optimal feature subsets. As a result, a feature niching based differential evolution (FNDE) is proposed for FS on high-dimensional data. Specifically, the population is divided into several parts from the perspective of dimension using feature niching technique. In addition, the ranking-based selection strategy is used to choose the elite individuals based on their fitness rankings, so that high-quality solutions are preserved and poor ones eliminated. The experimental results show that the proposed FNDE achieves higher classification accuracy while selecting fewer number of features compared with other state-of-the-art methods on 8 benchmark high-dimensional datasets.

Keywords: Feature niching · Differential evolution · Rank-based selection · Feature selection · High-dimensional data

1 Introduction

As data collection methods improve and sources become more diverse, we now have access to more information, which has led to the rise of high-dimensional data [8]. A large number of features contain a lot of redundant information, which negatively affects classification accuracy and increases the training time

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of models. Therefore, the effective reduction of data dimensionality is necessary to address the challenges caused by high-dimensional data [27].

Typically, feature selection (FS) is a crucial step in data preprocessing, aimed at selecting the most relevant features that significantly impact model performance while removing redundant, irrelevant, or noisy ones. By reducing data dimensionality, FS enhances model training efficiency, decreases computational costs, mitigates overfitting risks, and improves model interpretability. As a vital technique in data mining, it helps optimize datasets by retaining only the most informative features [13]. Generally, FS is regarded as an NP-hard problem. For a dataset with n features, the number of possible feature subsets is 2^n [25]. As the number of features increases, the quantity of feature subsets is dramatically grows, which usually increases exponentially, especially in high-dimensional data, where it becomes particularly significant.

Meta-heuristic algorithms, which do not require prior knowledge or assumptions about the solution space, are widely employed in FS [15]. Among them, evolutionary algorithms (EAs) are particularly prominent due to their adaptability and effectiveness in solving complex optimization problems. EAs are population-based optimization methods inspired by the process of natural evolution, where candidate solutions are gradually improved through iterative operations such as mutation, crossover, and selection. This allows EAs to explore the solution space efficiently and converge toward optimal or near-optimal solutions [18]. Existing EAs used for FS include genetic algorithms (GAs) [11], particle swarm optimization (PSO) [14], and differential evolution (DE) [12], among others. Notably, DE, which consists of mutation, crossover, and selection [16], stands out in FS because of its simplicity, efficiency, and robust performance. It has received significant attention for its ability to handle multi-objective optimization problems by generating a diverse set of feasible solutions through population-based searches in multiple directions. This makes DE particularly well-suited for FS problems that aim to balance competing objectives, such as maximizing classification accuracy while minimizing the number of selected features.

Niching technique has significant advantages in solving multi-objective optimization problems because of its capability to find multiple feasible solutions. These techniques aim to maintain population diversity by dividing the population into niches, each focusing on a specific region of the solution space. This division allows different subpopulations to explore distinct areas, reducing the risk of premature convergence and ensuring a more thorough exploration of the solution space. Nowadays, niching mainly include crowding [20], fitness sharing [24], restricted tournament selection [4], and speciation [10], and others. Incorporating niching into EAs [7,9] allows multiple feasible solutions to be obtained in a single run by maintaining stability within the population, while the exchange of evolutionary information is limited within the niches. In addition, the population is divided into several subpopulations by niching from the perspective of individual [5]. However, the coding length is equal to the dimensionality of the

original dataset, and the state of a feature is only affected by the coding of the current bit.

In this paper, a feature niching based DE (FNDE) is proposed to solve FS on high-dimensional data and remove redundant features while maximizing the classification accuracy. In particular, the population is divided into several subpopulations using the feature niching technique based on dimensions. The elites are then preserved in an archive, and the optimal feature subsets are obtained by ranking these individuals based on their fitness.

2 The Proposed Method

This section presents the details of the proposed FNDE. Firstly, the process of feature niching technique is introduced to acquire niches. Further, the ranking-based selection strategy is discussed to obtain the optimal feature subsets. Finally, the general framework of FNDE is described.

2.1 Feature Niching Technique

The solution space refers to the set of all possible solutions in a FS problem. It includes all potential feature combinations, regardless of their relevance or redundancy. As mentioned in many existing FS studies [21,26], niching is an effective technique to obtain feature subsets in the extensive solution space, which helps to avoid getting trapped in local optima. In general, the niching technique shows distinct advantages on low-dimensional data due to the variety of ways to partition niches. Thus, a feature niching technique is introduced to partition the solution space and reduce the encoding length of individuals when handling high-dimensional data. The pseudocode of feature niching technique is presented in Algorithm 1. The process begins by randomly generating a population with D dimensions, where each individual represents a solution in the feature space. To evaluate the importance of each feature, mutual information is calculated between each feature and the target label. Mutual information is a measure that quantifies the dependence between two variables, helping to assess how much information a feature provides about the label. After the mutual information is computed for all features, the features are sorted in descending order based on their importance. This sorting step is crucial as it enables the partitioning of features into multiple subpopulations, where each subpopulation contains a mix of highly informative features. By distributing features in this way, the algorithm maintains diversity within each subpopulation, ensuring that different subpopulations focus on various aspects of the solution space. This approach reduces redundancy and preserves the most important features, ensuring their effective utilization in the final selected feature subsets.

An example of the feature niching technique is given in Fig. 1. In this example, the 12 features are ranked in descending order based on their mutual information with the label. The features are then divided into four niches, each containing a mix of highly correlated and weakly correlated features. Features within each

niche are mutually exclusive, meaning no feature is repeated across niches. This helps to maintain diversity within each niche and ensures that different aspects of the feature space are explored.

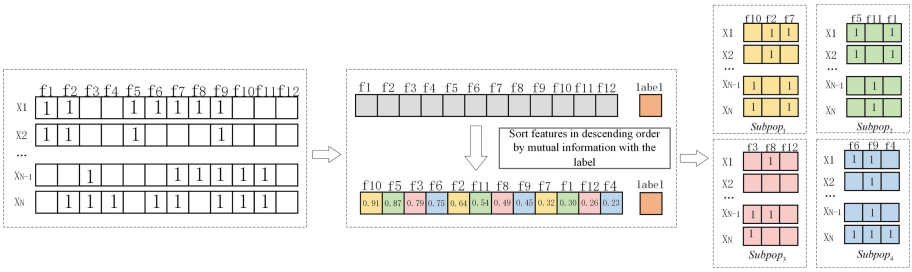

Fig. 1. An example of feature niching technique

Algorithm 1. Feature Niching Technique

Input: *Data*: training set; P: the population; G: the number of niches
Output: Several *subpopulations*: $subpop_1$, $subpop_2$, $subpop_3$ ... $subpop_G$

```
F_score ← Calculate the correlation between each feature and label
SortedFeature ← The sequence of features is generated by sorting the F_score in descending order
for i = 1, 2, ..., G do
    sf_i ← {}
    j = i
    while j < |SortedFeature| do
        sf_i ← SortedIndividual_j
        j ← j + G
    end while
end for
subpopulations ← The individuals in each niche are made up of the corresponding features in the sf
```

2.2 Ranking-Based Selection Strategy

An individual in the subpopulation represents only a part of the complete solution. To form a complete solution, individuals from each subpopulation must be combined with individuals from other subpopulations before calculating their fitness. The new population is generated using two combination methods simultaneously: one method combines individuals within each subpopulation according to the previous sequence, while the other method uses random combinations to ensure diversity in feature sets. This is crucial for effectively exploring the

solution space. Once the new population is generated, the ranking-based selection strategy is applied to select the optimal solutions. The process begins by removing any duplicate individuals from both the archive and the newly generated population to preserve diversity. Next, the remaining individuals are ranked according to their overall performance, considering both feature count and accuracy. Those with higher accuracy and fewer features are ranked higher. Finally, the top N individuals are selected from the remaining individuals. These elites are then preserved in the archive. Algorithm 2 details the main steps of the ranking-based selection strategy.

Algorithm 2. Ranking-based Selection Strategy

Input: A: the archive; O: the new population ; N: the population size
Output: A': the updated archive; *gbest*: the updated global optima in the population

1: $A' \leftarrow A \cup O$ and remove duplicated individuals
2: **for** $i = 1, 2, \ldots, |A'|$ **do**
3: $R_i^1 \leftarrow$ Calculate the ranking of the classification error rate for *i-th* individual
4: $R_i^2 \leftarrow$ Calculate the ranking of the number of selected features for *i-th* individual
5: Calculate the fitness of *i-th* individual by R_i^1 and R_i^2
6: **end for**
7: $A' \leftarrow$ Sort all individuals in A' by the ascending order of fitness
8: **if** $|A'| >$ N **then**
9: $A' \leftarrow$ Preserve the first N individuals in A'
10: **end if**

2.3 General Framework of FNDE

As in most existing FS methods [2,19], the datasets are divided into training and test sets. Then, the optimal feature subsets are obtained based on the training set using the proposed FNDE and are applied to the test set to evaluate their effectiveness. The process of feature niching technique is first implemented to divide the population from the perspective of dimension and allocate them to different niches. The details of ranking-based selection strategy are provided to choose the elites to be placed into the archive in each iteration, which serves as the output when the specified termination criterion is met. The pseudocode of the general framework of FNDE is provided in Algorithm 3. After training, the feature subsets are used to assess the classification accuracy of the proposed FNDE on the test set.

3 Experiment Results and Discussions

In this section, 8 high-dimensional datasets were used to examine the performance of the proposed FNDE for FS. The details of the datasets are shown in

Algorithm 3. Framework of FNDE

Input: *Data*: training set; *Maxgen*: maximum number of generations; G: number of niches; N: the population size
Output: The *archive*

archive ← {}
P ← Initialize a population
archive ← *archive* ∪ P
$subpopulations^0$ ← Feature Niching Technique (*Data*, P, G)
$t \leftarrow 1$
while $t <$ *Maxgen* **do**
  gbest ← Find the global optima in the P $subpop^{t-1}$
  for $subpop_g^{t-1} \in subpopulations^{t-1}$ **do**
    V_g ← Mutation ($subpop_g^{t-1}$, *gbest*)
    C_g ← Crossover ($subpop_g^{t-1}$, V_g)
    $subpop_g^t$ ← Choose the better individuals between C_g and $subpop_g^{t-1}$
  end for
  P' ← Combine individuals from each subpopulation into complete solutions
  archive ← Ranking-based Selection Strategy (*archive*, P', N)
  $t++$
end while
return the optimal feature subsets in the *archive*

Table 1, listed in ascending order of the number of features, with dimensions ranging from 2308 to 12600.

Each dataset was randomly divided into a training set and a test set in a proportion of 70% and 30% [1]. KNN was used as a classifier for all algorithms. The number of nearest neighbors of KNN was set as 5 to balance classification accuracy and search efficiency. To avoid FS bias during the training process, KNN with 5-fold cross-validation was used to calculate the classification error rate, and the classification results of the unseen test set will be reported.

Table 1. Description of Datasets

Dataset	#Features	#Instances	#Classes
SRBCT	2308	83	4
DLBCL	5469	77	2
Brain Tumor1	5920	90	5
CNS	7129	60	2
Adenoma	7457	36	2
Prostate	10509	102	2
Leukemia2	11225	72	3
Lung Cancer	12600	203	5

3.1 Comparison Algorithms

To evaluate the effectiveness of FNDE, 5 algorithms were utilized for comparison in the paper, i.e., SparseEA [19], MF-CSO [6], VGS-MOEA [3], NDEDA [21], and DMBDE [23]. Among them, SparseEA, MF-CSO, and VGS-MOEA are EMO-based FS methods, among which the former three are mainly used to address high-dimensional FS. NDEDA and DMBDE are DE-based FS methods that combine with niching technique.

Table 2. Parameter Settings of Different Algorithms

Algorithms	Parameters
SparseEA [19]	Mutation rate = 1/Dimension, $CR = 1$
MF-CSO [6]	$r_i, r_j, r_k \in [0, 1]$, $g_1 = 0.1$, $g_2 = g_3 = g_4 = 0.45$, $p_{\max} = 0.5$
VGS-MOEA [3]	$p_c = 0.9$, $p_m = 1/D$, $\alpha = 10$, $k = \min\{10 \times \log_2 n, n\}$
NDEDA [22]	Threshold for selecting features $\theta = 0.6$. Scale factor = 0.5, $CR = 0.5$
DMBDE [23]	Threshold for removing features $\delta = 0.1 \times \text{Score}_{\max}$. Scale factor = 0.5, $CR = 0.5$
FNDE	$CR = 0.5$, $w = 0.2$

3.2 Parameter Settings

For the above comparison algorithms, the parameters recommended in the corresponding papers were used for setting, as shown in Table 2. For fairness, each algorithm is independently run 30 times on all datasets. The population size N is set to 300, and the maximum number of generations is fixed as 100. The experiments were carried out on an Intel Core-i5-6500 CPU 3.20 GHz with 8 GB RAM, using MATLAB R2021b or PyCharm 3.3. Among them, SparseEA is carried out using PlatEMO [17], and the other algorithms are implemented according to the code given by the authors.

3.3 Performance Analysis of FS

Tables 3 and 4 show the mean value of 6 algorithms in terms of the classification accuracy and number of selected features. FNDE achieves the highest classification accuracy on 6 out of all datasets, while NDEDA achieves the highest classification accuracy on other datasets, FNDE exhibits no significant difference in classification accuracy compared to NDEDA and FNDE selects fewer features than NDEDA. SparseEA performs in an outstanding ability in terms of reduction of data dimensionality. Nevertheless, these algorithms separate the relationships between features to focus solely on independent features, as disregarding feature combinations and excessive sparsity of selected features may cause no features are selected in an individual. More importantly, FNDE notably surpasses SparseEA in terms of classification accuracy on 8 datasets. It is worth

noting that FNDE outperforms DMBDE on all datasets with fewer selected features and higher classification accuracy. Although MF-CSO and FNDE show similar number in terms of selected features, FNDE exhibits better classification accuracy on all datasets, with significantly superior performance on all datasets.

Table 3. Mean value of the classification accuracy (%) on test sets

Dataset	SparseEA	MF-CSO	VGS-MOEA	NDEDA	DMBDE	FNDE
SRBCT	89.18±3.38 ↓	89.83±1.98 ↓	91.95±3.28 ↓	97.59±3.55 ≈	85.72±6.27 ↓	93.81±2.14
DLBCL	84.16±2.04 ↓	91.59±1.93 ≈	89.89±2.85 ≈	89.24±5.97 ≈	88.20±3.23 ↓	91.64±2.51
BrainTumor1	80.78±4.01 ≈	78.87±4.45 ≈	74.49±3.99 ↓	80.00±4.22 ≈	83.06±4.95 ≈	83.10±4.88
CNS	61.52±6.74 ≈	63.52±7.77 ≈	53.48±4.28 ↓	61.92±8.88 ≈	61.26±5.22 ≈	64.12±5.22
Adenoma	84.13±3.68 ↓	92.30±5.75 ≈	92.35±4.36 ≈	98.18±4.07 ↓	93.36±2.52 ≈	93.96±6.59
Prostate	89.47±2.64 ≈	87.06±3.77 ↓	87.60±2.25 ↓	83.20±6.64 ↓	82.67±5.28 ↓	89.49±4.18
Leukemia2	83.00±3.47 ↓	88.51±2.19 ↓	89.25±1.96 ≈	91.82±2.03 ≈	88.73±1.58 ≈	90.25±4.02
Lung Cancer	86.70±2.90 ↓	90.06±3.04 ≈	87.65±1.79 ↓	94.82±4.09 ≈	90.20±2.51 ≈	92.54±3.14

Table 4. Mean value of the number of selected features on test sets

Dataset	SparseEA	MF-CSO	VGS-MOEA	NDEDA	DMBDE	FNDE
SRBCT	9.3±0.3 ↑	91.7±26.7 ↓	42.0±12.0 ↓	356.9±37.8 ↓	139.8±23.5 ↓	25.7±6.2
DLBCL	9.5±1.8 ↑	30.7±12.0 ≈	37.2±14.5 ≈	946.6±11.9 ↓	353.3±77.3 ↓	45.5±24.8
BrainTumor1	7.8±0.9 ↑	157.6±41.5 ↓	9.9±2.4 ↑	1143.1±84.3 ↓	368.2±78.1 ↓	59.2±23.6
CNS	7.9±1.0 ↑	172.3±119.0 ↓	9.6±4.3 ↑	1678.7±132.1 ↓	503.6±151.2 ↓	66.2±27.4
Adenoma	8.8±0.0 ↑	24.0±8.1 ≈	21.0±2.4 ↑	1219.8±23.9 ↓	482.3±14.7 ↓	27.5±11.5
Prostate	8.7±0.4 ↑	126.1±74.8 ↓	25.1±24.9 ↑	2203.6±164.1 ↓	403.3±125.6 ↓	78.4±48.2
Leukemia2	14.0±8.0 ↑	117.0±50.7 ↓	51.4±15.8 ≈	2171.3±71.1 ↓	566.4±121.0 ↓	72.5±48.4
Lung Cancer	9.5±1.6 ↑	422.74±207.8 ↓	17.5±6.3 ↑	2791.3±194.3 ↓	460.2±66.5 ↓	153.8±65.9

3.4 Efficiency of FNDE

To clarify the efficiency of different algorithms, the average running time on all datasets is presented in Table 5, and the maximum number of evaluations is the same. It indicates that MF-CSO exhibits the lowest running time on all datasets, and the running time of SparseEA and VGS-MOEA are similar, with each under 15 min. However, the unselected features are continuously removed from the dataset in the iteration, the potential combination of different features is ignored with the excessive contraction of solution space, and the essence of FS is against to maintaining the classification accuracy. For the remaining 3 algorithms, as the dimension of datasets increases, the running times also increased to seek for more possible feature subsets. Compared to NDEDA and DMBDE, FNDE requires significantly less time on all datasets, and it is more than 4/5 less than that of NDEDA. In general, while FNDE require more time compared to

some algorithms with incomplete search on the solution space, sacrificing some computation overhead to improve the quality of feasible solutions is deemed acceptable.

Table 5. Running time of different algorithms (Min)

Dataset	SparseEA	MF-CSO	VGS-MOEA	NDEDA	DMBDE	FNDE
SRBCT	8.71	5.82	8.41	98.05	46.04	45.51
DLBCL	10.2	6.14	8.98	226.73	77.32	51.32
BrainTumor1	10.51	6.30	8.88	240.97	86.66	56.21
CNS	10.58	6.29	9.49	283.7	94.7	58.08
Adenoma	10.50	6.31	8.61	300.3	76.2	49.9
Prostate	12.06	6.61	9.46	434.2	115.11	69.86
Leukemia2	12.24	8.50	9.78	472.17	145.11	94.52
Lung Cancer	13.75	8.02	11.13	507.9	154.33	91.47

4 Conclusion

In this paper, a FS method based on feature niching DE, called FNDE, which is proposed to handle high-dimensional data. The proposed FNDE mainly includes feature niching technique and ranking-based selection strategy. Among them, the solution space is divided into several parts by feature niching technique, which makes the search range broader and more comprehensive. The ranking-based selection strategy maintains many superior solutions. FNDE was compared with 5 algorithms on 8 high-dimensional datasets ranging from 2308 to 12600 dimensions. The experimental results show that FNDE outperforms other state-of-the-art algorithms and achieves the satisfactory classification accuracy while selecting a smaller number of features. FNDE can be effectively applied in the early stages of systems dealing with high-dimensional data, such as those in machine learning, image processing, and genomics. In these stages, FNDE improves efficiency and performance by selecting the most relevant features, reducing computational complexity and enhancing model quality.

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Evaluating the Performance of Open-Source LLMs in Local RAG Systems: A Practical Study on Low-Carbon Data Applications

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Abstract. People often underestimate their carbon emissions, but once they become aware of the environmental impact of their actions, it can lead to significant behavioral changes. Numerous studies have highlighted the importance of improving low-carbon literacy, especially through personalized information, which is more effective in driving behavior change than generalized information. Chatbots can play a key role in delivering tailored messages, providing guidance on food choices, building materials, appliance use, transportation options (e.g., car sharing), and other behaviors that influence an individual's carbon footprint. This study aims to develop a simple, scalable, and easily maintainable low-carbon activity database using open-source software, operating locally to reduce reliance on paid APIs. Implemented on the Ollama platform, the system can be deployed on consumer-grade computers with acceptable response times. By utilizing a Retrieval-Augmented Generation (RAG) approach, the system converts data into a vector database and uses a large language model for retrieval. Carbon emission data from four sources were used, and the system's retrieval performance was tested with 100 prompts in English, Traditional Chinese, and Simplified Chinese. The results showed that database size had the largest impact on recall rates, especially when irrelevant or similar data were included. Model selection was also critical, particularly in multilingual contexts where proper language settings played a significant role. The study found that chunk size had a minor influence on results, but the choice of language, while important, was less critical than database size. These findings underscore the need for careful database design and model selection, especially in multilingual RAG systems, and suggest further research to address challenges like scalability and adaptation to diverse data types.

Keywords: Carbon Emission Database · Retrieval-Augmented Generation (RAG) · Multilingual Natural Language Processing · Large Language Models (LLMs)

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1 Introduction

People generally tend to underestimate the carbon emissions they generate [1]. Once they become aware of the environmental impact of their actions, it can significantly influence their attitudes and even their behaviors [2, 3]. Many studies have reported that increasing carbon awareness can enhance low-carbon literacy, which, when empowered, can extend its influence [4]. As individuals' self-efficacy improves, they seek to broaden their impact on society, contributing more substantially to low-carbon efforts [1, 5]. Literature also suggests that providing personalized information is more effective in changing individual behaviors than offering generalized information to the public [6]. Therefore, enabling the general public to easily understand the carbon reduction potential of various activities would greatly support the promotion of low-carbon initiatives. Life Cycle Assessment (LCA), widely used in academia, is a rigorous, complex, and highly demanding process [7]. The ISO 14064 standard for greenhouse gas inventory is primarily aimed at carbon credit calculation, offsetting, or hotspot evaluation [8], which may be overly complex for the general public's needs. Consequently, various carbon footprint calculators have been developed to meet these demands, offering non-expert carbon data evaluations that are convenient for educational and outreach purposes [9]. This study aims to establish a simple, expandable, reliable, and easily maintainable AI-driven low-carbon coefficient database.

The emergence of large language models (LLMs) has opened up new opportunities for technological applications and provided a broad scope for future innovation. These models, particularly in semantic understanding, significantly outperform other knowledge systems. If these powerful language models could be integrated with efficient data retrieval systems and deployed through user-friendly interfaces to provide timely and accurate information, such a system could have a greater impact than traditional advocacy strategies [6, 10, 11]. This study aims to develop a small, efficient database that uses a friendly interface to provide the general public with the low-carbon information they need, helping them understand the specific contributions of various behaviors to carbon reduction. While there is extensive literature on RAG (Retrieval-Augmented Generation) technology [12–14], and some applications in local systems [15, 16], practical cases of its application in low-carbon coefficient retrieval are still relatively rare. We aim to create a database that can respond quickly to user needs, retrieving relevant data and providing the most useful information, allowing users to make informed choices. The main objective of this study is to evaluate the construction process of a low-carbon activity coefficient database designed for the general public and to explore the challenges that may arise during this process and how to overcome them. We hope that this database can be easily installed, deployed, and maintained even in resource-constrained NGOs or academic institutions. In this study, we considered several core questions: First, can the system be entirely based on open-source software without the need for long-term software subscription fees or API usage costs? Second, does the quality of the retrieved results meet our design requirements? Third, in the absence of high-quality databases, can the database be expanded by gradually collecting reliable data sources while maintaining its credibility and data source integrity? Based on these questions, we constructed a small, locally deployed large language model retrieval system using the Ollama platform, which can be easily installed on consumer-grade computers and

provides acceptable response times. The principle of the RAG retrieval system is to convert data or documents into a vector database using an embedding model, then combine this with a large language model to perform retrieval and deliver optimal results [7, 15]. Factors influencing the performance of the RAG retrieval system include the choice of embedding model [17], the setting of chunk size [18], the degree of overlap, and the volume of data. These factors are particularly complex in multilingual environments. Additionally, data quality is a critical factor. Since Traditional Chinese is considered a mid-to-low resource language globally, we also aim to test whether retrieval performance differs when using different languages in a database primarily constructed in English.

2 Research Methodology

2.1 Preparation

To enable local deployment and reduce reliance on paid APIs, this study utilized the Ollama large language model platform. We employed the RAG example programs provided on the official Ollama website (https://github.com/ollama/ollama/tree/main/examples), which integrate the Private GPT module (developed by Iván Martínez) and modified by the Ollama website, Zylon by PrivateGPT, 2023; https://github.com/zylon-ai/private-gpt). These example programs are based on the LangChain framework and use models such as Llama 2 Uncensored, with the default embedding model being all-MiniLM-L6-v2. The programs include an ingest module capable of converting various data formats (e.g., CSV, PDF) into a vector database. We fine-tuned PrivateGPT to process 100 prompts in batches, with these prompts designed based on the relevant documents within the datasets to test the retrieval performance across different vector databases.

This study employed four different sources of carbon emission data (see Table 1). The first source (D1) is a commonly referenced dataset on food carbon emissions, which we believe is most likely to be integrated into the system's database in the future. After simplification, the dataset only includes the food name and the corresponding carbon dioxide emissions (kg/kg). The second dataset (D2) was obtained from an open data website. Although this dataset provides detailed descriptions of food items and exhibits regional cultural characteristics, we anticipate facing several challenges in processing this dataset. We retained the nutritional content section, which includes 503 data entries. The third and fourth datasets (D3 and D4) originate from Japan and Germany, respectively, and document the annual carbon dioxide emission reductions associated with various low-carbon lifestyle choices [19, 20]. Our tests included 100 prompts designed based on the documents within the four datasets (see Table 2), aiming to observe the results of the RAG retrieval system. To test the effectiveness of different languages, we used English as the keyword for standard answers and conducted tests in English, Traditional Chinese, and Simplified Chinese, collecting feedback from the large language model for subsequent analysis.

Given the platform's ease of use and stable performance, we modified and utilized this program as the testing platform. The hardware used was a desktop computer equipped with an RTX 3090 GPU and 32GB of memory.

Table 1. The datasets applied in this study

Datasets	Documents	Description	source
D1	38	Foods and their CO2e Kg/kg	Hannah Ritchie (2020) - "You want to reduce the carbon footprint of your food? Focus on what you eat, not whether your food is local" Published online at OurWorldInData.org. Retrieved from: 'https://ourworldindata.org/food-choice-vs-eating-local' [Online Resource]
D2	503	Foods' CO2e and nutrients' information	The Big Climate Database https://denstoreklimadatabase.dk/en/download
D3	57	Carbon emission reduction for various lifestyle including dietary, mobile, housing and consumption	[20] Koide, R., Kojima, S., Nansai, K., Lettenmeier, M., Asakawa, K., Liu, C., & Murakami, S. (2021). Exploring carbon footprint reduction pathways through urban lifestyle changes: a practical approach applied to Japanese cities. Environmental Research Letters, 16(8), 084001
D4	63	Carbon emission reduction for various lifestyle including dietary, mobile, housing and consumption	[19] Sköld, B., Baltruszewicz, M., Aall, C., Andersson, C., Herrmann, A., Amelung, D., Barbier, C., Nilsson, M., Bruyère, S., & Sauerborn, R. (2018). Household preferences to reduce their greenhouse gas footprint: A comparative study from four European cities. Sustainability, 10(11), 4044

2.2 KPI

In the RAG system, the most commonly used KPI metrics are precision@K and recall@K. Precision@K measures the accuracy of the retrieval results, while recall@K assesses the quantity of relevant results retrieved. Since our database initially contained fewer than ten entries, we adopted two additional metrics for evaluation. The first metric is recall, which indicates whether a relevant document is retrieved among the ten candidates selected by the RAG system. This metric is critical to our database's original design, which aims to provide users with a list of suggested options. The second metric is precision, which measures whether the system can correctly answer the questions by

Table 2. Examples of prompts, answer and searched results setups

data	ID	prompt	answer	searched_result
D1	4	Apple's carbon footprint	Apple	['Apples', 'Apple juice; plastic bottle', 'Apple', 'Apple juice; from concentrate; plastic bottle']
D1	17	What is the carbon footprint of other legumes?	Other Pulses	['Other Pulses']
D2	26	How many kilograms of CO2e does Thousand Island Salad Dressing have?	Salad dressing; thousand island	['Salad dressing; thousand island']
D2	29	What is the carbon footprint of blackcurrants?	Currant; black	['Currant; black']
D2	30	What is the carbon footprint of 1kg of hamburger buns?	Burger bun	['Burger bun']
D2	38	How much carbon emissions does producing mayonnaise create?	Remoulade; average values	['Remoulade; average values']
D3	51	How much carbon emissions can be reduced by using car sharing?	Ridesharing	['Ridesharing']
D3	52	What is the carbon footprint reduction of insulated windows (kg CO2e/year)?	Window Insulation	['Window Insulation']
D3	65	How much carbon emissions can be saved each year by living near your workplace?	Living Close to Workplace	['Living Close to Workplace']
D3	75	What is the reduced carbon footprint of a vegetarian diet (kg CO2e/year)?	Vegetarian Diet	['Vegetarian Diet']
D4	76	How much carbon emissions can be reduced by switching to solar collectors for heating and hot water?	Shift to thermal solar collector for heating and hot water	[' Shift to thermal solar collector for heating and hot water']
D4	77	How much carbon emissions can be reduced by buying at least 30% more locally produced food?	Buy at least 30% more locally produced food	['Buy at least 30% more locally produced food']

(*continued*)

Table 2. (*continued*)

data	ID	prompt	answer	searched_result
D4	94	How many kilograms of CO2 emissions can be saved each year by switching to a smaller, greener car?	Change to a smaller AND more eco-friendly car	[‘Change to a smaller AND more eco-friendly car’]
D4	95	If the indoor temperature increases or decreases by 1 °C, how many kilograms of carbon dioxide equivalent will be reduced each year?	Lower inhouse temperature by 1 °C	[‘Lower inhouse temperature by 1 °C’]

finding the corresponding carbon emission parameters in the documents. In addition to these metrics, we also used MRR and MDCG to evaluate the RAG system’s performance, but we still consider the recall rate to be the most crucial metric since it aligns with the primary goal of our database design. Our questions were divided into four major categories, with 25 questions in each category, and we also calculated the accuracy for each of these categories (Q1-Q4).

2.3 Trial Runs and Data Collection

After completing the initial planning, we began the trial runs. A total of 32 runs were conducted, with each condition collecting 3 to 9 sets of data (as three languages were tested, the number of datasets was a multiple of three). At the beginning of the study, we only vectorized a single copy of each of the four databases and then adjusted the chunk size and different parameters of the embedding models. Each run included tests in English, Traditional Chinese, and Simplified Chinese, with English used as the standard for correct answers. As the trial runs progressed, we simulated potential issues that might arise in the actual operation of the database, such as the gradual increase in the number of documents. We first attempted to expand the D1 database by 20 times, keeping the data items and corresponding values unchanged but rearranging the order randomly. We then increased the size of the D2 database and cross-increased the data in the C and D databases. Finally, we conducted tests where all databases were expanded by 20 and even 60 times. All trial data and KPI calculations were collected using Python programs written with the assistance of Github Copilot (https://github.com/) and Codeium (https://codeium.com/). The researchers subsequently used Python to write programs, Microsoft Excel to organize the data, and SPSS to perform statistical analysis and testing.

3 Results and Discussion

The factors affecting the search efficiency of RAG include parameters such as chunk size, the size of the database (number of documents), and the type of embedding model. For multilingual data, the embedding model’s support for multiple languages is also a

critical factor, as well as the influence of language itself, overlap, retrieval strategies, and other factors. We first broadly examine potential influencing factors. The research aims to achieve a high recall rate (for 100 prompts) and hit rates for each set of questions. The hit rates are categorized into four groups, Q1 to Q4.

Initially, ANOVA was used to analyze the effects of embedding model type, database size, prompt language, and chunk size on the overall recall rate and the sub-categories Q1-Q4. In most cases within our study, the paraphrase-multilingual-MiniLM-L12-v2 model (hereafter referred to as the paraphrase model) outperformed LaBSE. ANOVA results showed that different embedding models had significant effects on Q2 and Q4. Specifically, the paraphrase model exhibited a significantly higher recall rate for Q2 (mean = 0.3944, N = 129) compared to LaBSE Q2 (mean = 0.1281, N = 69), whereas for Q4, LaBSE (mean = 0.5901, N = 69) outperformed the paraphrase model (mean = 0.5901, N = 129). The ANOVA effect sizes (eta-square) for Q2 and Q4 were 0.384 and 0.066, respectively [21]. From Table 2, it is evident that Q2's relevant documents were more detailed, categorizing various foods into 500 types, thus requiring more specific and regionally varied language. Such vocabulary presents a challenge to embedding models. On the other hand, Q4's relevant documents often consisted of longer sentences, introducing more variables during judgment.

The influence of different prompt languages significantly impacted Q1 and Q2, with English outperforming both Traditional and Simplified Chinese, with effect sizes of 0.056 and 0.127, respectively. Regarding chunk size, it significantly impacted the accuracy of Q2 and Q4. A chunk size of 200 resulted in significantly lower accuracy (mean = 0.0800, N = 18) compared to other groups (mean = 0.3016–0.3456). However, for Q4, a chunk size of 200 (mean = 0.8267, N = 18) significantly outperformed a chunk size of 700 (mean = 0.5436, N = 39). Eta-squared and Epsilon-squared indicate that the effect of chunk size was minimal (<0.01) [18, 21].

Considering the impact of database size, increasing the size of D1 significantly reduced the recall rate for Q1-Q4. Detailed data show that increasing D1's size had minimal impact on Q1 but significantly affected Q2 (from 0.37 to 0.27), Q3 (from 0.83 to 0.49), and Q4 (from 0.96 to 0.53), thus reducing the recall rate. In contrast, D2 had a significant effect on all Q1, Q3, and Q4 in the ANOVA analysis. When the number of documents in D2 increased from 503 to 10,653, the accuracy rates for Q1 (from 0.74 to 0.49), Q3 (from 0.71 to 0.22), and Q4 (from 0.78 to 0.27) were significantly diluted. Additionally, increasing D3 by 20 times significantly reduced Q4's recall, and similarly, increasing D4 by 20 times nearly significantly reduced Q3's recall rate ($p = 0.067$). We hypothesize that a larger data volume leads to a reduction in recall numbers; therefore, we tested different conditions with increased data volumes, with results presented in Fig. 1.

Figure 1 shows the significant effect of database document count on the recall rate. In this study, D2 had a relatively large database size with 503 documents, approximately 10 times the amount of other databases. For analysis, we divided the database load into three groups: low, medium, and high. This allows us to evaluate potential future scenarios. A low load indicates that each database only contains one file for embedding (documents no. = 661). The medium load involves increasing the databases D1, D3, and D4 to 21 times their original size (documents no. = 1421–3821). The third is the

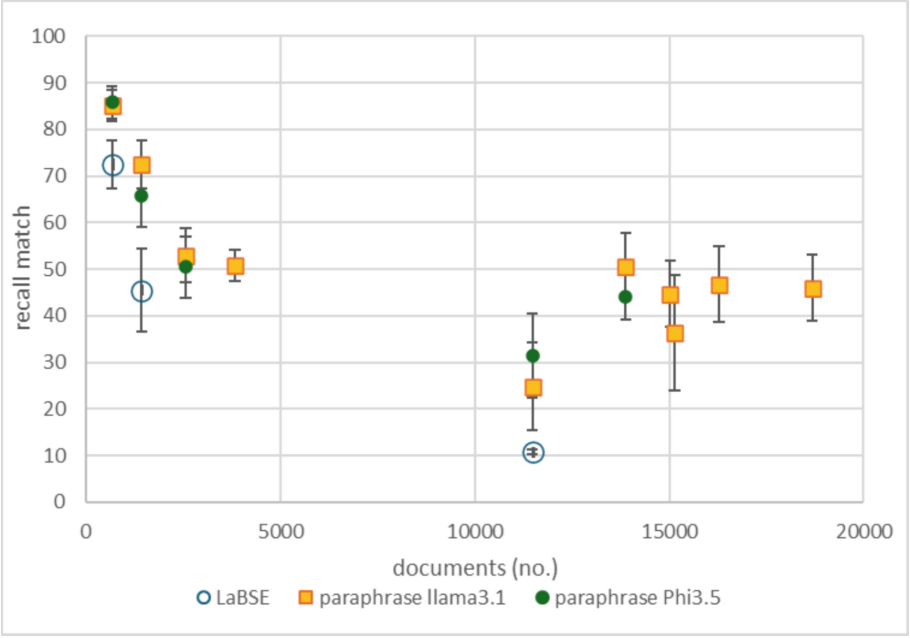

Fig. 1. The variations of recall match numbers from 100 prompts under various amounts of documents; legends mean operated by different embedding models and LLMs.

high-load condition, which includes increasing D2 by 21 times and all other databases by 21 times or more (documents no. $\geq$ 11,481).

Under low-load conditions, the model significantly affects recall and Q1, Q2, Q3 results, with the paraphrase-multilingual-MiniLM-L12-v2 model outperforming LaBSE by a considerable margin. The chunk size of 200 performed significantly worse in many areas, with a significantly lower recall rate than chunk size 700; it also performed worse in Q2 accuracy compared to chunk size 500 and in Q3 accuracy compared to chunk size 700. There was no significant difference in the impact of different languages under these conditions.

Under medium load, the model significantly affects recall and Q1, Q2, Q3 results, with the paraphrase model again outperforming LaBSE. English prompts also performed significantly better than Traditional Chinese (Q1, Q2) and Simplified Chinese (Q1). Under high load, the model significantly affects recall and Q1, Q2, Q3, Q4 results across the tested conditions, with the paraphrase model outperforming LaBSE. English prompts continued to perform better than Traditional Chinese (Q1, Q2) and Simplified Chinese (Q1, Q2). Interestingly, not all high data volume conditions negatively impact recall; the significant influence mainly occurs when an excessive amount of irrelevant data is introduced. In Fig. 1, the lowest recall point is due to the significant impact of the increased non-relevant information (D2) relative to Q3 and Q4. After adding additional D3 and D4 data, the recall rate gradually recovered to a stable level. In practical applications, as databases expand over time, they should trend toward a normal distribution, making extreme conditions rare.

Preliminary testing suggests that the locally deployed open-source solution presented in this study is quite feasible. Although the recall rate may not be outstanding under some conditions, this study is just a starting point, and there is still considerable room for practical improvements, such as using different pre-trained embedding models [22]. Additionally, employing LLM combined with RAG in the entire LCA process could enhance the system [7]. Table 3 lists cases where none of the conditions in this study successfully identified the target, indicating that some conditions failed to find the target due to improper setting of ground-truth responses [12]. However, this does not impact the study's objectives and findings. On the other hand, cultural differences in specific terms might have caused significant retrieval failures in any condition, as seen in Table 3's ID 17, where translation between Chinese and English could lead to discrepancies. Moreover, ID 95 (Table 2) showed that temperature units could be corrupted during file conversion, causing comparison failures. Such terms should be filtered out before entering the system to avoid retrieval issues. Additionally, examining the original files revealed that sometimes RAG retrieves too many types of candidates, leading the generative LLM to fail in identifying the correct answer, often opting to respond with "unknown." There were also several instances where matches failed to strictly recognize differences in case sensitivity, singular/plural forms, and differences between 'and' and '&'. Such problems could be reduced with more advanced LLMs, although local deployment may have caused models to be quantized, reducing context and impairing semantic understanding [23]. Nevertheless, advanced RAG strategies and search methods in the

Table 3. The cases where none of the conditions in this study successfully identified the target

ID	Data	Prompt	Relevant_documents
17	D1	What is the carbon footprint of other legumes?	['Other Pulses']
38	D2	How much carbon emissions does producing mayonnaise create?	['Remoulade; average values']
48	D2	What is the carbon footprint of beef pasta?	['Lasagne; beef; ready meals']
50	D2	How much carbon is produced by producing sweet peas?	['Sugar pea; snow pea; mangetout']
51	D3	How much carbon emissions can be reduced by using car sharing?	['Ridesharing']
69	D3	What is the carbon footprint reduction value of ridesharing?	['Ridesharing']
70	D3	What is the carbon footprint reduction (kg CO2e/year) of following the Food Guide?	['Light Vehicle']
71	D3	What is the carbon footprint reduction value of going vegetarian?	['Food Guide Snacks/Drinks']
94	D4	How many kilograms of CO2 emissions can be saved each year by switching to a smaller, greener car?	[' Change to a smaller AND more eco-friendly car ']

literature could help mitigate these issues [14]. Language barriers may be less significant than initially thought. Therefore, utilizing open-source resources and deploying them locally to establish a low-carbon behavior information database is highly feasible [15].

4 Conclusion and Recommendations

Based on the findings of this study, the following conclusions are drawn:

1. **Feasibility of Using Open-Source Resources and Consumer-Grade Computers to Build a Low-Carbon Behavior Information Database:** This study confirms the feasibility of using open-source software, without relying on paid APIs, and employing only consumer-grade home computers to build a low-carbon behavior information database. Despite the use of uncleaned, unprocessed data, acceptable retrieval results were obtained, even in tests involving over ten thousand entries. This demonstrates the practical viability of the system. Moving forward, further refinement is necessary to enhance system performance, particularly in handling larger, more diverse datasets.
2. **Impact of Database Size on Retrieval Performance:** The study highlights that database size has the most significant effect on the retrieval system's recall rate. Particularly when irrelevant or redundant data are present in large quantities, retrieval efficiency declines. As the database size increases, the recall rate for specific queries decreases, indicating that careful selection of data sources and optimization of database design are essential for improving performance.
3. **Choice of Model and Language:** The paraphrase-multilingual-MiniLM-L12-v2 model deployed in this study performed well, and other models such as Llama 3.1 and Phi 3.5 also delivered satisfactory results. The choice of model and language settings is critical, especially in multilingual environments, where specific adaptations, such as for low-resource languages like Traditional Chinese, are required to ensure system reliability.
4. **Minimal Impact of Chunk Size:** In this study, chunk size had minimal impact on system performance compared to other factors, such as database size and model choice. However, fine-tuning chunk sizes in specific use cases may still improve precision, which could address some limitations related to data chunking in diverse datasets.
5. **Importance of Data Quality:** For terms that are prone to semantic mistranslation or have significant cultural differences, there should be a supervisory mechanism to filter them before they enter the system to avoid retrieval failures.

This study represents a preliminary feasibility test, demonstrating considerable practical potential. Several directions are worth pursuing in future research. First, in practical use, the sources of increased carbon emission factors are likely to be diverse. In the future, it will be natural to add semantically similar data, transforming it from an obstacle to an asset. Consideration should be given to the impact of LLMs' large language model contexts, introducing more effective strategies suitable for the search and integration of small, diverse carbon footprint databases. Future work will involve collecting large-scale databases and evaluating their acceptance of different document formats and their effectiveness.

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Data Selection for Close-Domain Data in Medical Continual Pretraining: A Case Study on Data Selection via Importance Resampling (DSIR)

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Abstract. Continual pretraining for domain-specific tasks faces significant computational challenges, particularly with unlabeled and closely related domain data. Identifying and ranking data that effectively supports the learning of domain-specific language patterns remains a complex task. This study addresses this gap by proposing a case study on Data Selection via Importance Resampling (DSIR), leveraging n-gram distribution for improved performance and time efficiency compared to existing data selection methods. The findings show that DSIR selects clinical data from PubMed, covering close-domain topics such as biomolecular science, medical ethics, and medical procedures. This selection leads to improvements in ICD-10 classification, a complex medical task, with increases in micro F1 and precision at 8 by approximately 5.17% and 3.71%, respectively, over random selection. Additionally, the study highlights that the KL divergence of selected data strongly correlates with end-task performance, potentially serving as an early indicator of data selection quality for continual pretraining.

Keywords: Data Selection · Medical Pre-trained Model · Sampling Important Resampling · KL divergent · ICD-10

1 Introduction

Large-scale datasets, such as The Pile [5] and FLAN [3], are essential resources for training language models due to their wide domain coverage and diversity. However, continually pretraining on the entirety of these datasets often leads to computational inefficiencies and suboptimal performance on specific tasks. In 2020, Suchin et al. [6] demonstrated that domain-adaptive pretraining, a focused second phase of pretraining on a specific domain, can effectively enhance model performance while reducing computational costs. While existing research has

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been devoted to separating distinct domains with clear differences in n-gram distributions and representations from mixed datasets, the challenge of extracting highly specialized data from closely related domains remains largely unexplored. This study addresses this gap through a case study on PubMed Central, a dataset characterized by its close-domain distribution encompassing biomolecular science, medical ethics, institutional medical articles, and clinical data, such as medical procedures, symptoms, and diseases. The inherent similarity in language patterns within these subdomains makes the selection of clinical data challenging. By focusing on this specific problem, this paper aims to advance understanding and methodology in close-domain data selection.

This study examines the strengths and limitations of existing methods for selecting data from close-domain distributions, as discussed in the literature. The study adopts Data Selection via Importance Resampling (DSIR) [16], a method known for its ability to balance computational efficiency and model performance. To address the challenge of isolating clinical data from PubMed Central, which includes various closely related subdomains, DSIR is employed and compared with two baseline methods: random selection and BM25 [13], a keyword-based approach widely used in information retrieval. The experiments use WangchanBART [12], a bilingual Thai-English BART model, for pretraining with the selected data, which is then fine-tuned using synthetic ICD-10 data [7]. Model performance is measured using micro F1 score, precision at 8, and AUC micro F1, offering a thorough evaluation of each data selection method's effectiveness in a complex medical classification task.

In addition to evaluating DSIR for data selection, this study explores its broader utility in optimizing the fine-tuning process for large language models. Fine-tuning models like BART often requires extensive experimentation with hyperparameter settings, resulting in high computational costs. The study investigates the use of KL divergence, a metric proposed by Xie et al. [16], as a potential early indicator of end-task performance. This work aims to establish whether KL divergence can reliably predict model effectiveness, enabling early stopping during fine-tuning and reducing the time and computational resources required.

The remainder of this paper is organized as follows. Section 2 reviews related work, focusing on data selection methods and the concept of KL divergence. Section 3 outlines the methodology, including data preprocessing, the DSIR framework, and experimental design. Section 4 presents the results and analysis, comparing DSIR, random selection, and BM25 in terms of performance metrics and computational efficiency, while also including a KL divergence analysis. Section 5 concludes the paper by summarizing the findings and suggesting future work.

2 Related Work

The challenge of selecting data for domain-specific pretraining has been studied using different approaches, each addressing specific issues in the process. N-gram based data selection methods rely on linguistic patterns to filter data, focusing on

efficiency and alignment with target domains. On the other hand, Representation based data selection methods use deeper semantic features and model outputs to identify data that can improve performance in specific tasks. While both approaches provide useful perspectives, they also have their own limitations, which are discussed in the following sections as a basis for the proposed method.

2.1 N-Gram Based Data Selection

This section discusses the traditional selection method based on perplexity as well as DSIR [16], a recently proposed method. In 2010, Moore and Lewis [10] addressed the challenge of using all available data to build an auxiliary language model for machine translation. They proposed that model performance could be enhanced by selecting training data that aligned with the intended application output. The study identified a subset of training data for language model construction by comparing the perplexity between an in-domain language model and a non-domain-specific language model. Specifically, it distinguished between newswire data (Gigaword) and parliamentary data (Europarl), calculating the perplexity of each sentence in the Gigaword corpus for both models. The results demonstrated that the perplexity difference scoring method significantly outperformed random selection and in-domain perplexity scoring alone.

However, one limitation of the study was the absence of an evaluation metric to assess performance on the end task, such as the BLEU score in machine translation. Additionally, a concern raised in A Survey on Data Selection for Language Models [2] was the high computational cost of this method, particularly with modern, large-scale language models. Implementing this approach traditionally involved training two separate language models, which could be resource-intensive. Although pre-trained language models can be leveraged, calculating the perplexity of every unigram in a dataset containing 4,000,000 instances remained time-intensive.

In 2023, Xie et al. [16] proposed Data Selection via Importance Resampling (DSIR), addressing challenges in domain-specific dataset curation by overcoming traditional heuristic classification methods' limitations in selecting data aligned with target distributions. By utilizing hashed n-grams (unigrams and bigrams) grouped into 10,000 buckets and incorporating an importance weight to quantify dataset feature alignment, the approach enhanced data interpretability through computationally efficient, probability-based calculations. The method was applied to The Pile [5], a 735 GB dataset of non-domain-specific data, focusing on selecting data for four domains: Computer Science, Biomedical, News, and Reviews. Specifically in the Biomedical domain, the researchers targeted two datasets: the ChemProt [17], an expert-annotated dataset of approximately 2,000 instances designed for chemical-protein interaction extraction, and the RCT [4], a larger dataset of around 170,000 instances involving sequential sentence classification of PubMed abstracts with sentences labeled as background, objective, method, result, or conclusion.

The DSIR method performed comparably to expert curation across eight target distributions when selecting data for continued pretraining in specific

domains. Moreover, the approach outperformed heuristic classification and domain-adaptive pretraining checkpoints in terms of end-task performance. However, a limitation of this study was its reliance on n-grams, which provided only superficial features compared to neural model features that incorporate semantic information. Although extracting such features would provide more detailed insights, it was significantly more computationally expensive. A related challenge with this work, as well as with other domain-specific data selection methods, was determining how well the target dataset represented the entire domain, and the precise methodology for collecting such data remained unclear.

Another key finding of this study was that KL reduction was highly correlated with downstream performance. KL reduction was defined as the difference between the KL divergence of random selection and that of other selection methods, based on the assumption that effective data selection methods should outperform random selection. Consequently, random selection was used as the baseline. A relationship was observed between KL reduction and the F1 score of the downstream task, averaged across eight tasks in four domains. While the performance did not increase linearly, the general trend indicated that greater KL reduction corresponded to better performance.

2.2 Representation Based Data Selection

This section discusses the recently proposed method by Xia et al. [15] in 2024 and the study by Aharoni and Goldberg [1], which propose a broader view of utilizing representation for selecting data. Although the method proposed by Xia et al. [15] was primarily designed for the instruction-tuning phase, which covered a narrower data range compared to continual pretraining for the domain-specific phase, utilizing influence estimation for data selection was particularly noteworthy. Unlike heuristic classification, which considered input features such as n-grams or input representations, this approach focused on the final gradient after the input passed through the model, directly minimizing loss on target tasks. Furthermore, the study evaluated a wide range of benchmarks across tasks and languages, demonstrating significant performance improvements over existing methods, including random selection, BM25 [13], and DSIR.

In addressing the challenge of developing specific language model capabilities with limited and insufficient labeled data, Wang et al. [14] explored an innovative approach to data selection using the influence function derived from Taylor expansion, suggesting that gradients from the validation set could reflect specific skills. By utilizing FlanV2 [3], a comprehensive dataset spanning 473 datasets across diverse domains including law, medicine, mathematics, engineering, and economics, the research proposed a method to select data that aligns with target task gradients, suggesting that the method could implicitly select domain-specific data. However, a significant gap remained in understanding how many gradients were necessary to represent most of the data in a domain during the pretraining phase, and extracting all gradients from a model required substantial computational resources, posing a challenge for real-world implementation.

In 2020, Aharoni and Goldberg [1] proposed that trained language models like BERT, RoBERTa, and GPT-2 implicitly learned the structure of sentences from different domains. They conducted experiments by extracting hidden-state sentence representations and using PCA to reduce these representations to two dimensions. The preliminary results of their study showed that some domains, such as law and medicine, could be explicitly separated, while others, such as law and IT, tended to overlap.

Their study leveraged this property, along with similarity measures and classifiers, to distinguish in-domain data from raw data. However, a concern with this method was that sentence-level representations were not feasible for a dataset consisting of 4,000,000 instances. Consequently, document-level representations were considered as an alternative. Furthermore, it was necessary to test whether sentence-level representations could still effectively differentiate between biomedical and medical data.

3 Methodology

In this study, the methodology for evaluating data selection methods in domain-specific pretraining is illustrated in Fig. 1. The process begins with deduplicated data sourced from PubMed Central. The data selection phase involves three approaches: a variation of Data Selection via Importance Resampling (DSIR), BM25, and random selection. The selected subsets of data are then analyzed using Kullback-Leibler (KL) divergence to identify the subset with the highest divergence relative to the target distribution. This most KL-divergent dataset is subsequently employed for continual pretraining using WangchanBART.

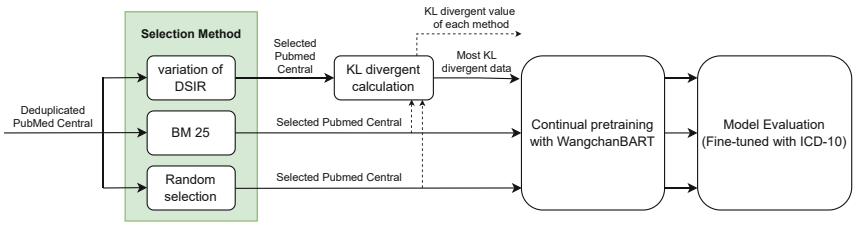

Fig. 1. Pipeline for Data Selection and Evaluation in Domain-Specific Pretraining

To establish a baseline, random selection is initially implemented. The BM25 method is applied using a keyword-based subset of ICD-10, as running BM25 with the entire English ICD-10 dataset requires significant computational resources, with runtime scaling linearly with the number of query terms. For DSIR, three experimental cases are conducted: the top-k False case and the top-k True case. These configurations are tested to evaluate their KL divergence.

Finally, the pretrained models are fine-tuned and evaluated on the ICD-10 classification task to comprehensively assess the effectiveness of each data selection method. This end-to-end process enables a robust evaluation of the proposed approaches in domain-specific continual pretraining.

3.1 PubMed Central from The Pile and Data Preprocessing

The Pile [5] is an 825 GB dataset consisting of diverse text designed for training large-scale language models. It is created to address the need for high-quality, varied data that enhances general cross-domain knowledge and downstream performance of language models. The Pile includes 22 subsets from academic and professional sources, such as GitHub, ArXiv, OpenWebText, and BookCorpus. Notably, it also includes PubMed Central, which comprises 165 GB of data from PubMed publications starting in 2008.

PubMed is a repository of biomedical literature managed by the U.S. National Institutes of Health's National Library of Medicine. It is one of the world's most widely accessible resources for biomedical literature. PubMed Central includes more than 30,000 journals, in addition to other types of articles. It covers a wide range of topics, such as biomolecular research, medical ethics, institutional medical articles, medical procedures, symptoms, and more.

Data from The Pile has already been cleaned and had all explicit content removed. Therefore, the data is only preprocessed by deduplication using MinHashLSH [11]. MinHashLSH is a technique for identifying and removing duplicate or near-duplicate documents in large datasets. By representing documents as sets of tokens, MinHash generates compact hash signatures that approximate their Jaccard similarity. These signatures are stored in a hash table using LSH, which ensures that similar documents are mapped to the same bucket with high probability. This approach reduces redundancy in document collections while maintaining scalability and computational efficiency.

3.2 Synthetic ICD-10

The International Classification of Diseases (ICD) [7] represents diagnostic and procedural information from medical visits. ICD codes are a standardized system for encoding information systematically and internationally, enabling their use in tracking healthcare statistics, quality outcomes, and billing.

As patient histories are confidential, it is challenging to obtain real patient history data. Therefore, a synthetic dataset is used for fine-tuning in this study. This dataset includes 10,339 entries with three columns: the full patient history in Thai, the full patient history in English, and the ICD-10 code derived from each history. Each patient record contains details on age, gender, underlying diseases, present illness, physical examination, lab or imaging studies, and the final diagnosis.

The dataset is created using prompts to GPT-3.5 and Gemini Pro 1.0, referencing real cases from the Infectious Disease Association of Thailand. It is then manually reviewed by doctors to ensure medical accuracy. On average, each

entry contains about six ICD-10 codes. The average length of patient histories is approximately 3,319 words in Thai and 3,617 words in English per document. Notably, about 3.21% of the Thai histories include English words, primarily clinical terms. This dataset covers approximately 6,235 distinct diseases.

3.3 Pre-trained Model (WangchanBART)

WangchanBART is based on the BART [9] model. BART is pre-trained using a denoising autoencoder approach, where it corrupts input sequences (e.g., token masking, deletion, or sentence shuffling) and trains the model to reconstruct the original text. This approach enables BART to handle a wide range of natural language processing tasks, leveraging both its encoder-decoder architecture and rich contextual representations.

WangchanBART [12] is a bilingual BART model trained on English and Thai datasets. However, the amount of Thai data is relatively low compared to the English data, which may result in the model sharing some representations between the two languages. It was trained on various data sources, including social media data, news articles, Wikipedia, and web-crawled data.

3.4 DSIR Framework: Data Selection for Medical Continual Pretraining

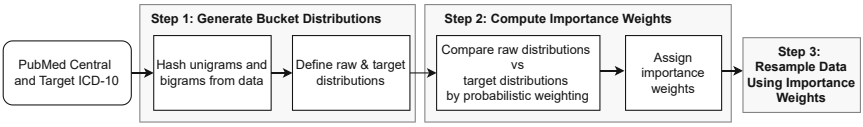

Fig. 2. DSIR Framework

To effectively select close-domain data for continual pretraining, the DSIR framework employs a probabilistic approach that leverages hashed n-grams and importance weights. This method ensures computational efficiency while aligning the selected data with the target distribution. The framework, as illustrated in Fig. 2, consists of three key steps as outlined below.

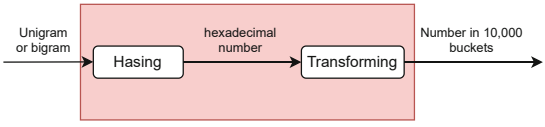

Fig. 3. Hashing process in DSIR

Step 1. Generate Bucket Distributions: The distributions for both the raw data (PubMed Central) and the target data (full English ICD-10) are obtained by encoding each unigram and bigram using hashlib.sha256 to generate a hexadecimal value. This value is then transformed into a decimal number and hashed into 10,000 buckets using the modulo operator (%). As illustrated in Fig. 3, this process involves hashing every unigram and bigram in the dataset to ensure efficient mapping into the defined buckets. Subsequently, the probability of each bucket in both distributions is calculated, as depicted in Fig. 4.

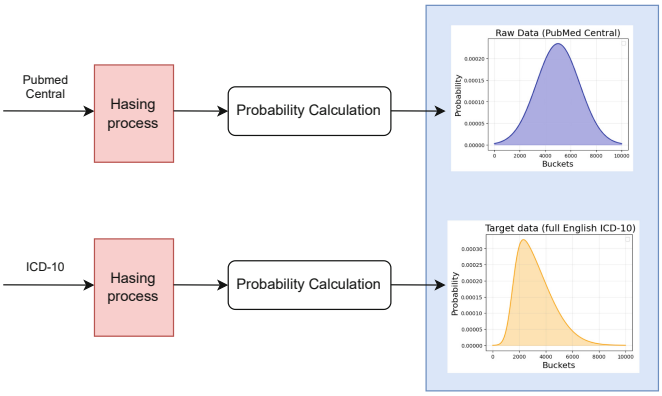

Fig. 4. Raw and target data distibution example

Step 2. Compute Importance Weights: Each data point is hashed into 10,000 buckets, and the importance weights are computed. The probability of each hashed data point is then calculated based on the probabilities of each distribution. Specifically, this is achieved by raising the probability of each bucket in the distribution to the power of its occurrence and then multiplying the resulting probabilities across all buckets to obtain a scalar value for each distribution. The importance weight W_i, where i refers to a specific bucket generated from the hashed n-grams in the dataset, is computed using $W_i = \frac{\mathbf{P}}{\mathbf{Q}}$,where $\mathbf{P}$ represents the hashed distribution of the target dataset, and $\mathbf{Q}$ represents the hashed distribution of the raw dataset. As a result, the first value (from the target distribution, $\mathbf{P}$) is expected to be large, and the second value (from the raw distribution, $\mathbf{Q}$) is expected to be small. Thus, the larger the importance weight, W_i, the higher the probability of selection.

Step 3. Resample Data Using Importance Weights: The data is resampled based on the importance weights as probabilities, ensuring that each example is selected only once by using a without-replacement approach.

In the DSIR framework, four variables are defined. First, the target distribution is the English ICD-10 dataset, as the selected data should have a

word distribution similar to this dataset. Two heuristics are used to define the target distribution: the full English ICD-10 dataset and a keyword-based ICD-10 dataset. The keyword-based ICD-10 dataset is derived by extracting English words from the Thai ICD-10 dataset, typically consisting of English medical terms, such as procedures and diseases. Second, the raw distribution is the deduplicated PubMed Central dataset, as this is the dataset used for selecting data. Third, the number of samples to be resampled is set to 5 percent of 4.5 million instances, which amounts to 225,000 instances. Although no exact percentage is commonly used, selection methods often select 3–5 percent of the data.

Lastly, the top-k parameter is set to both True and False. The original paper suggested setting top-k to False to avoid selecting data close to the mode. However, due to the diverse nature of medical data, the 225,000 instances may not provide sufficient information. To address this, performance results for the True case are also provided to assess the trade-off between close-to-mode data and diverse data.

3.5 Evaluation Pipeline and Framework

After pretraining WangchanBART with medical data, the model is fine-tuned for downstream tasks. The downstream task is created using synthetic data and manually verified for medical accuracy by a doctor. The dataset consists of 10,000 instances, where the input is patient medical records, and the target is the ICD-10 code, which may include more than one code.

The micro-average F1 score and AUC micro F1 are used as evaluation metrics because this is a multiclass classification task with class imbalance. Precision at 8 is also reported, as ICD-10 should not include diseases that the patient does not have (false positives). All selection methods are processed on an Intel(R) Xeon(R) Gold 6248 CPU @ 2.50 GHz. All training and evaluation processes are implemented using the PyTorch framework. The DSIR framework is reproduced from the code provided in the original paper [16], with adjustments made only to the top-k parameters, the number of selected data, target data, and raw data. Evaluation metrics are computed using the scikit-learn library.

4 Results and Discussion

The evaluation of data selection methods focuses on two key aspects: their impact on downstream performance and the relationship between data alignment and model effectiveness. The first subsection examines the performance of various methods in ICD-10 classification tasks, while the second subsection explores the role of KL divergence as a metric for assessing data alignment and its correlation with end-task performance.

4.1 ICD-10 Classification Performance

To compare end-task performance by data selection methods, Table 1 demonstrates that the DSIR data selection method consistently outperforms random selection and BM25 across all evaluation metrics. Specifically, for the English-language ICD-10 dataset, DSIR achieves a 5.17% improvement in Micro F1 Score and a 3.71% improvement in Precision at 8 compared to random selection. Furthermore, DSIR attains the highest AUC Micro F1 score (0.9364, +0.83 increase over random selection), reflecting its robustness in capturing relevant features for classification tasks. These results highlight the overall effectiveness of DSIR in both computational performance and its ability to enhance critical evaluation metrics.

Table 2 highlights the KL divergence values and selection times for each data selection method. DSIR with the "Full-English ICD-10" heuristic (Topk = False) achieves the lowest KL divergence at 0.410, indicating the most aligned data distribution with the clinical target. This alignment suggests that DSIR effectively minimizes the gap between the biomedical and clinical data distributions, enabling the model to learn clinically relevant features more efficiently.

Table 1. Comparison of End-Task Performance by Data Selection Methods

Selection Method	Micro F1 Score	Precision at 8	AUC Micro F1
Random Selection	0.4741	0.3908	0.9287
BM25	0.4802 (+1.29%)	0.3949 (+1.05%)	0.9282 (−0.05%)
DSIR	**0.4986** (+5.17%)	**0.4053** (+3.71%)	**0.9364** (+0.83%)

Table 2. Comparison of KL Divergence and Selection Time Across Methods

Method	Heuristic	KL Divergence (Raw Data: 0.586)	Selection Time
Random Selection	–	0.584	45 min
BM25	Keyword-based ICD-10	0.473	16 h
DSIR (Topk = False)	Full-English ICD-10	**0.410**	**3.5 h**
	Keyword-based ICD-10	0.503	3.5 h
DSIR (Topk = True)	Full-English ICD-10	0.486	3.5 h

Moreover, BM25, despite being keyword-based, achieves a KL divergence of 0.473, which represents a modest improvement over random selection (0.584). However, BM25's significantly longer selection time (16 h) underscores its computational inefficiency compared to DSIR, which completes selection in just 3.5 h. In contrast, DSIR with the "Keyword-based ICD-10" heuristic exhibits a slightly higher KL divergence of 0.503, likely due to its narrower focus on keyword

matching. These findings emphasize DSIR's capability to balance distribution alignment and computational efficiency, with the "Full-English ICD-10" heuristic emerging as the most optimal approach for data selection in this study. In summary, DSIR outperforms in all evaluation metrics except for the keyword-based heuristic, which focuses on capturing general patterns rather than just keywords. Additionally, DSIR requires less computational time.

4.2 KL Divergent Analysis

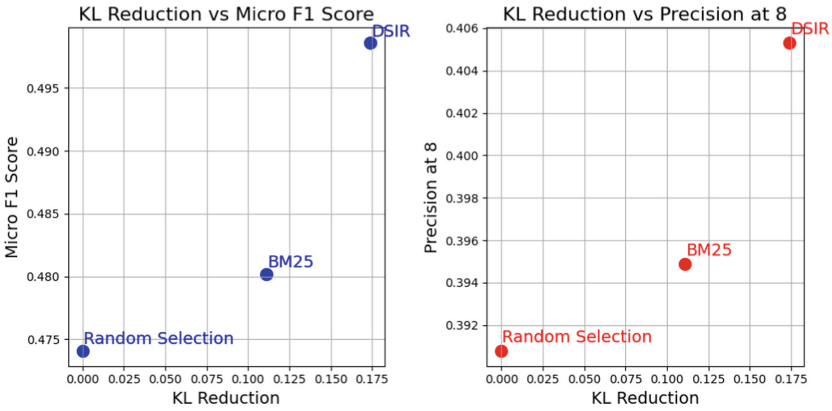

Fig. 5. Relationship Between KL Reduction and End-Task Performance

To analyze the relationship between data selection methods and end-task performance, KL reduction is defined in Equation (1). It measures the difference between the KL divergence of the data selected by each method and the KL divergence of the data selected randomly. The KL divergence of random selection is typically higher because it produces a data distribution that is less similar to the target distribution compared to the other methods. Figure 5 shows that KL reduction is related to end-task performance across different data selection methods. It highlights a strong positive correlation, where greater KL reduction is associated with higher Micro F1 scores and Precision at 8.

$$KL\,reduction = KL(Rand \mid Target) - KL(Selection\,method \mid Target) \quad (1)$$

where *Rand* represents the data obtained through random selection, and *Target* refers to the target data. In the left plot, DSIR achieves the highest KL reduction (0.175), aligning with its superior Micro F1 Score of 0.4986, which surpasses both BM25 (0.4802) and random selection (0.4741). Similarly, in the right plot, DSIR demonstrates the highest Precision at 8 (0.4053), further emphasizing the strong link between data distribution alignment (measured by KL reduction) and model accuracy.

These results highlight that KL reduction is directly proportional to end-task performance, emphasizing that aligning the selected data with the target

distribution enhances the model's ability to generalize and perform effectively on clinical tasks.

5 Conclusion

Close-domain data often contains numerous shared words, leading to overlaps in n-grams and word representations. This overlap makes data selection challenging in such datasets. Currently, there is no research specifically dedicated to data selection for these types of datasets. This study explores existing data selection methods and implements DSIR, which is recognized for its efficiency in both computational time and performance. Additionally, the study examines the relationship between KL divergence and end-task performance. The results demonstrate that DSIR outperforms both random selection and BM25 across all metrics. However, a limitation of this method is that it has only been applied in the medical domain, necessitating further exploration in other domains.

To enhance DSIR's performance, it is crucial to align the selected data distribution with the target distribution, which can be quantified using KL divergence. One approach involves identifying buckets with significant gaps after an initial DSIR run and performing a second DSIR iteration, where a constant is added to the selection probability for data in these buckets. However, it is important to carefully choose an appropriate constant. The study also highlights that KL divergence can serve as an early indicator of data selection performance in continual pretraining. By calculating this metric in advance, performance can be assessed quickly, saving time that would otherwise be spent waiting for end-task results, especially when fine-tuning large models like BART, which can be time-intensive. Based on these findings, future work could explore applying DSIR to other domains, such as extracting political news from Babel Briefings [8], a dataset with 4.7 million headlines across 30 languages, to assess DSIR's adaptability across diverse domains.

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Improving Domain-Specific Data Question Answering with Deep and Cross-Lingual Transfer Learning

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Abstract. In recent years, there has been significant development in the use of machine learning for various tasks involving natural language interpretation. Improving user experience is a top priority for many online services, including search engines and social media platforms. One such service is answering users' questions. Domain-specific data, such as sacred data, can present unique difficulties for natural language processing because of their intricate structures. In this paper, we employ domain-specific sacred data and a question-and-answer dataset to evaluate and contrast the optimal existing pre-trained deep transformer models for sacred data question answering. Based on the variations of the pre-trained deep transformer models, BERT models, we propose a multilingual domain and task-specific model. To improve the model predictions, we also undertake post-processing and cross-lingual transfer learning steps. On the official test set, our proposed model receives a Partial Reciprocal Rank (pRR) score of 62% recording promising results.

Keywords: Question Answering · Cross-Lingual · Transfer Learning · Sacred Text · Domain-Specific · Deep Learning

1 Introduction

The advent of intelligent systems has led to the development of a range of technologies that have replaced many tasks previously performed by humans, such as information retrieval, classification, generation, decision-making, and natural language understanding (NLU). One such task is automatic question answering (QA), and there is a lot of work being done to build intelligent QA systems [1, 2]. A typical QA approach involves input resources that include a question and a passage, with the output being an answer directly extracted from the text. QA is an important aspect of human cognitive development at any stage of life, and as technology has matured, there has been an abundance of intelligence-based QA systems that play a crucial role in human development. Popular examples of QA systems include search engines, which are widely used by internet

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users. There are also domain-specific QA systems that have a limited scope and are designed to handle specific problems or scenarios, such as healthcare QA systems that only answer health-related queries [3, 4]. QA systems can be divided into two types: abstractive and extractive. Abstractive QA systems learn knowledge from training data and generate an answer to a given query, while extractive QA systems extract the answer from the given context and question [5]. In this research, we focus on extractive QA systems. QA systems have been developed for a wide range of languages, but the most mature QA systems are for the English language. The largest dataset for training QA systems, SQUAD [6], is also in English.

In this work, we investigate QA systems for domain-specific data, sacred texts, and take a specific case with the Arabic language, specifically for the closed and low-resource [7] task of the sacred Quran question answering. While there are existing systems for the Arabic language [8], there is currently no state-of-the-art QA model for the Quran. This is due to the unique characteristics of the Arabic language used in the Quran, which differs from contemporary Arabic, and the scarcity of data for developing QA systems for the Quran. In our proposed approach, a question posed may have multiple possible responses within the Quranic text. To accommodate this, the system is designed to generate several anticipated answers, which are then prioritized so that the most likely accurate responses appear at the top of the list, allowing the user to verify these first before exploring further down the list. The system supports both factual and non-factual questions. Factual questions typically include terms like "who," "when," "where," and "how long/many," while non-factual questions often contain words such as "why," "describe," and "evidence." Additionally, questions can be categorized as single-answer or multiple-answer types. In some cases, multiple answers to a single question may be found within the same verse or across different verses in the Quran. There are also instances where a question has a single-word answer, but understanding this answer requires resolving co-references to previous verses.

Specifically, we propose a methodology for building a QA system for the Quran while maintaining the constraint of data scarcity. We will compare our methodology with other QA methods. We also demonstrate that by using a language model pre-trained on domain-specific tasks, it is possible to achieve similar results to a large ensemble of language-specific models. These models have the advantages of being faster and more efficient during inference, as well as requiring fewer computational resources for training. We will leverage transformer-based models to develop and evaluate our QA system, with our key contributions being:

- Achieving promising results on Quranic QA tasks, our research demonstrates XLM Roberta's ability to accurately interpret and respond to complex questions within a specialized domain.
- By enhancing inference speed, we have improved the model's capacity to deliver answers more quickly and efficiently, making it suitable for practical applications where rapid and accurate responses are critical.
- Our findings reveal that task-specific pre-trained models can outperform language-specific models for certain NLP tasks, offering valuable insights into the relative strengths and limitations of these models and advancing our understanding of their optimal application in NLP.

2 Related Work

Advancements in Arabic QA systems have been limited due to a paucity of datasets and resources for NLP designed exclusively for Quran QA. As a result, many Arabic QA studies use information retrieval-centric strategies, such as applying rules to select responses and rating relevant paragraphs, phrases, and named entities. This section focuses on research that has used deep learning methods and transformer models to address Arabic QA reading comprehension problems. Several researchers have converted existing English QA datasets into Arabic to address the lack of substantial reading comprehension datasets in the language relying mostly on Bidirectional Encoder Representations from Transformers (BERT) [11]. For instance, Mozannar et al. [8] used the transformer model Multilingual BERT (mBERT) [12] to translate the English SQuAD 1.1 QA dataset [6] into Arabic, creating a dataset of 1,395 questions called the Arabic Reading Comprehension Dataset (ARCD). With a BERT-based model, they obtained a 0.50 F1 score on the ARCD dataset and a 0.48 F1 score on the Arabic-SQuAD dataset [8].

To enhance understanding of Arabic, Antoun et al. [13] developed AraBERT, a BERT model that has been pre-trained for the language. The model was fine-tuned for QA tasks, where it is used to locate a section of text that includes the answer to a posed question. AraBERT's performance was found to be significantly higher than that of Google's multilingual BERT, and the model also achieved state-of-the-art results on the majority of Arabic NLP tasks. The Arabic Question-Answer dataset (AQAD) was introduced by Atef et al. [14] to measure reading comprehension in Arabic. The dataset was created using over 17,000 questions and Arabic Wikipedia articles and was designed to be similar to the popular SQuAD dataset. They used mBERT trained on various languages, including Arabic, as well as Arabic fastText embeddings and the BiDAF model, to evaluate the AQAD dataset. They found that mBERT achieved a 33 Exact Match (EM) score and a 0.37 F1 Score, while the BiDAF model obtained a 32 EM score and a 0.32 F1 score.

Antoun et al. [15] developed ARAELECTRA, a set of text discriminators for pre-training in Arabic language comprehension. In order to fine-tune the model on reading comprehension tasks, a linear classification layer was built on top of ELECTRA's output, which has a structure and layering comparable to a BERT model. Several Arabic NLP tasks, such as the Typologically Diverse Question Answering Dataset (TyDiQA) and the Arabic Reading Comprehension Dataset (ARCD) [8], were used to assess ARAELECTRA's performance. The model outperformed the competition with scores of 71.22 on the ARCD dataset and 86.68 on the Arabic TyDiQA dataset [16], demonstrating its cutting-edge capability.

Clark et al. [16] introduced the TyDiQA question-answering dataset, which includes 11 languages with a range of typological similarities and differences. The dataset included a passage selection task, a minimal answer span challenge, and a gold passage task for responding to questions. The gold passage (GoldP) challenge, in which only the gold answer paragraph is provided instead of the whole Wikipedia page, can be used to evaluate reading comprehension datasets. mBERT was trained on all of the gold language variants in TyDiQA, and the F1 score for the Arabic language was 81.7%.

ElKomy and Sarhan [17] presents a QA system for the Holy Quran in Arabic. The team, called TCE, uses an ensemble of BERT-based models for prediction and a post-processing step to improve the performance of the ensemble. In the field of question answering, the use of ensembles of models has been shown to improve performance by combining the predictions of multiple models. BERT is a popular choice for NLP tasks such as QA due to its ability to capture contextual information in text. The TCE system was evaluated on the Quran QA 2022 dataset (QRCD), a collection of questions and answers about the Holy Quran. The authors found that TCE was able to achieve strong performance on the dataset, outperforming the baseline model and other state-of-the-art approaches by achieving 56.6% of pRR.

Malhas and Elsayed [9] introduced the QRCD dataset as the first Qur'anic Reading Comprehension Dataset. To enhance the performance of the model, they proposed a new pre-trained model, CLassical-AraBERT, which is based on AraBERT and is further pre-trained on approximately 1B words of classical Arabic (CA) dataset. Additionally, they introduced a new evaluation metric, pAP (partial average precision), which takes into account partial matching in the evaluation process for both multi-answer and single-answer MSA questions. Using this new model and metric, they obtained an F1 score of 49.68 and a pAp of 53.97. This is the closest to our work, and we were inspired by this work (and many of the teams in [18]) and most of the work presented in this paper built on the foundations of it.

Table 1: Distribution of Q-P answer triplets in the QRCD by question type, with some unique cases where answers may appear multiple times in the same Quranic text [9].

Type of Questions	No. of Q-P Pairs	No. of Q-P answer triplets
Single-answer	939	949
Multi-answer	154	388
Total	1093	1337

3 Data

Following [9], we use the Arabic Holy Quran QRCD dataset which was made available as a component of the OSACT5 Quran QA 2022 Shared Task,1 a challenge focused on Arabic QA about the Arabic Holy Quran. This dataset (Table 1) is the first substantial one made available to a broad audience. It consists of a question, answer, and context triplets, where each question may have multiple answers. The dataset is in Json line format, with separate files for the train, validation, and test datasets. A sample dataset is shown in Fig. 4. It follows the same pattern as SQUAD 1.0 but is not the same size. SQUAD 1.0 has 100k unique questions, while this dataset has 118 unique questions that are repeated for different contexts and answers.

The dataset is split into three parts: training, validation, and testing. The training set consists of 710 unique question-passage pairs, the validation set contains 109 pairs,

and the test set includes 274 pairs. This distribution ensures that the model is trained on a substantial amount of data while still having separate datasets to validate and test its performance effectively. The scarcity of data for QA for the Quran, as well as the repetition of questions with different contexts and answers, presents a challenge for creating a generalized QA system. Additionally, the answers to the questions are just a span from the context, and the dataset includes the starting span of the answers. This is because it is an extractive QA task.

4 Proposed Approach

This research addresses the complex task of extractive Question Answering (QA) for the Arabic Holy Quran, focusing on the challenge of accurately identifying the start and end tokens that span the answer within a given context when presented with a question-context pair (q, c). Extractive QA in this context is particularly challenging due to the intricate linguistic features and the rich semantic content of the Quranic text [18].

To tackle this problem, and following [9], we leverage BERT-based models, which have established themselves as state-of-the-art in various natural language understanding tasks. BERT, or Bidirectional Encoder Representations from Transformers, is pre-trained using two key tasks: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). These pre-training tasks enable BERT to learn the subtleties and intricacies of language, making it highly effective for downstream tasks. BERT-based models, due to their versatility, can serve as powerful language models and be fine-tuned for a wide range of NLP tasks, such as sentiment analysis, text classification, and in this case, extractive QA. Fine-tuning involves adding task-specific layers on top of the pre-trained BERT model to adapt it to the specific needs of the QA task.

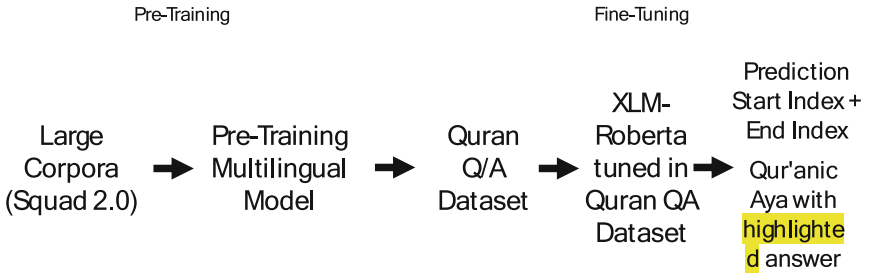

Fig. 1. Followed approach: A multilingual model is first trained on high-resource data (SQuAD) and then fine-tuned on the Quran QA dataset, utilizing Cross-Lingual Transfer Learning [9, 17, 18].

We employ BERT-based models that are either monolingual, specifically trained on Arabic, or multilingual, encompassing Arabic along with other languages. We explore and compare three distinct learning approaches to optimize the performance of our Quran QA system. The first approach involves fine-tuning a BERT model that has been pre-trained exclusively on Arabic language data, leveraging the linguistic of Arabic

and adapting it to the specific characteristics of the Quranic text. The second approach involves transfer learning from a BERT model that has already been fine-tuned on a general Arabic language QA dataset, further fine-tuning it on our Quranic QA dataset [9]. This method capitalizes on the prior knowledge the model has gained from a broader Arabic QA task, potentially enhancing its performance on the more specialized Quranic text. The third approach, cross-lingual transfer learning, uses a multilingual BERT model pre-trained on a variety of languages, including Arabic, and fine-tuned on the English SQuAD 2.0 dataset. We then adapt this model to the Quranic QA task, exploring the potential of leveraging knowledge from English QA tasks to improve performance on Arabic Quranic text.

Each of these approaches is designed to assess the effectiveness of different pre-training and fine-tuning strategies for enhancing the performance of BERT-based models in the specific context of Quranic QA. By comparing these methods, we aim to identify the most effective strategy for accurately and efficiently extracting answers from the Quranic text, thereby contributing to the broader field of NLP and enhancing the capabilities of AI systems in understanding and processing religious texts. The overarching methodology and the comparative analysis of these approaches are visually summarized in Fig. 1, providing a clear framework for how each approach is applied and evaluated in the context of our research.

5 Experiments and Results

We utilized Transformer-based models, known for their state-of-the-art performance in NLP tasks, to address the Quranic QA challenge. Implemented in Python using the Hugging Face Transformers library[1], the problem was approached as a classification task. The model received a question and context as input, separated by a SEP tag, and predicted the start and end tokens in the context as the answer.

Given the distinct nature of Quranic Arabic, we experimented with various Arabic monolingual BERT variants, including AraBERT [13], CAMeLBERT-CA [27], and QARiB [26], to fine-tune the model for this specific task. These models, originally pre-trained on modern and classical Arabic texts, were chosen to assess their effectiveness in understanding and extracting answers from Quranic text.

AraBERT, trained on Arabic Wikipedia and news articles, served as one of the base models. CAMeLBERT-CA, specifically trained on classical Arabic, was selected due to its relevance to Quranic language. QARiB, which focuses on diverse Arabic datasets, was also fine-tuned for the task.

In each case, we fine-tuned the pre-trained models by adding a final layer that learned to predict the position of the start and end tokens, adapting the models to accurately handle Quranic QA. In our experiments, we employed transfer learning and cross-lingual transfer learning to enhance the performance of our Quran QA system.

For the transfer learning approach, we fine-tuned a language model on a larger Arabic QA dataset before applying it to our smaller Quran QA dataset. Specifically, we used the SOQAL [8] dataset, which includes Arabic Reading Comprehension and a translation of

[1] https://huggingface.co/docs/transformers/.

the English SQuAD dataset, to fine-tune the AraELECTRA [15] model. AraELECTRA, a smaller yet effective model from the ELECTRA family, has shown strong performance in Arabic NLP tasks. By leveraging the knowledge acquired from the larger SOQAL dataset, we aimed to improve the model's accuracy in answering Quranic questions.

Additionally, we explored cross-lingual transfer learning using the multilingual XLM-Roberta [35] model, which is trained on 100 languages, including Arabic and English. We fine-tuned this model on the English SQuAD dataset and then tested it on the Quran QA dataset. The model was evaluated in both zero-shot and few-shot settings, allowing us to assess its ability to transfer knowledge from English to Arabic and handle the specific linguistic characteristics of Quranic text.

Both approaches aimed to maximize the effectiveness of the limited Quranic QA data by leveraging pre-trained models and large, resource-rich datasets.

5.1 Evaluation

An effective extractive QA system for Quranic passages should not be limited to providing only a single correct answer, as multiple correct answers may exist for a given question. Ideally, the system should return all possible correct answers. When a mix of correct and incorrect answers is returned, partial credit should be given. To accurately evaluate such systems, a ranking-based measure that allows for partial matching of answers is necessary. The system's predicted answers can be compared against the "gold" answers in the dataset to measure its performance. Malhas and Elsayed's [9] proposed method enables a comprehensive evaluation of the system's performance, allowing for meaningful comparisons with other QA systems. For single-answer questions, we followed the standard practice [9, 17, 36] and utilized evaluation metrics following those used in SQuAD [6], namely the F1@1 score and Exact Match (EM). The F1@1 score measures the overlap between the predicted tokens of the top answer among the five predicted answers and the best matching answer in the ground truth. On the other hand, EM is another standard metric, which is a binary metric that determines whether the topmost predicted response exactly matches any of the ground truth answers. These metrics are crucial for assessing the accuracy and relevance of QA systems. While the F1@1 score provides a detailed evaluation of the predicted answers, the EM metric offers a more straightforward binary measure of accuracy. The F1@1 score is calculated as:

$$F_1@1(R) = m_{r1}$$

where R is the ranked list of predicted answers returned by the system, and r_1 is the predicted answer at the top of the list. The EM score is calculated as:

$$EM\,(R) = \left\{ \begin{array}{l} 1\,if\ m_{r1} = 1 \\ 0\,otherwise \end{array} \right\}$$

For questions with multiple answers, we employ the Reciprocal Rank (RR) metric. The RR is defined as:

$$PR(S) = \frac{1}{k}$$

where k represents the rank among the five predicted answers that exactly matches the ground truth. As the rank of the exact match answer decreases, the score subsequently increases, and vice versa. We use a variant of Reciprocal Rank called Partial Reciprocal Rank (pRR). The pRR metric is designed to give partial credit to systems that return correct answers at lower ranks and reward those that return a larger fraction of the correct answer span.

5.2 Results

The results of our experiments using various Transformer-based models on the Quranic QA task are summarized in Table 2. Our approach focuses on leveraging both monolingual and multilingual models, as well as employing transfer learning techniques to enhance performance on this domain-specific task. The results demonstrate that our proposed approach outperforms other models across all evaluation metrics, including Partial Reciprocal Rank (pRR), Exact Match (EM), and F1 score. Specifically, our approach achieved a pRR of 0.62, an EM of 0.31, and an F1 score of 0.59. This represents a significant improvement over the other models evaluated, such as AraBERT, CAMeLBERT-CA, and AraELECTRA.

Table 2. Performance comparison of various models on the Quranic QA task, highlighting the effectiveness of the proposed approach.

Model	pRR	EM	F1
AraBERT	0.50	0.21	0.47
CAMelBERT-CA	0.47	0.18	0.45
AraELECTRA	0.49	0.16	0.46
TCE (1st)	0.56	0.26	0.50
QQA (2nd)	0.55	0.24	0.51
Proposed	**0.62**	0.31	0.59

AraBERT, a monolingual Arabic model, achieved a pRR of 0.50, an EM of 0.21, and an F1 score of 0.47. While AraBERT demonstrated solid performance, it was surpassed by models that incorporated transfer learning or multilingual training. CAMeLBERT-CA, specifically trained on classical Arabic, slightly underperformed compared to AraBERT, with a pRR of 0.47, an EM of 0.18, and an F1 score of 0.45. This suggests that while CAMeLBERT-CA is well-suited for classical Arabic, the Quranic Arabic still present challenges. AraELECTRA, known for its efficiency and smaller model size, also performed competitively with a pRR of 0.49, an EM of 0.16, and an F1 score of 0.46. Its performance is consistent with its design, balancing model size and effectiveness.

The transfer learning approach demonstrated its effectiveness in improving performance on the Quranic QA task. By fine-tuning AraELECTRA on a larger Arabic QA dataset before applying it to the Quranic dataset, we achieved results that were competitive but did not outperform the cross-lingual approach. The XLM-Roberta model,

fine-tuned on the English SQuAD dataset and subsequently tested on the Quran QA dataset, demonstrated the highest performance among all tested models. This cross-lingual transfer learning approach leveraged the vast amount of English training data to enhance the model's ability to understand and process Quranic Arabic, leading to a significant improvement in all metrics. When compared to other approaches, such as TCE and QQA [36], which achieved pRR scores of 0.56 and 0.55, respectively, our approach demonstrates superior performance. The notable improvement in EM and F1 scores further highlights the effectiveness of our method in accurately identifying and extracting answers from Quranic text, particularly in a domain with linguistic complex as those found in Quranic Arabic.

5.3 Performance Analysis

The model demonstrates strong performance on both the test and validation sets, as evidenced by the close similarity in the range and mean pRRs. This consistency suggests that the model generalizes well across different datasets (Fig. 2). The histogram of pRR (Fig. 3) scores on the test set further illustrates that the model accurately predicted the best answer for a significant number of examples.

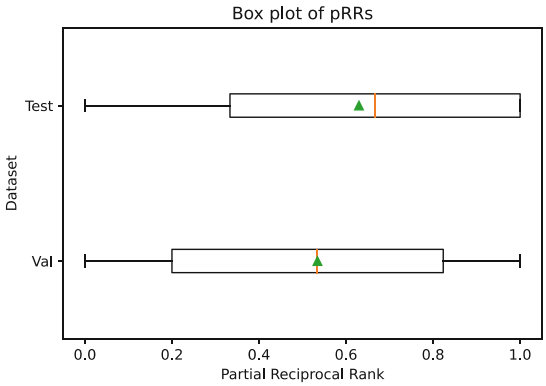

Fig. 2. Box plot of pRRs for test and validation sets, showing similar range and mean values, indicating strong generalization.

In our performance analysis, we evaluated several examples of successes and failures to better understand the strengths and limitations of the optimized XLM-Roberta model, which achieved the highest performance on the QRCD dataset. This examination provides insights into the model's capabilities and identifies areas for improvement that can be addressed in future iterations. The QRCD dataset presents a mix of questions, including those with single and multiple answers, as well as factual and non-factual queries. While the XLM-Roberta model successfully handled many of these challenges, as demonstrated by the successful cases, it also encountered difficulties in certain scenarios. One notable challenge is the Quran's tendency to provide evidence-based responses to non-factoid questions. For instance, in Table 3 (column 3), the model failed to generate

Table 3. Successes and failures examples (similar Ayat to from [9] and [18]) of the proposed model emphasizing strengths in evidence-based answers and challenges with multi-answer queries.

	1	2	3	4	5
مقطع قرآني Quranic Passage	ولقدْ ذرأنا لجهنّمَ كثيرًا مِنَ الْجنِّ والإنسِ ۖ لَهُمْ قُلُوبٌ لَا يَفْقَهُونَ بِهَا ولَهُمْ أعيُنٌ لَا يُبْصِرُونَ بِها ولَهُمْ آذانٌ لَا يَسْمَعُونَ بِها ۚ أُولَٰئِكَ كالأنْعامِ بَلْ هُمْ أضَلُّ ۚ أُولَٰئِكَ هُمُ الْغافِلُونَ	ومِمَّنْ حَوْلَكُم مِّنَ الْأَعْرَابِ مُنَافِقُونَ ۖ ومِنْ أَهْلِ الْمَدِينَةِ ۖ مَرَدُوا عَلَى النِّفَاقِ لَا تَعْلَمُهُمْ ۖ نَحْنُ نَعْلَمُهُمْ ۚ سَنُعَذِّبُهُم مَّرَّتَيْنِ ثُمَّ يُرَدُّونَ إِلَىٰ عَذَابٍ عَظِيمٍ وَآخَرُونَ اعْتَرَفُوا بِذُنُوبِهِمْ خَلَطُوا عَمَلًا صَالِحًا وَآخَرَ سَيِّئًا	الرَّ ۚ كِتَابٌ أَنزَلْنَاهُ إِلَيْكَ لِتُخْرِجَ النَّاسَ مِنَ الظُّلُمَاتِ إِلَى النُّورِ بِإِذْنِ رَبِّهِمْ إِلَىٰ صِرَاطِ الْعَزِيزِ الْحَمِيدِ اللَّهِ الَّذِي لَهُ مَا فِي السَّمَاوَاتِ وَمَا فِي الْأَرْضِ ۗ وَوَيْلٌ لِّلْكَافِرِينَ مِنْ عَذَابٍ شَدِيدٍ الَّذِينَ يَسْتَحِبُّونَ الْحَيَاةَ الدُّنْيَا عَلَى الْآخِرَةِ وَيَصُدُّونَ عَن سَبِيلِ اللَّهِ وَيَبْغُونَهَا عِوَجًا ۚ أُولَٰئِكَ فِي ضَلَالٍ بَعِيدٍ وَمَا أَرْسَلْنَا مِن رَّسُولٍ إِلَّا بِلِسَانِ قَوْمِهِ لِيُبَيِّنَ لَهُمْ ۖ فَيُضِلُّ اللَّهُ مَن يَشَاءُ وَيَهْدِي مَن يَشَاءُ ۚ وَهُوَ الْعَزِيزُ الْحَكِيمُ	كُتِبَ عَلَيْكُمْ إِذَا حَضَرَ أَحَدَكُمُ الْمَوْتُ إِن تَرَكَ خَيْرًا الْوَصِيَّةُ لِلْوَالِدَيْنِ وَالْأَقْرَبِينَ بِالْمَعْرُوفِ ۖ حَقًّا عَلَى الْمُتَّقِينَ فَمَن بَدَّلَهُ بَعْدَمَا سَمِعَهُ فَإِنَّمَا إِثْمُهُ عَلَى الَّذِينَ يُبَدِّلُونَهُ ۚ إِنَّ اللَّهَ سَمِيعٌ عَلِيمٌ فَمَنْ خَافَ مِن مُّوصٍ جَنَفًا أَوْ إِثْمًا فَأَصْلَحَ بَيْنَهُمْ فَلَا إِثْمَ عَلَيْهِ ۚ إِنَّ اللَّهَ غَفُورٌ رَّحِيمٌ	يَا أَيُّهَا الَّذِينَ آمَنُوا إِن تَنصُرُوا اللَّهَ يَنصُرْكُمْ وَيُثَبِّتْ أَقْدَامَكُمْ وَالَّذِينَ كَفَرُوا فَتَعْسًا لَّهُمْ وَأَضَلَّ أَعْمَالَهُمْ ذَٰلِكَ بِأَنَّهُمْ كَرِهُوا مَا أَنزَلَ اللَّهُ فَأَحْبَطَ أَعْمَالَهُمْ أَفَلَمْ يَسِيرُوا فِي الْأَرْضِ فَيَنظُرُوا كَيْفَ كَانَ عَاقِبَةُ الَّذِينَ مِن قَبْلِهِمْ ۚ دَمَّرَ اللَّهُ عَلَيْهِمْ ۖ وَلِلْكَافِرِينَ أَمْثَالُهَا ذَٰلِكَ بِأَنَّ اللَّهَ مَوْلَى الَّذِينَ آمَنُوا وَأَنَّ الْكَافِرِينَ لَا مَوْلَىٰ لَهُمْ
السؤال Question	هل أشار القرآن إلى العضو الذي يعقل به الإنسان؟ Does the Quran refer to the organ that makes sense of a person?	ماهي الآيات التي تلمح إلى العذاب؟ What are the verses that allude to the torment?	مالدلائل على أن القرآن ليس من تأليف النبي محمد (ص)؟ What is the evidence that Quran was not written by Prophet Muhammad (PBUH)?	مالآيات التي تتحدث عن موضوع الوصية؟ What are the verses that talk about the commandment?	مالدلائل على أن القرآن ليس من تأليف النبي محمد (ص)؟ What is the evidence that Quran was not written by Prophet Muhammad (PBUH)?
Gold Answer	لهم قلوب لا يفقهون بها	سنعذبهم مرتين ثم يردون إلى عذاب عظيم	كتاب أنزلناه إليك	كتب عليكم إذا حضر احدكم الموت إن ترك خيرا الوصية للوالدين والأقربين بالمعروف حقا على المتقين فمن بدله بعدما سمعه فإنما إثمه على الذين يبدلونه	ذلك بأنهم كرهوا ما أنزل الله
Predicted Answer	لهم قلوب لا يفقهون بها	سنعذبهم مرتين ثم يردون إلى عذاب عظيم	وما أرسلنا من رسول إلا بلسان قومه ليبين لهم	كتب عليكم إذا حضر احدكم الموت إن ترك خيرا الوصية للوالدين والاقربين بالمعروف حقا على المتقين	يا أيها الذين آمنوا إن تنصروا الله ينصركم ويثبت أقدامكم. والذين كفروا فتعسا لهم وأضل أعمالهم ذلك بانهم كرهوا ما أنزل الله
	لقد ذرأنا لجهنم كثيرا من الجن والإنس لهم قلوب لا يفقهون بها ولهم اعين لا يبصرون بها		ليبين لهم فيضل الله من يشاء ويهدي من يشاء	سميع عليم. فمن خاف من موص	ذلك بانهم كرهوا ما أنزل الله أفلم يسيروا فالأرض فينظروا
			إلا بلسان قومه ليبين لهم ليبين لهم فيضل الله من يشاء ويهدي من يشاء	فأصلح بينهم فلا إثم عليه	

an answer that included evidence supporting the claim that Prophet Muhammad (PBUH) did not author the Quran. In contrast, the same table (columns 1 and 2), presents examples where the model successfully generated responses that correctly provided evidence for this claim. Other issues arose from partial failures, such as missing components of multi-answer questions, omitting essential words or phrases, or including non-essential elements in the predicted answer span. These shortcomings are illustrated in Table 3 (column 5). However, the model demonstrated its strength in handling multi-answer queries, as shown the table (column 4), where it correctly predicted all necessary answers. Overall, this analysis sheds light on both the successes and limitations of the XLM-Roberta model in addressing Quranic QA, guiding future efforts to enhance its accuracy and reliability. Moreover, Fig. 4 shows a comparison of inference times between the proposed model and the TCE. The TCE model took around 37.4 s on the CPU to make a single prediction, while our model only took 1.51 s, making it much faster. On the GPU, the TCE model took 4.61 s, whereas our model completed the task in just 0.166 s, showing a noticeable improvement.

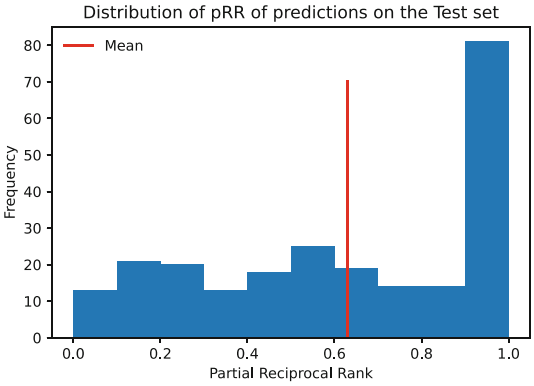

Fig. 3. Histogram of pRR scores on the test set, highlighting that the model predicted the best answer for approximately 69 examples.

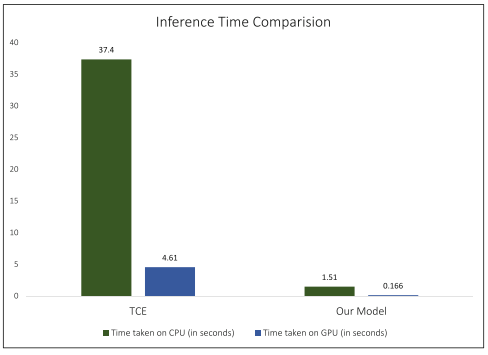

Fig. 4. Inference time comparison between the proposed approach and the TCE [17] model.

6 Conclusion

In this study, we developed and evaluated a Quranic QA system using advanced Transformer-based models, with a focus on optimizing performance through transfer learning and cross-lingual approaches. Our findings demonstrate that the proposed model, particularly the optimized XLM-Roberta, outperformed other models in handling the unique linguistic and contextual challenges presented by Quranic text. The model's ability to accurately predict answers, especially in cases requiring evidence-based reasoning, underscores its potential for broader applications in religious text analysis and other domain-specific NLP tasks. However, the evaluation also revealed some limitations, particularly in dealing with multi-answer queries and partial predictions. These shortcomings highlight areas for future improvement, suggesting the need for enhanced fine-tuning techniques, more sophisticated answer span prediction methods, and the integration of additional linguistic and contextual features specific to Quranic Arabic.

Looking ahead, several avenues for future research can be pursued. First, expanding the dataset with more diverse and complex question types could improve the model's robustness. Second, incorporating more advanced pre-training tasks tailored to Quranic text, such as tasks that focus on theological and historical context, could further enhance the model's understanding and performance. Additionally, exploring hybrid models that combine Transformer-based approaches with rule-based systems or knowledge graphs may offer new ways to address the challenges of multi-answer and partial predictions.

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Leveraging Co-Occurrence Graphs and NLP for Enhanced Job-Skill Matching

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Abstract. The rapid growth of the IT industry is driven by emerging technologies such as Artificial Intelligence, Cloud Computing, Big Data Analytics, and the Internet of Things. Current job-matching platforms, such as LinkedIn and Indeed, often struggle to extract hidden or unmentioned skills from job descriptions, reducing the efficiency of job matching. This paper proposes a method that combines Co-occurrence Graphs and Natural Language Processing (NLP) techniques to improve job-skill matching. By analyzing relationships between frequently co-occurring skills in job descriptions and utilizing SpaCy's Entity Ruler for skill extraction, the method enhances its ability to match candidate profiles with job postings. Evaluation metrics, including Precision at K (P@K), Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (NDCG), demonstrate the effectiveness of this approach. The results indicate the potential of the proposed method to provide more accurate and relevant job-skill matching.

Keywords: Job Matching · Co-occurrence Graph · Natural Language Processing · Skill Extraction

1 Introduction

The Information Technology (IT) sector remains one of the fastest-growing fields, driven by the integration of emerging technologies such as Artificial Intelligence (AI), Cloud Computing, Big Data Analytics, and the Internet of Things (IoT). In 2024, the demand for expertise in these areas has surged, transforming how organizations function. AI plays a key role in strategic decision-making and enhancing customer experiences, creating a growing need for AI specialists [1, 2]. Cloud technology enables organizations to scale efficiently, while Big Data Analytics is essential for generating insights that fuel innovation and guide decision-making [3, 4].

Digital platforms like LinkedIn and Indeed have become essential tools for connecting job seekers with employers, leveraging advanced algorithms to match profiles with relevant job opportunities. LinkedIn, for example, uses machine learning to analyze user

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data and job listings, aligning skills with employer needs [5]. However, these systems face challenges in identifying missing skills from job descriptions, often resulting in inaccurate matches. Studies have shown that algorithms struggle to detect skills not explicitly listed, reducing match accuracy [6]. Addressing the challenge of extracting implicit, complex skills from job postings remains a crucial area for improvement.

This research proposes a method that uses Co-occurrence Graphs combined with Natural Language Processing (NLP) techniques to enhance the job-matching process. Co-occurrence Graphs analyze relationships between frequently co-occurring skills in job descriptions, enabling more precise identification of relevant skills. The NLP tool "SpaCy", with its Entity Ruler, is applied to extract these skills from job descriptions. This approach aims to improve the alignment between a candidate's skills and job requirements by leveraging contextual relationships between skills, without relying on complex transformer models.

A Co-occurrence Graph is a network that maps relationships between words or skills that frequently appear together in contexts like job descriptions or resumes. Co-occurrence Graphs are useful for identifying patterns of related skills across different job roles. Previous research has applied Co-occurrence Graphs to study labor market dynamics and identify in-demand skills. For instance, Giabelli and colleagues developed the GraphLMI system, which uses Co-occurrence Graphs to explore labor market trends and effectively highlight similarities between skills and occupations [7]. This method improves the accuracy of skill-to-job matching, resulting in better recommendations for relevant skills.

The proposed system integrates Co-occurrence Graphs with NLP techniques, such as SpaCy's Entity Ruler, to improve skill extraction and matching. This approach addresses the limitations of existing platforms that struggle to identify and match implicit skills in job descriptions.

2 Methodology

In today's competitive IT job market, finding roles that match a candidate's skills and qualifications can be challenging. While platforms like LinkedIn and Indeed offer numerous job listings, manually searching through hundreds or thousands of postings is time-consuming. Additionally, the rapid evolution of required skills and technologies in IT industry adds to the difficulty. This study aims to automate the process of finding job opportunities that align with a user's profile. It leverages web scraping, natural language processing (NLP), and Co-occurrence Graphs to improve the accuracy of profile matching. This study is structured into four main stages: (1) Extracting job data from LinkedIn and Indeed, (2) Analyzing text with NLP to extract key information from job descriptions, (3) Matching profiles using a Co-occurrence Graph, and (4) Evaluating the system's performance, as shown in Fig. 1.

2.1 Extracting Job Data from LinkedIn and Indeed

The first step is to collect IT job postings from LinkedIn and Indeed. Web scraping automates navigation of these platforms, extracting relevant job information and storing

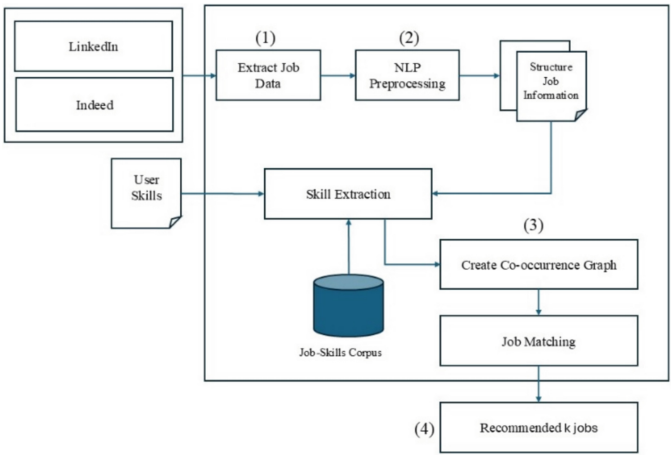

Fig. 1. The proposed framework.

it in a structured format for analysis. The main libraries used are Selenium, Requests, and BeautifulSoup.

1. Selenium automates interactions with the browser, simulating user actions such as searching for jobs, scrolling through pages, and selecting job postings. In this study, Selenium handles logging into LinkedIn and Indeed, searching for jobs with keywords like "software engineer", "network administrator" and "cloud architect" and extracting a wide range of postings [8].
2. Requests makes HTTP requests to retrieve job details from the LinkedIn and Indeed APIs. By sending GET requests, it gathers essential information such as job descriptions, seniority levels, qualifications, and skills, which are key data for text analysis.
3. BeautifulSoup parses HTML content, enabling accurate extraction of job-related data, including job titles, job IDs, and descriptions. Its flexibility with different HTML structures ensures that all necessary data is correctly captured and formatted for analysis [9].

These tools collect comprehensive job data, including job title, company name, seniority level, location, job description, number of candidates, and posting time, as shown in Fig. 2.

2.2 Text Analysis with Spacy

The text analysis phase uses Spacy, a powerful NLP library, to extract and analyze skills and key information from job descriptions and user data. Spacy's trained models handle tasks like part-of-speech tagging, Named Entity Recognition (NER), and lemmatization, which are crucial for identifying technical skills and related terms in IT jobs. The goal is to standardize job descriptions and user data into a common format [10].

- Tagging and Parsing: The first step in NLP is tagging each word in job descriptions and user data, followed by parsing to analyze sentence structure. This ensures phrases

Fig. 2. Example of extracted LinkedIn job data, highlighting job title, company name, location, seniority level, posting time, number of candidates, and job description.

like "cloud computing" or "cybersecurity" are treated as single entities, not separate words.

- Entity Recognition and Custom Entity Ruler: Entity Recognition and Custom Entity Ruler: Entity recognition helps identify and classify important entities such as skills, tools, and technologies in job descriptions. However, general models may misclassify technical skills, such as categorizing "Machine Learning" as an organization. To improve accuracy, we developed a custom entity ruler containing 2,136 unique IT skills extracted from frequently occurring skills in LinkedIn and Indeed job descriptions. The entity ruler uses a JSON file to define these IT-related skills and terms, as shown in Fig. 3.

This customization ensures important IT skills in job postings are accurately identified and matched with the user's profile.

```
{"label": "SKILL", "pattern": [{"LOWER": "python"}]} {"label":
"SKILL", "pattern": [{"LOWER": "cloud"}, {"LOWER": "computing"}]}
{"label": "SKILL", "pattern": [{"LOWER": "cybersecurity"}]}
```

Fig. 3. Entity Recognition and Custom Entity Ruler: Tagging and parsing job descriptions, with an entity ruler ensuring accurate classification of skills like "Python," "cloud," and "cybersecurity".

2.3 Profile Matching Using a Co-occurrence Graph

A Co-occurrence Graph matches user profiles with relevant job postings by highlighting the relationships between skills in job postings and user resumes. In this graph, nodes represent individual skills, while edges indicate how frequently these skills co-occur in job postings, helping to identify clusters of commonly sought-after skills.

- Graph Creation: The graph is generated from skills extracted from LinkedIn and Indeed job descriptions. Each node represents a specific skill, and the edges show connections based on how often skills co-occur in postings. For example, in Fig. 4, "DevOps" and "Cloud Services" are closely linked because they frequently appear together in IT job listings.

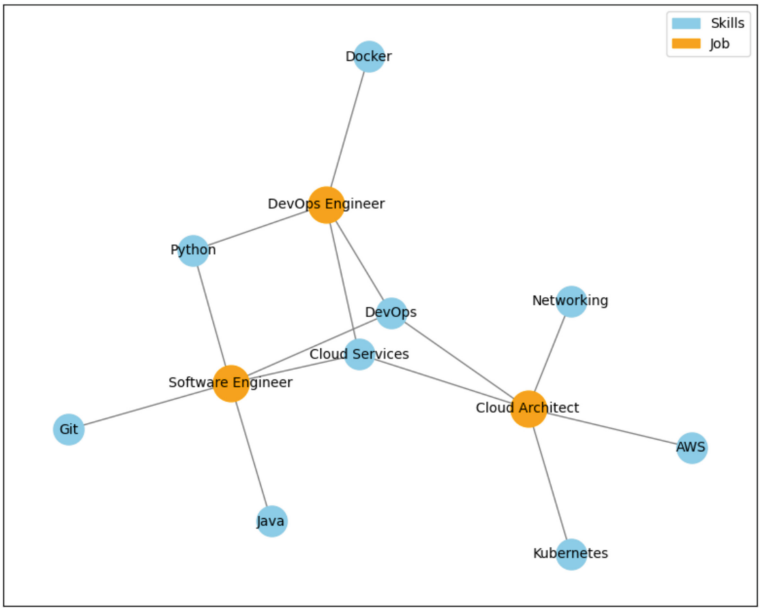

Fig. 4. Example of Co-occurrence Graph of skills and job titles.

- Profile Matching: After creating the Co-occurrence Graph, the system matches user profiles by analyzing the graph. It compares the skills in a user's resume with the nodes in the graph, considering the proximity and relevance of the user's skills to those in job postings. The result is a list of job postings that closely match the user's profile. Figure 5 shows a Co-occurrence Graph with User Profile Matching, illustrating the connections between skills (blue), job titles (orange), and the user profile (green). The user profile node is linked to relevant skills such as AWS, DevOps, and Python, aligning with the DevOps Engineer role. Each job title, such as Cloud Architect or Data Scientist, is connected to a distinct set of skills, demonstrating the typical requirements for each position

- Matching Score Algorithm: The matching score for each job is calculated based on the proximity of the user's skills to the skill clusters in the graph. Jobs with a higher overlap between the user's skills and the desired skill clusters receive a higher score. The algorithm is designed to weigh both the relevance and importance of each skill.

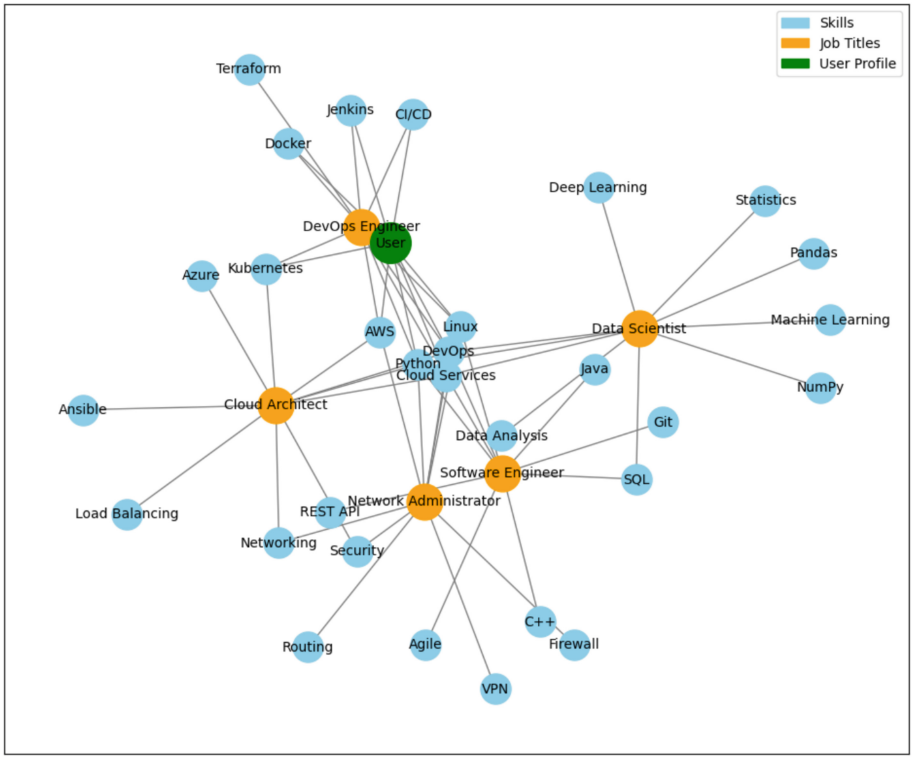

Fig. 5. Example of Co-occurrence Graph with user profile matching.

3 Experiments and Results

This section presents two experiments conducted to evaluate the method's performance. The first experiment measures the method's ability to extract skills from job descriptions by comparing its output to a ground truth dataset. The second experiment assesses the accuracy of matching candidate resumes to job descriptions using two techniques: Cosine Similarity and Co-occurrence Graphs.

3.1 Skill Extraction Performance

The first experiment evaluates the precision, recall, and F1 score of the method for extracting relevant skills from job descriptions. These performance metrics are adopted

from previous related studies [11, 12], which applied them to assess the performance of text classification and extraction models. The extracted skills are compared against a manually created ground truth dataset of 200 job descriptions to measure the accuracy of the skill extraction process. The experiment employs three evaluation metrics, detailed as follows:

1. Precision: The proportion of correctly extracted skills relative to all extracted skills.
2. Recall: The proportion of actual skills in the job descriptions that were correctly extracted.
3. F1 Score: The harmonic mean of precision and recall, which provides a balanced metric for assessing the system's performance.

The average results are 0.9900 for precision, 0.8035 for recall, and 0.8823 for the F1 score. To improve the recall score while maintaining high precision, we propose several enhancements. First, expanding our entity ruler with common skill variations and synonyms found in job descriptions (e.g., "ML" for "Machine Learning", "JS" for "JavaScript"). Second, implementing fuzzy matching to capture slight variations in skill descriptions. Third, using word embeddings to identify semantically similar skills that might be expressed differently. These improvements could help capture more valid skills without significantly impacting precision.

3.2 Comparison of Matching Performance

The second experiment aims to compare the effectiveness of two techniques: Cosine Similarity and Co-occurrence Graphs, in matching candidate resumes with job descriptions. The goal is to determine which method provides more accurate and relevant matches by utilizing multiple evaluation metrics, which are detailed as follows:

1. Precision at K (P@K): Measures the proportion of relevant matches within the top K results, as shown in Eq. (1), where K represents the number of top-ranked results considered for evaluation.

$$P@K = \frac{\text{Number of relevant matches in top K}}{K} \tag{1}$$

2. Mean Reciprocal Rank (MRR): Evaluates how quickly the first relevant result appears in the ranking, as shown in Eq. (2), where N is the total number of test cases, r is the rank position of the first relevant match in the i-th query or test case, and I is the index of the test case (ranging from 1 to N).

$$MRR = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{r_i} \tag{2}$$

3. Discounted Cumulative Gain (DCG): Measures the relevance of results in a ranking, assigning higher weight to results at the top, as shown in Eq. (3), where N is the total number of ranked results, i is the rank position in the result list (1-based index), and rel is the relevance score of the result at position i in the ranking.

$$DCG = \sum_{i=1}^{N}\frac{2^{\text{rel}_i} - 1}{\log_2(i+1)} \tag{3}$$

4. Normalized Discounted Cumulative Gain (NDCG): Assesses the quality of the ranking by giving more weight to relevant results that appear earlier in the list, as shown in Eq. (4), where IDCG is the Ideal Discounted Cumulative Gain, representing the maximum possible DCG for a perfect ranking.

$$NDCG = \frac{DCG}{IDCG} \tag{4}$$

The experiment was conducted using 625 job descriptions collected from LinkedIn and Indeed, using stratified sampling across 10 most popular job categories: Data Science, Java Developer, Python Developer, DevOps Engineer, Network Security Engineer, Database, Web Designing, Testing, ETL Developer, and Business Analyst. We matched 100 candidate resumes, distributed proportionally across these categories, with job descriptions using both Cosine Similarity and Co-occurrence Graph methods. Table 1 shows the comparison results, demonstrating that the Co-occurrence Graph consistently outperforms Cosine Similarity across all evaluation metrics, indicating better accuracy and ranking quality in job-skill matching.

Table 1. Comparison results between Cosine Similarity and Co-occurrence Graph.

Evaluation metrics	Cosine Similarity	Co-occurrence Graph
P@3	0.8000	0.9000
P@5	0.8400	0.8800
MRR@ 3	0.8833	1.0000
NDCG@ 3	0.8970	0.9920
MRR @ 5	0.8833	1.0000
NDCG @ 5	0.9092	0.9771

4 Conclusion

This research demonstrates the effectiveness of using Co-occurrence Graphs combined with NLP techniques to improve job-skill matching. Current job-matching platforms, such as LinkedIn and Indeed, face limitations in identifying hidden or unmentioned skills in job descriptions, leading to less efficient matches. By integrating SpaCy's Entity Ruler for skill extraction and using Co-occurrence Graphs to analyze frequently co-occurring skills, the proposed method significantly enhances the matching process. The experimental results show that the Co-occurrence Graph approach outperforms traditional methods such as Cosine Similarity across various metrics. While graph-based methods show promising results, they may face computational complexity challenges in real-world applications. Future work should focus on optimizing graph processing algorithms and implementing efficient data structures to handle large-scale deployments. This could include techniques such as graph partitioning, parallel processing, and dynamic

graph updates to maintain system performance as the dataset grows. Despite these challenges, this research offers a promising solution for more accurate and relevant job-skill matching, benefiting both job seekers and employers in the rapidly evolving IT sector.

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Risk Estimation with Active Labeling

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Abstract. We consider a setting where, for a given model, with a given labeling budget, we need to accurately evaluate its risk repeatedly, when the set of items considered for the risk evaluation, as well as the loss function, change over time. This is a natural setting in a non-stationary environment, such as that of large retail chains, whose catalogue changes year-round. Since evaluating risk often requires human judgement, the cost can increase dramatically over time. We propose a new estimator that minimize the labeling cost by reusing all available labels when possible and by actively selecting items to be labeled in an optimal way. We show that an optimal sampling profile can be derived, efficiently and at scale, as the solution of an optimization problem. We show how this approach with only a small added computational and storage cost, can efficiently reduce the labeling work required to measure the risk of a model in a non-stationary environment in a production system. We extend these results to the F_α measure and weighted risks. The presented approach is related to the Horvitz-Thompson estimator, importance sampling and active learning and provides a scalable, robust solution for risk evaluation in non-stationary environment that cannot be achieved with either Horvitz-Thompson estimator nor importance sampling.

1 Introduction

While evaluating the risk of a model is a well-studied problem, the systematic use of models in modern, complex systems, provides an interesting twist to it. Nowadays, a model that might have been conceived, and thus evaluated on a given dataset might be used in an environment where the data distribution changes overtime. At the same time, such systems are usually large scale, and thus require a careful management of the resources that are needed for estimating the model risk. This presents a new challenge: how to evaluate model risk in a changing context, at the lowest cost possible?

An example can be found in the eCommerce and the digital advertisement industries, where models can be used to extract metadata from products in a large catalog. Since catalogs changed daily, it's necessary to measure the risk of such model regularly. Since evaluating risk requires human judgment, it can be very expensive to repeat risk measurements daily.

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In this paper we present a framework that optimally leverages all data previously collected to estimate the model risk, in a non-stationary environment and selects new data to be collected, given a data collection cost, in order to minimize the variance of the risk. The cost minimization is obtained by simply solving a computationally efficient optimization problem, where the data necessary to solve the optimization problem can be compactly stored in a distributed fashion. Therefore, this technique can be scaled to very large dataset. We also extend the results to include F_α measure and weighted risks.

2 Problem Setting

We begin with the simple case where the risk estimation is over a fixed set of items. We will later introduce a framework that work under both changes in items, and changes in risk estimators over time.

Let X be the finite feature space that contains K items x_i for i in $1, \ldots, K$ and Y be the label space; an unknown distribution of the labels $p(y|x)$ is defined over $\mathrm{X} \times \mathrm{Y}$. Let $p\,(y|x;\theta)$ be a θ-parametrized model and let $f_\theta : \mathrm{X} \to \mathrm{Y}$ with $f_\theta(x) = \arg\max_y p(y|x;\theta)$ be the corresponding hypothesis. We are interested in estimating the risk

$$R_K := \frac{1}{K} \sum_{i=1}^{K} \int \ell(f_\theta(x_i), y) p(y|x_i) dy \tag{1}$$

of f_θ with respect to $p(y|x_i)$ where $\ell : \mathrm{Y} \times \mathrm{Y}$ is a loss function that measures the difference between the prediction and the true label. Since $p(y|x_i)$ is unknown, we typically use an empirical risk

$$\bar{R}_n = \frac{1}{n} \sum_{i=1}^{n} \ell(f_\theta(x_i), y_i), \tag{2}$$

where instances x_i are selected uniformly at random, and y_i are drawn from $p(y|x_i)$. In general, this is an expensive process that requires a human labeler to produce y_i. Therefore, we refer to n as the budget used for this risk estimation.

There are two main problems with using the risk estimator from Eq. 2. First, if different risk estimates are required either for different loss functions or for different subset of items, the cost of labeling a new sample of items can quickly become prohibitive. This is usually the case in environments where the set of items under evaluation is changing over time. The second problem is that the estimator does not leverage the model estimate $p(y|x;\theta)$ when sampling items to be labeled. Roughly speaking we would like to sample more often items where the model is more uncertain about the correct label.

We address both problems by introducing a new *randomized* estimator of the risk:

$$\hat{R}_{K,q} = \frac{1}{\sum_{i=1}^{K} \frac{Z_i}{q_i}} \sum_{i=1}^{K} \frac{Z_i}{q_i} \ell(f_\theta(x_i), y_i), \tag{3}$$

where Z_i are independent Bernoulli random variable with probability q_i. We call the estimator in Eq. 3, the *weighted risk estimator*. Later on, we will propose different ways to set the q_i's, and will study the resulting properties of the weighted risk estimator.

Notice that for a given choice of q_i's, the expected number of labels to be queried is $\sum_{i=1}^{N} q_i$. We refer to $B = \sum_{i=1}^{N} q_i$ as the labeling budget.

The estimate $\hat{R}_{K,q}$ given by Eq. 3 is a random variable as it depends both on the random variables Z_i, and on $p(y|x)$. Our goal is to find probabilities q_i that minimize the expected deviation from the risk R_K for a given budget.

Finally notice that the weighted risk estimator is closely related to the theory of importance sampling [15] and the Horvitz-Thompson estimator [10]. There are, however, important differences that make this approach better suited to address the challenge of a fast changing environment. Specifically, unlike the Horvitz-Thompson estimator, the denominator is a random variable. We will discuss the differences and the advantages of the estimator in Eq. 3 in Sect. 8.

3 Optimal Sampling Profile with Budget Constraint

We cast the optimal sampling profile problem as a minimization problem over the variance of the estimator with a budget constraint (see Sect. 7 for a cost minimization formulation). In Sect. 4.1, we will provide formal guarantees of the asymptotic behavior of the weighted estimator in a more general setting and provide a strong basis for the approximation made in this section, but for now we will proceed in a more informal way.

The variance $Var[\hat{R}_{K,q}]$ is technically not well defined because $\mathbb{P}(\sum_{i=1}^{K} \frac{Z_i}{q_i} = 0) > 0$ but we can arbitrarily assume that $q_1 = 1$ so that $\mathbb{P}(\sum_{i=1}^{K} \frac{Z_i}{q_i} = 0) = 0$. A close form solution of the variance is not available but we can approximate the ratio of two random variables using a Taylor expansion to obtain

$$Var\left[\hat{R}_{K,q}\right] \approx \frac{1}{K^2} \sum_{i=1}^{K} \frac{1}{q_i} \mathbb{E}\left[(\ell_i - R_K)^2\right] - (\mathbb{E}[\ell_i] - R_K)^2, \tag{4}$$

where $\ell_i = \ell(f_\theta(x_i), y_i)$ and the expectation is over $p(y|x)$. With this approximation the sampling probabilities q_i are the solution of the optimization problem

$$\begin{aligned} &\underset{q_i}{\text{minimize}} \sum_{i=1}^{K} \frac{c_i}{q_i} \\ &\text{subject to} \sum_{i=1}^{K} q_i = B, \quad 0 < q_i \leq 1,\ i = 1, \ldots, N, \end{aligned} \tag{5}$$

where $\ell_i = \ell(f_\theta(x_i), y_i)$ and $c_i = \mathbb{E}\left[(\ell_i - R_K)^2\right]$. This problem is convex and can be solved efficiently at scale(see Sect. 5) but the quantities c_i are unknown because R_K and $p(y|x)$ are not available.

As proposed in [19] we replace R_K with $R_{K,\theta} = \frac{1}{K}\sum_{i=1}^{K}\int \ell(f_\theta(x_i), y)\, p(y|x_i;\theta)dy$ and $p(y|x)$ with $p(y|x;\theta)$. If $p(y|x_i;\theta)$ is not available, we can set the c_i's to be the same arbitrary constant and as expected obtain a uniform optimal sampling profile. Even in this trivial case, this approach is still valuable in a non stationary situation as shown in Sect. 4.

3.1 Computing c_i's in two special cases

For the $\ell_{0/1}$ loss (see [19] for a derivation) $c_i \approx (1 - p(f_\theta(x_i)|x_i;\theta)(1 - 2R_{K,\theta})) + R^2_{K,\theta}$. In this case we have higher value of c_i and therefore higher sampling probabilities when the points are more uncertain (closer to the decision boundary). It is also interesting to notice that if the model has a value of $R_{K,\theta} = \frac{1}{2}$, the sampling profile becomes flat again because the model does a poor job of predicting the labels.

For squared loss for regression, assuming that the model is such that $p(y|x_i;\theta) = N(f_\theta(x), \sigma_x^2)$, $c_i \approx (3\sigma_x^2 - 2R_{K,\theta})\sigma_x^2 + R^2_{K,\theta}$ (see [19] for a derivation).

Algorithm 1. Risk estimation

Input Model parameter θ, loss function ℓ and budget B
Output Risk estimate

1: Compute c_i as in Section 3.1 or $c_i = 1$ if $p(y|x,\theta)$ is not available
2: Compute optimal sampling by solving problem 5
3: **for** $i = 1, \ldots, K$ **do**
4: Draw $z_i \sim \text{Bern}(q_i)$
5: **if** $z_i = 1$ **then**
6: Query label y_i
7: **end if**
8: **end for**
9: **return** $\frac{1}{\sum_{i=1}^{K}\frac{z_i}{q_i}}\sum_{i=1}^{K}\frac{z_i}{q_i}\ell(f_\theta(x_i), y_i)$

Algorithm 1 describes the step needed to estimate the risk of a model for a given budget B.

3.2 Confidence Interval

It is essential to provide a confidence interval for the estimate $\hat{R}_{K,q}$. We use the empirical estimate of the variance of $\hat{R}_{K,q}$, $S_K^2 = \frac{1}{\left(\sum_j \frac{1}{q_j}\right)^2}\sum_j \frac{1}{q_j^2}\left(\ell(f_\theta(x_j), y_j)) - \hat{R}_{K,q}\right)^2$, where j's are the n indices that were selected for the estimation ($Z_j = 1$). Using the variance estimate we can then easily compute a symmetric confidence interval around our estimate. This expression is closely related to the one of the self-normalized importance sampling variance estimate (see [15]).

3.3 Convergence

To justify the approximation in Eq. 4, we look at the limiting distribution for $K \to \infty$ when $K \to \infty$ and $q_i > \epsilon$ for all i's and the losses are bounded: $|\ell_i| \leq D$ where ϵ and D are two positive constants.

Theorem 1. *Let $\hat{R}_{q,K}$ be as defined in Eq. 3. Suppose that $q_i > \epsilon > 0$, the loss function is bounded $|\ell_i| \leq D$, $\lim_{K\to\infty} Var\left[\sum_{i=1}^{K} \frac{Z_i}{q_i}\ell_i\right] = +\infty$ and the limits*

$$\begin{aligned} &\lim_{K\to\infty} R_K, \quad \lim_{K\to\infty} \frac{1}{K}\sum_{i=1}^{K} Var\left[\frac{Z_i}{q_i}\ell_i\right], \\ &\lim_{K\to\infty} \frac{1}{K}\sum_{i=1}^{K} Var\left[\frac{Z_i}{q_i}\right] \end{aligned} \tag{6}$$

exist. Then

$$\begin{aligned} &\sqrt{K}\hat{R}_{K,q} \xrightarrow{d} \\ &\mathbb{N}\left(R, \lim_{K\to\infty} \frac{1}{K}\sum_{i=1}^{K} \frac{1}{q_i}\mathbb{E}\left[(\ell_i - R)^2\right] - (\mathbb{E}[\ell_i] - R)^2\right), \end{aligned} \tag{7}$$

where $R = \lim_{K\to\infty} R_K$.

Notice how this theorem not only shows the asymptotically normal behavior of the estimator but shows how the variance approximation used is in the limit the correct variance.

To simplify notation we write $\ell_i = \ell(f_\theta(x_i), y_i)$ and $W_i = \frac{Z_i}{q_i}$ where Z_i are Bernoulli random variable with probability q_i.

We first prove

Lemma 1. *Let $F_{q,K}$ be defined as $F_{q,K} = \frac{1}{K}\sum_{i=1}^{K} W_i\left(\alpha + \beta\ell_i\right)$, where $\alpha, \beta \in \mathbb{R}$, and let R_K be defined as in Eq. 1. For any $\alpha \neq 0$ and $\beta \neq 0$, if $q_i > \epsilon$ and the loss function is bounded $|\ell_i| < D$, then,*

$$\frac{F_{q,K} - (\alpha + \beta R_K)}{\sqrt{Var\left[F_{q,K}\right]}} \xrightarrow{d} \mathbb{N}(0,1) \tag{8}$$

for $K \to \infty$ provided $\lim_{K\to\infty}\ KVar\left[F_{q,K}\right] = +\infty$.

Proof. Since $|W_i\left(\alpha + \beta\ell_i\right)| \leq \frac{1}{\epsilon}(|\alpha| + |\beta|D)$, the terms $W_i\left(\alpha + \beta\ell_i\right)$ are bounded. Simply applying Corollary 2.7.1 in [12], we get the desired result.

Proof. Notice that condition 6 from Theorem 1 implies

$$\lim_{K\to\infty} KVar\left[F_{q,K}\right] = +\infty,$$

and therefore under the conditions of Theorem 1, we can use the results of Lemma 1.

Since we know that

$$\Sigma_{\alpha,\beta} = \lim_{k\to\infty} \frac{1}{K} \sum_{i}^{K} Var\left[W_i\left(\alpha + \beta\ell_i\right)\right]$$

exists, Lemma 1 implies that

$$\sqrt{K}\left(F_{q,K} - (\alpha + \beta R_K)\right) \xrightarrow{d} \mathbb{N}(0, \Sigma_{\alpha,\beta}). \tag{9}$$

As a direct application of Cramer-Wold Theorem(5.1.8 in [12]) and result 9 we get that

$$\sqrt{K}\left(\frac{1}{K}\sum_{i=1}^{K}(W_i\ell_i, W_i) - (R_k, 1)\right) \xrightarrow{d} \mathbb{N}((0,0), \Sigma) \tag{10}$$

where

$$\begin{aligned}\Sigma &= \lim_{K\to\infty} \frac{1}{K} \sum_{i}^{K} \begin{bmatrix} Var\left[W_i\ell_i\right] & Cov\left[W_i\ell_i, W_i\right] \\ Cov\left[W_i\ell_i, W_i\right] & Var\left[W_i\right]. \end{bmatrix} \\ &= \lim_{K\to\infty} \frac{1}{K} \sum_{i}^{K} \begin{bmatrix} Var\left[W_i\ell_i\right] & Cov\left[W_i\ell_i, W_i\right] \\ Cov\left[W_i\ell_i, W_i\right] & Var\left[W_i\right]. \end{bmatrix}\end{aligned}$$

Equation 10 implies that $\frac{1}{\sqrt{K}}\sum_{i=1}^{K}(W_i\ell_i, W_i) \xrightarrow{d} \mathbb{N}((R,1), \Sigma)$, where $R = \lim_{K\to\infty} R_K$.

Applying the multivariate Delta method to the function $g(x,y) = \frac{x}{y}$ with $x = \frac{1}{K}\sum_{i=1}^{K} W_i\ell_i$ and $y = \frac{1}{K}\sum_{i=1}^{K} W_i$, we conclude that

$$\sqrt{K}\hat{R}_{K,q} = \frac{1}{\sqrt{K}}\sum_{i=1}^{K} g(W_i\ell_i, W_i) \xrightarrow{d} \mathbb{N}(R, \nabla g^T \Sigma \nabla \text{ g})$$

Since

$$\begin{aligned}\nabla g^T \Sigma \nabla g &= [1, -R]^T \Sigma_K [1, -R] \\ &= \lim_{K\to\infty} \frac{1}{K} \sum_{i}^{K} Var\left[W_i\ell_i\right] \\ &- 2R \lim_{K\to\infty} \frac{1}{K} \sum_{i}^{K} Cov\left[W_i\ell_i, W_i\right] \\ &+ R^2 \lim_{K\to\infty} \frac{1}{K} \sum_{i}^{K} Var\left[W_i\right] \\ &= \lim_{K\to\infty} \frac{1}{K} \sum_{i=1}^{K} \frac{1}{q_i} \mathbb{E}\left[(\ell_i - R)^2\right] - (\mathbb{E}[\ell_i] - R)^2,\end{aligned}$$

we finally get

$$\sqrt{K}\hat{R}_{K,q} \xrightarrow{d} \mathbb{N}\left(R, \lim_{K\to\infty} \frac{1}{K}\sum_{i=1}^{K} \frac{1}{q_i}\mathbb{E}\left[(\ell_i - R)^2\right] - (\mathbb{E}[\ell_i] - R)^2\right).$$

4 Non-stationary Scenario

We now consider the situation where the set of items over which the risk is estimated changes over time. Let X_t denote the set of items at time t defined as $\mathrm{X}_t = \{x_i | i \in \mathrm{K}_t\}$ where K_t is the set of indices of items at time t with size K_t. We also assume that the set of labels Y is not changing over time. At time t, we want to measure the risk of a model using a loss function ℓ_t that depends on t for X_t using a given budget B_t. We also want to leverage all the labels that were acquired at previous times.

Let $q_{i,t}$ denote the probability that item x_i is sampled at time t. We let $z_{i,t}$ denote the outcome of the Bernoulli trial with probability $q_{i,t}$. Every time an item x_i is sampled with probability $q_{i,t}$, we store $q_{i,t}$ and if the item is selected, we query and store the label y_i.

If we sample an item at time t with probability $q_{i,t}^{\mathrm{inc}}$, and at time $t-1$ it was sampled with probability $q_{i,t-1}$ we have that the overall probability of the item being sampled at time t is

$$q_{i,t} = q_{i,t-1} + (1 - q_{i,t-1})q_{i,t}^{\mathrm{inc}}. \tag{11}$$

So at time t we can achieve a overall probability $q_{i,t} \geq q_{i,t-1}$ by sampling it with probability $q_{i,t}^{\mathrm{inc}} = \frac{q_{i,t} - q_{i,t-1}}{1 - q_{i,t-1}}$. At time t, we would like to find the optimal probabilities $q_{i,t}^{\mathrm{inc}}$ so that the $q_{i,t}$ minimize the variance of the estimator. This mechanism reuses the previously collected set of labels and their sampling probabilities in their entirety.

At time t, it might be possible to estimate the risk for a given loss ℓ_t without using any additional budget as long as $q_{i,t-1} > 0$ for all $i \in \mathrm{K}_t$. Equation 3 with probabilities $q_{i,t-1}$ would produce our estimate. The given budget B_t can still be used to reduce the variance of the estimate.

To minimize the variance of the estimator at time t for ℓ_t over the $q_{i,t}^{\mathrm{inc}}$ for given $q_{i,t-1}$ and budget B_t using Eq. 11, we solve the problem

$$\begin{aligned} &\underset{q_{i,t}^{\mathrm{inc}}}{\text{minimize}} \sum_{i\in\mathrm{K}_t} \frac{c_i}{q_{i,t-1} + (1 - q_{i,t-1})q_{i,t}^{\mathrm{inc}}} \\ &\text{subject to} \sum_{i=1}^{K} (1 - q_{i,t-1})q_{i,t}^{\mathrm{inc}} = B_t, 0 < q_{i,t}^{\mathrm{inc}} \leq 1, \quad i \in \mathrm{K}_t. \end{aligned} \tag{12}$$

Notice how the constraint on the budget is not in the form $\sum_{i=1}^{K} q_{i,t}^{\mathrm{inc}} = B_k$ to account for the existing probabilities $q_{i,t-1}$. Problem 12 is similar to problem 5 and can also be solved efficiently as described in Sect. 5.

The advantage of this approach is that at any given time t, we optimally reuse all the available labels independently of the risk we are trying to estimate and the items in the current set X_t. This is achieved at the added cost of solving problem 12 which has a linear complexity in the size of X_t and at the cost of storing a set of probabilities $q_{i,t}$ which takes space or order $\mathcal{O}(K_t)$. Notice that to solve problem 12, only the $q_{i,t-1}$ were used and therefore previously used probabilities can be discarded, so that the storage requirement is only linear in the total number of items. Both these added cost are negligible compared to the cost of acquiring more labels in most scenarios of interest. Algorithm 2 describes the steps needed to compute the risk at time t.

Algorithm 2. Non-stationary risk estimation at time t

Input Model parameter θ, loss function ℓ_t, budget B_t, probabilities $q_{i,t-1}$, and $z_{i,t-1}$, all labels collected up to time $t-1$

Output Risk estimate

Compute optimal incremental sampling $q_{i,t}^{\text{inc}}$ by solving problem 12
for $i \in \mathrm{K}_t$ **do**
  Update $q_{i,t} = q_{i,t-1} + (1 - q_{i,t-1})q_{i,t}^{\text{inc}}$
  Draw $z_{i,t}^{\text{inc}} \sim \text{Bern}(q_{i,t}^{\text{inc}})$
  Update $z_{i,t} = \max(z_{i,t}^{\text{inc}}, z_{i,t-1})$
  if $z_{i,t} = 1$ and $z_{i,t-1} = 0$ **then**
    Query label y_i
  end if
end for
return $\frac{1}{\sum_{i=1}^{K_t} \frac{z_{i,t}}{q_{i,t}}} \sum_{i=1}^{K_t} \frac{z_{i,t}}{q_{i,t}} \ell(f_\theta(x_i), y_i)$

4.1 Convergence for the Non-Stationary Case

We are interested in analyzing the asymptotic behavior for $K_t \to \infty$, to give a formal justification to the approximation Eq. 4 and to guarantee the soundness of the proposed estimator. For simplicity, we consider the case where a single new item is added at each time t, a new set of probabilities $q_{i,t}$ is used, and the loss function does not change. It is fairly straightforward to extend this result for the case where the loss function is changing as well.

Theorem 2. *Let* $\hat{R}_{q,t}$ *be* $\frac{1}{\sum_{i=1}^{t} \frac{Z_{i,t}}{q_{i,t}}} \sum_{i=1}^{t} \frac{Z_{i,t}}{q_{i,t}} \ell_i$, *where the* $Z_{i,t}$ *are Bernoulli distributed random variable with probability* $q_{i,t}$. *Suppose that* $q_{i,t} > \epsilon \not> 0$, *the loss function is bounded* $|\ell_i| < D$, $\lim_{t\to\infty} Var\left[\sum_{i=1}^{t} \frac{Z_{i,t}}{q_{i,t}} \ell_i\right] = +\infty$, *and the limits* $\lim_{t\to\infty} R_t$, $\lim_{t\to\infty} \frac{1}{t} \sum_{i=1}^{t} Var\left[\frac{Z_{i,t}}{q_{i,t}} \ell_i\right]$,

$\lim_{t\to\infty} \frac{1}{t}\sum_{i=1}^{t} Var\left[\frac{Z_{i,t}}{q_{i,t}}\right]$ *exist. Then*

$$\sqrt{t}\hat{R}_{q,t} \xrightarrow{d} \mathbb{N}\left(R, \lim_{t\to\infty}\frac{1}{t}\sum_{i=1}^{t}\frac{1}{q_{i,t}}\mathbb{E}\left[(\ell_i - R)^2\right] - (\mathbb{E}[\ell_i] - R)^2\right),$$

for $t \to \infty$*, where* $R = \lim_{t\to\infty} R_t$*.*

As with Theorem 1, this theorem not only shows the asymptotically normal behavior of the estimator but shows how the variance approximation used is in the limit the correct variance.

To simplify notation we write $\ell_i = \ell(f_\theta(x_i), y_i)$ and $W_{i,t} = \frac{Z_{i,t}}{q_{i,t}}$ where $Z_{i,t}$ are Bernoulli random variables with probability $q_{i,t}$. We first prove the following lemma:

Lemma 2. *Let* $F_{q,t}$ *be defined as*

$$F_{q,t} = \frac{1}{t}\sum_{i=1}^{t} W_{i,t}\left(\alpha + \beta\ell_i\right),$$

where Z_i *are Bernoulli distributed random variable with probability* q_i*,* $\alpha, \beta \in \mathbb{R}$*, and let* R_t *be defined as in Eq. 1. For any* $\alpha \neq 0$ *and* $\beta \neq 0$*, if* $q_i > \epsilon > 0$ *and the loss function is bounded* $|\ell_i| < D$*, then,*

$$\frac{F_{q,t} - (\alpha + \beta R_t)}{\sqrt{Var\left[F_{q,t}\right]}} \xrightarrow{d} \mathbb{N}(0,1)$$

for $t \to \infty$ *provided* $tVar\left[F_{q,t}\right] \to \infty$*.*

Proof. Since $|W_{i,t}\left(\alpha + \beta\ell_i\right)| \leq \frac{1}{\epsilon}(|\alpha| + |\beta|D)$, we have

$$\begin{aligned}
&\mathbb{E}\left[\left|\sum_{i=1}^{t} W_{i,t}\left(\alpha + \beta\ell_i\right) - \mathbb{E}\left[W_{i,t}\left(\alpha + \beta\ell_i\right)\right]\right|^3\right] \\
&\leq \frac{2}{\epsilon}(|\alpha| + |\beta|D)\mathbb{E}\left[\left|\sum_{i=1}^{t} W_{i,t}\left(\alpha + \beta\ell_i\right) - \mathbb{E}\left[W_{i,t}\left(\alpha + \beta\ell_i\right)\right]\right|^2\right] \\
&= \frac{2}{\epsilon}(|\alpha| + |\beta|D)\sum_{i=1}^{t} Var\left[W_{i,t}\left(\alpha + \beta\ell_i\right)\right] \\
&= \frac{2}{\epsilon}(|\alpha| + |\beta|D)tVar\left[F_{q,t}\right].
\end{aligned}$$

Since $tVar\left[F_{q,t}\right] \to \infty$ we have that

$$\begin{aligned}
&\left(\mathbb{E}\left[\left|\sum_{i=1}^{t} W_{i,t}\left(\alpha + \beta\ell_i\right) - \mathbb{E}\left[W_{i,t}\left(\alpha + \beta\ell_i\right)\right]\right|^3\right]\right)^2 \\
&= o\left(\left(tVar\left[F_{q,t}\right]\right)^3\right)
\end{aligned}$$

Using Theorem 2.7.2 in [12], we get the desired result.

The rest of the proof proceeds in the same way as for the statinoary case and we omit it here for brevity.

5 Solving Problem Efficiently

We consider here the problem

$$\begin{aligned} &\underset{x_i}{\text{minimize}} \sum_{i}^{K} \frac{c_i}{a_i + (1-a_i)x_i} \\ &\text{subject to } \sum_{i=1}^{K}(1-a_i)x_i = B, 0 \le x_i \le 1,\ i = 1, \dots, K, \end{aligned} \tag{13}$$

where $c_i > 0$ and $0 \le a_i \le 1$ for $i = 1, \dots, K$. Problem 13 is a generalization of both problem 5and problem 12, is convex (see [5]) with KKT conditions (see supplementary material for derivation)

$$B = \sum_{i=1}^{K}(1-a_i)x_i, \quad x_i = \begin{cases} 1 & \nu < c_i \\ \frac{\sqrt{\frac{c_i}{\nu}} - a_i}{1-a_i} & c_i \le \nu \le \frac{c_i}{a_i^2} \\ 0 & \nu > \frac{c_i}{a_i^2} \end{cases} \tag{14}$$

for $i = 1, \dots, K$ where $\nu \in \mathbb{R}$ is the equality constraint multiplier.

Using bisection over ν, we can compute the solution efficiently with a complexity that is linear in the number of items K. Moreover the required storage to solve the problem is again of order $O(K)$. Specifically while solving the problem is only necessary to store the coefficients a_i and c_i in memory that have size K each. However, it is not necessary to store the x_i separately but it is enough to store their weighted sum $\sum_{i=1}^{K}(1-a_i)x_i$. Therefore this problem can be solved efficiently for extremely large K.

6 F_α-measure and Weighted risk

We can extend the risk defined in Eq. 1 to what we call a generalized risk similar to the one introduced in [17], defined as

$$G_K = \frac{\sum_{i=1}^{K} \int \ell(f_\theta(x_i), y) w(x_i, y, f_\theta) p(y|x_i) dy}{\sum_{i=1}^{K} \int w(x_i, y, f_\theta) p(y|x_i) dy}, \tag{15}$$

where $w(x_i, y, f_\theta)$ is a weight function that assign different weights to each item. Clearly for $w(x_i, y, f_\theta) = 1$ we obtain the standard risk defined in Eq. 1. Moreover if w is only a function of x_i, G_K is simply a weighted risk estimate where each item is assigned a weight. As we will see, G_K can also be seen as a generalization of the F_α measure (see [16]).

If we define

$$\bar{G}_n = \frac{\sum_{i=1}^{n} \ell(f_\theta(x_i), y_i) w(x_i, y_i, f_\theta)}{\sum_{i=1}^{K} w(x_i, y, f_\theta) p(y|x_i)}, \tag{16}$$

where the x_i are sampled uniformly at random from the overall population, it can be shown that $\bar{G}_n$ is a consistent estimator of G_K (see [17]). For a binary classifier the F_α-measure [16] is defined as $F_\alpha = \frac{n_{tp}}{\alpha(n_{tp}+n_{fp})+(1-\alpha)(n_{tp}+n_{fn})}$ where n_{tp} and n_{fp} are the number of true and false positive and n_{fn} is the number of false negative. As shown in [17], the F_α measure is a special case of $\bar{G}_n$ for $w(x_i, y, f_\theta) = \alpha f_\theta(x) + (1-\alpha)y$ and $\ell = 1 - \ell_{0/1}$. So estimating G_K can be thought of as generalization of measuring F_α for a given model.

In the same spirit of previous sections, both to try to reduce the variance of the estimate of G_K and to be able to efficiently reuse labels acquired over time we can define a new estimator of G_K:

$$\hat{G}_{K,q} = \frac{1}{\sum_{i=1}^{K} \frac{Z_i}{q_i} w(x_i, y_i, f_\theta)} \sum_{i=1}^{K} \frac{Z_i}{q_i} w(x_i, y_i, f_\theta) \ell(f_\theta(x_i), y_i), \tag{17}$$

where the Z_i are independent Bernoulli variables with probability $q_i > 0$ and y_i are drawn from $p(y|x_i)$. It is easy to extend the results from the previous sections, including the convergence results, for this new estimator. In particular, the variance $Var\left[\hat{G}_{K,q}\right]$ is approximately:

$$\frac{\sum_{i=1}^{K} \frac{1}{q_i} \mathbb{E}\left[(w_i \ell_i - w_i G_{K,q})^2\right] - (\mathbb{E}[w_i \ell_i] - \mathbb{E}[w_i] G_{K,q})^2}{\left(\sum_{i=1}^{K} \mathbb{E}[w_i]\right)^2},$$

where $w_i = w(x_i, y_i, f_\theta)$. So to find the optimal probabilities q_i we can solve the same problem 12 where $c_i = \mathbb{E}\left[(w_i \ell_i - w_i G_K)^2\right]$. As before, we need to use an approximation for c_i by replacing $p(y|x)$ with $p(y|x, \theta)$ (see [17]).

For a weighted risk estimation, we have that $w(x_i, y_i, f_\theta)$ depends only on x_i, and we can write $c_i = w_i^2 \mathbb{E}\left[(\ell_i - G_K)^2\right]$. If $p(y|x, \theta)$ is not available, we can further simplify to $c_i = w_i^2$.

7 Budget Minimization for a Target Variance of the Estimator

So far, we have casted the problem as a variance minimization with budget constraint. It is straightforward to cast the problem as a budget minimization with a constraint on the variance to obtain

$$\begin{aligned} &\underset{q_{i,t}^{\text{inc}}}{\text{minimize}} \sum_{i=1}^{K} (1 - q_{i,t-1}) q_{i,t}^{\text{inc}} \\ &\text{subject to} \sum_{i \in \mathrm{K}_t} \frac{c_i}{q_{i,t-1} + (1 - q_{i,t-1}) q_{i,t}^{\text{inc}}} \leq V, \\ &\quad 0 < q_{i,t}^{\text{inc}} \leq 1, \quad i \in \mathrm{K}_t, \end{aligned}$$

where $c_i = \mathbb{E}\left[(\ell_i - R_K)^2\right]$, $V = \sigma_{\max}^2 K^2 + (\mathbb{E}[\ell_i] - R_K)^2$, and $\sigma_{\max}$ is the maximum allowed standard deviation of the estimate. As in the previous section, we practically replace R_K with $R_{K,\theta}$ and $p(y|x)$ with $p(y|x;\theta)$. This problem can also be solved very efficiently but we omit the discussion here due to space constraints.

8 Related Work

The proposed estimator is related to the Horvitz-Thompson estimator [10] and importance sampling but has fundamental differences.

8.1 Horvitz-Thompson

The Horvitz-Thompson (HT) estimator has been used in many areas including reinforcement learning [21], multi-arm bandits [1], active learning [2–4,11,20]. The HT estimator is unbiased but in this setting has in general a higher variance and can easily produce estimate of the risk that are inconsistent [6,15,19]. The Neyman allocation [8,14] can be used in this context to pick the probability profile. In [13], the HT estimator is used to evaluate future aggregate queries as accurately as possible. A similar optimization problem is formulated to minimize the variance of the estimate. The optimization problem obtained is similar to the one presented here but constants and constraints are different, producing different sampling profiles. This is because the HT estimator is different than the proposed one.

8.2 Importance Sampling

There is also a close connection between the presented approach and importance sampling [9] applied to risk estimation as, for example, in [17–19]. These approaches, as shown in this section, are very similar when the risk estimation is done once for a given budget. We are not familiar with any extension of the importance sampling approach to the non-stationary scenario where both the loss functions and the items under measurements change over time.

When using importance sampling for risk estimation, items in $\mathcal{X}$ are sampled with replacement using a pre-selected distribution D until the budget is completely used. For each sampled item x_i, labels are queried and then the risk is estimated as $\mathrm{R}^{\mathrm{is}} = \frac{1}{\sum_{i=1}^{n} \frac{1}{d_i}} \sum_{i=1}^{n} \frac{1}{d_i} \ell(f_\theta(x_i), y_i)$ where n is the budget, d_i is the probability of sampling item x_i under the distribution D with $\sum_{i=1}^{K} d_i = 1$. Notice how the importance sampling estimate R^{is} is identical to the one used in Algorithm 1 where d_i is replaced by q_i. The optimal distribution D that is obtained by minimizing the variance of the asymptotic limit of the estimator (see [19]) is such that $d_i \propto \sqrt{\mathbb{E}\left[(\ell_i - R)^2\right]}$. Looking at the KKT conditions 14 of problem 5for the stationary case ($a_i = 0$), we see that if the budget is small compared to K $q_i \propto \sqrt{c_i} = \sqrt{\mathbb{E}\left[(\ell_i - R)^2\right]}$ which is the same dependency of the

importance sampling probability distribution. Therefore for a budget B small compared to the total number of items K, the importance sampling estimator and $\hat{R}_q$ coincide.

In the non-stationary case, to our knowledge, there is no simple way to extend importance sampling to deal with pre-existing labels and changing items. To understand the reason, consider a situation where we have K items from which we sample n_1 items with replacement using some distribution D_1 and later we sample n_2 items using distribution D_2. In general, there is no distribution D such that sampling $n_1 + n_2$ items with replacement from it is equivalent to the two sampling procedures performed earlier. This makes it hard to store all the previous sampling that occurs over time and to cast the problem by factoring in all the previous history.

On the other hand with the kind of sampling performed in Eq. 3, no matter how many samplings are performed over time, we can define an equivalent sampling by simply compounding all the individual item probabilities as in Eq. 11. As shown in Sect. 4, this makes the extension to the non-stationary case with multiple loss functions natural. Moreover, in order to store the sampling history, our proposed approach requires to store one single number per item ($q_{i,t}$) separately from all other items, so that the information recorded is local to each item. This "decoupling" between items enables an easy parallel implementation.

8.3 Active Learning

There is also a close connection between this work and active learning. This approach to risk estimation is the "dual" problem of active learning. In active learning, the goal of the selection process is to minimize the variance of the predictions or the variance of the model parameters while in this work we seek to minimize the variance of the risk estimate. In [2,7,11], similar to the approach taken here, an active learning strategy is derived under the assumption that the model provides a good approximation of $p(y|x)$. In [11], the Horvitz-Thompson estimator is used as the base to derive the sampling strategy. In active learning is also common to compensate for the bias incurred in training by the active learning sampling, by using importance weighting as in [2–4,11,20].

9 Empirical Results

In this section we look at the performance of the proposed approach under three different scenarios and we benchmark the time required to solve Problem 12 We mainly focus our attention on the non-stationary case, since, as discussed in Sect. 8, the proposed approach is, by all practical means, identical to the one in [19]. The reader should therefore look at the results in [17,19] as a reference for this case.

9.1 Non Stationary Sampling Profile and Uniform Sampling

In this example we consider a situation where at time t, n_t items are added following the sequence $[10000, 2000, 2000, 2000, 2000, 2000, 2000, 2000]$. We have an initial set of 10000 items followed by smaller set of items at each time. The available budget at each time is $[100, 20, 50, 30, 20, 40, 30, 60]$, and we measure the risk of a model with no available $p(y|x, \theta)$. The synthetic data was created such that the actual accuracy of each batch of items follows the sequence $[0.9, 0.7, 0.6, 0.99, 0.6, 0.8, 0.7, 0.65]$. At each iteration we use the budget by solving Problem 12, we then sample and measure the absolute deviation between the risk estimate and the true risk over 1000 sampling iterations. Figure 1 shows on the continuous line the average absolute deviation of the risk where the confidence interval are the standard error together with the total budget used for each time step. In the case of uniform sampling, we find the required budget that achieves 'similar' average absolute deviation as the active sampling. For the given budget we show in Fig. 1 with a dashed line the average absolute deviation from the true risk which by construction is close to the active strategy and the required total budget at each time step. The total budget for the uniform sampling grows roughly quadratically compared to the linear growth of the active approach. This is expected as explained using an idealized scenario in Sect. 9.3 Notice that in this example the obtained sampling profile for the active sampling is not uniform. Nevertheless, our proposed approach leverages all existing data to minimize the variance of the next required measure.

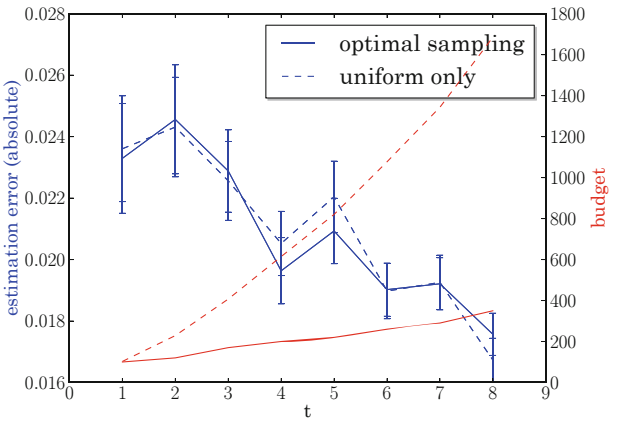

Fig. 1. Non stationary risk estimation. Weigthed risk estimate compared to uniform sampling over time. The errors are in blue and the budgets are in red. (Color figure online)

9.2 Reusing Labels and Uniform Sampling

In this example we look at what happens when the sampling probabilities $q_{i,t}$ are selected to be identical in order to satisfy the budget constraint. Specifically, we compare three different ways of computing the risk. The baseline is uniform sampling of items with the risk computed as in Eq. 2. In the second approach the risk is computed using Eq. 3 but $q_{i,t}$ are all the same such that $\sum_i q_{i,t} = B_t$. The third approach is the one described in this paper where risk is computed using Eq. 3 and the budget is used by solving Problem 12.

We compute the $\ell_{0/1}$ risk for a model that doesn't provide any estimate of $p(y|x_i;\theta)$ and therefore the c_i in 12 are set to 1. The model has a true risk on the first 400k items of 0.1 and of 0.4 on the next 200k items. It therefore has an overall risk of 0.2 on the total 600k items. Notice that the specific of the model in this case are irrelevant because we are neglecting $p(y|x_i;\theta)$ and to perform the simulation is enough to know the true risk of the model on the data.

Risk is measured on the initial 400k items by sampling each items with a probability $q_{i,1} = \frac{2000}{400000}$ so that in expectation the budget used is 2000. We then measure risk on the entire 600k items using a budget B_2. In one case we discard the previously collected labels and sample uniformly to use the budget. In the second case we sample items with probabilities $q_{i,2} = \frac{B}{600000}$ and then use Eq. 3 to compute the risk. In the third and optimal, case we solve Problem 12 and use Eq. 3. Figure 2 shows the average absolute error between the actual risk of the model and the estimate over 100 iterations for different budgets B_2. The bars correspond to the standard error. The optimal sampling profile outperforms both the uniform sampling and the weighted estimator from Eq. 3 where the sampling probabilities are selected suboptimally to be all the same.

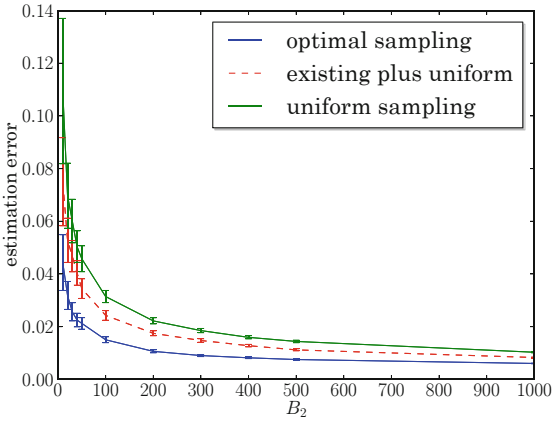

Fig. 2. Binary classifier risk estimation with non stationary scenario. Average absolute estimate error vs. extra budget B_2.

9.3 Idealized Scenario Comparison

To understand the differences in labeling cost between the proposed estimator and the classic approach to risk estimation, we construct an idealized scenario. We will see how the classic approach with uniform sampling has a quadratic labeling cost but the presented approach has only a linear one.

Let us assume that at regular time intervals a fixed amount of items n is added and a fix budget $B = B_t$ is available for labeling. We also assume that the model is not providing $p(y|x, \theta)$ and therefore each item looks the same from the point of view of the model under evaluation. It is easy to see that by solving Problem 13 at each time interval, we will always have a flat sampling profile with $q_i = \frac{B}{n}$ for all time intervals. On the other hand, if at each time interval, we set out to perform a uniform sampling that will give the same estimate variance as the proposed approach, at time t we would need to sample Bt items and therefore at time t the total budget used would be $\frac{t(t+1)}{2}B$. This is because we have to drop all the labels acquired up to time $t-1$ when evaluating risk at time t.

In this example, it might look easy to reduce the budget used during the uniform sampling by sampling uniformly only from the new items and then using all the labels obtained for risk estimation. In practice though, it would be very hard to do. One of the reason is that the budget might not be enough to obtain an equal sampling rate on the new items. Moreover, if risk estimates are performed regularly on multiple subsets of the entire set of items, it becomes impossible to reuse all labels by simply trying to obtain an effective uniform sampling on all items.

9.4 Timing Optimization Problem Solver

For this benchmark we implemented bisection algorithm presented in the paper in Python using Numpy. The actual implementation is only a few lines of code. We run it on a laptop with Intel Core i7 with 2.6G and 16G of memory. Our implementation uses a single core. Table 1 shows the time in seconds required to solve the problem for different sizes. As expected, we have roughly a linear dependence over the size of the problems and we quickly solve extremely large problems even with this simple implementation on standard hardware. Notice that this benchmark doesn't include the time required to compute the coefficients c_i that strongly depends on the kind of model under consideration.

Table 1. Solver timing

Number of items	Time (secs)
10K	0.0046
100K	0.041
1M	0.53
10M	7.2
100M	80.9

10 Conclusions

We have introduced a new estimator for computing the risk when labels are costly and we extended it for weighted risk and F_α measure estimation. This estimator has similar properties to the one introduced in [17,19] but has two main advantages compared to it. First, it works in a non stationary situation where multiple estimations are performed for different loss functions and where the items under measurement change over time. In this case, it provides a much more effective way to manage data labeling that can save money and time compared to both uniform sampling and the importance sampling approach presented in [17,19]. Second, it can be implemented efficiently because it requires little extra storage, and computational power and is intrinsically parallelizable. It can therefore be applied to problems where billion of items need to be evaluated.

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Basic Theories of Machine Learning and Emerging Application Technologies

Using Machine Learning Techniques to Discriminate Good and Poor Sleepers in Virtual Reality Environment

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Abstract. Stressors in modern life have led to a concerning increase in insomnia, poor sleep quality, and nightmares, underscoring the urgent need for innovative solutions to alleviate the associated psychological and societal burdens. This study bridges neuroscience and technology by integrating electroencephalogram (EEG) and heart rate variability (HRV) data to evaluate the effectiveness of relaxation techniques under two distinct conditions: closed-eye relaxation and immersive virtual reality (VR) environments. These physiological markers were selected for their relevance in understanding brainwave activity and autonomic nervous system regulation, which play critical roles in sleep quality and emotional well-being. Employing Few-shot learning approaches combined with machine learning algorithms such as k-nearest Neighbors (KNN) and Support Vector Machine (SVM), this study sought to address the common limitation of small sample sizes in clinical research. Results revealed that SVM consistently outperformed KNN in classification accuracy and F1 scores. Furthermore, cross-validation and principal component analysis (PCA) demonstrated that VR-based relaxation techniques significantly enhance the ability to differentiate between individuals with good and poor sleep. The study emphasizes the critical interplay between brainwave adaptability and autonomic nervous system regulation, revealing that even without external stimuli, individuals with poor sleep are prone to generating complex emotional and anxious responses through cognitive and attentional processes. This highlights the importance of targeted interventions for improving sleep quality. This research contributes to the scientific community by presenting a novel, interdisciplinary approach that combines VR technology with physiological monitoring and machine learning. It offers insights into the potential of VR-based therapies in advancing personalized treatment for insomnia and poor sleep.

Keywords: Electroencephalogram · Heart rate variability · Few-shot learning · Support vector machine · Insomnia · Virtual Reality Environments

1 Introduction

In the contemporary fast-paced society, insomnia has become a widespread concern that affects public health. Its ramifications extend beyond mere sleep disturbances, often manifesting in a myriad of health and behavioral challenges. Individuals suffering from

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insomnia frequently experience reduced attention spans, impaired memory functions, and compromised problem-solving skills, ultimately jeopardizing both personal safety and economic productivity. Consequently, addressing insomnia and increasing the complexities surrounding sleep quality have become imperative, prompting researchers to take advantage of cutting-edge technologies such as machine learning, classification techniques, and virtual reality within the realm of computer science. These tools offer promising avenues for unraveling the intricacies of sleep patterns and devising innovative interventions to mitigate the adverse effects of insomnia. Our research aims to investigate the intricate interplay between brainwave ratios, heart rate (HR), and heart rate variability (HRV) as they relate to sleep quality and stress levels. By shedding light on these physiological parameters, our objective is to contribute to the development of effective strategies to improve sleep quality and alleviate insomnia, thus promoting general well-being in individuals struggling with sleep-related disorders.

The study mentioned in [1], elucidates the disparate physiological pathways to relaxation between conventional approaches and the virtual reality environment (VRE). While the act of closing the eyes triggers an increase in alpha waves, facilitating relaxation, immersion in VRE stimulates changes in the autonomic nervous system (ANS) through sensory input, similarly fostering a state of relaxation. Virtual Reality (VR) technology offers the unique advantage of crafting customized environments tailored to specific objectives, providing users with a safe and controlled space to explore relaxation techniques. Additionally, by reviewing physiological markers such as heart rate and brainwave patterns, researchers can gain a deeper understanding of emotional and stress-related manifestations. This comprehensive understanding contributes significantly to the elucidation of sleep problems, providing information on the intricate interplay between physiological responses and mental states during relaxation and sleep. The use of VR technology not only enhances our understanding of relaxation processes, but also offers promising avenues for developing innovative interventions aimed at improving sleep quality and addressing insomnia. By bridging the gap between traditional relaxation methods and modern technological advances, this research has immense potential to revolutionize approaches to the treatment of sleep-related disorders in the modern era.

In the realm of sleep medicine, insomnia is caused by various factors, including physiological symptoms, insomnia triggered by associating mental disorders, and primary insomnia. A prevalent notion posits a strong link between insomnia and anxiety-driven emotions and mental states. However, chronic exposure to stress can disrupt the brain's self-regulation, hindering individuals from attaining optimal physiological states amidst daily rhythm changes. Delving into how people adapt to various circumstances via scenario design, leveraging these insights to forecast those grappling with sleep disturbances, and proposing effective interventions constitute a scientifically sound and economically feasible approach. This endeavor is considered critical to exploring insomnia and sleep quality, offering potential advances in understanding and addressing sleep-related challenges. By comprehensively investigating the interplay of physiological and psychological factors, researchers aim to develop customized solutions that enhance overall well-being and mitigate the burden of sleep disorders on individuals and societies alike.

In this study, our objective was to elicit physiological and psychological responses from participants through a VRE-designed scene and to collect physiological data at different stages, including heart rate and brain waves. We employ machine learning analysis to help identify potential traits associated with susceptibility to poor sleep, helping in the early detection and improvement of poor sleep among sufferers, thus preventing the development of insomnia cycles and reducing the phenomenon of medication abuse.

2 Related Work

2.1 The Regulation Mechanism and Features of Sleep

Various environmental factors influence sleep quality, including stress, health conditions, and the consumption of substances like caffeine. Sleep is a complex physiological process that is regulated comprehensively, regionally, and locally by cellular and molecular mechanisms in the brain. Sleep regulation plays an intrinsic role in many behavioral and physiological functions, including development, energy conservation, regulation of the immune response, cognition, alertness, disease, and psychological states [2]. There is a balance between arousal-promoting molecules and sleep-promoting molecules, likely regulated by brain mechanisms, thus influencing the wakefulness-sleep state. Through brainwave studies of sleep and brain activity, sleep spindles have been identified as transient signals on the electroencephalogram (EEG) of mammalian brains, associated with oscillations of thalamic neurons, believed to be closely related to sleep behavior [3].

We understand that some physiological data can provide insight into the human emotional states, especially in identifying responses to stress and the arousal of anxiety emotions. By assessing and identifying EEG data, we can understand brain changes induced by stress and physiological characteristics of individuals with sleep disorders. In [4], specific EEG patterns related to stress were explored, noting reduced activity in the 11–12 Hz upper alpha (UA) band, indicating changes in neurodynamic during stress. Furthermore, increased EEG amplitudes in the 19–22 Hz range signify increased brain activity correlated with stress intensity, deepening our understanding of neurophysiological changes during stress. The study also emphasized anxiety-related EEG activity, identifying activity in the 23–36 Hz range that represents an active anxious brain state. Increased activity in this range corresponds to higher anxiety levels, which helps to develop a nuanced understanding of the relationship between EEG signals and stress-related psychological states. Recent EEG-based stress assessment research has focused on the correlation between EEG patterns and personality traits, emphasizing the importance of the UA band. The UA band (11–12 Hz) has become crucial in stress assessment [5].

According to research findings in [6], it is believed that Theta activity during wakefulness and slow-wave activity during sleep are crucial components of the common steady-state sleep process. In the study in [7], insomnia patients exhibit significantly lower Theta power values and higher Beta power values. In the study [8], the performance of individuals with sleep disorders was tested by comparing their ability to adjust EEG patterns to the environmental demands of good sleepers and poor sleepers. The research showed a significant decrease in the Theta values from “eyes open at rest” to

"sensory attention" in good sleepers, while poor sleepers did not exhibit such performance. In the EEG of good sleepers, a better adaptability to environmental demands was observed.

In the article [9], a review of seven studies found that insomnia patients exhibit elevated Beta brainwave activity during sleep onset, consistent with polysomnography (PSG) data. This suggests possible overactivation of the central nervous system (CNS) during sleep, indicating a close association between insomnia and enhanced CNS activity during sleep onset or throughout sleep. The results of the study in [10] indicate a negative correlation between the Alpha/Beta ratio during stress, suggesting that power ratios can differentiate the data characteristics of brainwaves for stress assessment. When participants are exposed to stressful conditions, changes in the stress response lead to a decrease in Alpha and an increase in Beta activity, resulting in a decline in the Alpha/Beta ratio.

Studies linking EEG patterns with individual characteristic responses to various situations indicate EEG as a valuable tool for objectively assessing stress tendencies [11]. Alpha power, Theta power, and Beta power may be EEG factors associated with dysfunctional arousal insomnia. Such findings underscore the importance of understanding the neurophysiology of insomnia and its implications for targeted interventions. Based on these studies, we can use different ratios of Alpha, Theta, and Beta activity in various contexts to identify the evolution of brain response modulation when individuals face situational demands. For example, a decrease in the Alpha/Beta ratio may indicate a greater arousal or stress response, while changes in the Theta/Beta ratio could reflect changes in cognitive processing or relaxation levels. By monitoring these ratios in real time, we can gain valuable insights into individuals' adaptive responses to different environmental demands.

2.2 Machine Learning and Deep Learning of Physiological Data

In recent years, advancements in chip computing power have significantly bolstered deep learning, effectively supporting research across various aspects of life [12, 13], particularly in clinical medicine and healthcare [14, 15]. This progress has greatly enhanced our understanding of and response to issues related to life and health. The application of physiological data, including EEG, SCR, and HRV, is a crucial task in the field of human-computer interfaces for emotion recognition. In particular, with recent advances in machine learning and deep learning, there is a more precise analysis of emotion recognition methods. However, extracting discriminative features from these physiological signals for emotion recognition remains a challenge, and integrating useful information from both channels, related rate, and time aspects is an important issue. In the article of [16], the concept of brain rate is introduced, defined as the weighted mean frequency of the EEG spectrum. Similar to blood pressure, heart rate, and temperature, standard indicators of bodily activity, it is suggested to use brain rate as a preliminary measure for general mental activation or arousal levels. This innovative approach aligns with the use of specific biofeedback parameters such as the Theta-Beta ratio and the relative Beta ratio. Brain rate serves as a broader multi-band biofeedback parameter, expanding the understanding of mental states and providing a comprehensive measure of mental activation. The study in [17] inputs EEG data into a support vector machine

(SVM) to classify low, medium, and high levels of stress. The results indicate the feasibility and effectiveness of applying the Theta/Beta power ratio of brain signal and SVM classifier for three-level stress classification, achieving an overall classification accuracy of 90%. The analysis confirms that the Theta/Beta power ratio can serve as a characteristic of stress. This contributes to a more nuanced and effective assessment of neurofeedback interventions and identifying emotions. The study in [18] investigates the influence of the band power of the Alpha/Beta and Theta/Beta ratios, in conjunction with other characteristics, on classifying two levels of human stress based on EEG signals. A classification model is formulated using the derived clustering model with Naïve Bayes, demonstrating the highest accuracy at 95%, surpassing the performance of the other four common machine learning algorithms (SVM, Logistic, IBk, and SGD) evaluated using WEKA. In the study of [19], the researchers proposed an Attention-based Convolutional Recurrent Neural Network (ACRNN) to extract more discriminative features from EEG signals, with the accuracy of emotion recognition. ACRNN employs a channel attention mechanism to adaptively assign weights to different channels and utilizes a Convolutional Neural Network (CNN) to extract spatial information encoded in EEG signals. Subsequently, to explore the temporal information of EEG signals, attention integration is extended into Recurrent Neural Network (RNN) to re-encode importance based on the inherent similarity of EEG signals. Extensive experiments were conducted on the DEAP and DREAMER databases. The experimental results indicate that the proposed ACRNN outperforms most other methods, including CNN + RNN (CNN-RNN), the channel-wise attention mechanism + CNN + RNN (A-CNN-RNN), and CNN + the RNN + extended self-attention mechanism (CNN-RNN-A).

2.3 Few-Shot Learning for Clinical Physiological Data

In clinical research, EEG and HRV have long been the primary methods for identifying certain health conditions in patients. However, due to limitations in the number of subjects in clinical studies, the application of machine learning techniques often requires special consideration.

In the review of the literature discussed in [20], publications from 1988 to 2018 were reviewed to understand the various EEG classification methods used previously. The overall effectiveness of various machine learning methods and key features were examined, ranging from Naive Bayes to decision trees/random forests and to SVM. Supervised learning methods showed higher average accuracy compared to unsupervised learning methods, including K-Nearest Neighbors (KNN). The authors suggest that, while each method has limitations in its application, combining the methods correctly can lead to higher overall classification accuracy. Recent literature surveys on EEG signal analysis have proposed multiple learning models and various artificial neural network algorithms, such as recurrent neural networks, and vector quantization, to interpret seizure patterns in given EEG signal datasets. This issue can also be addressed using other models such as SVM, Adaptive Neuro-Fuzzy Inference Systems (ANFIS), adaptive learning, and time-frequency analysis. However, in reviewing published papers on epileptic EEG analysis, dimensionality reduction and selection have been identified as a crucial and intriguing topic for EEG analysis using machine learning methods [21].

However, the rapid advancement of machine learning has greatly improved the performance of EEG-based seizure detection. Since only a small amount of labeled seizure data is available for training, labeling requires extensive input from neurologists. Research [22] introduces a novel approach based on event-related EEG feature extraction and pattern recognition, utilizing Empirical Mode Decomposition (EMD) and SVM. Initially, EEG signals are decomposed into Intrinsic Mode Functions (IMFs) using EMD. Then, the coefficient of variation and the fluctuation index extracted from the IMFs are used as features, followed by classification using SVM for identification. The experimental results demonstrate that the algorithm achieves a sensitivity of 97.00%, a specificity for interictal and ictal EEG of 96.25%, a sensitivity of 98.00%, and a specificity of 99.40%. In the study presented in [23], a two-step semi-supervised seizure detection system was proposed to reduce labeling costs by making full use of unlabeled data. The proposed neural network training strategy achieves more robust and accurate decision boundaries by imposing consistency through dual predictions on the same unlabeled data.

Emotion detection has consistently been a significant research topic and plays a crucial role in establishing human-machine interfaces. In current work, emotions are often detected using EEG and HRV signals. In the study of [24], Independent Component Analysis (ICA) and machine learning techniques such as SVM and Linear Discriminant Analysis (LDA) were used to classify EEG signals into seven different emotions. The accuracy achieved by the two algorithms was calculated and compared, with SVM and LDA achieving average overall accuracy of 74.13% and 66.50%, respectively. Due to individual differences and non-stationarity in EEG signals, traditional machine methods struggle to classify brainwave emotions accurately, resulting in poor emotion recognition accuracy. In the study in [25], a single-source domain adaptive few-shot learning network (SDA-FSL) was proposed for the cross-subject recognition of emotions from EEG. By applying domain adaptation methods based on few-shot learning to the domain of EEG emotion recognition, the experimental results demonstrated that SDA-FSL outperformed other comparative methods and exhibited superior generalization performance in cross-dataset experiments.

The study in [26] aimed to identify children with autism, features extracted from EEG and eye-tracking were used as inputs for a machine learning method (SVM). The research results revealed that when selecting 32 features, the highest classification accuracy reached 85.44% for the combined dataset, with an AUC of 0.93. This provides an effective reference for the use of SVM in clinical research. The literature in [27] provides a review of EEG signal classification methods based on KNN and SVM algorithms, highlighting the importance of correctly adjusting the parameters and selecting the appropriate features from the EEG dataset to achieve good classification results. The use of KNN and SVM for EEG classification demonstrates success in distinguishing features within EEG datasets. In our study, we will employ KNN and SVM for machine learning to classify the physiological data. Our aim is to gain insight into the physiological regulation status of people experiencing poor sleep, with the aim of providing clinical assistance and interventions.

In an era of high stress, insomnia has become a prevalent public health issue. Insomnia often leads to various health and behavioral problems, including decreased attention,

memory impairment, and weakened problem-solving abilities. Chronic sleep deprivation can also lead to heart disease, high blood pressure, and stroke. Sleep quality poses a global health concern, and the use of modern technologies such as machine learning, classification techniques, and VR in computer science to improve sleep quality is emerging as a hot research topic. These modern technologies offer novel analytical solutions. In our study, we will make assumptions about brainwave ratios, HR, and HRV associated with sleep and stress. The aim is to explore the physiological characteristics of people experiencing sleep difficulties. Furthermore, we will attempt to classify the physiological data features of participants with and without sleep problems using machine learning methods.

3 Objectives and Hypotheses

3.1 Objectives

This research aims to use virtual reality technology to present a highly immersive sense of relaxation, evaluating changes in relevant physiological data (brainwaves, HR, and HRV) as participants transition from an anxious state to a relaxed environment. We have designed a natural beach environment under a starry sky as the primary setting, providing participants with a relaxed and tranquil virtual environment. We tried to employ machine learning to develop a classifier to identify individuals with poor sleep by collecting and analyzing physiological data. This classifier aims to predict and distinguish categories of people with poor sleep based on their physiological responses during virtual reality experiences. By studying the impact of these virtual scenes on users, our long-term objective was to contribute to the development of sleep assistance interventions and to provide support for preventive healthcare in the field of preventive medicine.

3.2 Hypothesis

Exploring the physiological and psychological adaptations of individuals to different environments has long been the goal of researchers who seek to understand and manipulate them. We attempt to identify brain regulation through various contextual stimuli, aiming to investigate differences in physiological changes induced in individuals experiencing different environments. We then use SVM to discern distinct physiological signal combination models for different individuals, such as good sleepers versus poor sleepers.

In the study, we will design several scenarios, particularly focusing on the conditioning of after-stress scenarios, divided into two modes: closed-eye relaxation and immersive VRE. As our objective is to differentiate the physiological characteristics of good sleepers from poor sleepers, both types of relaxation mode will involve two categories of participants. During these sessions, their EEG and HRV will be measured.

According to related research, Theta power, Alpha power, and Beta power have an impact on relaxation and sleep. Therefore, we will select Theta, Alpha, sensorimotor rhythm (SMR), and Beta power during conditioning into the features of SVM. Furthermore, our study focuses on regulatory mechanisms for the adjustment of brain waves in response to conditioning.

We hypothesize four study parameters: Alpha Ratio (AR), Alpha to Beta Ratio (ABR), Theta Ratio (TR), and high Beta Ratio (hBR). These models provide insight into various aspects of neurophysiological responses and adaptations during different states, such as baseline, stress, and relaxation with closed eyes or immersive VR experiences.

$$\text{Alpha Ratio (AR)} = \text{Baseline Alpha / Relax Alpha} \tag{1}$$

The Alpha Ratio (AR) quantifies the ratio of the Alpha power observed in the baseline state compared to the Alpha power recorded during a relaxed state, particularly when conditioning with closed eyes or immersed in a VRE. A decrease in AR signifies an amplification of the tendency toward relaxation.

$$\text{Alpha to Beta Ratio (ABR)} = \text{Relax Alpha / Relax Beta} \tag{2}$$

The Alpha-to-Beta Ratio (ABR) denotes the relationship between the Alpha power and the Beta power during a relaxed state. This metric serves as an indicator of neurophysiological arousal adaptation, where a higher ABR value signals a more substantial transition into a relaxed state.

$$\text{Theta Ratio (TR)} = \text{Baseline Theta / Relax Theta} \tag{3}$$

TR represents the ratio of Theta power from wakefulness (Baseline) to conditioning by closed eyes or VR-induced relaxation. Insomnia sufferers show a decreased waking Theta, with a heightened Theta during relaxation, indicating a stronger propensity to induce sleepiness. Studies suggest that insomniacs struggle to adapt Theta activity to sensory stimuli, lacking a clear trend of decrease. Consequently, we hypothesize higher TR values in good sleepers, reflecting superior sleep quality due to improved Theta modulation.

$$\text{high Beta Ratio (AR)} = \text{Stress hBeta / Relax hBeta} \tag{4}$$

The high Beta ratio (hBR) measures the proportion of high Beta power detected during stressful work conditions relative to high Beta power observed during a relaxed state, particularly when conditioning with closed eyes or immersion in a VRE. A reduction in hBR indicates difficulty in shifting toward a calm and tranquil state.

This study explores the dynamics of HRV and its implications for the regulation of ANS during different experimental stages. HRV analysis reveals that the low-frequency component (LF) of the LF/HF ratio reflects both the activity of the sympathetic and parasympathetic nervous systems, while the high-frequency (HF) predominantly reflects parasympathetic activity [28]. The LF/HF ratio, indicative of sympathetic-vagal balance, peaks during REM sleep and reaches its nadir during synchronous sleep.

This study aims to distinguish ANS regulation differences between individuals experiencing poor versus restful sleep, with potential predictive implications. Parameters such as the Heart Rate Difference (HRD) and the Heart Rate Variability Difference (HRVD) are established for future discussions. HRD, calculated by subtracting the conditioning heart rate from the baseline heart rate, signifies relaxation levels, while HRVD, derived from subtracting conditioning HRV from the baseline HRV, indicates ANS adjustments.

Furthermore, the Stress HRV Difference (SHRVD), obtained by subtracting conditioning HRV from stress-induced HRV, suggests variations in parasympathetic activation. These parameters serve as valuable tools to evaluate ANS activity and predict sleep quality.

Our experimental hypothesis is that people with poor sleep and those with good sleep, in addition to the potential differences in Alpha, Theta, and Beta, will also show significant differences in AR, hBR, TR, and ABR values. Furthermore, HR and HRV influenced by the autonomic nervous system are expected to show variations, and these variables are the focus of our discussion. We intend to utilize these physiological features to design a classifier of a semi-supervised machine learning approach, such as using the SVM. We will then compare and analyze the performance of this classifier against the classification method based on raw physiological data.

4 Experiment Design

4.1 Materials

Our study utilized Unity3D game engine for VR environments due to its robustness. Figure 1 depicts a serene beach scene at night with stars and a campfire, visually comforting. The background audio of waves and the crackling campfire provided soothing sounds. We employed Oculus Quest 2 as the Head-Mounted Display (HMD), featuring high-resolution displays and intuitive controllers, ensuring a seamless and captivating VR experience, and enhancing participant engagement.

Fig. 1. Virtual scenes utilized in this study: Sleep generally occurs at night, and the study created a virtual nighttime starry sky scene to offer participants an immersive experience designed to promote relaxation and induce sleepiness.

4.2 Participants

The experimental protocols were approved by the National Chengchi University Ethics Committee. Before the experiment begins, inform the participants of the experiment details to familiarize them with the process and request that they sign a consent form prior to the experiment. The study employs two relaxation methods: Closed-Eye Rest (NA) and Virtual Reality Immersion (VR). Participants are stratified into healthy good sleepers (control) and individuals with Poor Sleep (Experimental) within each method. Sixty participants, aged 20–50 years with at least high school education, were recruited, 15 per group. This design facilitates the comparison of relaxation efficacy between sleep quality groups, shedding light on interventions tailored to different sleep patterns.

4.3 Measurement of Physiological Data

We used the ProComp Infiniti™ 5.0.3 to collect relevant physiological data from study participants, which is a reliable device for measuring physiological signals. Specifically, we gathered EEG signals from the brain's Fz region of the brain and HRV data. The EEG signals were obtained by placing electrodes on the scalp to detect brain electrical activity of the brain. The EEG was defined as Theta (4–8 Hz), Alpha (8–12 Hz), SMR (13–15 Hz) Beta (13–21 Hz), and high Beta (19–36 Hz). HRV data was collected by analyzing heart rate fluctuations and assessing variability in consecutive heartbeat intervals. These physiological measurements provide direct insight into participants' relaxation states under baseline, stress-induced, closed-eye, or immersion conditions in virtual reality.

4.4 Psychological Rating Scale

In this investigation, participants will be grouped according to their Pittsburgh Sleep Quality Index (PSQI) scores. The PSQI questionnaire evaluates multiple dimensions of sleep, including subjective sleep quality, sleep onset latency, total sleep duration, disturbances during sleep, sleep efficiency, daytime dysfunction, and use of sleep medications [29]. With scores ranging from 0 to 21, exceeding 5 indicates disturbances in sleep quality, with higher scores correlated with poorer sleep quality. The original PSQI questionnaire exhibits good internal reliability, as demonstrated by its Cronbach's α coefficient of 0.83.

4.5 Procedure

The participants underwent three experimental stages: Baseline (preparation), Stressful (Problem-Solving), and Conditioning (regulation). Each stage lasted two minutes, totaling an approximately 6-min experiment. Physiological data was continuously recorded and monitored throughout, as shown in Fig. 2. In the baseline stage, participants opened their eyes and maintained relaxation, serving as a baseline measure. The stress stage involved the Digit Span Test, aiming to induce anxiety and increase brainwave activity and sympathetic nervous system activation. The participants remained with their eyes open to complete the task. During conditioning, participants were randomly assigned to close their eyes or immersed in VRE. The entire experimental process, including the completion of the PSQI, spanned approximately 10 min.

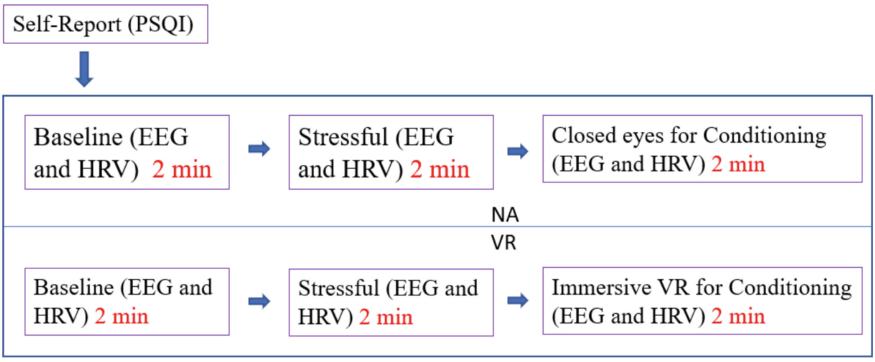

Fig. 2. The Experimental Procedure: After completing the PSQI questionnaire, participants were divided into two groups: a closed-eye relaxation group (NA) and a virtual reality immersion group (VR). The experimental process consisted of three phases: a baseline phase, a stress phase, and a relaxation phase.

5 Experimental Results

5.1 The KNN

In the evaluation of two datasets, NA and VR, a standard procedure was followed to assess the performance of a KNN classifier. The first step involved splitting each data set into training and testing subsets, with an 80–20 split ratio, 80% for training and 20% for testing purposes. Subsequently, the KNN classifier was implemented to classify the data in both datasets. The choice was made to set the value of K, representing the number of nearest neighbors considered during the classification, at 3. This parameter selection ensures that each prediction is based on the consensus of its three closest neighbors.

In the evaluation of the NA dataset, the KNN classifier yielded an accuracy of approximately 0.5, indicating that approximately half of the instances were correctly classified. The corresponding F1 score was approximately 0.4, reflecting a moderate balance between precision and recall. On the contrary, in the evaluation of the VR dataset, the KNN classifier demonstrated improved performance. Here, the accuracy score reached approximately 0.83, suggesting a higher percentage of correctly classified instances compared to the NA data set. Furthermore, the F1 score for the VR dataset was approximately 0.67, indicating a more favorable balance between precision and recall compared to the NA data set.

In summary, the evaluation process highlights the varying performance of the KNN classifier on different datasets. Although the NA data set exhibited modest performance, the VR data set showcased superior classification accuracy and a more balanced trade-off between precision and recall. Such insights provide valuable feedback for optimizing model parameters and refining classification strategies in future analyses.

5.2 The SVM and Cross-validation Scores

When analyzing the closed-eye model using SVM, the kernel was set to RBF with a regularization parameter ($C = 0.1$), resulting in a mean CV accuracy of 0.77 and a mean

CV F1 score of 0.79. In contrast, when the kernel was set to Linear with C = 0.1, the mean CV accuracy improved to 0.83, and the F1 score reached 0.82. Since we aim to evaluate the optimal performance of various relaxation modes, we selected the Linear kernel with C = 0.1 as the configuration for further discussions in the NA model analysis. However, in the analysis of VR Immersion, when the kernel was set to Linear with C = 0.1, the mean CV accuracy was 0.77, and the mean CV F1 score was also 0.77. In contrast, the RBF kernel configuration with C = 0.1 exhibited superior performance, achieving an accuracy of 0.80 and an F1 score of 0.80. Consequently, the RBF kernel configuration will be retained in subsequent discussions of the VRI model.

In the closed-eye model, we initialized the SVM classifier using the SVC function, setting the kernel function to linear and the C value to 0.1. The results of the analysis show a best accuracy of 1 and a best F1 score of 1. Cross-validation is employed to assess the model's generalization performance. The accuracy scores obtained from cross-validation, recorded as an array [1.00, 0.83, 0.67, 0.83, 0.67], reflect the accuracy of the model across different folds. The mean cross-validation accuracy, computed as 0.8, provides an aggregate measure of the model's performance across the folds. Similarly, F1 scores are calculated for each fold of cross-validation, resulting in an array [1.00, 0.86, 0.50, 0.80, 0.75], with a mean cross-validation F1 score of approximately 0.78. These metrics offer insight into the model's performance and its ability to maintain a balance between precision and recall. In the VR model, we initialized the SVM classifier using the SVC function, setting the kernel function to the radial basis function (RBF) and the C value to 0.1. The results of the analysis demonstrate a best accuracy of 1 and a best F1 score of 1. The accuracy scores obtained from cross-validation, recorded as an array [0.83, 0.67, 0.83, 0.67, 1.00], reflect the accuracy of the model across different folds. The mean cross-validation accuracy, computed as 0.8, provides an aggregate measure of the model's performance across the folds. The F1 scores are calculated for each cross-validation fold, resulting in an array [0.86, 0.67, 0.80, 0.67, 1.00], with a mean cross-validation F1 score of approximately 0.8, as shown in Table 1.

Table 1. The cross-validation score (accuracy and F1) in the SVM

The NA model
Cross-validation accuracy scores: [1.00 0.83 0.67 0.83 0.67]
Mean CV accuracy: 0.80
Cross-validation F1 scores: [1.00 0.86 0.50 0.80 0.75]
Mean CV F1: 0.78
The VR model
Cross-validation accuracy scores: [0.83 0.67 0.83 0.67 1.00]
Mean CV accuracy: 0.80
Cross-validation F1 scores: [0.86 0.67 0.80 0.67 1.00]
Mean CV F1: 0.80

Furthermore, this study employed Support Vector Regression (SVR) for the analysis and discussion of classification objectives, using Mean Squared Error (MSE) as the indicator to evaluate the performance of the regression model. Based on the results of SVR, the obtained MSE is 0.171 by closed eyes and 0.113 by VR in the study. MSE

measures the average of the squared differences between the predicted values and the actual observed values. The smaller the MSE, the closer the model's predictions are to the actual observed values, indicating better model performance. For comparison, we observe that the MSE in the VR model is smaller than that in the NA model, indicating that the VR model had much better performance.

5.3 The Feature Selection (SelectKBest) and Cross-validation Scores

In the closed-eye relaxation method (NA), classification of sleep quality by machine learning was performed using an SVM with a linear kernel and C = 0.1. This study uses the feature selection technique to choose the optimal features within the data set, to improve the efficiency and effectiveness of the model. It uses the significance of statistics to assess the importance of features and compares the top 4 and 5 most significant features separately. The analysis results indicate that the use of Relax SMR, Relax Beta, TR, and HRD as features for classification yielded a mean CV accuracy of 0.77 and a mean CV F1 score of 0.75. By including the SHRVD feature, the mean CV accuracy improved to 0.83 and the mean CV F1 score to 0.82, as shown in Table 2. The SHRVD feature represents the modulation of ANS of individuals from a stressed state to a relaxed state. Based on the results, we know that people with good sleep have better regulatory abilities.

Table 2. Accuracy and F1 score of the feature selection analysis in the SVM of the closed eye model

The SVM of NA	Kernel = 'Linear' and C=0.1	
Select Features	Mean CV Accuracy	Mean CV F1 Score
Relax Beta	63%	0.70
Relax Beta and TR	60%	0.65
Relax SMR, Relax Beta, and TR	63%	0.67
Relax SMR, Relax Beta, TR, and HRD	77%	0.75
Relax SMR, Relax Beta, TR, HRD, and SHRVD	83%	0.82
Relax SMR, Relax Beta, TR, HRD, HRVD, and hBR	80%	0.78

For the classification of sleep quality based on relaxation through virtual reality using machine learning, we used an SVM with an RBF kernel and a regularization parameter C set to 0.1. This study uses the feature selection technique to choose the optimal features within the data set. The analysis revealed that for the Relax Alpha%, the Mean CV accuracy achieved was 0.87, and the mean CV F1 score reached 0.86. In particular, upon incorporating additional features, Relax Theta, Relax Alpha, AR, TR, hBR, HRD, HRVD, and SHRVD, the accuracy and F1 score were reduced, reaching a score of 0.8. This outcome signifies the effectiveness of the Relax Alpha% that significantly improved the accuracy of the SVM model and F1 score, as shown in Table 3. The findings underscore the importance of these specific features in achieving optimal performance in the context of SVM machine learning models.

Table 3. Accuracy and F1 score of the Feature Selection Analysis in the SVM of the VR model.

The SVM of VRI	Kernel = 'RBF' and C=0.1	
Select Features	Mean CV Accuracy	Mean CV F1 Score
Relax Alpha %	87%	0.86
Relax Alpha and Relax Alpha %	73%	0.76
Relax Alpha, Relax Alpha %, and hBR	73%	0.76
Relax Alpha, Relax Alpha %, hBR, and HRD	76%	0.74
Relax Alpha, Relax Alpha %, AR, hBR, and HRD	76%	0.74
Relax Alpha, Relax Alpha %, AR, hBR, HRD, and SHRVD	76%	0.74
Relax Alpha, Relax Alpha %, AR, hBR, HRD, HRVD and SHRVD	77%	0.75
Relax Alpha, Relax Alpha %, AR, TR, hBR, HRD, HRVD and SHRVD	77%	0.75
Relax Theta, Relax Alpha, Relax Alpha %, AR, TR, hBR, HRD, HRVD, and SHRVD	80%	0.80
Relax Theta, Relax Alpha, Relax Alpha %, AR, TR, hBR, HRD, HRVD, SHRVD, and Relax HRV	80%	0.80

The NA model utilizes a specific set of features for classification. These features, namely 'Relax SMR', 'Relax Beta', 'TR', 'HRD', and 'SHRVD', are deemed relevant for the classification task at hand. Cross-validation is employed to assess the model's generalization performance. The accuracy scores obtained from cross-validation, recorded as an array [1.00, 0.83, 0.67, 1.00, 0.67], reflect the accuracy of the model in different folds. The mean cross-validation accuracy, computed at 0.83, provides an aggregate measure of the model's performance across the folds. Similarly, F1 scores are calculated for each fold of cross-validation, resulting in an array [1.00, 0.86, 0.50, 1.00, 0.75], with a mean cross-validation F1 score of approximately 0.82. These metrics offer insight into the model's performance and its ability to maintain a balance between precision and recall.

The VR model is based on a set of features carefully selected for classification. The feature is "Relax Alpha%" which is believed to contribute significantly to the task of

classification at hand. To gauge the model's generalization performance, cross-validation is employed. The accuracy scores obtained from this process are as follows: [0.83, 0.83, 1.00, 0.67, 1.00]. These scores depict the model's accuracy across different folds of the cross-validation. The mean cross-validation accuracy, computed as 0.87, serves as a consolidated metric, providing an overview of the model's performance across various folds. Similarly, F1 scores are calculated for each fold of the cross-validation process. The array of F1 scores is: [0.86, 0.80, 1.00, 0.67, 1.00]. The mean cross-validation F1 score, calculated as 0.86, provides information on the precision and recall balance in different folds, as shown in Table 4.

Table 4. The cross-validation score (accuracy and F1) via Features Select in the SVM

The NA model
Selected Features: Index (['Relax SMR', 'Relax Beta', 'TR', 'HRD', and 'SHRVD'])
Cross-validation accuracy scores: [1.00 0.83 0.67 1.00 0.67]
Mean CV accuracy: 0.83
Cross-validation F1 scores: [1.00 0.86 0.50 1.00 0.75]
Mean CV F1: 0.82
The VR model
Selected Features: Index (["Relax Alpha%'])
Cross-validation accuracy scores: [0.83 0.83 1.00 0.67 1.00]
Mean CV accuracy: 0.87
Cross-validation F1 scores: [0.86 0.80 1.00 0.67 1.00]
Mean CV F1: 0.86

In summary, the evaluation results underscore the effectiveness of the classification model trained on the NA and VR datasets. Both the cross-validation and test set results indicate high mean accuracy and mean F1 scores, highlighting the model's proficiency in accurately classifying instances and maintaining a harmonized precision-recall balance.

5.4 Dimensionality Reduction by PCA

The study conducted a principal component analysis (PCA) on the selected features Relax SMR, Relax Beta, TR, HRD, and 'SHRVD' in the closed eyes model, reducing them to 3 principal components, as shown in Table 5. Each principal component is a combination of weighted original features. Despite this dimensionality reduction, the classifier maintained an accuracy of 0.83 and an F1 score of 0.82. This suggests that the model performance remains quite good when these principal components are used for classification.

Each principal component is a linear combination of the original features, and the loadings indicate the weight of each original feature in the principal component. For example, in "Principal Component 1", the highest loading is for 'HRD' with a value of 0.997, indicating that 'HRD' has the highest contribution to the variation captured

Table 5. The loadings of each feature of the PCA in the NA model

Principal Component 1	Principal Component 2	Principal Component 3
HRD: 0.997	SHRVD: -0.943	Relax Beta: 0.785
SHRVD: -0.066	Relax Beta: 0.285	Relax SMR: 0.523
Relax SMR: 0.024	Relax SMR: 0.160	SHRVD: 0.327
Relax Beta: 0.007	HRD: -0.069	TR: -0.049
TR: -0.003	TR: -0.009	HRD: 0.003

by the first principal component. Conversely, SHRVD has a negative loading of -0.066, suggesting an inverse relationship with the first principal component.

This result demonstrates the application of SVM with the RBF kernel and C = 0.1, along with the features in dimensionality reduction in the VR model. PCA was employed here to transform the original feature, including "Relax Alpha%," into a principal component. These features were retained after dimensionality reduction. The accuracy reached 0.87 and the F1 score reached 0.86, indicating excellent performance of the SVM classifier in the reduced feature space, as shown in Table 6.

For example, in Principal Component 1, "Relax Alpha%" has the highest coefficient, implying its significant contribution to principal component 1. Overall, this result signifies that, by preserving the highest accuracy, PCA successfully transformed the original features into a new set of features that effectively represent the data.

Table 6. The loadings of each feature of the PCA in the VR model

Principal Component 1
Relax Alpha%: -1.0

6 Discussion and Limitations

In our comparison between KNN and SVM, we found that SVM achieved higher accuracy and F1 scores than KNN for both the NA and VR models. This suggests that SVM is more effective for classification in studies with a limited sample size.

SVM may encounter overfitting issues when the sample size is small because it seeks to find the maximum margin that separates the hyperplane during training. We employed some methods to address this problem, including kernel trick, regularization, cross-validation, and feature engineering. When regularization parameters (such as the C value) are adjusted to appropriate values, the complexity of the model can be controlled, thus preventing overfitting. Proper selection of kernel functions can better separate data in high-dimensional space, thus reducing the need for a large number of samples. The closed-eyes model used linear and C = 0.1, and the VR model used RBF and C = 0.1. Evaluating the model's performance through cross-validation helps better understand its performance on different subsets of data, thereby avoiding overfitting more effectively.

Through cross-validation scores analysis, we observed an instance of an F1 score of 0 occurring in the cross-validation of the closed-eye dataset. This occurrence is typically attributed to the possibility of a data imbalance. Initially, we employed a simple K-fold cross-validation, dividing the data set into 5 equal parts. This method did not protect the distribution of classes within the dataset, which could lead to over-representation or under-representation of certain classes in specific cross-validation iterations. Later, we used Stratified K-Fold (n_splits = 5, shuffle = True, random_state = 42). Although it also divides the dataset into five equal parts, unlike the simple K-Fold, the Stratified K-Fold maintains a class distribution within each subset similar to that of the entire dataset. This ensures that each class has roughly the same representation in each cross-validation iteration, thereby providing a better representation of the entire dataset. Stratified K-Fold cross-validation is more suitable for data sets with class imbalance and is generally considered a more robust cross-validation method. This helps to improve the overall accuracy and recall. The best accuracy and F1 score of the VR model both reached 1.0, while the results of cross-validation showed that the model's mean accuracy was approximately 0.87, with a mean F1 score of 0.86. This indicates some fluctuation in performance between different subsets of the training data during cross-validation, but overall the model still performs well. However, the NA model only has a mean accuracy of 0.83, with a mean F1 score of 0.82, clearly not performing as well as the VR model. The analysis suggests that distinguishing between good and poor sleepers through immersion in VRE is more suitable.

We tried to utilize variables that had statistically significant differences, set as feature selections in SVM. Attempting to extract more useful features or performing feature selection to reduce the dimensionality of features can sometimes help to improve the model's performance. The SVM exhibits strong capabilities to handle high-dimensional data. Therefore, when dealing with a large number of features, SVM may be better equipped to capture complex relationships within the data. In general, increasing the number of features can be beneficial in improving accuracy and F1 scores, but requires careful consideration. For example, using the SVM method, we found that selecting only the Relax Alpha% feature in the VR model resulted in a mean accuracy of 0.87 and a mean F1 score of 0.86. However, when we included seven features - Relax Alpha, Relax Alpha%, AR, HRD, HRVD, and SHRVD - the mean accuracy decreased to 0.77, with a mean F1 score of 0.76. Subsequently, with the addition of Relax Theta and TR, both the average accuracy and the average F1 score improved slightly to 0.80. Therefore, we may need to perform tasks such as feature selection or dimensionality reduction to avoid the curse of dimensionality and enhance model efficiency in practical applications. Overall, the result showed that the NA model and the VR model performed well in the training set, being able to predict the results accurately.

In the selection of SVM features within the VR model, choosing one feature resulted in the best accuracy and F1 score. However, as more features were added, the accuracy and the F1 score decreased. When 9 features were used, an accuracy and F1 score of 0.8 was achieved. This could be attributed to several factors, such as overfitting, feature correlation, and feature importance. When the number of features increases, the model may capture noise or unimportant patterns in the training data, leading to a decrease in performance on the testing data. This could explain the observed decrease in performance

as the number of features increases. When the number of features reaches a certain threshold, the model starts overfitting, resulting in a decrease in performance. However, as the number of features further increases, the impact of overfitting may diminish, and performance may begin to improve again. Because of the correlation between features, adding multiple similar features may not improve the model's performance and may even lead to a decrease in performance.

In the case of overfitting, the model may perform well in the training set but poorly in new data. To enhance the generalization capability of the model and avoid overfitting, we adopted the PCA dimensionality reduction technique. The results indicate that we successfully reduced the feature space from five dimensions to three principal components, achieving classification in this reduced-dimensional feature space in the NA model. An effective reduction into 3 principal components was achieved, and the accuracy remains at 0.83 and the F1 score at 0.82.

With the EEG only for the detection instrument, using the Relax Beta feature in the closed-eye model results in an accuracy of only 63% and an F1 score of 0.7. Then, using the Relax Alpha% feature in the VR model achieves an accuracy of 87% and an F1 score of 0.86. If only HRV detection instruments were available, we found that in the closed-eye model, using the HRD and SHRVD features achieves an accuracy of 67% and an F1 score of 0.65. Similarly, in the VR model that uses the same HRD and SHRVD features, the accuracy reaches 73%, with an F1 score of only 0.69. According to the SVM analysis, we found that if the budget for research equipment is limited, it would be advisable to use EEG devices with VR models to classify subjects. However, if only HRV detection instruments were available, there would not be a significant difference in performance between utilizing the closed-eye mode and the VR mode in terms of the F1 score. Therefore, if VR is not available as an instrument, it is recommended to use physiological instruments for HRV detection as supplements.

Based on research findings, both the closed-eye mode and the VR immersion mode demonstrated excellent accuracy and F1 scores in classifying individuals with good or poor sleep quality. Through feature selection and PCA analysis, we observed that the primary influences lie in the mutual interaction and regulatory capacity between brainwaves and HRV. For example, due to the influence of sensory stimulation in the VR model, Relax Alpha% becomes a key factor. This indicates that during the VR-induced relaxation process, the elevated proportion of induced alpha waves in the EEG may affect the brain's inhibition of other brain wave activities, making individuals prone to enter a state of relaxation. However, PCA in the closed-eye model revealed that HRD and SHRVD constitute the highest loading components, suggesting that adaptations in the ANS in response to situational changes are crucial for sleep. Furthermore, poor sleepers induced variations in beta power ($t = -2.744$, $p = 0.013$ *) and SMR waves ($t = -2.37$, $p = 0.025$ *) can contribute to differences in sleep quality due to intricate internal cognitive processes and emotions in the absence of stimuli from external situations (NA). Because SMR brainwaves are associated with alert wakefulness and are closely related to individual attention levels [30, 31]. These findings imply the importance of SMR in cognitive functions and may have profound implications for understanding brain activity and attention control mechanisms.

7 Conclusions and Future Work

Our comparison between KNN and SVM indicates that SVM outperforms KNN in classification tasks, particularly in studies with limited sample sizes. This effectiveness stems from SVM's ability to handle high-dimensional data and capture complex relationships within the dataset. Employing techniques such as feature selection and dimensionality reduction further enhances efficiency and performance. Specifically, our SVM model achieved high accuracy and F1 scores, demonstrating its effectiveness in distinguishing between good and poor sleepers, especially in VR models. The variables retained after the "Selected Features combined with PCA for dimensionality reduction, highlight the robust performance of SVM in a reduced feature space and provide valuable insights into feature importance. However, caution is warranted to mitigate the potential issue of overfitting, particularly in models with a large number of features and a small sample size. By considering the aforementioned methods comprehensively, one can select appropriate techniques based on specific circumstances and data characteristics, or combine multiple approaches to address the challenge of insufficient samples in SVM.

In our study, we explored the significance of adapting and regulating brainwaves and the autonomic nervous system (ANS) in response to various situations related to sleep. Additionally, we observed differences and variations in Beta and SMR waves in the absence of external stimuli between people who sleep well and those who do not. This finding highlights the possibility that sleep quality may be linked to an individual's intrinsic cognitive processes and their level of external attention. Poor sleepers may experience diminished sleep quality due to complex cognitive processes or heightened sensitivity to external stimuli, which can lead to emotional disturbances and anxiety. In VR relaxation mode, which facilitates the transition from stress to a contextual scenario, individuals who sleep well tend to exhibit relatively high percentages of Alpha power. This increase in Alpha power encourages thoughts and behaviors associated with relaxation and anxiety suppression, aiding them in achieving a state of rest. In this study, we found that relaxation in a designed VRE scenario is more effective at distinguishing between good and poor sleepers compared to relaxation with closed eyes. However, since the study classified participants using PSQI rather than including those diagnosed with insomnia, future research could focus on clinical cases to enhance classification models, explore various feature selection techniques, and validate the study results in larger datasets to ensure generalizability and reliability.

This study establishes a foundational framework for integrating VR technology, EEG, HRV, and machine learning to investigate sleep quality and develop related interventions. To enhance the impact and application of these findings, several future directions are proposed.

In terms of clinical validation, future studies should include clinical cases of insomnia and other sleep disorders to refine classification models and strengthen the correlations between physiological markers and sleep quality. Incorporating objective measures, such as polysomnography (PSG), alongside subjective tools like the PSQI, will provide a more comprehensive assessment of sleep states. These efforts will elucidate the mechanisms underlying the relationship between VR interventions and physiological responses, thereby enhancing the translational value of this research for clinical applications.

In the realm of feature optimization, it is essential to explore advanced feature selection and extraction methods, including deep learning and ensemble techniques, to enhance the accuracy and efficiency of classification models. These approaches will not only mitigate potential overfitting in high-dimensional data but also improve the robustness of the models. Future research could concentrate on identifying key features that significantly influence sleep quality, thereby facilitating the development of dynamic and targeted analytical models.

In the realm of personalized therapeutics, VR relaxation modes can be customized to meet individual needs, providing a tailored approach to enhancing relaxation and improving sleep quality. Future research may focus on developing adaptive VR systems that can make real-time adjustments based on users' physiological states. These systems could optimize relaxation protocols to address the unique requirements of each user, ultimately enhancing intervention outcomes and user satisfaction.

In terms of scalability and generalizability, future studies should increase the dataset size and include multicenter trials that encompass diverse populations from various demographic and cultural backgrounds to ensure the reliability and applicability of the findings. This approach will validate the stability of the models and promote the standardization of VR-based relaxation protocols for use in both clinical and non-clinical settings.

In the context of longitudinal studies, investigating the long-term effects of VR-based relaxation interventions on sleep quality will yield valuable insights into their potential role in preventing sleep disorders. These studies could assess the sustained impact of such interventions on mental health, emotional regulation, and overall well-being, thereby enhancing our understanding of their broader implications.

In terms of integrating wearable technology, combining VR relaxation techniques with wearable devices for real-time monitoring of physiological markers, such as EEG and HRV could create new opportunities for at-home interventions. Wearable technology would offer continuous feedback to users, enhancing the convenience and accessibility of these interventions. This integration could lead to scalable, cost-effective solutions for improving sleep quality across larger populations.

In conclusion, this study serves as a foundation for the development of innovative, personalized approaches to addressing sleep disorders. By pursuing the proposed directions, future research can significantly enhance both the academic understanding and practical application of VR-based relaxation techniques and machine learning in the fields of sleep science and mental health.

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San Jose Urban Forest – An Open-Source Tree Canopy Surveying and Assessment Tool

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Abstract. The accurate monitoring of tree canopy coverage is crucial for social and environmental studies as well as city planning. Our goal is to provide a solution that utilizes available resources like yearly aerial imagery to generate tree data to track all tree canopy within a city such as San Jose, CA. By utilizing open-source software and publicly available data, we streamlined the process by eliminating the need for expensive equipment and time-consuming manual efforts associated with traditional methods.

The methodology involves the conversion of aerial imagery datasets into geo-referenced TIFF format through a resampling process. To optimize efficiency, we strategically select 2% of the images from the dataset for labeling, minimizing the manual effort required. The labeled images serve as the foundation for training our advanced model, enabling it to accurately identify and categorize individual trees.

The result is a highly efficient, cost-effective solution that significantly reduces manual labor while providing precise tree canopy coverage calculations for the entire city. Our approach not only improves accuracy but also enhances scalability for future endeavors such as adaptation to seasonal variations, monitoring tree growth patterns, and identifying areas susceptible to potential canopy loss or even potential adoption of our solution by other cities.

Keywords: Tree canopy · urban planning · machine learning · data mining · deep learning · computer vision · GIS

1 Introduction

The City of San Jose, in California, has been losing tree canopy at an alarming rate. From 2012 to 2018 the city lost around 11.8% of total tree canopy cover, equivalent to 2.7 square miles [5]. For comparison, the citywide tree canopy coverage went from 15.36% in 2012 to 13.54% in 2018, that 1.82% loss accounts for the 11.8% of total canopy cover mentioned earlier. If this tree canopy cover trend persists, it is projected that by 2030 the City's overall canopy coverage can be less than 10%.

The reasons why the City is losing trees at a rapid rate are yet to be fully determined, as there are multiple potential sources, such as pests or diseases, drought, and illegal tree

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removals to name a few. Recently, the shothole borer pest was identified in San Jose [2] and could have been killing trees in the past while staying undetected. Furthermore, in San Jose, removing a tree from private property often requires a permit which typically requires replacing it with a certain number of trees, the specifics of which depend on the size and species of the tree being removed and the property type; however, if planting replacements on the same property isn't possible, applicants can opt to pay an in-lieu fee, which the Department of Transportation (DOT) uses to plant a tree in another location [4]. Additionally, the City Auditor found that there were mass discrepancies in permits submitted towards the in-lieu fees, which resulted in significantly fewer trees being planted. The City currently does not have a verification process set in place to ensure the necessary trees were replanted and if trees were planted, the city has no way of verifying if they are suitable species or appropriate size to replace the initial tree with. Presently, the city relies on a complaint-based system for Code Enforcement to investigate illegal tree removals, leaving unreported cases to result in the loss of existing trees without the requirement for replacement.

The City of San Jose directly manages an estimated 300K public space trees and it last completed a tree inventory in 2014 [5]. Most cities often update their tree inventory when planting, maintaining, or pruning trees. However, because in San Jose private property owners are responsible for around 86% of street trees, the city does not have a reliable inventory system. As a result, the city does not have a reliable tree inventory or overall tree data. Conservatively, a citywide tree inventory would cost around $4 per tree which means a tree inventory for all 300K trees would cost more than $1 million.

To be able to manage its trees, the City will need data, better yet, City employees need reliable, up-to-date information, to make decisions. Frequently updated data to track trees can empower arborists to make more efficient and better decisions for the urban forest's well-being.

2 Related Work

2.1 Concepts

2.1.1 Tree Canopy

Tree canopy is the area of tree crowns that cover the ground surface. It consists of all the branches and leaves that cover the ground from a bird's-eye view perspective. This layer is essential for blocking sunlight, which provides shade and contributes to cooler, more comfortable, and livable urban spaces.

2.1.2 Tree Inventory

Trees are a valuable resource to a city and as such they need to be accounted for. Tree inventories are detailed records urban trees that includes data such as species, size, location, and condition. By tracking a city's public space trees, tree inventories help support environmental conservation and preserve the benefits afforded to local communities. Additionally, it is a dataset that can help with public engagement and policy development by providing valuable insights into a city's urban forest needs.

2.1.3 Urban Forest

Urban forest is a city's collection of trees. It encompasses all forms of urban greenery, including trees in parks, streets, and private properties. Essentially, an urban forest represents the integration of nature into the urban landscape, its nature that is interwoven into the urban landscape.

2.1.4 Urban Planning

A city's urban forest is its green infrastructure that needs to be managed, maintained, and tracked to insure sustainability. A good canopy cover provides numerous benefits, including reducing urban heat islands, improving air quality, enhancing aesthetic value. Simply put, areas with a lot of trees have higher property value.

2.1.5 Vendor Lock

Vendor Lock-In refers to a situation where an organization becomes dependent on a service or when it is too costly to change, essentially a customer is locked into a cycle with a vendor. For context, the US government spends $10 - $15 Billion annually on new software and maintenance/support of existing software [9]. Government agencies used more than $15 billion on noncompetitive IT contract spending from 2013 to 2017 [8]. Vendor lock is a big problem for government agencies, and they need to be mindful and consider the consequences for tax payers into the future. Some companies offering tree inventory services may seem inexpensive at first, however they typically require additional subscription fees to use proprietary software. Therefore, government agencies must look beyond upfront costs and consider ongoing expenses and reliance. To mitigate these issues, there is a need to extract tree information from publicly available resources such as remote sensing data from satellites or aerial imagery.

2.2 State-of-the-Art Approaches

2.2.1 DeepForest

The studies conducted in [14, 15] laid the groundwork for us to perform canopy detection on aerial RGB images in San Jose. Tree delineation is a crucial task when it comes to maintaining an inventory of forested landscapes. With the emergence of deep learning techniques pertaining to images and abundance of RGB aerial images, in both studies, a way to make tree crown delineation easier was developed. Their DeepForest package contains a pre-trained deep learning model based on the Resnet-50[1] architecture. The pre-built model was trained using a semi-supervised approach. The authors first used an unsupervised training method with LIDAR data before pivoting to a supervised technique with labeled RGB images to train the model. Our work builds on this pre-trained model through transfer-learning where we fine tune the model on our own aerial image dataset of San Jose.

[1] ResNet-50: https://tinyurl.com/hw7uekx5.

2.2.2 Individual Tree Detection in Large-Scale Urban Environments

Critical work has been done that includes identifying trees within San Jose's city limits [13]. However, this work is limited to only generating tree location points and creating a tree density map. Unfortunately, this is not ideal because capturing tree canopy measurements is imperative for urban planning especially for addressing issues like urban heat islands. For tree inventories, we need to capture as much information about a tree as possible, and a tree's canopy size is part of that.

The authors in [6] performed a comparative analysis of deep learning models for urban tree detection and geolocation using different types of image data to address the limitations of traditional inventory methods. They tested various convolutional neural networks for tree detection in ground-level images and canopy detection from aerial/satellite images. The results showed that YOLO5x[2] detects trees with a highest precision of 86.9%, whereas the DeepForest model achieved the highest precision of 79.6% for canopy detection.

The authors in [12] discuss the application of the DeepForest model for urban tree detection in Sofia, Bulgaria. The study highlights the challenges faced in urban environments, such as obstructions from buildings and overlapping tree crowns, and re-trained the DeepForest model to adapt to these conditions. The results showed that the re-trained DeepForest model could be successfully applied to urban settings, achieving a high F1-score for tree detection even in complex urban landscapes.

The final objective of the studies is to detect individual tree and tree canopies, but the approximation of the actual area covered by the tree canopies is not explored. Through our study, we would like to go one step further and cover this gap by also estimating how much physical area is covered by the tree canopy.

3 Proposed Solution

We propose a solution that utilizes available resources such as yearly aerial imagery to generate data on individual trees to track all of San Jose's Tree canopy, both in public and private spaces within the city limits. This solution, which follows a semi-supervised learning approach, is cost-effective by utilizing publicly available data and open source software.

Essentially, our solution is able to break down available aerial imagery datasets and describe them for future usage. From the aerial imagery, we are able to produce location information for individual trees and their size. The location information is derived from tree bounding boxes interpreted as coordinates. Additionally, from the bounding boxes, we are able to calculate the area and give an estimate for the total tree canopy coverage the City has in the given year's aerial imagery. This is significant because total tree canopy cover is the main measurement the City uses to track its tree canopy. Lastly, the finishing touch is converting the generated data to GeoJSON and Shapefile formats so that others, like city workers, can use the data on existing Geographic Information System (GIS) software. This is helpful because it means there is no learning curve or cost of adoption for a city like San Jose to use our solution. Ultimately, this solution can be a cost-effective way to track the City's tree canopy on a yearly basis.

[2] YOLOv5: https://pytorch.org/hub/ultralyticsyolo5/

3.1 Methodology

3.1.1 Download Aerial Imagery Using *Wms-Tile-Get*

We used aerial images to detect the tree canopy within San Jose's city limits. To obtain the data for this project, we utilized the *wms-tile-get* tool [10]. *Wms-tile-get* requires several inputs, such as the server's address, image size, zoom level, output path, and GeoJSON file. The GeoJSON file we used outlines the boundary limits of the city of San Jose, which is in the shape of a polygon. *Wms-tile-ge*t is able to download images from a Web Map Service (WMS) server in a slippy map format. The zoom level and image size are crucial factors. If the image is overly zoomed in, we will have more images to process, requiring significant computational resources and time. Conversely, if the image is too zoomed out, the model may struggle to clearly identify trees.

We downloaded San Jose's 2022 aerial imagery from the California Department of Fish and Wildlife's website [3] because it had WMS server capabilities. Due to the limitations in resolution, a zoom level of 18 seemed to be the optimal choice. As a result, we gathered a total of 33,768 image tiles, each sized 256 x 256 pixels.

3.1.2 Resampling Tiles

After downloading the images from the server, we need to process them for further use in our machine learning model. Each image is converted from PNG format to geo-referenced TIFF format [11]. Additionally, we resize the images and introduce some overlapping. Resizing images from 256 x 256 to 1312 x 1312 enables them to offer higher spatial resolution.

When processing the images, it's important to note that edges may behave differently than central regions, potentially leading to edge effects where the model may not perform as well near the borders. To address this, we provide an overlap of 96 pixels to handle the edge effect and ensure that the model learns to recognize object boundaries effectively. After resampling, there will be a total of 1844 images for each year. We will also generate a metadata file containing the *"nodata"* ratio for each resampled image.

3.1.3 Tile Selection for Labeling

The DeepForest algorithm can be fine-tuned for San Jose's urban cover using supervised learning, which requires a labeled training set as input. After resampling, we obtained a total of 1844 images. Randomly selecting images to form a training set may not be representative of the entire dataset. To better capture the diversity of the dataset, we employed an unsupervised learning algorithm. Specifically, we used k-means clustering to group the images based on their characteristics. This approach ensures that the selected images for training are representative of the entire dataset. The "*nodata"* ratio represents the proportion of an image that contains no information. Selecting images with a high *"nodata"* ratio is not useful for training the algorithm, as these images contain mostly zero values. Therefore, we filtered the images to exclude those with a *"nodata"* ratio above 30% before applying the k-means algorithm to form the training set.

Gist descriptors are vector representations of the characteristics of an image, capturing features such as color, texture, and shape. DeepForest expects an 832-component

gist descriptor for each image [16]. To reduce the computational resources required by the k-means algorithm with 832-dimensional gist descriptors, we performed Principal Component Analysis (PCA) to reduce the dimensionality of the gist image descriptors (Table 2).

Table 1. Histogram Statistics

"nodata" Ratio	Number of Images
≤ 0.25	1344
≤ 0.3	1375
≤ 0.4	1445
> 0.3	469

Table 2. The selection of principal components using PCA

Number of Principal Components	Variance Captured
3	31.5%
12	55.9%
24	71.2%
64	91.25%

Table 3. K-means selection results

First Level Clusters	Number of Images	Second Level Clusters
0	816	17
1	559	12

We use the *train_test_split* method in the *TrainingSelector* class from the *DetecTree* library [1], which internally computes gist descriptors for each image and then uses k-means clustering combined with PCA to select the images for the training set. Finally, from Table 2, we can see the 64 components capture more than 91% of the variance in the data. This process first creates two primary clusters, splitting the dataset into 817 images in cluster-0 and 558 images in cluster-1. We then apply k-means clustering again to each of these primary clusters, setting the number of clusters to 2% of the data in their respective first-level clusters. An image nearest to each second-level cluster centroid is then selected from all the second-level clusters (Table 3).

Finally, we got 29 images from 1375 images for the train set.

3.1.4 Labeling

With our 29-image training set defined, we proceeded to label all the tree crowns we could find using a tool called LabelMe[3]. The process resulted in the creation of a JSON file for each image containing coordinates for each bounding box. We utilized a repository called *UrbanTree* to help us collate all the coordinates and their corresponding image file names into a single CSV such that the data was compatible with DeepForest's requirements.

Finally, DeepForest's *split_raster* function segmented all 29, 1312x1312 images into 450x450 images ready for processing by DeepForest's pretrained model.

3.1.5 DeepForest

As mentioned previously, we fine–tuned DeepForest's pre trained deep learning model using our curated dataset. After this, we applied our tuned model to all the aerial images we had of San Jose.

3.2 Testing

To measure the effectiveness of our fine-tuned model we measured the box recall and precision on our validation dataset. Due to the unique nature of our labels, the DeepForest library recommended we use an IntersectionOverUnion (IoU) metric. The IoU metric measures the area of overlap of two bounding boxes over the total union area. A representation of this metric can be seen in Fig. 1. DeepForest recommends using a 0.4 threshold rather than the conventional 0.5 because it is more useful for ecological purposes.

After running the evaluation method DeepForest provides, our overall box recall was 0.65 and our precision was 0.47 (Table 4). While these scores may seem relatively low it's worthwhile to keep in mind that state-of-the-art LIDAR algorithms possessed a 0.71 box recall. Hence, it's fair to say that the model trained on RGB imagery performs fairly well given its limitations.

We also decided to run a comparison with the pre-trained model to see how well the fine-tuned model did. As you can see from Table 4, our fine-tuned model did almost twice as much better than the pre-trained version.

3.3 Results

Our model efficiently detects large trees in urban areas. Although it may not capture all trees, it can detect canopy at a much higher rate than the base model. The effectiveness of our model in identifying trees is illustrated in Fig. 3, contrasting with the performance of the base model which only detected 2 trees in Fig. 2 top edge.

Based on the information generated by our model, we calculated there was a total of around 21,000 acres of tree canopy across the City of San Jose in 2022. The City of San Jose [5] reported an estimated 12,889 acres of tree canopy cover in 2018. Although this may look like a drastic difference, our model performed well considering our limitations and when compared to notable industry leaders. For example, Google, which has a

[3] LabelMe – https://github.com/labelmeai/labelme.

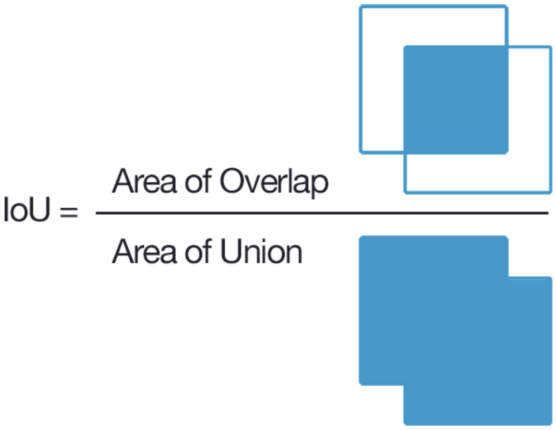

Fig. 1. Representation of the IoU Metric[4]

Table 4. Box recall and precision for the pre-trained and our fine-tuned model

	Pre-Trained Model	Fine-Tuned Model
Box Recall	0.38	0.65
Box Precision	0.35	0.47

similar tree canopy tool [7], estimates that the City of San Jose has 81 square kilometers

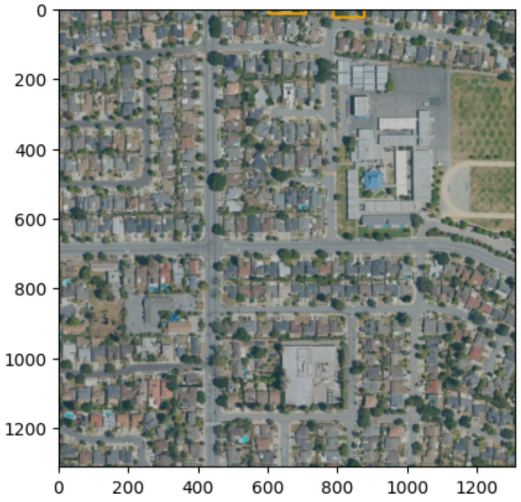

Fig. 2. Base-Trained Model - 2 detections on top edge

[4] https://deepforest.readthedocs.io/en/latest/evaluation.html

(equivalent to 21,015.5 acres) of tree canopy. Based on Google's estimates, our model is off by around 5%. Furthermore, our solution goes a step further. While Google's solution simply highlights tree canopy areas as seen in Fig. 4, our approach gives information about individual trees such as location and area as seen in Fig. 3. Finally, to make our solution easily accessible to all, we converted our model's generated output to a Shapefile and GeoJSON format that can be used by any mapping tool. We converted our data to the desired output by utilizing Python's GeoPandas and Shapely library packages.

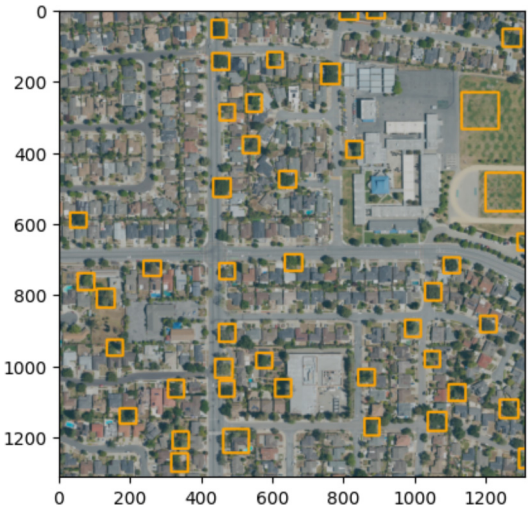

Fig. 3. Fine-tuned Model

4 Evaluation

We argue that the reason for such drastic improvement with our fine-tuned model lies in the fact that it was able to learn the representation of tree canopies within more urbanized areas, which perhaps the pre-trained model lacked. Since our training set was diverse and represented a wide array of landscapes in San Jose, our solution was able to generalize better. Hence, our box recall rate almost doubled and our box precision improved too. With that being said, our model still has a few shortcomings. Currently, it is trained on aerial images from a single year of data which means that it performs well on images from the trained year, but poorly on aerial images from a different year. Furthermore, there were a few misclassifications of tree canopies, however, we envision that these issues can be addressed by improving our solution with more robust features.

Fig. 4. Tree Canopy - Google Environmental Insights Explorer

5 Future Work

In our future work, we will gather aerial imagery with better resolution than the one provided by the California Department of Fish and Wildlife [3]. With the current resolution, it was difficult to manually label trees to fine-tune the model. Having higher resolution imagery will provide more information that can enhance our model's ability to identify trees accurately.

In order to increase the robustness of our model we wish to implement data augmentation techniques. We can add to our existing data set by randomly rotating, shifting, flipping and zooming in on images. This helps us generate a larger data set to train our model on, increasing its ability to generalize even more. Furthermore, those transformations add noise to each of the images making the model more resilient to actual variations in testing sets.

Furthermore, we will develop map layers for different years. By comparing these layers, we can identify changes in San Jose's urban forest over time, highlighting areas of growth and loss. This will allow us to highlight changes in San Jose's urban forest by each year. This information can help city workers focus preservation efforts on critical areas. It can also assist in identifying the root causes of canopy loss in San Jose, aid in establishing a tree inventory, and help save crucial resources.

6 Conclusion

This work presents a cost-effective solution for tracking a city's urban forest and its tree canopy. Our model was able to efficiently detect medium to large trees within the city limits of San Jose, estimating around 21,000 acres of tree canopy in 2022, closely aligning with Google's estimates. Furthermore, our solution provides detailed information on individual trees and outputs the generated data in Shapefile and GeoJSON formats for easy integration with mapping tools.

To enhance the accuracy and efficiency of our model, we manually labeled the most representative tiles within the City of San Jose. By employing k-means clustering and PCA, we reduced the computational resources required and streamlined the data preparation process. This approach allowed us to cluster similar regions and focus on the most representative samples, minimizing the amount of manual labeling needed. By prioritizing the most representative tiles, we maximized the return on investment of our labeling efforts, ensuring that our data preparation would be impactful to the model's training. This strategy contributed significantly to the model's performance, facilitating more accurate and efficient detection of large trees in urban environments.

Despite its success, the model has limitations, such as dependency on specific-year aerial images and some misclassifications. Future work will address these by using higher resolution imagery, applying data augmentation techniques, and developing annual map layers to track changes over time.

In summary, our work offers a way to track a city's urban forest and its tree canopy with currently available public resources. There is no cost of adoption and avoids the risk of vendor-lock. Tree data is crucial for Smart Cities to enable efficient urban forest management and contribute to overall sustainability of urban environments. With our solution, the City of San Jose can track its urban forest and tree canopy on a yearly basis, helping identify critical areas for preservation. Ultimately by providing tree data, we can support data-driven urban planning decision-making for any city in need of such option.

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Towards a Machine Learning-Based Approach to Predicting Stock Price Volatility and Its Associated Risk in Egypt

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Abstract. In emerging stock exchange markets, the ability to predict stock price volatility is critical for managing financial risk. This paper presents a novel approach to forecasting volatility by employing a hybrid machine learning model. We introduce a two-tiered hybrid model which integrates the robust predictive power of eXtreme Gradient Boosting (XGBoost) with a corrective step using linear regression to refine predictions further. This model is designed to address the challenges posed by the inherent volatility in emerging markets. Our methodology leverages historical stock data of the top 30 performing stocks in the Egyptian stock exchange listed under the EGX30 index to predict volatility, employing a machine learning framework that surpasses traditional methods in terms of accuracy. The results demonstrate that our model achieves a mean absolute percentage error (MAPE) of less than 10% for 20 out of 30 stocks analyzed, with an overall MAPE of 9.35%. Additionally, the model consistently delivers a mean directional accuracy of 82.1% across the index constituents. These outcomes not only provide a significant improvement over existing benchmarks but also contribute to the literature by offering a refined approach to managing the unpredictability of volatile markets through innovative machine learning techniques.

Keywords: Financial Risk Management · XGBoost · Time Series Analysis · Egyptian Stock Market · Stock Volatility

1 Introduction

One of the first stock exchange market has existed since 1682 when exchange rules and regulations were first established in Frankfurt, Germany. Back then and until the mid 1800s, communicating trans-Atlantic trades would take weeks

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to be locked in [1]. Nowadays, viewing stock prices and locking in trades happen in real-time, whether through traditional brokers or using mobile applications. Such advances in financial technology within the stock trading market have highlighted the importance of risk quantification and management for traders.

One of the well-established and widely adopted metrics to assess the risk associated with a stock is its price volatility. Volatility quantifies the dispersion of a stock's returns over a specific period of time. It serves as a numerical measure to summarize the likelihood of encountering deviations and fluctuations in the returns distribution. Typically, volatility is assessed using the standard deviation of returns, which provides insight into the extent to which returns deviate from their mean. An increase in volatility signifies a higher probability of experiencing substantial fluctuations in returns, both positive and negative. That being said, there was little evidence, back in 1990, that the level of stock return volatility has increased since the early 1980s [7]. Consequently, understanding and accurately measuring volatility is essential for stock trading associated risk assessment and management, as it reflects the underlying uncertainty and potential variability in the performance of financial assets.

An efficient stock market was defined as one where the pricing of its constituents is unaffected by revealing the information set to all market participants [2]. This means that all available information is already reflected in the stock prices. In other words, if the market is efficient, new information is quickly and accurately incorporated into stock prices, leaving little room for unexpected price changes. As a result, predicting future price movements becomes more difficult because any new information that could influence prices is random and unpredictable [4]. Later, in 1994 and 2005, researchers repeatedly demonstrated that portfolios composed of stocks selected by experts consistently fail to outperform the market. Additionally, professional investment managers generally do not outperform their index benchmarks, providing evidence that, for the most part, market prices reflect all available information [3,5,6]. Given that market prices reflect all available information, which changes in real-time, it is essential to quantify risk at a comparable pace.

Nonetheless, researchers alongside traders and brokers have always attempted to beat the market in order to realize gains on their investments in stocks. In this process, one must assess and quantify the risk associated with the investment through metrics such as the aforementioned volatility.

Time-series statistical and machine learning approaches to predicting volatility include the Exponential Weighted Moving Average (EWMA), AutoRegressive Integrated Moving Average (ARIMA), and the Generalized AutoRegressive Conditional Heteroskedasticity (GARCH) models. Machine learning-based models to predict volatility include using deep neural networks (DNN), Long-Short Term Memory (LSTM) and Convolutional Neural Network Long-Short Term Memory (CNN-LSTM) hybrid models that are typically used in time-series forecasting.

In this paper, we begin by reviewing existing literature in the domain of forecasting volatility in the statistical analysis domain and in the machine learning

realm. We then present our methodology beginning with our proposed feature engineering steps. Secondly, we propose a prediction framework based on-top of eXtreme Gradient Boosting (XGBoost) and Linear Regression that outperform the models surveyed in literature. We finally further discuss our results. Our main contributions in this paper are two-fold. First, we propose a novel hybrid-model to be used in stock price volatility prediction. Second, we verify that our model outperforms benchmarking models that are commonly used to address this problem.

2 Related Works

2.1 Defining Volatility

Nowadays, information spreads much more quickly than a decade ago, leading to an increase in the rate at which the price changes in response. As a means of quantifying the risk associated with such changes, we use the volatility of a stock's closing price.

Additionally, volatility plays a crucial role in shaping the bid-ask spread; stocks with greater volatility typically exhibit wider spreads between the buying and selling prices set by market makers. Considering that investors often prefer risk-averse strategies, volatility can serve as a key indicator of the risk involved in owning a particular stock [8]. A prevalent method for measuring stock market volatility is the standard deviation of daily returns. This statistic, which quantifies the variability of returns, is instrumental in analyzing how volatility evolves over time. The formula for calculating the standard deviation of returns is given as follows:

$$\sigma = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(r_t - \bar{r})^2} \tag{1}$$

where:

- σ is the standard deviation of returns, representing volatility.
- T is the total number of observations within the window.
- r_t is the return at time t.
- $\bar{r}$ is the average return across all observations [8].

The heightened variability in stock prices, regardless of direction, leads to increased stock volatility. Concurrently, as volatility intensifies, the likelihood of experiencing larger gains or losses in returns also rises.

2.2 Statistical Techniques

In modeling volatility, time series statistics may be used to find the best forecast of the volatility and other machine-learning-based techniques also exist.

Early research extensively explored the use of Autoregressive Moving Average (ARMA) models, along with the Autoregressive Integrated Moving Average model (ARIMA) for time series analysis. ARMA models generally necessitate that the data exhibit stationarity-meaning its statistical properties, such as mean and variance, do not change over time-a condition that stock data often fails to meet. Consequently, ARIMA has become more prevalent for analyzing stock time-series data, given its applicability to non-stationary datasets [12].

One application for the ARIMA model discussed applying it to predict prices on both the New York Stock Exchange and the Nigerian Stock Exchange [12]. Their experiment revealed that ARIMA is effective at forecasting stock prices over short periods. Moreover, in Bangladesh, a similar emerging market to its Nigerian counterpart, ARIMA was used to predict the stock price of specific indices, obtaining an accuracy of 82.1% on average [19]. Researchers have also obtained 36.05% mean absolute percentage error (MAPE) using the Apple stock data (AAPL) when utilizing a SARIMA model, the seasonal variant of the ARIMA [20].

The autoregressive conditional heteroskedasticity (ARCH) model represents a significant advancement in the field of volatility modeling and has become exceedingly popular across various academic and practical applications. Widely utilized in analyzing financial data from multiple countries, ARCH and its numerous derivatives leverage conditional variance to enhance the precision of return variability forecasts [8]. Subsequently, expansions on the ARCH model introduced the Generalized ARCH (GARCH) model. GARCH models the volatility as an amalgamation of long-term average variance and the immediate past expected variance, while also considering the magnitude of the most recent observed changes [9]. For longer-term time series predictions, GARCH has shown superior performance compared to ARIMA [13].

Although ARCH and GARCH models have been widely successful, they do not fully account for certain characteristics evident in financial and economic data [8]. ARCH and GARCH models, while effective in modeling volatility, struggle to capture certain characteristics of financial data, such as fat tails and sudden market shifts, as they assume a constant variance structure. Machine learning techniques, by contrast, can model complex, non-linear patterns and adapt to evolving market conditions, potentially offering more accurate predictions in volatile environments. Notably, early observations by researchers such as Nelson and Fama highlighted that unconditional price or stock returns often display fatter tails compared to those predicted by a normal distribution [10,11].

That being said, empirical evidence shows that the behavior of asset returns in the real-world changes randomly over time. This led the authors to examine time-variant models for measuring volatility [8]. Hence, the need for machine learning-based techniques arose, as these soft computing methods overcome the limitation of the discussed models.

2.3 Machine Learning-Based Techniques

Soft computing approaches encompass machine learning algorithms and deep learning techniques. These methodologies are favored due to their ability to model complex patterns and make predictions from large, nonlinear datasets, characteristics often found in real-world applications. For example, deep neural networks (DNN) were used to predict the price changes of commodities with an accuracy of 46% [14].

In another literature, Xiong et al. attempted using a long-short term memory (LSTM) model incorporating Google domestic trends in order to predict Standard & Poor's (S&P500) index volatility with a mean absolute percentage error (MAPE) of about 25% [15]. Similarly, Wang et al. created a vector derived from the textual embedding of news topics pertinent to specific securities. This vector, in combination with historical stock prices, served as input for their LSTM model, which achieved an accuracy rate of up to 69% [12].

Moreover, Shah et al. used DNN and LSTM models to predict the volatility of India's BSE Sensex, a free-float stock market index based on its 20-year historical closing price [16]. Using DNNs and LSTMs, they forecasted daily stock market volatility over various time window sizes. The DNN model achieved a directional accuracy peak of 60.6%, while the LSTM model recorded an root mean square error (RMSE) of 200 [16]. For a specific stock, Tech Mahindra Ltd, the models recorded an RMS error of 6.33, approximately 1% of the average data value. Despite the challenges these models faced in adapting to rapid shifts over a 7-day forecasting horizon, enhancing the LSTM with additional data attributes demonstrated potential for detecting trends and providing more reliable long-term predictions for volatile stock datasets [16].

Recent studies highlight the effectiveness of combining machine learning techniques with alternative data sources for stock market prediction. Wang et al. used Latent Dirichlet Allocation (LDA) to analyze financial news and applied a Support Vector Machine (SVM) for predicting market trends, improving accuracy during volatile periods by integrating news sentiment [21]. Similarly, Mohapatra et al. introduced a hybrid model combining XGBoost, LSTM, and CNN-based feature selection, which enhanced prediction accuracy by focusing on key technical indicators [22]. These approaches align with recent trends discussed in this review, where hybrid models and non-traditional data sources outperform traditional models like ARIMA in volatile market environments.

While predicting gold price volatility, other researchers employed prediction using LSTM networks complemented by the pre-trained VGG-16 Convolutional Neural Network (CNN). The LSTM was used to generate an embedding for the data and this embedding was processed and passed into the VGG16 CNN-pretrained network to output the forecast [17]. The model they proposed surpassed the performance of six other benchmark models that the researchers selected for comparison. However, the model incurred a significant computational cost due to its complexity. Additionally, a major limitation of their approach was the reliance on 50 years of training data, a duration that is typically not available for most stocks currently listed [17].

Combining deep learning models like LSTM with techniques such as CNNs or wavelet transforms offers several benefits for forecasting stock market volatility. LSTMs are excellent at capturing sequential dependencies in time series data, while CNNs excel at extracting features from structured data, such as trends or patterns within stock prices. This combination can enhance prediction accuracy by leveraging the strengths of both models. Similarly, wavelet transforms are useful for decomposing complex time-series data into different frequency components, allowing models like LSTM to focus on key patterns at multiple scales. However, their complexity can lead to challenges in interpretability, making it difficult for practitioners to understand how predictions are made. Additionally, the performance of these models can be sensitive to hyperparameter tuning, and overfitting can occur if not carefully managed.

Another hybrid implementation was addressed by Wang et al., utilizing a hybrid of discrete wavelet transform coupled with an ARIMA model and tree-based machine learning classifier [18]. In their paper, the proposed DWT-ARIMA-XGBoost hybrid model begins by employing a discrete wavelet transform (DWT) to decompose the dataset into its approximation and error components. Subsequently, the ARIMA model is applied to the approximation data, while the enhanced XGBoost-based model addresses the error data. The final predictions are then synthesized through wavelet reconstruction. Comparative experimental analysis conducted on 10 stock datasets revealed that the DWT-ARIMA-XGBoost model produces lower errors than the other models: ARIMA, XGBoost and DWT-ARIMA-XGBoost respectively [18].

In summary, traditional statistical models like ARIMA and GARCH are well-established for predicting stock price volatility and provide clear interpretability with their parameter-driven structures. However, they require the data to meet certain assumptions, such as stationarity, and may struggle to capture the complexity of non-linear relationships in financial data. In contrast, machine learning approaches, such as LSTM and hybrid models, excel in handling large, non-linear datasets and adapting to evolving patterns in stock markets but often come with higher computational costs and reduced interpretability compared to traditional models.

Given the comprehensive literature review performed, we identified that there is little literature addressing volatility in emerging markets. To the best of our knowledge, no literature exists addressing the volatility predictions within the Egyptian stock market with high accuracy. Hence, we propose our methodology addressing the constituents of a major Egyptian index, EGX30.

The proposed machine learning methodology for the Egyptian stock market could offer insights or improvements over existing models used in developed markets by addressing the unique challenges faced by emerging markets. For instance, markets like Egypt often exhibit higher volatility, less liquidity, and more abrupt price movements compared to developed markets, making them harder to model using traditional statistical approaches. By leveraging machine learning models that can handle non-linear relationships and adapt to rapid shifts, the proposed approach could offer more precise volatility predictions in

such environments. Additionally, in future work, these models can integrate alternative data sources, such as macroeconomic factors or news sentiment, which are often more influential in less developed markets. This flexibility may lead to better-tailored predictions than those derived from models built for developed, more stable markets.

3 Methodology

3.1 Preprocessing and Feature Engineering

Historical daily closing prices for all 30 stocks were obtained and sorted according to the trading date for the past trading year. For each individual stock from the index, let p_t represent the closing price at time t. The daily percentage change r_t at time t, representing the investment return, is given by:

$$r_t = \frac{p_t - p_{t-1}}{p_{t-1}} \tag{2}$$

The rolling standard deviation of the percentage changes over a window of n days, where $n = 5$ in this case to represent a trading week, is defined as:

$$\sigma_{\text{weekly rolling},t} = \sqrt{\frac{1}{n-1} \sum_{i=t-n+1}^{t} (r_i - \overline{r})^2} \tag{3}$$

where $\overline{r}$ is the mean of returns over the window from $t - n + 1$ to t.

The feature engineering and generation process for each trading day in a single stock's historical dataset involves the following steps:

- **Lagged Volatility Features:** For each trading day, we compute 262 features derived from the volatility values of a little over one trading year. This is achieved by using the rolling standard deviation of the daily returns in Eq. 3. The resulting volatility measure for each day is then lagged to generate 262 distinct features, capturing the historical volatility trends over the trading year.
- **Previous day closing price:** providing insight of the most recent market position of the stock.
- **Temporal Features:** Additional features are integrated to provide temporal context:
  - An integer representing the day of the trading week, thereby capturing the influence of the weekly cycle on market behavior.
  - An integer denoting the week of the year, offering insights into longer-term seasonal patterns in trading activity.
  - An integer for the month of the year, which helps in capturing monthly variations and trends within the financial market.

These combined features are intended to enrich the model's input with both historical volatility data and temporal identifiers, which are crucial for more accurate predictions of market movements.

3.2 Modeling

In this study, we employ XGBoost to construct a model utilizing the predefined features. XGBoost was chosen as it prevents overfitting through regularization, which enhances model generalizability. Moreover, XGBoost allows for parallel processing and tree pruning, significantly improving computational efficiency and prediction accuracy. This model is specifically designed to forecast a daily figure representing a weekly rolling volatility for each subsequent day of the year. To reduce the effects of short-term fluctuations and enhance the representation of weekly trends, we compute a rolling average of the predicted values over a 5-day window.

That being said, the observed residuals are shown in Fig. 1. Our analysis across the stocks listed in the EGX30 index reveals a correlation coefficient ranging from -0.7 to -1 between the actual weekly rolling volatility and the prediction errors. This strong negative linear relationship indicates that increases in weekly rolling volatility are consistently associated with increases in the error of our model's predictions. These observations highlight the challenges of achieving accurate forecasts during period of high volatility. This is further illustrated in Fig. 2 which shows the strong negative correlation for one stock between the actual weekly average volatility and the errors produced from our XGBoost model. This is the case for almost all given stocks in the EGX30 indicator.

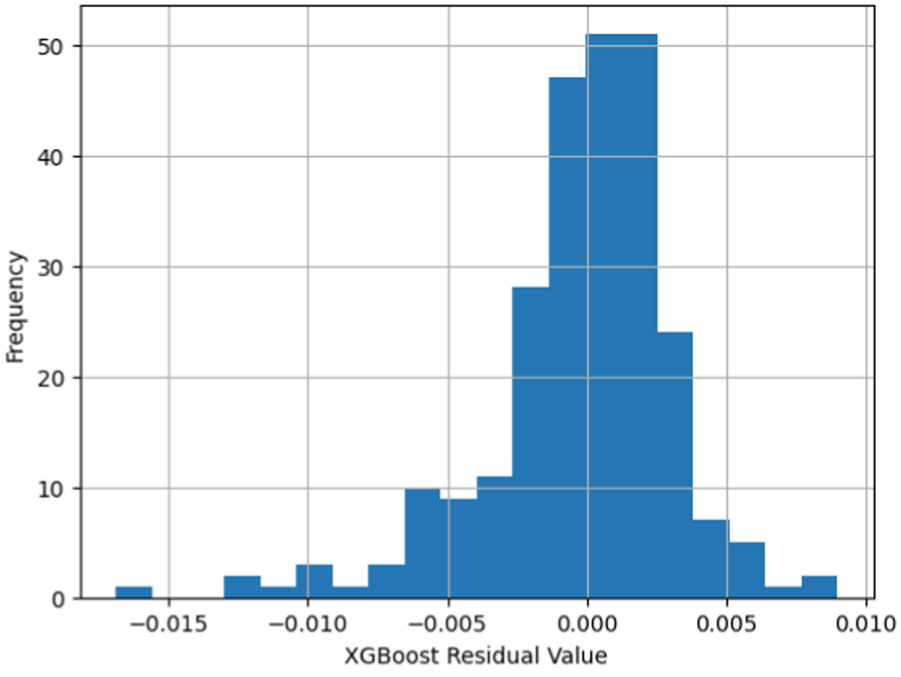

Fig. 1. Residuals resulting from the XGBoost model

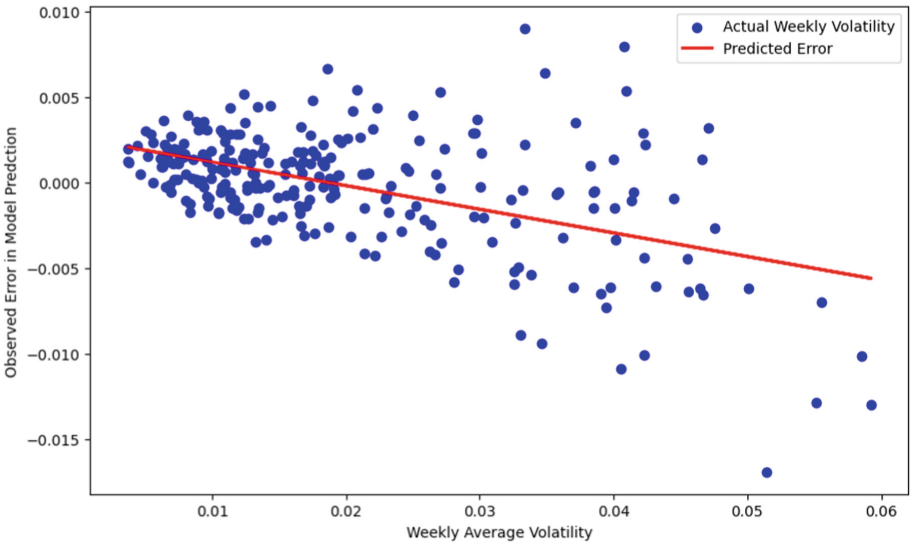

Fig. 2. Simple Regression Line Predicting Error observed in XGBoost predictions using the weekly average volatility of one stock in EGX30

Observing the residuals and the strong negative linear correlation observed between the actual daily volatility and the prediction errors of our initial XGBoost model, we decided to incorporate a corrective predictive step post the XGBoost prediction. This layer involves training a simple linear regression model to estimate the prediction errors based on the historical volatility of the stocks.

Specifically, the linear regression model predicts the weekly average error, denoted as $\hat{e}_{weekly}$, using the actual historical volatility $\sigma_{\text{weekly},t}$ as the predictor. The regression model is formulated as follows:

$$\hat{e}_{weekly} = \beta_0 + \beta_1 \sigma_{\text{weekly},t} + \epsilon$$

where β_0 and β_1 are coefficients estimated by the regression, and ϵ represents the error term.

Upon obtaining $\hat{e}_{weekly}$, we adjust our initial XGBoost volatility predictions by adding this predicted error term. Consequently, the revised weekly volatility prediction, denoted by $\hat{v}_w$, for the stock is given by:

$$\hat{v}_w = \hat{v}_{\text{XGB},w} + \hat{e}_w$$

where $\hat{v}_{\text{XGB},w}$ is the weekly volatility prediction from the XGBoost model.

This two-tiered approach, outlined in Algorithm 1, aims to refine our predictions by compensating for systematic errors identified in the initial model, thereby enhancing the accuracy of our forecasts for weekly stock volatility.

Algorithm 1. Weekly Stock Volatility Prediction Algorithm

procedure PREDICT_WEEKLY_VOLATILITY(*stock_data*)
Calculate historical daily returns r_t
Compute rolling standard deviation $\sigma_{\text{weekly rolling},t}$
Generate 262 lagged volatility features
Include temporal features: day, week, month
Train XGBoost model, obtain $\hat{v}_{\text{XGB},w}$
Apply smoothing using the rolling window average
Train Linear Regression model to predict residuals
Adjust XGBoost predictions by predicted residuals
return adjusted weekly volatility $\hat{v}_w$
end procedure

Further, we quantify the accuracy of our model's predictions by evaluating the prediction errors, where each data point is a daily representation for the 5-days volatility. The following metrics are used to assess the performance of our model, where v_i is the actual weekly rolling volatility at day i, $\hat{v}_i$ is the predicted weekly rolling volatility at day i, and n is the number of days:

- **Mean Absolute Percentage Error (MAPE):**

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^{n} \left| \frac{v_i - \hat{v}_i}{v_i} \right| \tag{4}$$

- **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |v_i - \hat{v}_i| \tag{5}$$

- **Mean Directional Accuracy (MDA):**

$$\text{MDA} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1} \left(\text{sign}(\hat{v}_i - \hat{v}_{i-1}) = \text{sign}(v_i - v_{i-1}) \right) \tag{6}$$

where $\mathbf{1}(\cdot)$ is the indicator function that returns 1 if the condition is true and 0 otherwise, and $\text{sign}(\cdot)$ is the sign function.

- **Root Mean Square Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (v_i - \hat{v}_i)^2} \tag{7}$$

- **Coefficient of Determination** (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (v_i - \hat{v}_i)^2}{\sum_{i=1}^{n} (v_i - \bar{v})^2} \tag{8}$$

where $\bar{v}$ is the mean of the actual weekly volatilities.

4 Findings and Results

In this section, we present the performance of our hybrid model, applied to the EGX30 index constituents. We achieved MAPE of less than 10% for 20 out of 30 stocks, with an overall mean MAPE of 9.35%. Using the same data, our proposed model outperforms all other benchmarking models utilized as shown in Table 2. These benchmarking models include the main 4 approaches discussed in literature for other markets. Table 1 summarizes the performance metrics for the top-4 performing stocks sorted according to MAPE ascendingly. In our experiments, all individual stock MAPE, except one, were below 20%. Excluding this outlier, the mean MAPE improves to 8.00%. It is important to mention that upon further investigation, we discovered that this particular outlier stock underwent a stock split during the study period, which caused its price to drop from around EGP500 to EGP60. Moreover, our model consistently achieved a daily mean directional accuracy of 82.1% across the index. In Fig. 3, one can observe the year-to-date accuracy of our hybrid model's prediction for a stock.

Table 1. Summary of Final Model Performance Metrics For Top-4 Performing Stocks in MAPE

Stock	MAPE	MAE	MDA	RMSE	R^2
Stock 1	3.108%	0.000844	77.78%	0.001117	0.708
Stock 2	3.158%	0.001032	100%	0.001150	0.911
Stock 3	3.415%	0.000863	66.67%	0.001157	0.899
Stock 4	3.575%	0.000733	77.78%	0.001083	0.865

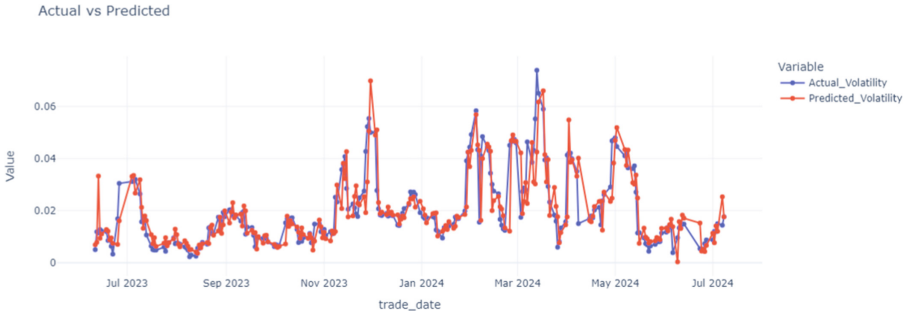

Fig. 3. Actual volatility versus predicted volatility for a stock year-to-date

Furthermore, in Table 3, we can observe the improvement in MAPE due to the usage of the linear regression model proposed as predictive error-correction step.

Table 2. Benchmarking Model Results

Model	MAPE	MAE	RMSE
Multiple Linear Regression	41.13%	0.00363	0.00494
ARIMA	31.44%	0.00264	0.00393
LSTM	94.10%	0.00710	0.00920
K-Nearest Neighbors (KNN)	66.10%	0.00575	0.00796

Table 3. Comparison of Evaluation Metrics Before and After Corrective Linear Regression Step For Some EGX30 Constituents

Stock Index	Before Corrective Regression					After Corrective Regression				
	MAPE	MAE	MDA	RMSE	R^2	MAPE	MAE	MDA	RMSE	R^2
Stock 10	17.87%	0.0032	78.6%	0.0037	0.914	7.32%	0.0018	82.1%	0.0022	0.971
Stock 11	11.80%	0.0027	75.0%	0.0034	0.896	6.44%	0.0018	80.4%	0.0026	0.939
Stock 12	10.37%	0.0029	80.4%	0.0034	0.891	6.81%	0.0019	87.5%	0.0023	0.950
Stock 13	7.76%	0.0029	71.4%	0.0038	0.794	6.82%	0.0026	80.4%	0.0034	0.842
Stock 14	9.09%	0.0031	80.4%	0.0037	0.899	7.22%	0.0024	82.1%	0.0030	0.933

Given that XGBoost utilize gradient boosting, we can retrieve importance scores observed in Fig. 4 for each feature from the boosted trees. The XGBoost model's feature importance plot was scrutinized to identify the most significant predictors in forecasting financial metrics. Notably, the features *volatility_lag_1* and *volatility_lag_5* emerged as critical. These features are indicative of:

- *volatility_lag_1*, *close_price* and *volatility_lag_2* capture immediate market reactions and daily price fluctuations.
- *volatility_lag_5* and *volatility_lag_15* reflect the influence of market conditions over a standard trading week, capturing weekly cyclical effects.
- *week_of_year* allows the model to incorporate annual cyclical trends, such as fiscal policies, holiday effects, and economic cycles, which can significantly impact financial markets.

The inclusion of the *day_of_week* and *week_of_year* features underscores the model's enhanced capacity to recognize both short-term daily variations and broader patterns influenced by the calendar year, facilitating a more nuanced understanding of periodic market behaviors. These temporal features collectively enrich the model's predictive accuracy, accommodating fluctuations in trading activity and economic conditions that vary throughout the year.

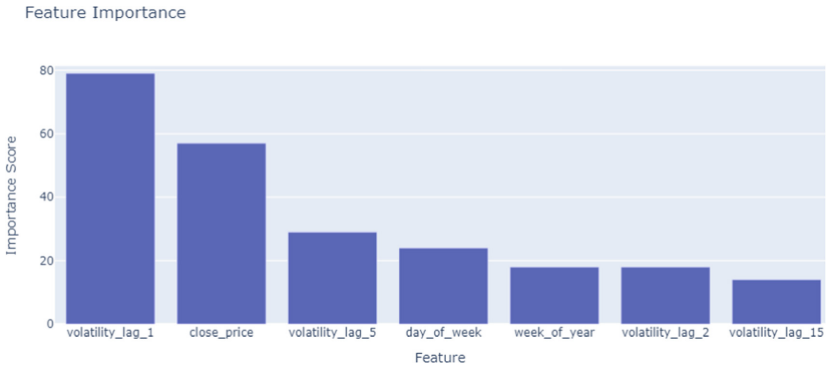

Fig. 4. XGBoost-based feature importance scores

5 Conclusion

In this study, we introduced a hybrid model combining XGBoost with a corrective linear regression step to predict stock price volatility in the Egyptian stock market. Our approach leveraged the strong negative linear correlation between weekly average volatility and XGBoost residuals to refine predictions, resulting in a significant improvement in error rates. The hybrid model achieved a MAPE of less than 10% for 20 out of 30 stocks, with an overall average MAPE of 9.35% and a mean directional accuracy of 82.1%. Key features such as previous day volatility, 5-days ago volatility, and the week of the year were identified as crucial predictors, capturing both immediate market reactions and longer-term trends.

Future work will focus on enhancing the model by incorporating additional data sources, such as other stock market indicators, macroeconomic factors pertinent to Egypt's economy and public's sentiment towards the market and its constituents. This comprehensive approach aims to further improve the accuracy and robustness of volatility forecasts quantifying the stock-associated risk. Our findings demonstrate the potential of advanced machine learning techniques in financial risk management, offering valuable insights for traders and analysts in emerging markets such as Egypt.

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Lift and Shift of Model Code Using Machine Learning Microservices with Generative AI Mapping Layer in Enterprise SaaS Applications

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Abstract. Prior research demonstrated with the Minerva platform that there is a need for easy-to-do machine learning (ML) frameworks. Additionally, SaaS applications are closely related when they form an application suite, which brings forth the need for an ML framework that can facilitate the "lift and shift" of ML model code in similar needs in multiple enterprise applications in a suite. This study recommends a portable ML microservice framework, Minerva (also known as contain second generation), a microservices-based container framework for Applied Machine Learning, as an efficient way to modularize and deploy intelligent microservices in both traditional "legacy" SaaS application suites and cloud environments, especially in the enterprise domain. This study identifies that there is an impetus to innovate quickly for machine learning features in enterprise SaaS applications. The study highlights the real-world implementation of Minerva for "lift and shift," doubling innovation speed with the same resources. It evaluates ML model code reusability across applications, resulting in 1.15 to 2X faster adoption compared to previous methods in a marketing application suite. Secondly, while a layered design accelerated innovation through porting existing models to related business suites, generative AI methods, while promising, have not yet yielded significant gains with smaller models already optimized with no code solutions.

Keywords: Machine Learning · Microservices · Generative AI · Enterprise SaaS · Code Migration · Oracle Cloud Infrastructure · Docker

1 Introduction

[39] demonstrated that there is a need for easy-to-do machine learning (ML) frameworks. Recently, with the advent of ML, there has been a growing need for closely related Marketing SaaS applications. Marketing products like Responsys1 and Eloqua2 belong to a suite that is closely knitted in theme but still serve different market segments: Business to Business (B2B) versus Business to Customer (B2C). This differentiation is evident from the list of common marketing goals in both products, as listed by [23]:

- B2B1:

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1. Personalized customer story and content recommendations.
2. Integrated, targeted advertising.
3. Cross-channel campaign configuration.

- B2C 2:

1. Produce inspiring and personalized email messages simply and efficiently.
2. Targeting and segmentation.
3. Cross-channel experience.

1 https://www.oracle.com/cx/marketing/b2c/

2 https://www.oracle.com/cx/marketing/b2b/

From the above, it can be asserted that both B2B and B2C suites have common needs for personalization, targeting, and cross-channel experiences. Hence, ML innovations in one B2C product, like Responsys, can carry over to B2B, such as in Eloqua. Secondly, there is a need for Portable ML Models in SaaS applications. Philipp et al. (2020) paraphrased a 2018 assertion by Zhao identifying a need for the lift and shift, "sharing of ML models by dividing them into microservices." Further, the researchers emphasized the necessity of having such services common across businesses: "Containerized microservices can further be integrated into different applications within a business" [25]. They identified lift and shift (a pseudonym for interoperability) as a goal of recent efforts: "Interoperability is further obtained by enabling availability to any software client via REST (representational state transfer) APIs and Extensible Markup Language (XML) or JavaScript Object Notation (JSON) data formats." This further strengthens the need to build model interoperability within SaaS application suites.

The main purpose of this study is to propose a portable (lift and shift) ML microservice framework as an efficient way to address portability, quickness, and the challenge of ML models. Portability is notably difficult to achieve in the latest ML platforms: "This makes the management of AI/ML models, e.g., in terms of portability and maintainability, an extremely difficult endeavor that relies on craftsmanship (learned through many failures), rather than a predictable engineering approach" [34]. This study identifies a lift and shift layer in the design of microservices to suit the needs of enterprise application suites.

Research Questions

The study addresses the following research questions:

- **RQ1:** How does the integration of machine learning accelerate innovation in SaaS application suites?
- **RQ2:** What are the development and deployment efficiencies of ML microservices system architecture?
- **RQ3:** How effective are "Lift and Shift" development methodologies in enhancing the portability of ML model code?
- **RQ4:** What viable solutions can be implemented to address the capacity scaling limits of the proposed ML microservices?

2 Related Work

An extensive survey of literature (i.e., peer-reviewed journals and conference articles) was conducted. The initial search yielded over 400 matches, which were then narrowed down to studies that specifically address the application of microservices in enterprise or SaaS applications. Only one study was found that relates to the use of ML microservices in legacy SaaS applications, particularly focusing on accelerating ML deployments in related products. This dissertation focuses on this gap in the literature, as no comprehensive studies were found on model portability in ML microservices within an application suite (Table 1).

Industry has recognized these issues by developing various ML platforms such as MLFlow [36], FBLearner [7], Michelangelo [12], and TFX [4] for ML lifecycle management. These platforms primarily operate in cloud environments, while many legacy SaaS applications still run on native machines in data centers. Van den Heuvel and Tamburri [34] argued that it is difficult to achieve portability of models and AI/ML solutions within current ML Ops frameworks.

Current methods and approaches to AutoML and ML-Ops provide little support for portability and transparency, which are essential for explainability [18]. Additionally, they lack support for key operational considerations such as latency, performance, reliability, deployability, and other essential dependability properties [1].

The proposed model in this study builds on and extends the capabilities of existing platforms. Due to the value delivered to SaaS application suites by running the solution in an "embedded" fashion, it is essential to address security and contractual concerns. Similar to Databricks' MLFlow [36], the model proposed in this study supports polyglot library capabilities by developing generic REST APIs that can work with any ML library or algorithm. However, this model differs by parceling both model and frameworks into simpler project modules/files within container images, rather than using complex deployment repositories. This approach facilitates portability and enables the broader "lift and shift strategy," which is central to this study.

Other recent ML platforms like Facebook's FBLearner [7], Uber's Michelangelo [12], and Google's TFX [4] have attempted to solve similar problems but remain confined to their own ecosystems. Apache PredictionIO offers a centralized solution using Hadoop and Spark MLib but lacks polyglot support and decentralization.

Generative AI has emerged as a promising approach for optimizing SaaS applications, as highlighted in recent studies [2, 10, 26, 29]. Generative AI has proven effective in various fields, including education [6, 16, 32] and marketing, which is the primary focus of this dissertation. Additionally, it has shown potential in optimizing workflows in software development [12, 15, 38] and the financial industry [31]. However, the existing literature on generative AI in software development is often limited to descriptive surveys without quantifiable results.

This study integrates generative AI technology, specifically ChatGPT, to perform migration tasks within the proposed solution. The results are quantified, contrasting with previous studies that have focused on descriptive analyses. This dissertation further demonstrates that generative AI can lead to cost savings by automating repetitive software development activities, significantly improving innovation speed.

Finally, recent dissertations have not adequately addressed ML code portability using microservices. [5] discussed slow application responses and resource usage in microservices, while [21] focused on the development process for ML software. [11] explored resource efficiency in cloud providers but did not address portability concerns.

3 Research Methodology

3.1 Methodology

The methodology for this research primarily involved three evaluations to verify the key claims of the study, as summarized in Table 2:

- **MicroservicesEval:** Evaluation of ML microservices for (a) development and (b) deployment efficiencies.
- **LiftAndShiftEval:** Evaluation of lift and shift development efficiencies of ML microservices portability.
- **CapacityEval:** Assessment of the capacity scaling limits of the proposed ML microservices.

Table 1. Literature Survey Results

Term	Number of Results
Microservices	1000 +
AND Machine Learning	20 +
AND Cloud SaaS Applications	5 +
AND On-Premise SaaS Applications	2 +
AND Model Portability	1 +
AND Generative AI	0 +

The first evaluation involved setting up experiments with and without ML microservices, as described in Sect. 4. The development time (devtime) and deployment time (deptime) metrics were measured in the relevant case study dataset.

The second evaluation focused on using the lift and shift methods proposed in Sect. 5 and measured development time per manpower (devtimeperman) relative to the baseline metric. Finally, the capacity of the proposed microservices was evaluated in terms of requests (results) per second (rps) using a relevant case study dataset.

3.2 Sampling

The following sampling methodology was used in alignment with the approach outlined by Emmel [8] for qualitative analysis of MicroservicesEval and LiftAndShiftEval.

Purposeful Sampling for Case Study: Samples were generated using purposive sampling to capture the challenges of describing, analyzing, and interpreting the approach. This methodology aligns with realist principles, which consider empirical facts independent of perceptions, theories, or constructions. The study probes into these aspects, presenting the sample both as an empirical fact and an evaluation. For quantitative analysis in CapacityEval, the same sample set as used in MicroservicesEval was employed to ensure consistency and enable valuable comparisons.

3.3 Experimental Design

MicroservicesEval (Research Purposes1 & 2: Faster innovation efficient deployment).

Ablation study for data tuning layer This experimental design centered around the data tuning layer, a concept analogous to embeddings utilized in fine-tuning model transferability in deep learning. Ali (2020) detailed the successful evaluation of transfer learning. At its heart, the challenge lay in replicating the base case benefits or features observed in App1. An ablation study provided a mechanism to evaluate the impact of the data tuning layer. According to Wikipedia, "Ablation is the removal of a component of an AI system. An ablation study investigated the performance of an AI system by removing certain components to understand the contribution of the component to the overall system" (author's emphasis). Figure 2 illustrated one such ablation study wherein App1 operated without a data tuning layer, in contrast to app2, which incorporated the layer to achieve a specific desired outcome. Ali (2020) suggested employing a black box methodology to gauge the efficacy of end-to-end translation systems. Typically, the baseline would be the off-the-shelf full implementation as in Web app1. This baseline would be compared with the various versions of the transfer learning scheme of the data tuning layer in app2. If a particular version of data tuning in app2 retained the quality gains of baseline full implementation, then, echoing a conclusion, "it suggested that the simplified version had captured some of the information being transferred. Since information may be transferred redundantly, the claims were limited to sufficiency rather than exclusivity." Thus, app1 exhibited full implementation, and app2 showed limited data-layer-tuned implementation.

Table 2. Claims and Evaluations

Acronym	Research Purpose	Methodology
Research Purpose 1	Faster innovation	MicroservicesEval.a
Research Purpose 2	Efficient deployment	MicroservicesEval.b
Research Purpose 3	Portable	LiftAndShiftEval.a&b&c
Research Purpose 4	Challenges	CapacityEval

Ablation of Microservices Design.

An ablation study of Microservices architecture also revealed its efficacy as shown in Fig. 1(a & b). Figure 1.a depicts an implementation without ML microservices, and Fig. 1.b depicts the proposed approach. As in Fig. 2.a, the enterprise application was built as several sub-systems: UI, DB, CORE, PLATFORM, etc. in a data center. It generated the ML model within the App's DB with tight integrations within its system.

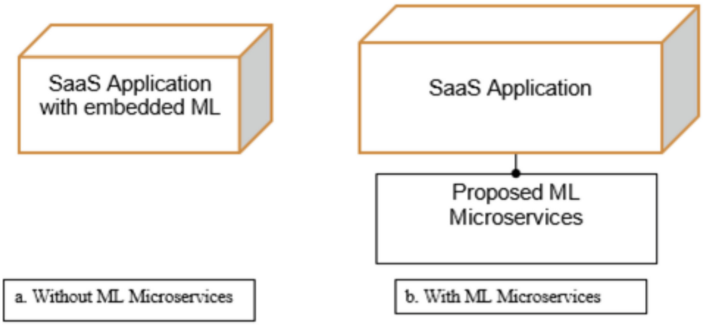

Fig. 1. MicroservicesEval Experimental Design

In Sect. 3.1.b, the proposed ML microservices were added as a docker container at the bottom of the system, implemented as highlighted in Sects. 4 and 5. In short, the solution had three layers: core, abstraction (stub), and app. The "core" layer handled interaction with other systems, process management within the container, concurrency controls, security mechanisms, and configurations. The "abstraction" stub wrapped the algorithm/model, handled dynamic loading of projects ("apps"), versioning, exceptions, and callbacks. The "App" layer implemented the abstractions and coded the model, algorithm, and logic using the best-of-breed ML libraries. The proposed microservices interacted with the enterprise application's sub-systems like UI, core, etc. for training and predictions.

LiftAndShiftEval (ResearchPurpose3: Portable).

This experiment involved leveraging a model artifact ("app") App1 built in Enterprise Application1 and moving the same to Enterprise Application2 without significant modification (more than a sprint: 2 weeks of development effort). The enterprise application1 had a shared storage space that the ML microservice could leverage for intermediate storage of the resulting trained model's artifact. The proposed microservices in Enterprise Application2 had a new component called the data layer that interacted with the enterprise application's sub-systems like UI, core, etc., for training and predictions. Figure 2 illustrated the experimental design. The experiment built a model app1 on Enterprise Application 1 and subsequently ported it to Enterprise Application 2 with minimal modifications to the structure of model app1. Central to the model was the incorporation of Minerva, equipped with a data tuning layer for the second application. App1, with an inherent model 1, was trained and developed on SaaS Application 1. For the discussion, the notion of app portability was equated with model portability. This portability notion

could be measured in terms of code portability and/or input features portability. The study delved into the intricacies of the data layer; however, in abstraction, it acted as a map, converting inputs from SaaS Application 2 to the acceptable inputs (interface) for the model in SaaS Application 1. This mapping was as generic as the underlying constructs of the data tuning layer.

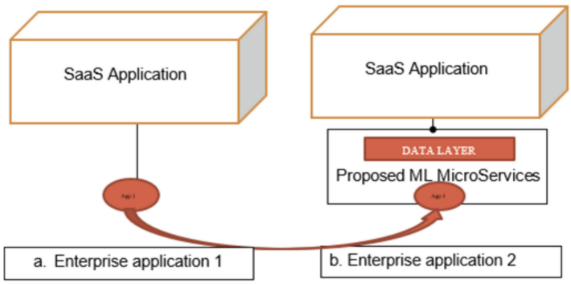

Fig. 2. LiftAndShiftEval Experimental Design

CapacityEval (Research Purpose 4: Challenges).

This evaluation involved two main sequences of activities. First, a training request was generated and sent to the ML microservices (as shown at the bottom of Fig. 3) using a simulator, after the data was preloaded into the database (DB). In the second sequence, predict requests were periodically sent from the simulator to the ML microservices to conduct performance evaluations.

3.4 Saas Application Dataset Sample

The SaaS application products chosen were from a realistic setting in a large enterprise corporation from the Marketing cloud suite. The applications set included newer generation (new gen) applications that ran on the latest "newgen" cloud infrastructure and older "legacy" applications that were developed a decade ago and ran on legacy cloud or infrastructure. Table 3 lists the chosen products.

Table 3. Sample SaaS Products

SaaS Application	Sample Case Study Details
Product 1 (Rsys)	Legacy SaaS app in Marketing
Product 2 (Elq)	Legacy SaaS app in Marketing

Baseline Implementation for App1 and Comparison with Minerva-Enhanced Version.

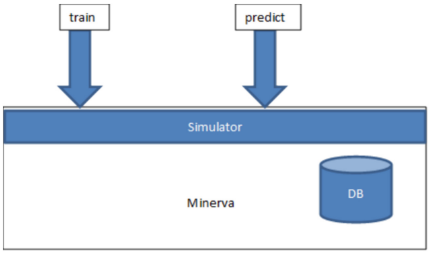

Fig. 3. CapacityEval Experimental Design

The baseline "off-the-shelf" implementation for App1 used traditional ML deployment methods, tightly integrating models within the application's core environment. This setup was limited by static configurations, making updates and porting challenging. In contrast, the Minerva-enhanced version adopted a microservices approach, encapsulating models in Docker containers, which improved flexibility and scalability. With dynamic loading and a modular data-tuning layer, Minerva facilitated quicker deployment and fine-tuning across different applications. Thus, Minerva demonstrated significant gains in innovation speed and efficiency over the baseline.

ML App Dataset Samples.

Each app was evaluated using realistic customer datasets (ds0, ds1, ds2, ds3) as shown in Table 4, which displays the characteristics of the datasets used.

Table 4. Sample Datasets

SaaS Application	Sample Case Study Details
ds0 (Joann)	Retail vertical customer for Product 1
ds1 (DSW)	Retail vertical customer for Product 1
ds2 (Farfetch)	Retail vertical customer for Product 2
ds3 (ATT)	Telecom vertical customer for Product 3

Innovation Case Study Apps.

Faster Innovation of Intelligent App Samples. The faster innovation claim was evaluated by studying the development of three different case study apps: appRFM, appSLO, and appFFC, as outlined in Table 5. The study assessed the efficacy of the proposed solution in realistic scenarios. These apps were used to build marketing intelligence into large enterprise SaaS applications, namely Product 1, Product 2, and Product 3, and were

evaluated using datasets ds0, ds1, ds2, and ds3 (Tables 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, and 16).

Table 5. Innovation Case Study Apps

Sample Innovation Apps	Details
appRFM	An ML feature app to perform intelligent marketing segmentation
appSLO	An ML feature app to perform marketing creative optimization
appFFC	An ML feature app to perform marketing customer persona identification

Table 6. Evaluation Metrics.

Experimental Design	Description
MicroservicesEval.a	Duration of app development in weeks for the proposed solution (Weeks per person)
MicroservicesEval.b	Duration of app deployment in release weeks for the proposed solution (Weeks per person)
LiftAndShiftEval.suite.a	Standardized development time per resource (unit manpower) in weeks per person
LiftAndShiftEval.context.b & genAI.c	Standardized development time per resource (unit manpower) in weeks per person
CapacityEval	Number of requests handled by the studied app concurrently (Requests Per Second)

3.5 Evaluation Metrics

MicroservicesEval and LiftAndShiftEval were evaluated based on the metrics of development time (devtime) and deployment time (deptime), representing the duration in weeks. CapacityEval measured the number of prediction requests handled per second, also known as throughput or Requests (Results) per Second (RPS).

To account for variations in manpower during experimentation and development, the metrics were normalized by man- power/resources in the team experimenting. The derived metric, devtimeperman, was used for comparison in MicroservicesEval and LiftAndShiftEval (Table 6).

Table 7. Mapping Structures

File	Content
model-input.json	Consolidation of all possible input fields for multiple app models. Defines subset or superset of inputs for the model
model-mapping.json	Mapping data to model attribute names. Defines internal model names
model-mapping-schema.json	Document Type Definition (DTD) for inputs. Used for input validation

Table 8. Releases of the Development of Intelligent Features (High Code).

Needs	Q2 FY'19 19A-B[1] Release	Q3 FY'20 20A[2] Release
RFM Segmentation*	X	
SubjectLine Predictions[1]		X

[1]Non-microservices deployment
[2]Minerva Deployment

Table 9. Implementation of the Development of Intelligent Features (High Code)

Needs	Q2*FY'19 19A-B Release	Q3 FY'20 20A[2] Release
RFM Segmentation* (dev^{RFM})	8/100 wpp	
Subject Line Predictions[1] (dev^{SL})		4/100 wpp

* Non-microservices deployment (RFM):
6 weeks (2 sprints) * 2 releases * 2 resources / 300 lines of code = 24/300 weeks per person (wpp)
1 Minerva Deployment (SL):
3 weeks (2 sprints) * 1 release * 2 resources / 300 lines of code = 12/300 weeks per person

Table 10. P-value of devtime of Minerva Development of Intelligent Features

Improvement over non microservices	P-value
devRFM vs devSL	0.019 (significant +)

Table 11. Implementation of the Deployment of Intelligent Features

Needs	Q2 FY'19 19A-B Release	Q3 FY'20 20B Release	Q4 FY'20 20C Release
RFM Segmentation	X		
Subject Line Predictions		X	
Intelligent Switch Fatigue Analysis			X X

Table 12. Deployment Time (deptime) of Intelligent Feature.

Needs	Q2 FY'19 19A-B2 Release	Q3 FY'20 20B3 Release	Q4 FY'20 20C3 Release
RFM Segmentations[2]	1/300 wpp		
Intelligent Switch[3] Fatigue Analysis[3]			0.5/300 wpp 0.5/300 wpp
Subject Line Predictions[3]		0.5/300 wpp	

[2]Manual Deployment (depRFM):
3 weeks (1 sprint) * 1/3 releases * 1 resource / 300 lines of code = 1/300 weeks per person.
[3]Minerva Deployment (depFA, depISW, depSL):
3 weeks (1 sprint) * 1/6 releases * 1 resource / 300 lines of code = 0.5/300 weeks per person.
P-value of Deptime Improvement of Intelligent Features with Minerva Deployment in the Oracle CX Marketing Suite

Table 13. P-value of deptime Improvement of Intelligent Features

Improvement	P-value
depRFM vs depFA & depISW	0.56 (not significant)

4 Solution Architecture

The system architecture and its integrations are depicted in Fig. 4. In summary, the proposed ML microservices are encapsulated in a Docker container, named Minerva, and implemented as outlined in Sect. 4. The Minerva container comprises three distinct layers: Core, Abstraction (Stub), and App.

4.1 Lift and Shift Design (low/No Code)

Static Data Layer Design (No Code).

The current design embeds the data layer deeply within the apps/models, leading to tight integration. This approach introduces coupling issues that prevent the reuse of the same model across different products for experimentation. This tight coupling results

in significant development effort to train new models for different applications, subsequently extending the time-to-market (TTM) capabilities. This dissertation proposed universally abstracting the data layer. By doing so, the same model could be adapted for different projects or apps with only minor adjustments to the data layer specifications. And the results of this study's evaluations are presented in chapter 10. The anticipated benefits of this approach included:

- Reusability: Enabling the use of the same components across various projects.
- Decreased TTM: Shortening the time-to-market.
- Faster Experiments: Accelerating the experimentation process.
- Reduced Development Effort: Lowering the effort required for model development.
- Better Resource Utilization: Optimizing the use of resources.
- Maintainability: Simplifying the maintenance of models and code.

By segregating the Minerva Data Layer, the Minerva Platform's capabilities are augmented, fostering the reuse of a single model/app with different datasets. For instance, an SL Model utilized for push classification can be reused for email classification by merely altering the underlying data. Similarly, the SL model tailored for Responsys can be transitioned for use with Eloqua.

Static Data Layer Interface.

The static data layer interface is essential for robust data science applications, encompassing model mapping, input handling, and data validation:

Each data science application includes a model mapping definition file (e.g., model-mapping.js), which outlines the columns used by the model and the corresponding input mappings. Additionally, a model-input.json file specifies the expected inputs for the model. The model-mapping.json file is validated against a JSON schema for each request to ensure the integrity and correctness of the input data. Initial data validation rules apply to both numerical and categorical variables. For example, while a schema might validate a numerical attribute as a float, additional rules ensure these floats fall within a defined range.

4.2 GenAI Assistant Data Layer Design (Low Code)

The GenAI Data Layer leverages generative AI to optimize data layer transformations, enhancing model code portability across various enterprise applications. This layer primarily facilitates data migration tasks such as renaming tables, adding or dropping columns, and modifying existing columns.

Generative AI models, trained on the capabilities of ChatGPT and the GPT-3.5 framework, provide an automated and suggestive interface to perform these tasks. This system architecture, as illustrated in Fig. 4, streamlines data layer management and supports efficient transformations across different applications.

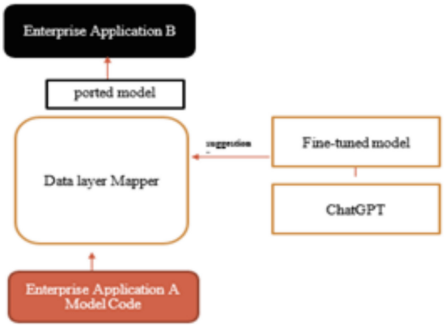

Fig. 4. Generative AI Model Mapper Layer

5 Results

5.1 Microservices Evaluation (MicroservicesEval.Dev.A) (High Code)

The first ML project ("app") (RFM) was successfully implemented using traditional methods across two Oracle calendar releases spanning a total of three two-month timeframes. The subsequent four ML projects ("apps"), termed "Intelligent Features" (Subject Line Predictions, Program Intelligent Switch, Predictive CLV, Fatigue Analysis), were built on top of Minerva and integrated in the next two releases over the following six months. Thus, deploying ML intelligence features in production accelerated at least two times faster compared to traditional approaches with similar resources.

Implementation of the Development of Intelligent Features using Minerva in the Oracle CX Marketing Suite.

Consumer-driven payloads are fundamental to the interaction with the subsystems. Figure 1 [39] illustrates the design and implementation of a generic train payload for batch training any model. The example specifically trains the Customer Lifetime Value (CLV) model within the CX product suite. Various job tracking information is passed in the top section of the JSON payload. The framework captures job statuses, which help in monitoring and debugging issues. Additionally, a polling API reveals the statuses of the ML jobs to the orchestrator. Training is triggered at a pre-determined frequency per account (aka customer) by the host application using an orchestrator. The data is made available by the data processing unit at the CLV_MODEL_INPUT table. The CLV algorithm then builds and stores the ML model in the ML_MODEL_STORED table in a database. The database ensures model security, versioning, and fault tolerance. An asynchronous prediction can be similarly triggered by changing the action type to 'predict', generating results in CLV_MODEL_OUTPUT for each account. The UI then displays these results to the end users.

5.2 Microservices Evaluation (MicroservicesEval.Dep.B)

Minerva was successfully adopted for at least four ML projects ("apps") to enhance connected intelligence in the Oracle CX product suite. According to Oracle, "Oracle CX Marketing (part of Oracle Cloud CX) is an integrated portfolio of digital marketing

solutions that help B2B and B2C marketers use real-time intelligence and customer signals to deliver personalized experiences. It accelerates business growth and brand loyalty".

The next four ML projects ("apps")—"Intelligent Features" (Subject Line Predictions, Program Intelligent Switch, Predictive CLV, Fatigue Analysis)—were built on top of Minerva and integrated in subsequent releases over the next six months. This expedited the deployment of ML intelligence features in production to at least twice the speed of traditional approaches with similar resources.

Implementation of the Deployment of Intelligent Features Using Minerva in the Oracle CX Marketing Suite

Deployment Time (Deptime) of Intelligent Features Using Minerva in the Oracle CX Marketing Suite

5.3 Lift-and-ShiftEval.Suite.A (no Code)

In this experiment, the same context lift-and-shift ablation is evaluated between the same SaaS application suite using Rsys vs. Eloqua for the RFM app in Minerva to demonstrate the portability of models within the same marketing suite (Table 14).

Implementation of Lift and Shift (Same Context) with Static Mapper Data Layer of Intelligent Features

Table 14. Implementation of no code Lift and Shift

Needs	Q3 FY'20 20B Release	Q2 FY'21 21B Release
Subject Line Predictions[1]	X	
Subject Line Predictions[2]		X

Table 15. Development Time (devtime) for No Code Lift and Shift

Needs	Q3 FY'20 20B2 Release	Q4 FY'20 21B3 Release
Subject Line Predictions1[2]	4/100 wpp	
Subject Line Prediction 2[3]		2/100 wpp

[2]Minerva Deployment (SL1) (devSL1):
3 weeks (2 sprints) * 1 release * 2 resources / 300 lines of code = 12/300 weeks per person.
[3]Minerva Deployment (SL2) (devSL2):
3 weeks (2 sprints) * 1/2 release * 1 project * 2 resources / 300 lines of code = 6/300 weeks per person.

Development Time (Devtime) for Lift and Shift (Same Context) Using Static Mappers of Intelligent Features with Minerva in the Oracle CX Marketing Suite

P-value of Lift and Shift (Same Context) Using Static Mappers of Intelligent Features with Minerva Deployment

Table 16. P-value of Lift and Shift

Improvement over high code	P-value
devSL1 vs devSL2	0.4 (not significant)

5.4 Capacity Evaluation (CapacityEval)

Table 17 presents Minerva's performance and availability metrics, reflected in requests per second (RPS) and Non-Availability percentages during predictions for batch sizes of 1 or 1000, across various CPU and Memory configurations of the container. The model employed was a neural network with two layers, using customer behaviors as inputs for predicting the best path in the intelligent switch feature in the Oracle CX Marketing B2C product. The findings indicate that larger machines offer better performance guarantees. Additionally, by adjusting Minerva's concurrency throttling (active processes limit) levels, one can achieve various levels of availability. For these tests, 2000 requests with a concurrency level of 5 were sent to the platform for calibration.

Table 17. Minerva's Performance and Guarantees

Configuration	RPS	Not Available	Available
5CONC-48G, -48CPU-150limit-Batch1000	0	100%	0%
5CONC-48G, -48CPU-85limit-Batch1000	7.19	16%	84%
5CONC-48G, -48CPU-95limit-Batch1000	6.47	7%	93%
5CONC-48G, -48CPU-95limit-Batch1	9.19	0%	100%
5CONC-32G, -32CPU-75limit-Batch1	0	100%	0%
5CONC-32G, -32CPU-60limit-Batch1	10.5	53%	47%
5CONC-32G, -32CPU-68limit-Batch1	7.8	37%	63%
5CONC-32G, -32CPU-71limit-Batch1	6	18%	82%
5CONC-32G, -32CPU-73limit-Batch1	5.5	11%	89%
5CONC-48G, -72CPU-500limit-Batch1000	10.4	0%	100%

5.5 Lift-And-ShiftEval.Context.B (Manual Code)

In this experiment, multi-context lift-and-shift ablation was evaluated between different SaaS application suites, such as Mailchimp vs. Rsys for the RFM app in contAIn, to demonstrate the portability of models across different marketing suites.

Increasing concurrency limits, as seen in Table 17, can lead to higher throughput by allowing more simultaneous processing of requests. However, this often results in increased CPU and memory usage, which may cause resource contention. While higher concurrency can boost RPS (requests per second), it may also lead to system instability

Table 18. Implementation of Intelligent Features RFM manual code.

Needs	Q3 FY'23 23B2 Release	Q4 FY'23 23D3 Release
RFM 1 minerva-like-contAIn Predictions[2]	X	
RFM 2 contAIn Predictions[3]		X

[2]Minerva-like-contAIn Deployment.
[3]contAIn Deployment.

Table 19. Development Time (devtime) for RFM manual code.

Needs	Q3 FY'23 23B2 Release	Q4 FY'23 23D3 Release
RFM1 [2] minerva-like-contAIn	6/300 wpp	
RFM2 [3] contAIn		1/300 wpp

[2]contAIn (Minerva-like) Deployment (RFM) (devRFM1):
$\approx$ 3 weeks (2 sprints) * 1 release * 1 project * 1 resource / 300 lines of code = 6/300 weeks per person.
[3]contAIn Deployment (RFM Mailchimp) (devRFM2):
$\approx$ 1 week (1 sprint) * 1 release * 1 project * 1 resource / 300 lines of code = 1/300 weeks per person.

Table 20. P-Value of devtime (Multi-Context) of Development Improvements

Improvement over high code	P-Value
devRFM1 vs devRFM2	0.05 (significant +)

Table 21. Implementation of RFM with Generative AI low code.

Needs	Q3 FY'23 23D2 Release	Q4 FY'23 23D3 Release
RFM 2 contAIn Predictions[2]	X	
RFM 3 contAIn Predictions[3]		X

[2]contAIn deployment RFMconstantcontact.
[3]contAIn deployment RFMmailchimp.

if resources are overwhelmed, resulting in latency spikes or higher non-availability percentages. Effective concurrency management is crucial to balance performance gains with resource limitations, ensuring the system maintains stability without excessive resource consumption.

Table 22. Development Time (devtime) of RFM with Generative AI low code

Needs	Q3 FY'23 23D2 Release	Q4 FY'23 23D3 Release
RFM2[2] contAIn	1/300 wpp	
RFM3[3] contAIn		(6/7)/300 wpp

[2]contAIn Deployment (RFMconstantcontact) (devRFM2):
≈ 1 week (1 sprint) * 1 release * 1 project * 1 resource / 300 lines of code = 1/300 weeks per person.
[3]contAIn Deployment (RFMmailchimp) (devRFM2):
≈ 6/7 weeks * 1 release * 1 project * 1 resource / 300 lines of code = (6/7)/300 weeks per person.

Table 23. P-Value of devtime (Similar-Context) of Development Improvements

Improvement over no code	P-Value
devRFM1 vs devRFM2	0.78 (not significant)

Implementation of Intelligent Features RFM in ContAIn in Multi-Context Marketing of CX Marketing and ContAIn Suite

Development Time (Devtime) for RFM in Minerva-Like-ContAIn vs. Third-Party App Mailchimp's Model Creation in ContAIn

P-Value of Devtime (Multi-Context) of Development Improvements Due to Minerva-Like-ContAIn in Multi-Context

5.6 LiftandShiftEval.GenAI.C

In this experiment, the lift-and-shift ablation with the Generative AI assistant ETL mapping layer was evaluated between similar context SaaS applications like Mailchimp vs. Constant Contact using contAIn. This demonstrates the generalization of model portability across different marketing suites.

Development Time (Devtime) of RFM in ContAIn in Similar-Context Marketing Suite with Generative AI

P-Value of Devtime (Similar-Context) of Development Improvements Due to Generative AI Layer in ContAIn

The implementation of the RFM model using Generative AI in contAIn for similar-context marketing suites, such as Mailchimp and Constant Contact, is summarized in Table 21. The table details the release timelines and deployment status of two prediction models: RFM 1 for Constant Contact and RFM 2 for Mailchimp. RFM 1 was integrated during Q3 FY'23, while RFM 2 was completed by Q4 FY'23, demonstrating the flexibility of the contAIn framework to support different marketing applications.

The development time for these models, presented in Table 22, shows that RFM 1 required approximately 1/300 weeks per person (wpp), whereas RFM 2, which involved more complex requirements, needed around (6/7)/300 wpp. This difference highlights

the additional development effort needed for RFM 2 due to the varying data specifications and context adaptations required. The impact of the Generative AI layer on development efficiency was not statistically significant, as shown in Table 23. The P-Value of 0.78 indicates that while the Generative AI layer facilitated model portability, its contribution to reducing development time was limited. This suggests that further enhancements to the AI-assisted ETL processes are necessary to achieve more substantial improvements in similar contexts.

6 Conclusion

This study proved that Microservices were essential when it came to putting machine learning (ML) models into older Software as a Service (SaaS) applications. Introducing Minerva and subsequent ContAIn ML framework is a big step forward in strategy meant to make ML models more flexible, reusable, agile, tech-stack independent, and quick to install in old SaaS systems. It is emphasized that Minerva is a ML system that can be changed and used to port and lift and shift models between closely related applications that form a enterprise suite. It works especially well for older applications that can't move right away to cloud servers or send their data to modern cloud machine learning platforms to be processed. This method helps with the problems that come with adding machine learning to older, single-piece SaaS apps by creating a link between standard business systems and the benefits of new machine learning technologies. It is also forward compatible with new breed cloud applications as demonstrated by experimentation. With AI and ML moving at a fast pace, this study looks at the speed of innovation with several experiments. It has been discovered that a layered design can help with acceleration of innovation by adopting and porting existing models from similar applications into a related business suite or application. Generative AI methods are an encouraging phenomenon in this porting but significant gains do not currently materialize with smaller models already optimized with no code solutions when measured with lines of code.

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Machine Learning-Driven Extended Creativity for Reshaping Traditional Artistic Pieces

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Abstract. This paper explores the transformative impact of Artificial Intelligence (AI) and Extended Reality (XR) within the framework of machine learning, focusing on their applications in creative industries. Specifically, it examines how AI-driven creativity and XR technologies are reshaping traditional artistic practices, particularly within the art community in Bangkok and its surroundings. By analyzing key variables such as cost, time, and technological capabilities of AI-generated art and XR devices, this research highlights how these technologies are revolutionizing the creative process and enhancing accessibility for artists. The study investigates the underlying machine learning models and soft computing methodologies that enable the integration of AI and XR into artistic media. Furthermore, it examines how XR enhances storytelling and audience engagement through advanced learning systems. The findings suggest that AI and XR technologies are merging to provide new avenues for creative expression, ultimately influencing the professional practices of artists. The paper concludes by outlining future research directions in machine learning and soft computing, emphasizing their critical roles in the evolution of artistic expression. This research offers a framework for understanding the coexistence of human creativity with AI and XR in the digital age, underscoring the potential for innovation and collaboration in the arts.

Keywords: Computer Learning · Generative AI Art · Extended Creativity · Metaverse · Human Creativity

1 Background

In today's rapidly advancing technological landscape, machine learning (ML) and related computational techniques are profoundly transforming human behavior and artistic practices. These technologies are becoming increasingly integrated into interdisciplinary fields [26–28], including journalism, media, and visual arts, creating new paradigms for how art is conceptualized, created, and consumed. Over the past decade, the revolution in Information Technology (IT) has positioned computers as indispensable tools for creative professionals. The fusion of hardware, software, and digital services has empowered digital artists to explore uncharted creative territories, fundamentally altering how art is produced and perceived.

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Traditional artistic methods, such as painting on canvas, are gradually being complemented—or in some cases, replaced—by digital tools that enhance efficiency and accessibility. Artists now leverage search engines, image-generation software, and prototyping tools to gather inspiration and refine their creative visions, reflecting the growing intersection of technology and art. These advancements have enabled artists to work across platforms, explore new media, and expand the boundaries of creativity in ways that were previously unimaginable.

However, despite these technological breakthroughs, certain aspects of creativity—particularly those tied to nuanced, emotive, and personal expression—remain inherently human. In 1968, Pablo Picasso remarked on the limitations of computers, pointing out their inability to generate original creative insights. Yet, as of 2022, Artificial Intelligence (AI) has evolved to surpass human capabilities in numerous domains, from diagnosing complex medical conditions to excelling in strategic games like chess and Go. This progression has extended to the arts, where modern AI systems, powered by deep learning and neural networks, have begun producing visually striking pieces of art. These developments represent a shift toward what some describe as "machine creativity", challenging traditional notions of artistic authorship and originality.

The emergence of AI platforms such as Midjourney, launched in 2022, has further showcased the artistic potential of these technologies. By transforming simple text prompts into high-quality visual outputs, AI tools are not only democratizing access to artistic creation but also sparking debates about the displacement of traditional creative processes. Many human artists express concerns over the potential loss of creative control, fearing that automation may diminish the value of their work. Yet, others see AI as a collaborative tool that can augment human creativity rather than replace it, offering new ways to conceptualize and execute artistic ideas.

At the same time, Augmented Reality (AR) is redefining the way art is experienced and consumed. AR combines digital content with physical environments, leveraging machine learning algorithms for real-time interaction and precision, as discussed by Seeliger et al. in 2024 [3]. This technology allows for the creation of immersive art experiences that bridge the gap between the digital and physical worlds, transforming static works into dynamic, interactive creations. Unlike AI, which focuses on automating the creation process, AR provides a platform for artists to expand the context of their work, creating multisensory experiences that transcend traditional artistic mediums. Notable applications include virtual exhibitions, interactive installations, and immersive storytelling, as described by Haahr et al. in 2025 [10].

For instance, AR technology has enabled viewers to engage with iconic works of art in novel ways. Imagine experiencing a traditional Thai painting where vibrant elements come to life through AR, allowing audiences to explore details and narratives beyond the canvas. Projects like these highlight the potential of AR to democratize art by making it more accessible, interactive, and globally relevant.

1.1 The Evolution of AI and AR in Art

The rise of AI and AR in artistic creation signals a paradigm shift. AI tools like Midjourney and DALL·E have made it possible for individuals with no formal artistic training to produce professional-quality images, raising important questions about authorship,

copyright, and the value of traditional artistic skills [19]. Concurrently, AR technologies—powered by machine learning algorithms—have unlocked possibilities for spatial storytelling and interactive design, blending the physical and digital realms in ways that expand the definition of art itself.

These technologies are reshaping not only the creative process but also the broader ecosystem of artistic production and consumption. For example, the relatively low cost and high efficiency of AI-generated art make it an attractive option for commercial applications, while AR installations offer unique opportunities for cultural institutions, galleries, and brands to engage audiences. However, such advancements also underscore the importance of balancing technological innovation with the preservation of human creativity.

1.2 Machine Learning and the Future of Art

This study delves into the transformative impact of machine learning and Soft Computing on artistic practices, with a focus on key variables such as cost, time efficiency, and the technological capabilities of AI and AR tools. By examining these variables, the research highlights how machine learning technologies are enhancing accessibility, streamlining production processes, and influencing the professional practices of artists.

Focusing on Bangkok and its surrounding areas, the study identifies a growing acceptance of AI and AR among local artists, entrepreneurs, and consumers. This adoption is driven by the increasing affordability of machine learning-powered tools and the recognition of their potential to complement traditional methods. For instance, in Bangkok's burgeoning creative hubs, AR has been integrated into public art displays and educational programs, while AI tools are used for rapid prototyping and digital storytelling.

As machine learning and soft computing continue to evolve, their role in shaping the future of art will likely deepen. Artists who embrace these technologies stand to benefit from enhanced creative possibilities, while those who resist may find it challenging to compete in an increasingly digital-first landscape. Nonetheless, the coexistence of human creativity, AI innovation, and AR interactivity offers a compelling framework for the future of artistic expression in the digital age.

2 The Proposed Framework

Figure 1 illustrates the conceptual framework of this research, divided into two main components: Machine Learning Artist Models and AR Technology Integration. The first section focuses on machine learning-driven artist models, with three key variables playing crucial roles:

1. Capabilities of Machine Learning Artist Models: This variable encompasses the diversity of machine learning models designed for creative tasks. It examines the range of output possibilities, such as image resolution and flexibility in inputs, including the ability to refine or modify existing images based on user preferences.

2. Cost of Machine Learning Models: This variable addresses the financial aspect of employing machine learning models for creative tasks, considering pricing structures, accessibility, and ease of use. It emphasizes the importance of making these models user-friendly and accessible to artists without requiring advanced programming skills, enabling broader adoption among creative professionals.
3. Time and Iteration: This variable concerns the time required to generate and iterate upon outputs from machine learning models. It explores how long it takes to produce an image that meets specific user requirements, often involving multiple iterations to achieve the desired level of detail and precision.

These three primary variables are analyzed as significant factors influencing the dependent variable, which is the impact of machine learning artist models on artists and creative practitioners in Bangkok and surrounding regions.

The second section focuses on the integration of AR Technology. The key variable under investigation is:

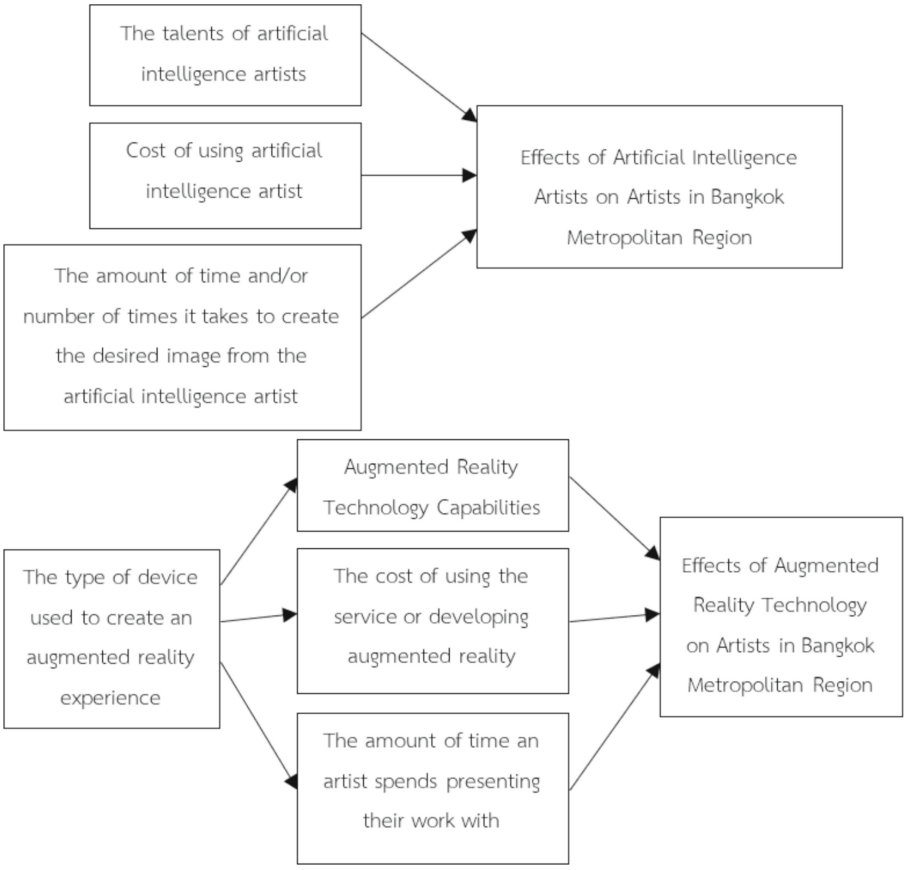

Fig. 1. The conceptual framework of the presented research

- Type of AR Device: This variable examines the choice of devices used for creating augmented reality experiences. The type of device—such as smartphones, projectors, or AR headsets—plays a significant role in determining the capabilities and impact of AR technology.

Additional considerations in the AR domain include:

- Cost of AR Technology: This variable assesses the financial implications of both developing and accessing AR tools, influencing the adoption and practical usage of AR within creative fields.
- Presentation Time: This variable highlights the time artists need to effectively showcase their work using AR technology, underscoring the efficiency and feasibility of integrating AR into creative processes.

We hypothesize that these variables in both machine learning and AR technology will significantly shape the dependent variable, which is the influence of these technologies on artists and creative practitioners in Bangkok and surrounding areas.

3 Artificial Intelligence Artist

The term "artist" encompasses individuals engaged in various creative fields, including painting, sculpture, architecture, literature, and more. The elusive origins of creativity, which manifest within an artist's mind, have been a longstanding subject of investigation. A study conducted by Beaty [1] in 2020 utilized brain imaging to explore the sources and processes underlying creativity, aiming to address questions such as what triggers creative thought and why some individuals exhibit higher levels of creativity than others. This research yielded three significant conclusions:

1. Creativity Requires Cognitive Effort: Contrary to being an automated process, creativity necessitates conscious cognitive effort from the brain. This effort is vital for breaking free from established knowledge frameworks and existing memories.
2. Involvement of the Brain in Future Imagination: The same brain regions responsible for crafting imaginative visions of the future are also pivotal in the creative process.
3. Enhanced Brain Network Communication: Creativity is closely associated with increased communication among networks in the brain that typically function independently. The study revealed that highly creative individuals exhibit significant interconnectivity within three specific brain regions: the default, control, and salience networks. This heightened interconnectivity fosters the generation of more innovative ideas.

As neuroscientists strive to unravel the complexities of the human brain, another group of researchers is focused on developing machines and algorithms capable of achieving human-like intelligence—a pursuit encapsulated by the term "Artificial Intelligence" (AI), which was first introduced in a 1955 paper by John McCarthy, a computer science professor at Stanford University. Notably, the initial mathematical model aimed at replicating the neural workings of the human brain dates back to 1944, conceived by Warren McCullough and Walter Pitts at the University of Chicago (Hardesty, 2017).

Today, Artificial Neural Networks (ANNs), designed to simulate the human brain's functions, represent a significant branch of artificial intelligence, as explained by Stork in 2024 [2].

The early development of neural networks focused on creating equations that mirrored human cognitive functions. However, modern neural networks employ techniques that differ substantially from the actual workings of the human brain. The main commonality lies in the complexity of their operations, which often defy straightforward explanations of how intelligent decisions or actions originate. This inherent limitation poses a significant challenge for contemporary neural networks. Despite their inability to clarify the origins of their intelligent outcomes, these networks demonstrate remarkable capabilities for analyzing, connecting, and inferring insights from vast datasets [24]. This intelligence has raised concerns about the potential displacement of professions centered on computation and analysis.

In 1995, Steve Jobs famously stated in an interview with Wired magazine that "creativity is just connecting things." This notion resonates strongly with the capabilities of neural network models, which excel at linking disparate pieces of information and knowledge. Thus, it is not surprising to see the integration of AI, particularly in the form of "Artificial Intelligence artists (AI artists)," within the creative domain. This integration is especially prominent in fields such as painting and image creation. Noteworthy AI artists like Midjourney, DALL-E 2, as explained by Ramesh et al. [4], and Stable Diffusion, as discussed by Rombach et al. [5], gained popularity in 2022 for their ability to generate images based on textual descriptions. These images range from lifelike photographs to intricate line art and unique styles reminiscent of famous artists or studios, even conjuring fantastical scenes that seem to spring from the depths of imagination.

In addition to generating images from text descriptions, these AI artists provide users with features that allow for repeated selection and editing of their preferred versions until the desired image is achieved, as explained by Heitzinger and Stork [6]. Today's leading AI artists also possess advanced capabilities, including super-resolution to enhance image detail and clarity, text-based enhancements for specific image areas, image inpainting and outpainting to fill in missing portions, and guidelines for element placement within images. Within just two months of Midjourney's public launch in July 2022, it sparked the emergence of new services and phenomena, including text input services for each AI artist (prompt generators) and instances where AI-generated works won awards, such as a $3.00 prize at the Colorado State Fair Fine Arts Competition, igniting controversy among competing artists.

The rise of AI artists has generated significant interest and popularity among users worldwide [21], propelling rapid advancements in AI-generated art and related services, as mentioned by Kawakami and Venkatagiri in 2024 [23]. Consequently, it is essential for those affected by this technological shift, particularly professional artists and painters, to understand the nature and capabilities of AI artists, as discussed recently by Jääskeläinen et al. in 2024 [22] and Jedrusiak in 2024 [25]. This understanding is crucial for making timely adjustments and ensuring the sustainable future of artistic careers in an evolving landscape shaped by machine learning.

4 Findings on User Perceptions of AI and XR

The AR and Machine Learning Integration.

AR has emerged as a significant application within the domain of machine learning, enabling the overlay of digital content onto physical environments in real-time. By leveraging machine learning algorithms [29, 30], AR ensures precise alignment of virtual graphics with the real world, optimizing positional and temporal accuracy, as presented previously in our works [11–13]. These systems depend heavily on machine learning for real-time data processing, encompassing image recognition from cameras, motion detection from sensors, and the analysis of environmental variables, alongside essential hardware and software components, as discussed by Battineni et al. in 2024 [18]. Current AR platforms primarily operate across smartphones, headsets, and projectors [7], each offering unique strengths and constraints for various use cases.

AR as a Tool for Artists.

For professional painters and artists, AR has transitioned from being a novel concept to a complementary tool, expanding the creative possibilities of traditional art forms. Unlike the challenges posed by AI-driven art generation, AR facilitates the digital augmentation of art, transforming static pieces into dynamic, immersive experiences. For instance, the works of Thailand's renowned artist Thawan Duchanee, including his paintings, have been adapted for augmented reality displays. This technological enhancement has contributed to the rise of the "Augmented Reality artist," a specialized role dedicated to developing interactive digital-physical art experiences.

Machine Learning Enhancements in AR Applications.

AR's growing capabilities have been amplified by advanced machine learning techniques, as introduced recently by Pierdicca et al. in 2024 [16]. For instance, Neural Style Transfer, as introduced by Gatys et al. [9], enables seamless integration of digital objects with real-world environments by harmonizing lighting and textures. Similarly, object detection models, as explained by Nie et al. [8], underpin practical AR applications like IKEA Place, which allows users to virtually place furniture in physical spaces for informed decision-making [17]. Machine learning techniques, such as Optical Character Recognition (OCR), further enhance AR functionalities, exemplified by applications like Google Translate, which interprets and translates text from physical environments into digital outputs.

Interviews and User Perceptions of AI and AR.

This study assessed the impact of AI and AR on art through interviews with 15 individuals spanning art creation, entrepreneurship, and consumption. It also analyzed feedback from 35 users of Midjourney, evaluating four key dimensions—meaningfulness, meeting expectations, academic value, and commercial value.

1. Meaningfulness: Most users rated this aspect between 2 and 4, reflecting mixed reactions. While some found the images meaningful, others felt ambivalent, suggesting the subjective nature of interpreting AI-generated art.
2. Meeting Expectations: User ratings predominantly ranged from 3 to 5, indicating that the generated images generally meet expectations but could benefit from customization for more precise alignment with user needs.

3. Academic Value: Scores varied widely, from 1 to 5, suggesting the academic utility of AI and AR-generated content is context-dependent and warrants further exploration to establish its value in scholarly domains.
4. Commercial Value: Scores for this dimension were evenly distributed between 1 and 5, offering no conclusive evidence of substantial commercial viability. A more detailed analysis of market trends and specific applications may be necessary to evaluate potential.

Aesthetic Appeal and Efficiency.

In the domains of aesthetics and efficiency, most users awarded scores of 4 to 5, indicating high satisfaction with the visual appeal and speed of AI-generated content. Interviews reinforced this finding, as a significant number of respondents agreed on the aesthetic appeal of text-generated imagery by Midjourney.

Implications for Machine Learning and Soft Computing.

The results highlight the dual role of machine learning and soft computing techniques in AR and AI. While these technologies offer scalability and efficiency, they are best utilized as tools to complement human creativity rather than replace it. Adaptive learning mechanisms and user feedback loops are essential for refining outputs and catering to diverse user expectations.

The Role of AR in Transforming Artistic Expression.

AR bridges the digital and physical worlds, offering artists a unique medium for immersive storytelling. By blending traditional art with interactive technology, AR enables artists to create experiences that engage audiences in innovative ways. The integration of AR into Thawan Duchanee's works demonstrates the potential of this technology to expand the boundaries of visual art, transforming passive observers into active participants.

The synergy between AI, AR, and human creativity will continue shaping the art world. AI-powered AR systems, as they evolve, have the potential to assist artists in generating, refining, and elevating their creative processes. However, the unique emotional depth and narrative complexity of human artistry remain irreplaceable. While AI accelerates content creation, it harmoniously coexists with human creativity, enriching the broader artistic landscape.

By embracing advancements in AI and AR, artists stand to gain a competitive edge, leveraging these technologies to innovate and redefine artistic expression. The findings underscore the importance of maintaining a balance between technological innovation and the irreplaceable essence of human creativity.

5 Summaries and Future Prospects

In this paper, we have explored the intricate interaction between machine learning, soft computing, and their influence on artists, particularly in the Bangkok region. We focused on three key variables that affect the integration of AI-generated art and how it impacts the work of human artists and painters [20]. Additionally, we have analyzed the role of AR in reshaping artistic expression and the technological advancements in recent years. The results reveal a significant shift in how ML and soft computing techniques are applied to artistic practices, enhancing both the creative process and audience engagement.

Looking ahead, future research will focus more specifically on examining the effects of AR and AI-driven art on artists and creators within niche domains. These areas of interest may include the application of machine learning in fields such as artistic gymnastics, as presented by Mazurova [14] and the convergence of AI and dance within the broader field of artistic computing, as explained by Castro et al. in 2022 [15]. By extending our research across diverse cities and creative industries globally, we aim to gain a deeper understanding of the nuanced ways in which ML and soft computing are transforming artistic domains.

Our ongoing investigation seeks to offer more comprehensive insights into how AI technologies continue to augment and reshape the creative landscape, not only in Bangkok but across specialized artistic domains worldwide. This future work will contribute to the evolving understanding of the relationship between human artists and AI, shedding light on the opportunities and challenges presented by these technological advancements in the creative industries.

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Towards AI/ML-Powered Hybrid Project Management Strategy for the Healthcare Sector

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Abstract. In the rapidly evolving healthcare sector, project management faces challenges due to the complexity and dynamic nature of its environment. Traditional methodologies including PMI PMBoK, and PRINCE2, combined with Agile, provide structured and adaptable approaches; however, the potential of the integration of artificial intelligence (AI) techniques is still underexplored. This study investigates how AI-driven tools, such as machine learning (ML) and predictive analytics, embedded with project management methodologies can improve project efficiency, decision-making, and resource management in the healthcare sector. Through a comprehensive literature review, we identify key AI technologies that augment task automation, real-time insights, and predictive capabilities within healthcare project management. This study presents statistical evidence from the literature on the percentage distribution of project management methodologies with key aspects including adaptability to AI, compliance, flexibility, stakeholder engagement and risk management. We discuss how a hybrid approach that leverages the strengths of PRINCE2/PMI, Agile, and AI can accelerate timelines, improve adaptability, and enhance stakeholder satisfaction. Despite challenges such as data privacy and compliance, this study presents a mapping of AI technologies and project management methodologies along with a conceptual design towards building a strategic framework that aligns AI advancements with organizational goals, optimizing healthcare project outcomes.

Keywords: Project Management · Artificial Intelligence (AI) · Machine Learning (ML) · Agile · Healthcare

1 Introduction

In the healthcare industry, managing projects can be particularly challenging due to the complex nature and the need to adapt quickly to new developments. Traditional project management methodologies, such as PMI PMBoK (Project Management Institute - Project Management Body of Knowledge) [1], PRINCE2 (PRojects IN Controlled Environments) [2] and offer well-defined structures for planning, control, and risk management. These methodologies provide a detailed framework with extensive documentation and clearly defined roles, making them suitable for projects that require strict adherence to processes and thorough oversight, especially in healthcare-related projects

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[3]. However, Agile methodologies [4] provide a more flexible approach, focusing on iterative progress and constant feedback. Agile encourages ongoing adjustments and collaboration, which is beneficial in fast-changing environments where requirements can shift frequently. Iterative sprints in Agile allow teams to adapt project goals based on real-time input, enhancing their ability to respond to new challenges.

Artificial Intelligence (AI) is revolutionizing a vast variety of interdisciplinary areas, with the main objective of improving standards and creating new possibilities for improving overall workflow. For example, AI has been implemented to improve supply chain management [5], smart factory-based operational efficiency and warehouse management [6] including health and medicine, in particular, cancer care [7] among other avenues such as pandemic forecast [8]. Integrating AI into project management approaches can bring additional advantages and reduce human errors [9]. AI technologies, such as machine learning and predictive analytics [10], can boost decision-making, streamline resource management, and deliver real-time project insights. By automating repetitive tasks, analyzing large datasets, and forecasting potential issues, AI can make project management more efficient and responsive [13].

By combining the structured methodologies of PRINCE2/PMI with the flexibility of Agile and the smart technology of AI, a powerful new approach to healthcare project management emerges. This hybrid model combines the detailed planning of PRINCE2/PMI, the adaptability of Agile, and the advanced capabilities of AI, creating a comprehensive strategy that addresses the unique challenges of healthcare projects while enhancing efficiency and agility [11, 12].

2 Literature Review

Artificial Intelligence (AI) is making project management easier and more efficient in several ways. For starters, AI can handle repetitive tasks like scheduling, tracking progress, and generating reports, which means project managers have less paperwork to deal with [14]. It also helps improve how we estimate the time and effort needed for tasks by analyzing past data, making planning more accurate [12]. AI can match the skills of team members to the tasks they're best suited for, ensuring that resources are used effectively [13]. It also helps in spotting potential risks early, so teams can address them before they become major issues [11]. Plus, AI provides real-time insights that help teams make better decisions on the fly [13].

When we bring AI into traditional project management methods like PRINCE2/PMI and Agile, we get a powerful combination. PRINCE2/PMI offers a structured approach with detailed planning and control, while Agile provides flexibility and adaptability through its iterative processes (Project Management Institute, 2021). By merging these methods with AI, we can enjoy the best of both worlds: the thorough planning of PRINCE2/PMI, the flexibility of Agile, and the smart technology of AI. This hybrid approach helps manage projects more effectively by improving planning, adapting to changes, and making better decisions based on real-time data [3]. The literature provides several dimensions of healthcare management where AI has been implemented and this study categorizes this into eight key areas (as depicted in Fig. 1) including diagnosis of diseases, medical image diagnosis, drug discovery, personalized medicine, medical robots, electronic health records, clinical trials and outbreak prediction.

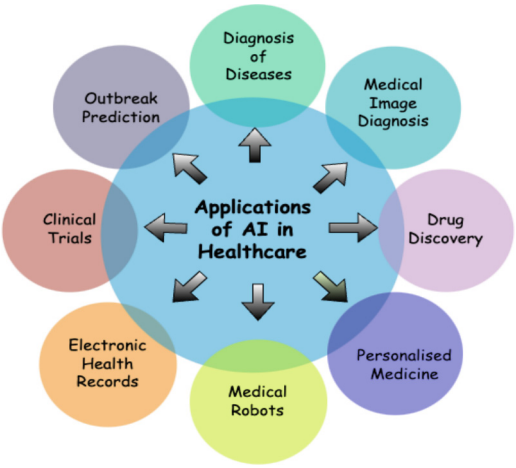

Fig. 1. Applications of AI in Healthcare Sector

2.1 Diagnosis of Diseases

AI and machine learning (ML) have revolutionized how diseases are diagnosed and treated [15]. By harnessing the power of computational statistics these technologies sift through intricate data sets to reveal patterns and links that might slip past the keen eyes of human experts. AI techniques have been effectively utilized in the diagnosis of diseases early on, without letting it become worse, e.g., catching the first hints of cancer [7]. Advanced AI-based imaging technology can tell the difference between harmful tumors and harmless ones incredibly well. IBM's Watson Genomics [16] stands out for its use of smart computing to quickly identify the genetic makeup of tumors which helps in diagnosing cancer swiftly. Similarly, MD Anderson [17] presents a machine learning tool with the aim of foreseeing the harsh side effects in patients with head and neck cancer who are receiving radiation therapy. This effort highlights the capability of AI to improve how treatments are approached. The literature suggests even though AI can't take over the role of a doctor's expert decision-making it can back up what doctors find in their exams. This makes it easier to get the diagnosis right.

2.2 Medical Image Diagnosis

Computers are now getting better at analyzing the images from medical tests with the help of AI-based Computer Vision approaches which have been making big progress in the field of radiology. When these smart programs look at X-rays PET MRI and CT scans they can be just as good as the doctors who have been doing it for years. In the realm of medical advancements machine learning has reached an impressive accuracy level of up to 89% when it comes to identifying breast cancer [18]. This achievement showcases the significant impact that such technological advancements could have on the way diagnoses are made. Similarly, Microsoft's "InnerEye" project [19] provides the advancement of tools aimed at analyzing medical images. This endeavor serves to

sharpen the accuracy of diagnoses. Big collections of medical images have made it possible to create smart ML tools.

2.3 Drug Discovery

The use of AI in creating drugs and in their production has completely changed these processes that used to take a long time and were often risky. AI/ML algorithms are now able to figure out which biomarkers will tell us what we need to know and confirm that the targets for treatment are correct doing all this with a greater level of efficiency. They have the ability to foresee different ways to treat illnesses, e.g., cancer treatment tailoring their approach to meet the unique requirements of each patient. Models at the cutting edge such as Seq2seq Fingerprint [20] use learning without direct supervision to uncover new paths in drug discovery by spotting trends and hints within the data that lead to breakthroughs in treatment. Similarly, AI/ML techniques are leading the way to tailor drug mixes specifically for illnesses e.g., acute myeloid leukemia [21].

2.4 Personalized Medicine

Utilizing the power of AI, personalized care has become a reality allowing medical treatments to be customized for each patient. This approach considers the distinct genetic makeup and medical history of the patients. By sifting through extensive data from Electronic Health Records and merging it with genetic details machine learning algorithms can pinpoint the treatment paths that are likely to work best. For instance, IBM Watson Oncology [22] taps into the history of patients to suggest various action plans. These are then looked over by doctors to see if they fit well. Using this tailored method boosts how accurate and quick treatments are especially in tricky areas such as cancer care.

2.5 Physical Robots

With the emergence of humanoid robotics, customized robots powered by AI can now assist in the operating room remotely operated by medical experts, e.g., prostate health issues. Now these robots are smartened up with AI brains [23]. This allows them to handle tricky jobs during surgery with a lot of accuracy making sure patients get better results.

2.6 Electronic Health Records

In the realm of healthcare, the introduction of AI has been a game-changer, especially in taking over admin duties. It cuts down on the mistakes people make by hand and makes keeping track of records smoother. On the other hand, tools like ML-driven OCR [24] and sorting out documents make handling data a breeze. Such systems are capable of examining records from past patients which allows them to offer initial steps to take. This aids greatly in making decisions regarding clinical care.

2.7 AI in Clinical Trials

During clinical trials the help of AI in choosing who gets to participate keeping an eye on the process as it happens and looking at the data afterwards makes everything work smoother and faster. When companies like "Medtronic's" [25] team up with IBM to work on diabetes and insulin data by using machine learning it shows us how these smart programs make it easier to manage information leading to stronger and quicker results from the trials.

2.8 Outbreak Prediction

During the COVID-19 crisis the skill of AI to foresee the future played a big role. It looked at fresh info from many places like social media and news sites to guess how fast the virus would move. Online portal such as ProMED-mail [26] showed how AI can keep an eye on new sicknesses and let health experts know fast. This helped the healthcare professionals to act quickly to stop the diseases from spreading. The fields of AI and ML are transforming healthcare in remarkable ways i.e., making diagnoses more precise improving how treatments are tailored to the individual and making the whole clinical and admin side of things run smoother.

3 Towards AI/ML-Powered PM Methodology for Healthcare

The prevalence of AI technologies is imminent, however, it's important to investigate the success possibilities of AI projects in the healthcare sector. The latest literature suggests that the integration of PRINCE2/PMI, Agile methodologies, and AI in healthcare project management is essential to ensure the successful implementation of AI projects. In order to bridge this gap, this study follows a thorough literature review strategy using online electronic databases such as ScienceDirect, Scopus, Web of Science, IEEE Xplore, and PubMed. This review focused on identifying relevant studies that explore the integration of AI with traditional project management frameworks like PRINCE2/PMI and Agile, and their healthcare applications. The selected studies are analyzed to understand the benefits and challenges of this hybrid approach. The search also included examples from healthcare projects that have successfully implemented AI in conjunction with PRINCE2/PMI and Agile methodologies.

Figure 2 presents the findings from our analysis of selected literature review of how AI is being integrated into traditional project management methodologies like PRINCE2 and PMI, as well as Agile approaches, within the healthcare sector. These visualizations summarize that Agile is the most frequently used methodology, with a significant number of papers using it, followed by Agile-Hybrid, followed by PMI/Prince2.

Figure 3 provides a percentage distribution of the dominance of Agile in the literature we reviewed. It also illustrates the adaptability and preference of Agile and Agile Hybrid methodologies across various dimensions, underscoring their ability to manage complex and dynamic projects. According to this comprehensive analysis, iterative and flexible project management approaches are becoming more popular, especially in the rapidly evolving healthcare project management.

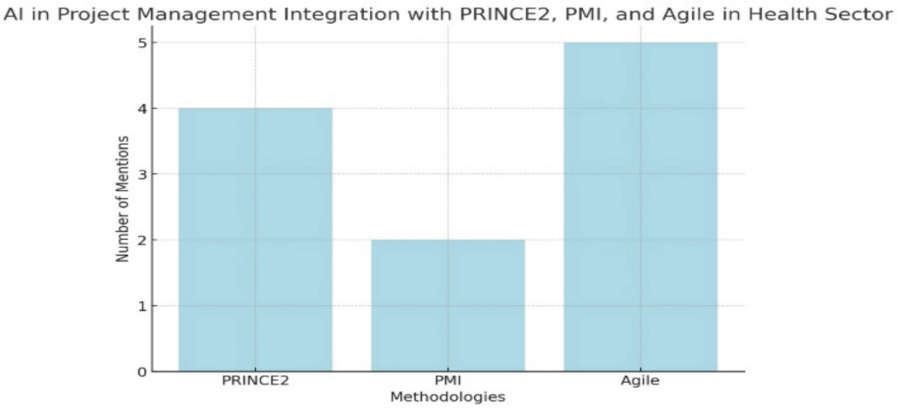

Fig. 2. Project Management Strategies in Healthcare Sector

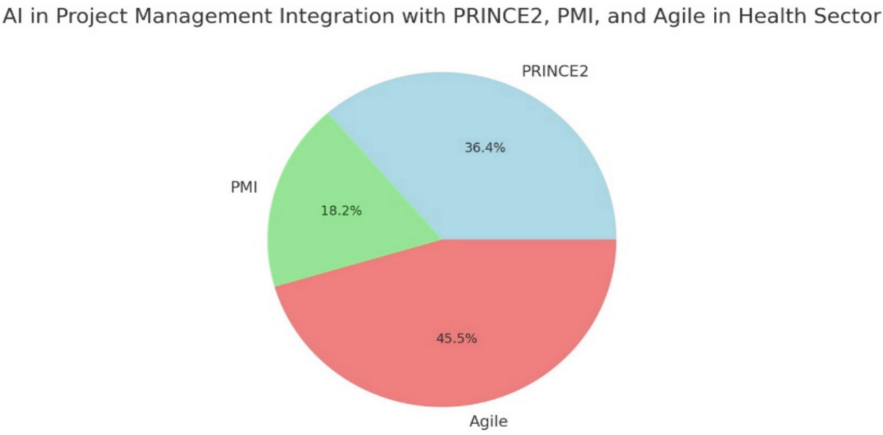

Fig. 3. Percentage Distribution of Project Management Methodologies

The observation via a pie chart in Fig. 3 breaks down the proportion of AI integrations among the three methodologies. Agile accounts for nearly half of all mentions, underscoring its adaptability and the growing preference for Agile in AI-driven projects within the healthcare sector. PRINCE2 and PMI, while still significant, represent smaller shares, indicating that these methodologies may require more adaptation to fully leverage AI.

This study also explored key trends within AI-driven healthcare projects, which are visualized in the radar chart in Fig. 4. The chart highlights AI adoption and personalized medicine as the most prominent trends, reflecting the sector's increasing reliance on AI to tailor treatments and improve patient outcomes. Other important trends include explainable AI, which addresses the need for transparency in AI decision-making, and stakeholder engagement, which remains crucial for gaining trust and ensuring compliance with regulatory standards.

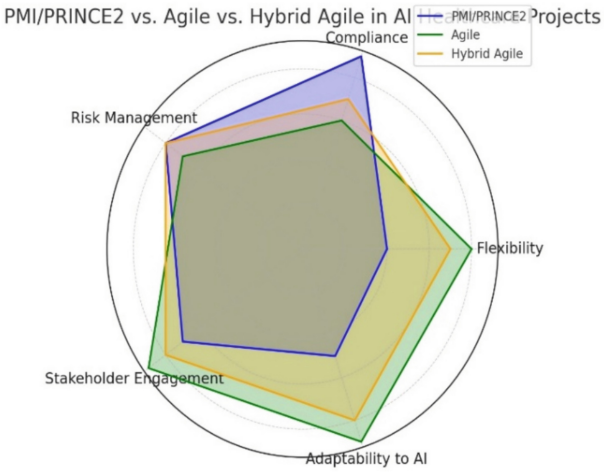

Fig. 4. Key Trends in AI-powered Healthcare projects

This study also provides a comparison of the effectiveness of different project management methodologies—PMI/PRINCE2, Agile, and a Hybrid Agile approach in healthcare projects. The results show that while traditional methodologies like PMI/PRINCE2 excel in areas such as risk management and compliance, Agile and Hybrid Agile methodologies are better suited for projects requiring flexibility and quick adaptation to new AI technologies. The Hybrid Agile approach, in particular, offers a balanced performance across all parameters, making it a strong candidate for managing AI-driven healthcare projects [27]. Table 1 presents synthesizes the findings to develop practical recommendations for combining AI with PRINCE2/PMI and Agile methodologies to improve methodologies and recommendations, aligning them with current trends and advancements in healthcare project management. This table illustrates how different project management methodologies (Agile, PRINCE2/PMI, and Hybrid) are applied in the healthcare sector, with a focus on project management activities, areas, AI techniques, and the resulting improvements in efficiency and patient care.

The findings of this study provide a mapping of AI and project management methodologies for the Healthcare sector in Fig. 5, which explains that the integration of AI techniques with project management methodologies like PMI, PRINCE2, Agile, and hybrid approaches presents a transformative potential for healthcare innovation. Predictive analytics, a core AI capability, enhances patient care by identifying potential risks early. This enables healthcare providers to make informed decisions, reducing complications and costs. The use of AI streamlines complex healthcare tasks, improving patient outcomes. As the PMI approach, particularly in risk management, ensures that healthcare projects mitigate risks related to patient safety, its comprehensive risk assessments and mitigation strategies enable healthcare projects to align with healthcare safety goals and meet their objectives effectively. Similarly, PRINCE2's emphasis on rapid iterative development allows healthcare providers to adapt quickly to feedback and evolving patient needs. The iterative nature of PRINCE2 enables projects to be flexible, making it easier for

healthcare teams to implement rapid changes while maintaining control over scope and objectives.

Table 1. Key PM Areas and Applied AI/ML Techniques in Healthcare Projects

PM Approach/ Methodology	Key PM Area	Key PM Activity	Applied AI/ML Techniques	Improvements Using AI/ML Techniques	Reference Studies
Agile	Team Collaboration	Performance Analysis	Predictive Analytics	Enhanced adaptability to changing healthcare requirements	[4]
	Flexibility	Spring Retrospectives	Natural Language Processing	Improved communication between software development and healthcare teams	[6]
		Product Backlog Feature	Sentiment Analysis	Automated prioritization of patient feedback	[28]
PMI	Scope Management	Detailed Planning	Generative AI Algorithms	Task Dependency Identification, and conflict resolution e.g., hospital staff and resources	[26]
		Work Breakdown Structure	Collaborative GUI based AI systems	Automation of WBS Trees and Work Allocation and Compliance with regulations	[15, 20]
	Risk Management	Qualitative and Quantitative Risk Analysis	Risk Prediction AI-based Risk Scoring	Optimizes healthcare resource usage, reduces overallocation, improves project efficiency	[12]

(*continued*)

Table 1. (*continued*)

PM Approach/ Methodology	Key PM Area	Key PM Activity	Applied AI/ML Techniques	Improvements Using AI/ML Techniques	Reference Studies
PRINCE2	Directing and Controlling	Rigorous Quality Control	AI-based Triggers and Automated Reporting	Improved healthcare quality compliance and maintained regulatory requirements	[29]
	Governance and Product Delivery	Setting Stage boundaries, track progress	AI scheduling Workflow automation	Automates task tracking, improves stage completion estimates, reduces delays	[2]
		Ensuring product delivery timelines	ML-based timeline predictive reporting	Enhanced product quality, automated defect detection, improved delivery time	[27]
Hybrid	Integration Management	Combining agile sprints with PMI/PRINCE2	AI-Driven Dashboards, ML-based visuals	Enhanced resource utilisation and along with a structured approach and compliance	[30]

As the aforementioned discussion unfolds that Agile project management, with its focus on clear governance and accountability, supports the efficient allocation of resources and enables healthcare teams to adapt to changing priorities. Hence, a hybrid methodology that combines the structure and oversight of traditional approaches like PMI and PRINCE2 with the flexibility of Agile can bring the actual advantage of the technological advancements of AI in the healthcare sector more effectively. These methodologies allow healthcare projects to innovate rapidly, deploying AI-driven tools to address complex healthcare challenges while maintaining robust project controls. This mapping of AI and project management methodologies illustrates a comprehensive

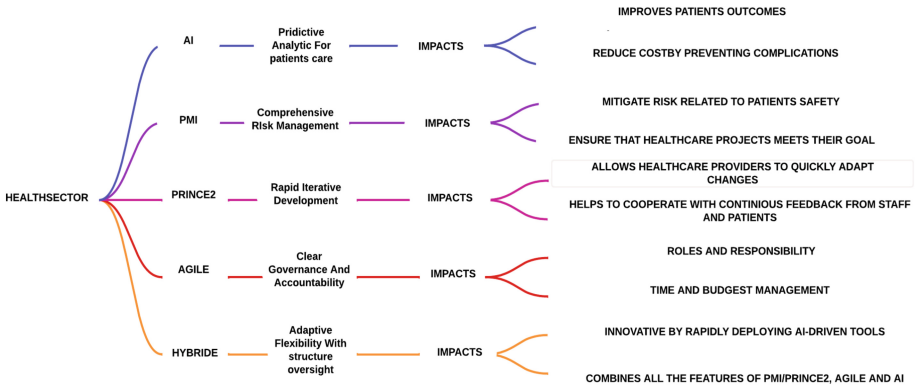

Fig. 5. Mapping of AI and PM Methodologies in Healthcare

approach to managing healthcare projects, leveraging the strengths of each methodology to improve patient outcomes, reduce costs, and increase the adaptability of healthcare systems. The fusion of AI with project management ensures a balanced approach to innovation, combining technological advancements with structured governance frameworks.

The integration of AI technologies into project management frameworks enhances efficiency, adaptability, and decision-making capabilities. Predictive analytics helps identify bottlenecks and delays, optimising scheduling and risk management, while natural language processing (NLP) automates stakeholder communication and documentation through sentiment analysis and feedback prioritisation. Machine learning (ML) supports resource allocation by creating optimised work breakdown structures and ensuring quality control. Additionally, computer vision tracks progress in real time, enabling visual monitoring and ensuring quality assurance. These techniques are mapped to methodologies like Agile, PMI/PRINCE2, and hybrid approaches. For instance, Agile benefits from predictive analytics for flexibility, PMI/PRINCE2 leverages AI-driven dashboards for compliance tracking, and hybrid approaches use AI for integration management. A practical example includes electronic health record (EHR) implementations [31], where predictive analytics streamlines prioritization to meet project goals within regulatory constraints.

Figure 6 represents a comprehensive AI implementation framework within the healthcare sector, illustrating the integration of various data sources like electronic health records, medical imaging, wearable devices, and patient surveys. This data is utilized to develop AI models leveraging techniques such as deep learning, natural language processing, predictive analytics, and computer vision. This framework highlights the central role of stakeholder engagement, including healthcare providers, patients, researchers, and administrators, to ensure AI-driven healthcare innovation. Key areas of implementation include robotic surgery, telemedicine, and mobile health platforms. These innovations undergo clinical trials, retrospective studies, and peer review, contributing to evidence-based healthcare improvements. The administrative support tools ensure seamless integration into healthcare operations, allowing AI to optimize care

Fig. 6. Implementation of AI/ML Techniques for Project Management in Healthcare Sector

delivery processes. This AI-based approach aims to enhance patient care, improve operational efficiencies, and foster innovation in medical practices by integrating real-world evidence into healthcare workflows.

Data privacy and compliance challenges in AI integration require a multi-faceted approach [23]. Sensitive data handling and collaborative sharing are primary concerns in healthcare projects due to strict privacy laws like GDPR and HIPAA. Mitigation strategies include federated learning to train AI models on decentralised data without transferring sensitive information, differential privacy to anonymise datasets, and encryption to secure data in transit and at rest. Regular audits and explainable AI frameworks enhance transparency while aligning AI development with standards such as ISO 27701 ensures compliance [18]. Practical applications include using federated learning in EHR systems to keep patient data local while enabling global AI model improvements, applying differential privacy in clinical trials to anonymise participant data, and

leveraging AI-powered dashboards for real-time compliance tracking [9]. These measures foster trust, safeguard sensitive data, and ensure responsible deployment of AI in healthcare projects.

In order to provide an impactful and applicable approach, this study endorse a forward-looking strategy, e.g., in a hospital resource management scenario [11], the implementation of AI-driven predictive analytics was instrumental in optimising staff allocation. The AI-based approach is effective is analyzing the historical patient admission patterns and real-time resource availability, enabling the hospital to anticipate demand surges and reduce patient wait times. Similarly, this study highlights the importance of data privacy and compliance, by adopting federated learning techniques to safeguard sensitive health data and implementing regular audits for regulatory compliance. To address technological constraints, we recommend the integration of adaptive learning algorithms that allow AI models to evolve with new data trends, and explainable AI frameworks to improve transparency and trust. These actionable insights not only tackle present challenges but also ensure the sustainable adoption of AI in healthcare project management as technologies evolve.

4 Conclusion

This study provides insights into how the integration of AI into healthcare project management methodologies can help cope with the complexities of the sector. By combining the structured approaches of PRINCE2/PMI with the flexibility of Agile, and empowering them with AI technologies, this study demonstrates how project efficiency, decision-making, and adaptability can be significantly improved. AI's ability to automate tasks, provide predictive analytics, and generate real-time insights can streamline workflows and enhance resource utilization, contributing to better project outcomes. However, challenges such as data security, technological dependence, and compliance must be carefully managed. The proposed hybrid approach, which blends traditional and Agile methodologies with AI, provides a comprehensive solution that addresses these challenges while unlocking new potential in healthcare project management. As healthcare continues to evolve, the strategic adoption of AI will be critical for improving project outcomes and achieving organizational goals.

As this domain holds huge potential, there is a scope of future work to focus on developing and testing AI-driven tools specifically tailored for healthcare project management, incorporating advanced features such as adaptive learning algorithms and real-time risk assessment models. Further exploration is also needed to address the ethical and regulatory challenges associated with AI integration, particularly in terms of data privacy and security in healthcare environments. Additionally, longitudinal studies that evaluate the long-term impact of AI-enabled project management on healthcare outcomes, patient satisfaction, and cost efficiency would provide valuable insights. Collaborative efforts between healthcare professionals, AI developers, and project managers are essential to ensure that AI systems are designed to meet the specific needs of healthcare projects while maintaining transparency and trust.

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Enhancing Textual Deception Detection: A Fused Handcrafted Feature Approach with Machine Learning Models

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Abstract. Deception in textual communication presents a persistent and significant challenge across numerous domains, including media, finance, and cybersecurity, where accurate identification of misleading information is critical. This paper introduces a novel approach to deception detection by proposing a machine learning model designed to address the inherent complexities of textual deception. The core of our approach lies in the development of a Fused Handcrafted Features vector, which integrates a diverse set of features: lexical patterns, statistical distributions, syntactic structures, and semantic relationships derived from WordNet-based resources. These features are carefully engineered to capture both surface-level and deep linguistic patterns indicative of deceptive behavior.

Our model is further enhanced by its ability to analyze sequential dependencies and identify subtle deceptive cues, especially in longer text segments, where traditional methods often struggle. Unlike conventional ML techniques that primarily rely on shallow feature representations, our model leverages advanced architectures to process the nuanced characteristics of deceptive texts. By fusing these handcrafted features with sophisticated ML algorithms, we achieve a robust framework capable of generalizing across diverse datasets.

The proposed approach is rigorously evaluated on three widely recognized benchmark datasets, demonstrating its superior performance compared to traditional methods. Our model achieves accuracy improvements ranging from 2% to 7.5%, underscoring its effectiveness in handling various textual deception scenarios. These results highlight the potential of integrating handcrafted linguistic features with machine learning models to address the multifaceted nature of deception.

This research not only establishes the efficacy of our proposed Fused Handcrafted Features based model but also opens new avenues for further exploration in real-world applications. By providing a comprehensive analysis of deception through a multi-faceted feature representation, this

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work lays a solid foundation for future advancements in deception detection technologies, particularly in applications where trust and integrity in textual communication are paramount.

Keywords: Deception Detection · Machine Learning · Natural Language Processing · TF-IDF · Textual Analysis · Lexical and Statistical Features · Syntactic Analysis · Semantic Features

1 Introduction

The rise of deceptive content across media, finance, and cybersecurity poses significant challenges, driven by recent global events such as the war in Ukraine, MENA region crises, and political turmoil Elhadad [1]. Deceptive practices, including misinformation, fraud, and phishing, threaten trust and decision-making, impacting industries like finance and politics Hancock et al. [2], Sarzynska-Wawer et al. [3]. As digital communication expands, the need for robust deception detection in text has become essential for safeguarding information integrity.

Advances in machine learning (ML) have improved deception detection by enabling the capture of lexical and semantic features, although early approaches were limited by their reliance on handcrafted features and scalability challenges Pérez-Rosas and Mihalcea [4], Sultana and Palaniappan [5]. Enhanced feature engineering and integration techniques have further advanced the field, improving the ability to identify complex patterns indicative of deception.

1.1 Challenges and Hybrid Approaches in Deception Detection

This work proposes a hybrid model that integrates linguistic features such as statistical measures (TF-IDF), similarity measures (syntactic and semantic-based), linguistic measures (POS tagging, readability, and complexity), and sentiment shifts. This architecture addresses deception detection challenges in complex, multi-sentence texts like reviews and legal documents.

1.2 Datasets and Contributions

Our model was evaluated on three benchmark deception detection datasets:

- **Deceptive Opinion Spam Dataset** Ott et al. [6], which contains truthful and deceptive hotel reviews.
- **Open Domain Deception Dataset** Pérez-Rosas and Mihalcea [7], which includes deceptive and truthful statements on various topics.
- **Deception in Future Activities Dataset** Kleinberg and Verschuere [8], which includes statements from participants predicting their future activities, labeled as deceptive or truthful.

These datasets ensure diverse coverage for robust evaluation. Our model demonstrates superior performance over traditional ML and DL models, with gains in Accuracy, Precision, Recall, F1-Score, and AUC. An ablation study confirms the model's robustness and applicability across domains, supporting real-world tasks like media monitoring, e-commerce reviews, and legal document analysis.

This paper is organized as follows: Sect. 2 introduces the problem of deception, Sect. 3 reviews related work, Sect. 4 details our hybrid architecture, Sect. 5 describes experiments and results, and Sect. 6 concludes the study and discusses future directions.

2 The Problem of Deception

Deception detection is complex, extending beyond word-level cues to involve intricate discourse patterns over entire documents or conversations. While transformer models like BERT capture contextual relationships, they may struggle with coherence in long texts. Combining BERT with sequential models like LSTM helps address this by capturing both context and temporal flow, making it effective for deception detection.

Another challenge is the diversity of deceptive language across domains and cultures. For instance, deception in product reviews differs from that in political or legal texts Ott et al. [6], and linguistic cues vary across cultures. Models trained on specific datasets may thus struggle with generalization. To address this, diverse domain-specific datasets such as the Deceptive Opinion Spam Dataset Ott et al. [6] and Open Domain Deception Dataset Pérez-Rosas and Mihalcea [7] are used as benchmarks.

This study addresses the following research questions (RQs):

- **RQ1:** What are the limitations of current AI methodologies for deception detection, and how can hybrid models improve them?
- **RQ2:** How can diverse datasets be curated to train models effectively, ensuring minimal bias and broad generalizability?
- **RQ3:** Which transformer-based architectures combined with sequential learning best enhance deception detection Accuracy?
- **RQ4:** How can model performance be optimized to balance Accuracy, computational efficiency, and scalability?
- **RQ5:** What future AI research directions could further improve deception detection, especially in context-specific applications?

Deception detection in text remains a complex problem. AI and NLP advancements, particularly ML combined with feature fusion, significantly improve scalability and accuracy. However, developing models that generalize across domains, languages, and cultures, while remaining interpretable, is essential for creating reliable real-world applications.

3 Literature Review

Deception detection has advanced with NLP and ML, enhancing model Accuracy and interpretability. Following the PRISMA protocol Moher et al. [9], this review covers recent advancements, especially in ML-based deception detection Elhadad et al. [10].

3.1 PRISMA-Based Selection

Using PRISMA criteria, studies from the last five years were selected for their focus on model Accuracy, generalizability, and interpretability. The selection involved:

1. **Identification**: Database searches with keywords like "deception detection" and "ML."
2. **Screening**: Exclusion of unrelated, outdated, or incomplete studies.
3. **Eligibility**: Inclusion of studies featuring novel ML algorithms for deception detection.

3.2 Traditional Approaches and Early ML Methods

Earlier approaches to deception detection were primarily rooted in linguistic analysis. Psychological and linguistic studies showed that deceptive communication often exhibits specific patterns, such as fewer self-references, more negative emotion words, and increased cognitive load markers Hancock et al. [11], Pennebaker et al. [12]. Based on these findings, rule-based systems, such as the LIWC framework, were developed to automatically identify deceptive cues based on lexical categories Newman et al. [13].

However, these static models were limited in their ability to generalize across different contexts and domains. As computational capabilities grew, researchers began adopting ML algorithms to automate deception detection. Early ML models, such as Support Vector Machines (SVMs), Random Forests, and Logistic Regression, used manually engineered features (e.g., word frequencies, syntactic structures, and sentiment) to classify text as deceptive or truthful Pérez-Rosas and Mihalcea [4], Ott et al. [6]. These models demonstrated notable success in certain domains, such as online reviews and product feedback, where patterns of deception could be more easily detected through repetitive linguistic structures.

Despite these advances, traditional ML models suffered from a lack of flexibility. They relied heavily on feature engineering, which required human expertise and often missed the subtle, context-dependent nature of deception. Moreover, these models struggled to process long-range dependencies in textual data, limiting their effectiveness in more complex, longer texts.

3.3 Challenges and Directions

Key challenges include building generalizable models across languages and domains and enhancing transparency. Explainable AI methods like SHAP and attention mechanisms are essential for trust Rudin [14], Lundberg and Lee [15], with hybrid models and cross-domain datasets as critical directions for improving deception detection effectiveness.

4 Proposed ML Model for Deception Detection

Detecting deceptive language in text is a challenging task, as deception often manifests through subtle cues scattered across different parts of a document. To address this complexity, we propose a machine learning architecture that integrates multiple types of linguistic features. We implement a feature fusion process that merges contextual embeddings with handcrafted linguistic features, enhancing the model's ability to identify both obvious and subtle signs of deception.

To ensure that our approach remains methodologically solid and novel, we report experimental results using alternative architectures and a variety of feature vectors. These experiments showed that the alternative models offered competitive performance. However, our proposed feature integration approach consistently outperformed them, particularly in maintaining coherence over longer text spans.

This section outlines the architecture, comprehensive data preprocessing pipeline, feature fusion strategy, and evaluation process. The proposed framework uses three datasets to predict deception in text, and it comprises two primary approaches: an ML approach and a proposed model. Figure 1 shows the bolck diagram of the proposed model. First and foremost, data is preprocessed, and all unnecessary tokens and symbols were removed. Secondly, each dataset is split into training and testing sets. Next, a conversion of texts into vectors. Following that, a process of calculating WordNet Similarity, Transformer Similarity, Sentiment-Related Features, obtaining POS tags, and calculating TF-IDF is done in the Feature Fusion Stage to obtain the Fused Feature Vector. The obtained Fused Feature Vector is used to train different ML models. Finally, the performance of the proposed model is tested using performance evaluation metrics. The detailed stages of our model are illustrated in detail in the following points.

4.1 Data Collection

The datasets used in this study were sourced from various domains known to contain deceptive content. These include:

- Deception in future activities Dataset: The data were collected through crowdsourcing in which participants were asked to provide a statement about their most significant non-work-related activity in the next seven days. Participants were asked the two following questions:

  - Q1: "Please describe your activity as specific as possible"
  - Q2: "Which information can you give us to reassure us that you are telling the truth?"

  The dataset contains 1640 statements (857 deceptive and 783 truthful) consisting of two answers per participant. We selected this dataset because, as a result of strong experimental control throughout the collection process, it may be considered high quality. As a result, each statement's ground truth is known, whereas it is frequently unknown in other datasets Markowitz et al. [16].
- Open Domain Deception Dataset: Here, the authors asked 512 participants through Amazon Mechanical Turk to provide seven truths and seven lies, on topics of their own choice. The dataset includes 512 truthful and 512 deceptive statements Brzic et al. [17].
- Deceptive Opinion Spam Dataset: The dataset used in our experiment was collected from online sources, which consists of a total of 1600 reviews from TripAdvisor, Yelp.com, Expedia, Hotels.com, Priceline, and Amazon Mechanical Turk. There are two types of reviews in the dataset, i.e., reviews representing positive sentiment and reviews representing negative sentiment Hagendorf [18].

Each dataset was preprocessed to ensure consistency and compatibility across different sources and The number of instances in each dataset is shown in Table 1.

Table 1. The number of instances in each dataset.

Datasets	Sets	Truthful	Deceptive	Total
Deception in future activities	Training set	656	656	1312
	Testing set	164	164	328
				1640
Open Domain Deception	Training set	2867	2867	5734
	Testing set	717	717	1434
				7168
Deceptive Opinion Spam	Training set	640	640	1280
	Testing set	160	160	320
				1600

4.2 Data Filtering

Data filtering ensures text is clean, structured, and enhanced with features for accurate deception detection. Our pipeline involves several stages to improve the model's ability to identify linguistic patterns.

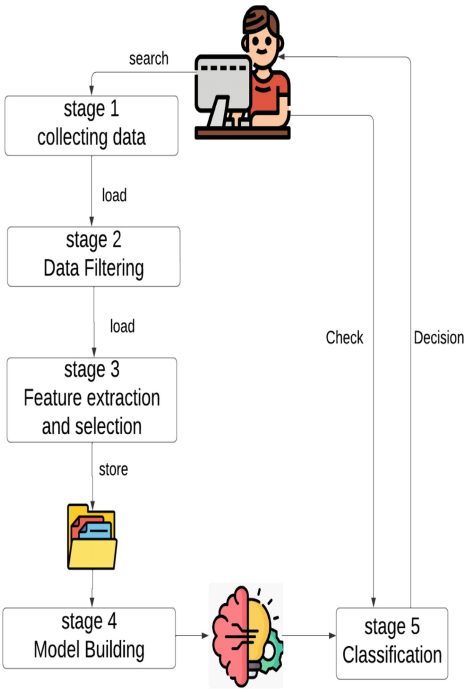

Fig. 1. The proposed system block diagram

1. Data Cleaning: Unnecessary elements like HTML tags, special characters, and stop words are removed, retaining only relevant content for analysis.

2. Tokenization: Using the BERT tokenizer, text is divided into tokens, handling complex and out-of-vocabulary words for both the input text and extracted features.

3. Lemmatization and Stemming: Words are reduced to their base or root forms (e.g., "running" to "run"), improving model consistency across word variations.

4. Handling Missing Data: Critical missing values are removed, and non-critical values are imputed to maintain dataset completeness and prevent bias.

5. Dependency Parsing: Identifies complex syntactic structures often found in deceptive statements by analyzing word relationships.

6. Coreference Resolution: Detects shifts in entity references (e.g., "I" to "we"), flagging inconsistencies that may signal deception.

7. Lexical Diversity and Repetition: Measures word variety, with repetitive language often indicating deception as it avoids new details.

4.3 Feature Extraction and Selection

Key linguistic features linked to deception are extracted, including:

- **TF-IDF (Term Frequency-Inverse Document Frequency):** Highlights significant words by weighing their frequency in the document versus the corpus.
- **PoS Tagging:** Identifies sentence structures, with deceptive statements often showing fewer self-references and more negative words Newman et al. [13].
- **Sentiment Shifts:** Tracks sudden sentiment changes, such as moving from criticism to exaggerated praise, often associated with deceptive behavior. Sarzynska-Wawer et al. [3]. In other words, it detects shifts in sentiment, as sudden positivity after negativity may indicate dishonesty.
- **Readability and Complexity Metrics:** Assesses text complexity, with deceptive content often being either overly simplistic or complex.
- **NER (Named Entity Recognition):** Flags inconsistent or vague references in entities like names or dates.
- **Lexical Diversity and Repetition:** Quantifies word variety to detect repeated phrases, common in deceptive statements.

Together, these techniques, combined with feature fusion and the ML model, provide a comprehensive understanding of text semantics and structure

Feature Fusion: After feature extraction, these handcrafted features are combined into a comprehensive feature vector. The Fused Handcrafted Features vector includes various linguistic and statistical features (e.g., TF-IDF scores, PoS tags, sentiment analysis). This comprehensive vector allows the model to leverage both contextual understanding and manually crafted linguistic features, enhancing its ability to detect deception in text.

4.4 Model Building

The model architecture integrates handcrafted linguistic features with machine learning techniques to capture both semantic nuances and long-range dependencies in the text.

1. Advanced Contextual Feature Extraction: Advanced feature extraction techniques are employed to capture contextual relationships within text. These methods analyze both local and broader patterns in language, enabling the identification of subtle shifts in tone, sentiment, and emotional content, which are critical for detecting deception.

2. Sequential Modeling for Long-Form Text: To process long-form text and capture evolving patterns of deception, the model integrates techniques that account for sequential dependencies across sentences and paragraphs. This sequential modeling is essential for detecting gradual changes in tone, sentiment, and narrative consistency-key markers of deception in extended discourse.

3. Integration of Feature Fusion: The Fused Handcrafted Features vector, which combines multiple linguistic features, is processed using advanced sequential modeling techniques. This approach enables the model to learn both immediate contextual relationships and long-term dependencies, enhancing its ability to detect complex deception patterns. For example, the model can track sentiment shifts over the course of a document and identify whether the tone of a review changes unexpectedly. Figure 2 illustrates the structured progression of the model combined with a feature fusion technique.

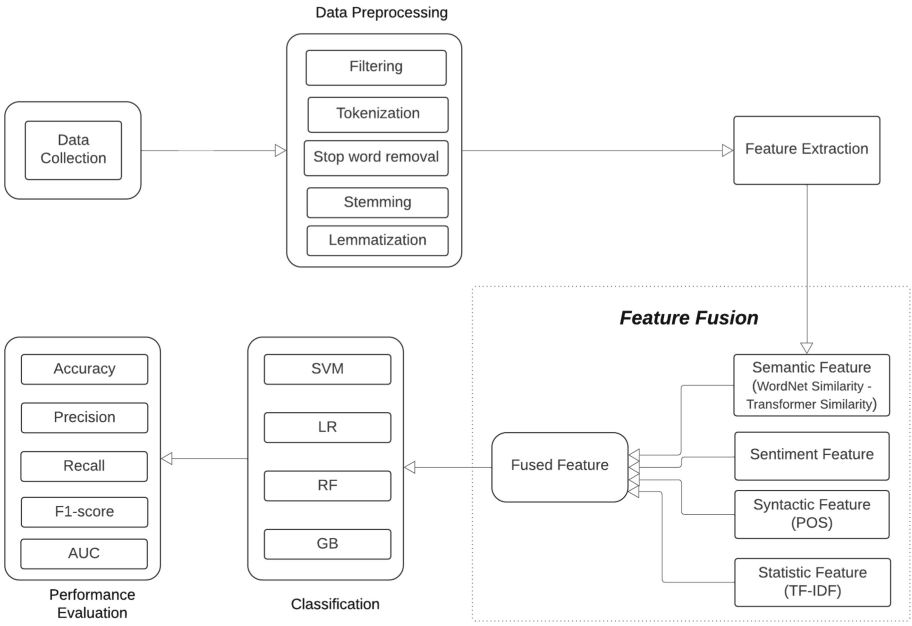

Fig. 2. Detailed Block Diagram of the Proposed Model

4.5 Training and Optimization Process

The training process is divided into multiple stages to ensure the model learns both general language patterns and domain-specific nuances:

- **Leveraging Large Text Corpora:** The model incorporates general language patterns derived from large text corpora. This approach enhances the model's ability to understand linguistic structures and semantic relationships, making it applicable across various domains.
- **Domain-Specific Training:** The model is trained on domain-specific datasets such as the Deceptive Opinion Spam Corpus Ott et al. [6] and the Open Domain Deception Dataset Pérez-Rosas and Mihalcea [4]. This process

allows the model to adapt to the specific linguistic characteristics of deceptive language in various contexts, such as product reviews or personal statements.

- **Feature Fusion Training:** After generating and extracting various handcrafted linguistic features, the Fused Handcrafted Features vector is processed using advanced machine learning algorithms for training.

4.6 Classification And Performance Results

The proposed model consistently outperforms traditional machine learning models across multiple deception detection datasets. On the Deceptive Opinion Spam Corpus, the model achieves an accuracy of 89.69%, significantly outperforming traditional models such as SVMs and Logistic Regression. The model also demonstrates strong generalization, achieving 64.83% accuracy on the Open Domain Deception Dataset.

This model sets a new standard in deception detection by effectively combining advanced machine learning techniques with handcrafted features, enabling robust performance across various domains and contexts.

5 Results and Discussion

The proposed ***Model*** demonstrates substantial performance improvements over standalone machine learning models across various deception detection datasets. This section provides a detailed analysis of the results, justifies the performance enhancements, highlights how the model addresses key limitations. Table 2, 3 and 4 show the detailed results.

Table 2. The detailed results of the Deception in Future Activities Dataset

Approach Models	Models	Features	Obtained Results				
			Accuracy	*Precision*	*Recall*	*F1-Score*	*AUC*
ML models	SVM	TF-IDF	74.30%	74.70%	74.30%	74.00%	74.57%
	RF		68.80%	68.80%	68.80%	68.80%	68.80%
	LR		73.30%	74.00%	73.39%	72.90%	73.80%
	GB		71.80%	72.08%	71.86%	71.80%	72.01%
	SVM	Word2Vec	62.39%	62.15%	62.39%	62.01%	62.20%
	RF		63.61%	63.63%	63.61%	63.18%	63.63%
	LR		61.77%	61.54%	61.77%	61.49%	61.58%
	GB		64.83%	64.81%	64.83%	64.55%	64.81%
	SVM	BOW	70.33%	70.40%	70.33%	70.35%	70.38%
	RF		70.94%	70.88%	70.94%	70.76%	70.90%
	LR		74.00%	73.98%	74.00%	73.97%	73.99%
	GB		70.94%	71.08%	70.94%	70.71%	71.04%
	SVM	Glove	69.41%	69.43%	69.41%	69.25%	69.42%
	RF		69.11%	69.10%	69.11%	68.96%	69.10%
	LR		67.58%	67.59%	67.58%	67.37%	67.59%
	GB		66.66%	66.80%	66.66%	66.68%	66.76%
	SVM	Proposed Fused Handcrafted Features	81.78%	82.63%	81.78%	77.04%	82.29%
	RF		80.68%	80.56%	80.68%	75.28%	80.61%
	LR		75.28%	76.41%	78.28%	77.05%	77.06%
	GB		79.98%	77.42%	79.98%	77.37%	78.33%

Table 3. The detailed results of the Open Domain Dataset

Approach	Models	Used Features	Obtained Results				
			Accuracy	*Precision*	*Recall*	*F1-Score*	*AUC*
ML	SVM	TF-IDF	62.06%	62.05%	62.06%	62.05%	62.05%
	RF		60.87%	61.30%	60.87%	60.79%	61.22%
	LR		62.40%	62.40%	62.40%	62.40%	62.40%
	GB		61.29%	62.04%	61.29%	61.00%	61.89%
	SVM	Word2Vec	55.09%	55.20%	55.09%	52.92%	55.19%
	RF		53.91%	54.00%	53.91%	53.86%	53.99%
	LR		53.56%	53.78%	53.56%	53.31%	53.76%
	GB		53.07%	53.29%	53.07%	52.80%	53.28%
	SVM	BOW	59.55%	59.87%	59.55%	59.50%	59.82%
	RF		62.41%	62.88%	62.41%	62.32%	62.78%
	LR		62.06%	62.44%	62.06%	62.00%	62.36%
	GB		60.80%	61.89%	60.80%	60.32%	61.68%
	SVM	Glove	58.08%	58.18%	58.08%	58.09%	58.17%
	RF		57.60%	57.79%	57.60%	57.49%	57.76%
	LR		58.50%	58.63%	58.50%	58.50%	58.61%
	GB		59.06%	59.13%	59.06%	59.04%	59.12%
	SVM	Proposed Fused Handcrafted Features	64.83%	64.83%	64.83%	64.83%	64.83%
	RF		63.22%	63.23%	63.22%	63.21%	63.23%
	LR		64.55%	64.55%	64.55%	64.54%	64.55%
	GB		63.85%	63.96%	63.85%	63.76%	63.94%

Table 4. The detailed results of the Dataset Deceptive Opinion

Approach Models	Models	Features	Obtained Results				
			Accuracy	*Precision*	*Recall*	*F1-Score*	*AUC*
ML models	SVM	TF-IDF	89.06%	89.18%	89.06%	89.05%	89.13%
	RF		85.90%	85.90%	85.90%	85.90%	85.90%
	LR		87.50%	87.67%	87.50%	87.50%	87.60%
	GB		85.30%	85.30%	85.30%	85.30%	85.30%
	SVM	Word2Vec	74.06%	74.06%	74.06%	74.04%	74.06%
	RF		71.56%	71.56%	71.56%	71.54%	71.56%
	LR		73.75%	73.79%	73.75%	73.70%	73.78%
	GB		73.44%	73.43%	73.44%	73.43%	73.43%
	SVM	BOW	84.37%	84.50%	84.37%	84.38%	84.45%
	RF		86.87%	87.12%	86.87%	86.83%	87.01%
	LR		87.81%	87.86%	87.81%	87.79%	87.84%
	GB		83.75%	84.40%	83.75%	83.81%	84.14%
	SVM	Glove	75.93%	76.26%	75.93%	75.93%	76.15%
	RF		76.56%	76.67%	76.56%	76.44%	76.63%
	LR		75.93%	75.99%	75.93%	75.84%	75.97%
	GB		73.75%	73.75%	73.75%	73.67%	73.75%
	SVM	Proposed Fused Handcrafted Features	89.69%	89.75%	89.69%	89.71%	89.72%
	RF		85.79%	85.73%	85.79%	85.72%	85.76%
	LR		86.79%	87.11%	86.79%	86.86%	86.97%
	GB		85.79%	85.94%	85.79%	85.83%	85.88%

5.1 Performance Evaluation

The proposed model was evaluated on three benchmark datasets: the ***Deceptive Opinion Spam Dataset*** Ott et al. [6], the ***Open Domain Deception Dataset*** Pérez-Rosas and Mihalcea [4], and the ***Deception in Future Activities Dataset*** Kleinberg and Verschuere [8]. Each dataset presents unique challenges, such as varying text lengths, context-specific deceptive cues, and different types of linguistic patterns.

5.2 Performance Gains: Justification of Enhancements

The model's integration of ML models and handcrafted features delivers substantial gains:

- **Advanced Feature Extraction for Contextual Understanding:** Utilizing advanced feature extraction methods enables the model to detect subtle language shifts critical to deception. For example, in the Deception in future

activities Dataset, the integration of handcrafted features improves accuracy by 7.5% compared to models relying solely on automated feature extraction.
- **Feature Fusion:** Handcrafted features (e.g., TF-IDF, POS tagging) enhance BERT's embeddings, capturing domain-specific cues for deception. In the Open Domain Dataset, fusion raised accuracy by 2.8%, highlighting the combined power of deep and surface-level features.

Table 5 shows the comparison with the state-of-the-art.

Table 5. Comparison with the State-of-the-Art Models

References	Models	Word Embedding	Datasets	Performance
Kleinberg and Verschuere [8], 2021	MLP	LIWC - PoS	Deception in future activities	Accuracy = 69%
Sultana and Palaniappan [5], 2020	SVM, MLP, RF, LR	TF-IDF	Deceptive Opinion Spam	Accuracy = 69.84% to 78.30%
Our Model	SVM	Fused Handcrafted Features	Deception in future activities	**Accuracy = 81.78%**
			Deceptive Opinion Spam	**Accuracy = 89.69%**

5.3 Addressing Key Model Limitations

The Proposed Model addresses common issues in deception detection as follows:

- **Contextual Understanding:** Advanced feature extraction techniques provide nuanced, bidirectional context, which is crucial for identifying subtle language shifts.
- **Surface-Level Linguistic Cues (Feature Fusion):** Handcrafted features like TF-IDF and POS tagging enhance the model's ability to detect deceptive language patterns by adding specificity and depth to the analysis.
- **Adaptability Across Domains:** The model generalizes well across domains (e.g., Deceptive Opinion Spam, Open Domain Deception) due to fine-tuning on multiple datasets.

5.4 Justification of Enhanced Performance

The model's improvements stem from:

- **Feature Fusion**: Adding TF-IDF, POS tagging, WordNet similarity, and sentiment analysis, capturing stylistic and grammatical cues.
- **Domain Adaptability**: Generalizes well across diverse domains, from opinion spam to open-domain deception.

Our model demonstrates strong deception detection across datasets by combining ML and handcrafted features.

Limitations: While robust, the model faces challenges in adapting to specialized domains without fine-tuning. It currently handles only textual inputs, limiting multi-modal deception detection.

6 Conclusion

In this paper, we introduced a novel ML Model that combines advanced feature representations and a rich fusion of handcrafted features for deception detection in textual data. The proposed approach was evaluated across three benchmark datasets: Deception in Future Activities, Open Domain Deception, and the Deceptive Opinion Spam Dataset. The model demonstrated significant performance improvements, outperforming traditional machine learning models in terms of accuracy, precision, recall, and F1-score.

By leveraging advanced contextual features and techniques for capturing long-term dependencies, the model effectively identified both subtle and overt deception cues. The feature fusion process, which included TF-IDF, POS tagging, WordNet similarity, semantic similarity, and sentiment analysis, further strengthened the model's capacity to capture surface-level linguistic cues, enhancing its robustness across various deception scenarios.

6.1 Key Contributions

The key contributions of this work are as follows:

- Implementing feature fusion to integrate embeddings and handcrafted features, enhancing detection of linguistic patterns.
- Extensive evaluation on diverse datasets, showcasing the model's robustness across various deception scenarios.

6.2 Addressing Model Limitations

While the model achieved strong results, it also addressed key limitations in traditional deception detection methods:

- **Long-range dependencies**: Advanced sequential analysis techniques effectively handle long-form texts.
- **Contextual understanding**: Contextual feature extraction captures nuanced shifts in meaning.
- **Linguistic cues**: Handcrafted features enhance domain-specific deception detection, addressing limitations of embedding-only models.

6.3 Future Work Directions

Future work will focus on incorporating reinforcement learning for real-time deception detection and enhancing the model's interpretability using techniques such as SHapley Additive exPlanations (SHAP) Lundberg and Lee [15]. This will make the model more adaptable and transparent, especially in high-stakes environments like fraud detection and legal investigations.

- **Interpretability**: Incorporate explainable AI methods like *SHAP* to clarify model decisions.
- **Multi-modal Detection**: Extend the model to handle audio, video, and other inputs for broader real-world applications.
- **Real-time Detection**: Optimize computational efficiency for rapid, real-time deception detection.

In summary, this model advances deception detection by integrating contextual and sequential analysis, achieving high performance on diverse datasets. Continued work on explainability and multi-modal capabilities will enhance its applicability across fields like fraud detection, media verification, and national security, addressing the growing demand for trustworthy AI in communication.

Our research contributes to AI-driven deception detection, paving the way for further exploration at the intersection of NLP and trust in digital communication.

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Machine Learning Models for Validating the Self-declaration Conformity Assessment: Risk Evaluation

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Abstract. Global trade faces the challenge of compromising product conformance and speeding up the conformity assessment (CA) process. Although the self-declaration CA ensures a fast CA process, it suffers from leakage of impartiality. In the present work, self-declaration CA was validated using machine learning (ML) models. Various ML models have been trained using the market surveillance data, and the highest performance models were selected. Different sources of risk have been investigated, including the risks of using inaccurate ML models, including redundant variables in ML training, leakage of learning in ML models, erroneous ML model predictions, and uncertainty of experimental results. A procedure including recommendations to minimize these risks has been proposed. ML models were applied to assess the electrical performance of air conditioning (AC) units which typically requires 6–8 h in the laboratory. The conformity of AC units could be predicted by ML models with an accuracy exceeding 98%. As a result, the productivity of the laboratory expanded from testing only one AC unit to testing an unlimited number of units per day, saving time, effort, and laboratory resources.

Keywords: Machine Learning Models · Feature Selection Analysis · Air Conditioner Test

1 Introduction

A product is considered safe when the risk affecting the safety and health of a person is within an acceptable range [1]. Therefore, standard procedures and international guidelines (e.g., [2–5] were established to ensure the product compliance before being placed in the market. As global trade continues to expand, a fast conformity assessment (CA) process is required without compromising its effectiveness. For high trustworthiness, the CA is conducted by an independent third party. However, this type of CA is costly and time-consuming. To address this issue, a self-declaration CA was proposed in which the manufacturer declares the conformity of his own products.

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The self-declaration CA offers a rapid process; but its trustworthiness is based only on the credibility of the manufacturer. Therefore, manufacturers are required by regulations to provide technical documentation that enhances the credibility of their self-declaring CA. Despite these regulations, it was reported that self-declaration CA may not always ensure the compliance of the product with essential requirements [6]. Therefore, a third-party CA in this case would be mandatory, especially for high-risk products or tests.

On the other hand, there are some products that are highly risky to be assessed through self-declaration CA and are difficult to manage by third parties. Air conditioning (AC) units significantly affect the energy efficiency strategy in a country [7]. Therefore, ISO 5151 [8] was established to assess the electrical efficiency of AC units through various parameters measured under certain conditions. However, testing more than one AC unit per day is a challenge because it typically requires 6–8 h. Furthermore, establishing a laboratory to test the AC performance is costly.

In the Kingdom of Saudi Arabia (KSA), the approval of AC units is based on the manufacturer's self-declaration. Nonetheless, the reliability of the manufacture self-declaration CA is continuously monitored through market surveillance in which samples are collected and tested by the national laboratory. The measurement results reported by the manufacturer are compared with the evaluations performed by the national laboratory using a suitable criterion established by ISO 5151 [8]. However, not all the products undergo testing in market surveillance, and non-compliant products are detected after they have been placed in the market. Therefore, the main objectives of the present study are to:

1. Using machine learning (ML) models to verify the manufacturer's self-declaration for the AC units before being placed in the markets.
2. Investigate the risks associated with applying ML to product CA.
3. Evaluate the uncertainty of the ML prediction and its effect on the CA.

2 Electrical Efficiency Test of AC Unit

The electrical efficiency test of AC unit is performed following the enthalpy method described by ISO 5151 [8]. The test is performed under specified conditions given in Table 1.

Table 1. Dry-air temperature settings in T1 and T3 conditions specified by ISO 5151.

Test Conditions	Indoor $\boldsymbol{T}$ °C	Outdoor $\boldsymbol{T}$ °C
$T1$	27	35
$T3$	29	46

A set of conditioning parameters is measured and monitored during the test. These parameters are defined in Table 2.

Table 2. Test condition parameters measured during the AC electrical efficiency test in accordance with ISO 5151.

Parameter	Symbol	Unit
Specific enthalpy of air entering/leaving indoor side	h_{a1}/h_{a2}	J/kg
Barometric pressure	p_a	kPa
Pressure at nozzle throat	p_n	kPa
Static pressure across nozzle	p_v	mmAq
Temperature at nozzle	T_n	°C
Air-volume rate	q_v	m^3/s
Specific volume of dry air portion of mixture at nozzle	v_n	m^3/kg
Specific heat of air	c_{pa}	kcal/kg °C
Specific humidity of air entering/leaving indoor side	W_{ia}/W_{i2}	kg/kg
Temperature of air entering/leaving indoor side, dry bulb	T_{a1}/T_{a2}	°C
Temperature of air entering/leaving outdoor side, dry bulb	T_{a3}/T_{a4}	°C
Temperature of air entering/leaving indoor side, wet bulb	T_{w1}/T_{w2}	°C

The output of the test includes four parameters that are used to assess the energy performance of AC unit: the total cooling capacity ϕ_{tci}(Btu/h), power input $P(W)$, current $I(A)$, and voltage $V(V)$. . The energy efficiency ratio (EER) in unit of (Btu/h)/W is evaluated as the ratio of the total cooling capacity to the power input. The compliance of AC unit demonstrates [9]:

1. EER (T1) $\geq$ 11.8((Btu/h)/W), and
4. EER (T3) $\geq$ 8.3((Btu/h)/W).

Further information about the test procedure and KSA national laboratory can be found elsewhere [8–10].

3 Pre-training Data Processing

3.1 Data Cleaning and Representation

During market surveillance, more than 800 reports from the National Electrical Efficiency Laboratory of SASO, KSA between 2016 and 2024 were collected. The reports included the manufacturer rated data, test results evaluated by SASO laboratory, and test conditions applied during the measurements. The data were examined to identify missing and extreme values. As these cases were found to be rare compared to the original data size, they were removed from the data. The rated results reported by the manufacturer compared to the values measured by SASO are illustrated in Fig. 1.

The EER results reported by SASO reveal that there are nonconformance AC units (EER $\leq$ 11.8(Btu/h/W). However, not all these nonconformance AC units were reported by the manufacturers during the self-declaration CA. This finding confirms

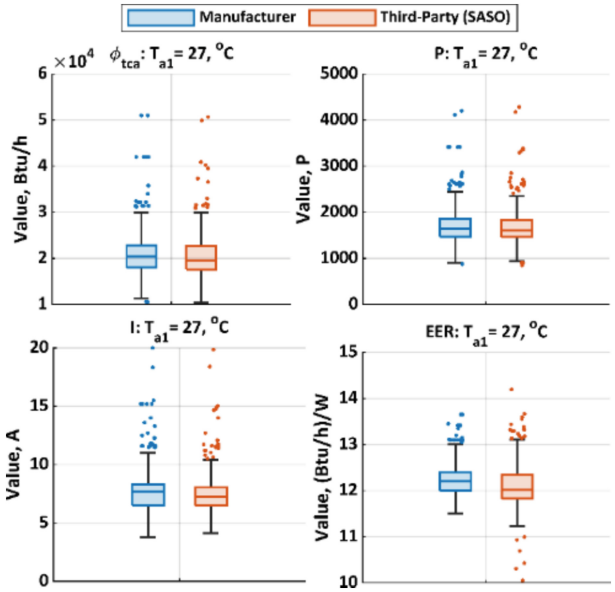

Fig. 1. Rated results reported by the manufacturers (boxes on left) compared to the measured results reported by SASO laboratory (boxes on right).

the need to verify the information provided by the manufacturer using other means. The test conditions applied during the measurements are presented in Fig. 2 for both *T*1 and T3 test conditions. The data was normalized using z-score to eliminate the effects of different units and scales. It should be noted that the extreme values detected by the box chart are attributed to the wide range of AC models and test conditions included in the data. These values are technically valid, and therefore, they were maintained in the data during the training stage of ML models.

3.2 Risk of Including Redundant Variables in ML Learning: Data Dimensional-Reduction Analysis

The input data included the AC model, test conditions defined by T_{a_1} and T_{w_1}, four rated variables reported by the manufacturers and the 15 test condition variables measured during the test. The AC model,T_{a_1}, and T_{w_1}, were defined as categorical variables. The target variables were set to the test output (ϕ_{tca}, P, $I\&EER$) reported by the third-party laboratory.

The input variables were subjected to feature selection analysis to eliminate redundant variables and increase the efficiency of the ML training process. The data-dimensional reduction analysis was performed using filter- and embedded-type methods [11]. The filter-type method was performed using F-test and minimum redundancy – maximum relevance (MRMR) algorithms [12].

The analysis results for ϕ_{tca} are presented in Fig. 3.

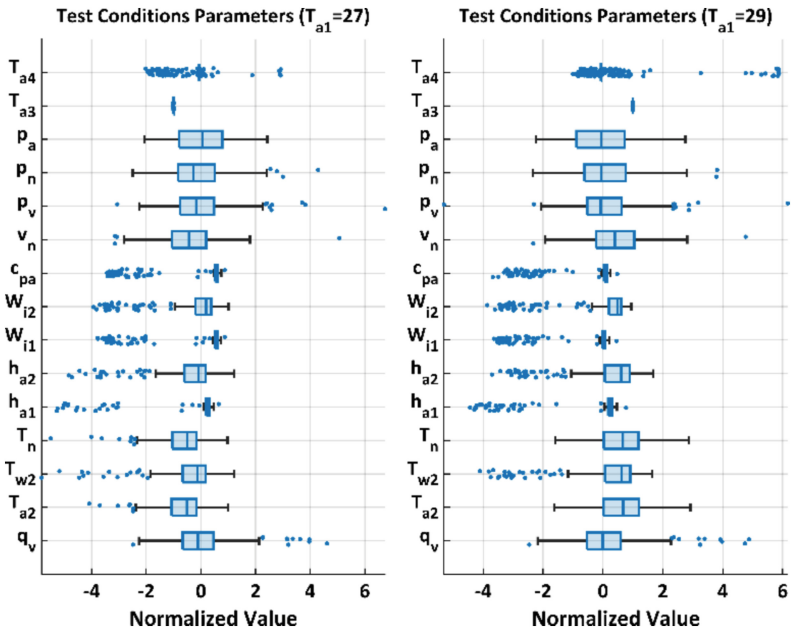

Fig. 2. The test conditions parameters measured for the indoor and outdoor test units. The data was normalized using z-score factor. The definition and unit of each variable was introduced in Table 2.

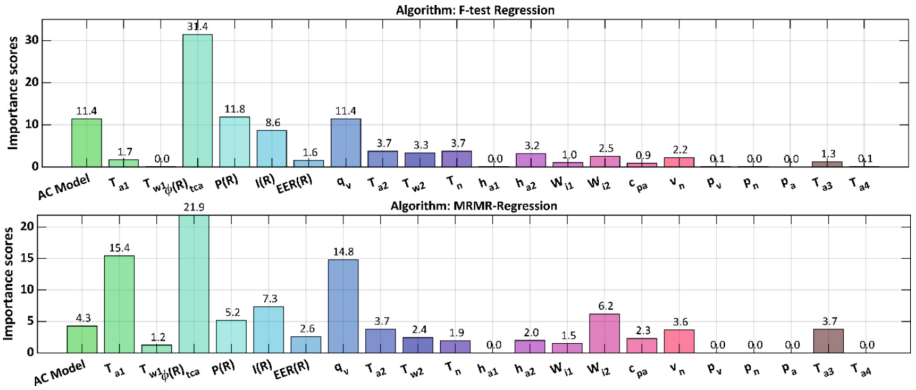

Fig. 3. Analysis results of filter-type feature selection analysis performed for the 22 input variables and the output variable ϕ_{tca}.

It has been shown that the most significant variables affecting the total cooling capacity ϕ_{tca} were the $ACModel$, T_{a1}, $\phi(R)_{tca}$, $P(R)$, $I(R)$, W_{i2}, and q_v. To validate this conclusion, the feature selection analysis was also performed using the embedded regression tree [13–15], ensemble of bagged decision trees [16], Gaussian process regression [17], stepwise linear and generalized linear regression [18]. It should be noted that in both

the regression tree and ensemble algorithm, the leaf sizes were determined throughout the hyperparameter optimization process, which involved evaluating the model's performance iteratively until the optimum loss function was obtained. Figure 4 shows the hyperparameter optimization process for the regression tree model, and the significant features affecting the output parameter ϕ_{tca} estimated by the ensemble algorithm. The analysis results of the regression tree were, on the other hand, identical with those estimated by the ensemble method.

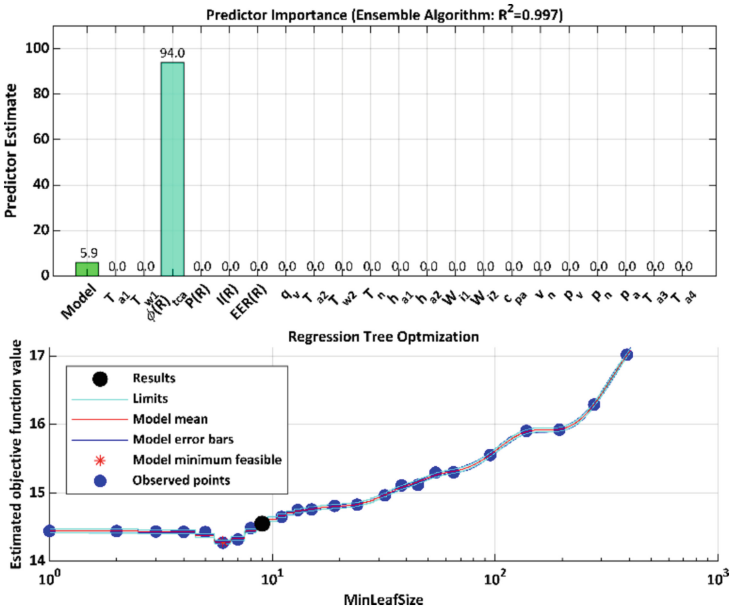

Fig. 4. Feature selection analysis performed by the embedded ensemble and regression tree algorithms.

The fitting using the stepwise algorithm, the formulated equation and the distribution of the model's error for the output parameter ϕ_{tca} are presented in Fig. 5. The stepwise algorithm could fit the data with $R^2 \sim 0.998$ using a formulated equation with the parameters denoted in Fig. 5. Excluding one of these parameters significantly affected the performance of the model, leading to a lower R^2 value. Therefore, all parameters included in the formulated equation were considered significant. The same procedure was also applied to the other target variables P and I. A comparison between the various methods was conducted to determine the significant features for each output variable. The feature was considered significant if it was selected by at least one method, while the other features were excluded from the ML model training process. The outcomes of the feature selection analysis are summarized in Table 3.

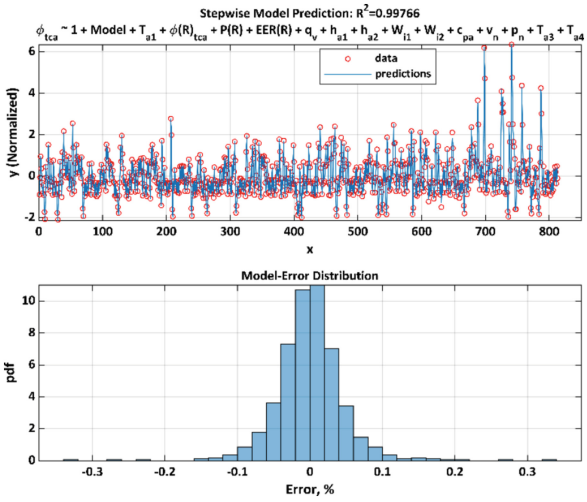

Fig. 5. Feature selection analysis performed using the stepwise algorithm. The model formulated equation and histogram of the model error are presented.

Table 3. Summary results of the feature selections analysis performed using filter- and embedded-type algorithms.

Target	Significant Features
ϕ_{tca}	AC Model, T_{a_1}, $\phi(R)_{tca}$, $P(R)$, $I(R)$, $EER(R)$, q_v, h_{a_1}, h_{a_2}, W_{i_1}, W_{i_2}, v_n, c_{p_a}, p_n, T_{a_3}, T_{w_3}
P	AC Model, T_{a_1}, $\phi(R)_{tca}$, $P(R)$, $EER(R)$, q_v, T_{a_2}, h_{a_1}, h_{a_2}, W_{i_1}, W_{i_2}, v_n
I	AC Model, T_{a_1}, $\phi(R)_{tca}$, $P(R)$, $EER(R)$, q_v, T_{a_2}, h_{a_1}, h_{a_2}, W_{i_1}, W_{i_2}, v_n

3.3 Risk of Lack of Learning in the ML Models Training

There is no unique ML model design or configuration that ensures optimum performance because it is problem dependent. Therefore, there is always the risk of choosing inaccurate ML models. To reduce this risk, it is essential to explore the performance of various ML models with different configurations to determine the most effective one [19]. In this study, the linear regression (LR) models, regression trees [13], Gaussian process regression (GPR) [17], support vector machines (SVM), kernel approximation [20], ensembles of regression trees [21], and artificial neural network (ANN) models were used. A set of 30 trials including these models with different configurations was performed to fit the data. The performance of each model was assessed using the R-squared and mean square error (MSE) parameters. In each trial, 10% of the data was used for testing. The results are presented in Fig. 6, where the maximum R-squared and minimum MSE values are also indicated.

The models with the highest performance were GPR, ANN, LR, and SVM. Moreover, the errors in the prediction results of the GPR and ANN models were almost identical

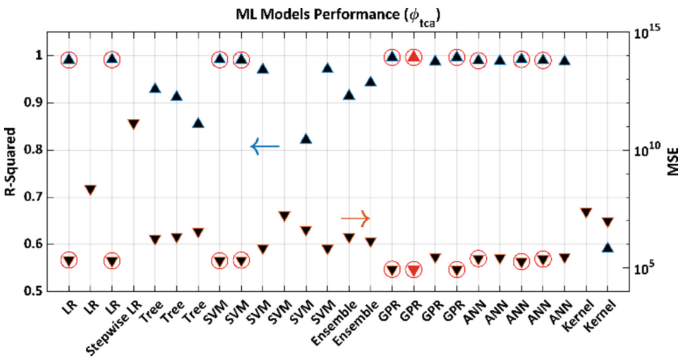

Fig. 6. Performance of various machine learning models for total cooling capacity ϕ_{tca}. The highest models' performances were indicated by a circle.

and were 10 times smaller than those of SVM and LR models. To minimize the risk of erroneous ML predictions, the predicted value was evaluated as the average of the ML models with the highest performance.

The epistemic uncertainty associated with the lack of learning in the ML model was introduced as the standard error of the ML model outputs [22]. The errors and uncertainties of the ML model output are presented in Fig. 7.

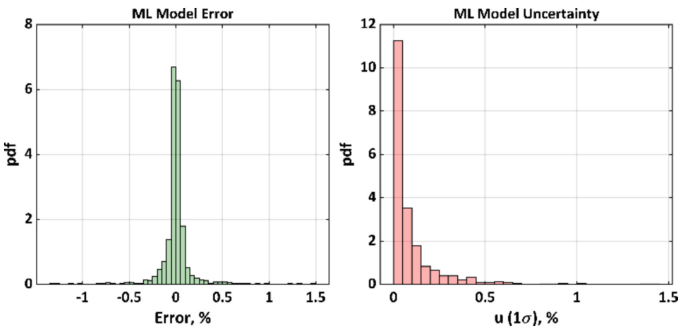

Fig. 7. The error and uncertainty associated with the ML models used to simulate the total cooling capacity ϕ_{tca} of air conditioning unit (AC).

The error has is randomly distributed around zero, with an uncertainty of less than 1.5%. The same procedure was applied to the other two target variables, P and I. Table 4 illustrates the ML models selected based on their performance for each target variable.

3.4 Risk of False Accept/reject CA Decision Made by ML Models

The specific consumer risk is the probability that a particular accepted item is nonconforming. By contrast, the producer's risk is the probability that a particular rejected item is conforming. These types of risks occur when the results approach the tolerance limits.

Table 4. ML models selected based on performance after various trials with different configurations.

Target	Selected ML Models	Accuracy Achieved
ϕ_{tca}	3-GPR models 3-ANN models	Error < 1.5 Uncertainty < 1.5
P	5-ANN models	Error < 1.0 Uncertainty < 1.0
I	5-ANN models	

The uncertainty associated with the results in this case increases the risk that the CA decision may be incorrect. The JCGM-106 [5] was established to evaluate the conformance probability p_c that an item fulfills a specified requirement. The product conforms to a specific requirement if the property value Y is within the conformity limits. Assuming a normal distribution for Y, the conformance probability is evaluated as:

$$p_c = \Phi\left(\frac{\mathrm{T_U} - y}{u}\right) - \Phi\left(\frac{T_L - y}{u}\right) : \Phi(\ddagger) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \exp\left(-\frac{t^2}{2}\right)dt \tag{1}$$

where T_L and T_U are the lower and upper tolerance, respectively, y is the estimated property value of Y, and u is the associated standard uncertainty. For single lower tolerance, where $T_U - \infty$, $\Phi(\infty) = 1$. While, for single upper tolerance, where $T_L - \infty$, $\Phi(-\infty) = 0$. The risk associated with the result's uncertainty can be expressed as:

$$\varphi_R = 1 - p_c \tag{2}$$

The uncertainty associated with the ML results can be estimated as [22]:

$$u_{ML} = \sqrt{u^2_{aleatoric} + u^2_{epistemic}} \tag{3}$$

The aleatoric uncertainty is associated with experimental measurements. This uncertainty component was evaluated in accordance with JCGM-100 [23] and was found to be ~2%. The epistemic uncertainty is associated with the lack of learning in ML models. This type of uncertainty can be expressed as the standard error of the results evaluated by the selected models.

To minimize the risk of false CA decisions owing to the uncertainty in the result, a minimum limit of conformance probability (maximum risk limit) should be set. Various limits were proposed by the International Laboratory of Accreditation Cooperation (ILAC) guide ILAC-g8 [4], with the recommendation of setting the maximum risk to 2.5% ($p_c(min) = 97.5\%$). To illustrate the uncertainty effect of the ML results, Fig. 8 presents a comparison between the CA performed by the third party and that performed using the ML models.

Eight (~1.5%) nonconformance AC units were falsely accepted and three (~1.1%) conformance AC units were falsely rejected by the ML models. To minimize the consumer risk, the acceptance threshold should be set at a level higher than that used by the third-party laboratory. However, it should be noted that minimizing the consumer

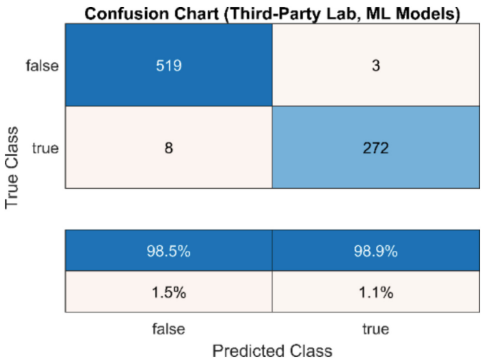

Fig. 8. Confusion chart between conformity assessment conducted by the third laboratory and ML models. The conformity probability limit was set at 97.5%.

risk increases the producer risk. Therefore, both consumer and producer risks should be minimized by improving the uncertainty associated with ML results. This can be achieved by improving either the uncertainty associated with the third party and/or the uncertainty associated with ML modeling, as follows:

1. Using experimental data from high accuracy laboratories.
2. Applying appropriate statistical procedures to improve the quality of data used in ML training.
3. Investigate the performance of various ML models and select the optimum modes.
4. Using more than one ML model to predict the results.
5. Including the uncertainty of the ML models in the CA process.

3.5 Risk of Reporting Incorrect Rated Values by Manufacturers

False acceptance is expected to occur in self-declaration CA, where the manufacturer may intentionally report incorrect results. These cases were observed in the comparison between the rated and third-party results (Fig. 1). However, these incorrect results did not influence the results estimated by the ML models. Figure 9 displays the clustering analysis using $k-$ means method [24, 25] for the rated EER versus ML results in comparison with the measured EER versus ML results. It should be noted that the optimum cluster number was determined based on an optimization process.

In the case of the rated EER versus ML results, there were scattered data around each cluster mean. It implies that the correlation between the rated results reported by the manufacturer and ML results is not always significant. In other words, a higher EER rated value does not necessarily lead to higher EER being estimated by the ML models. On the other hand, the clustering analysis of the EER results reported by the third-party laboratory versus the EER estimated by ML models show a clear positive correlation that approaches the unity. It implies that the predictions of ML models do not significantly depend on the information rated by the manufacturer but rather on the results of measurements conducted by the third-party laboratory, and the test conditions.

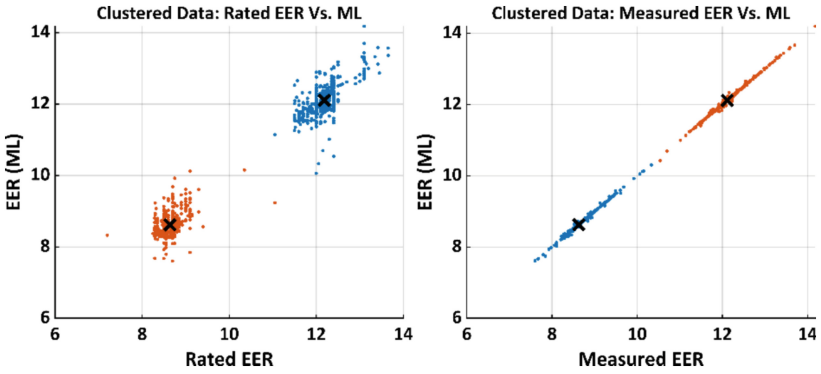

Fig. 9. Clustering analysis performed using k-means method for the rated EER Vs the ML results compared to the measured EER Vs. ML results.

4 Conclusion

ML models can be used to predict the conformity of products using data collected during the market surveillance process. The ML models can achieve an accuracy of more than 98%. The risk of using inaccurate ML models can be reduced by comparing the performance of various ML models before selecting the optimum one. While the risk of erroneous predictions made by ML models can be minimized by estimating the results using various ML models with high performance and not counting on only one ML model. On the other hand, the consumer conformity assessment risk can be minimized by increasing the minimum conformity probability limit compared to that used by the third-party laboratory. Furthermore, both the consumer and producer risks associated with the uncertainty of ML models can be minimized by improving the data quality, selecting high accuracy third parties, and using various high-performance ML models.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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Enterprise Credit Rating Framework Based on Risk Contagion Graph Neural Network

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Abstract. Enterprise credit rating is crucial for risk management in the complex business environment. However, existing models encounter challenges in adeptly handling heterogeneous relationships and efficiently addressing risk propagation. This study introduces a novel framework for enterprise credit rating, the Risk Contagion Graph Neural Network (RCGNN), designed to overcome these challenges. The framework comprises four distinct modules: enterprise feature embedding, risk contagion intra-aggregation, risk contagion inter-aggregation, and credit rating prediction. Through the integration of advanced graph neural networks and attention mechanisms, RCGNN effectively captures intricate risk propagation relationships among enterprises. Extensive experiments were conducted on a self-constructed dataset of Chinese listed companies, and the results demonstrate that RCGNN outperforms traditional machine learning methods and homogeneous graph neural network models in credit rating tasks.

Keywords: Graph Neural Network · Representation Learning · Credit Scoring

1 Introduction

Credit rating is crucial in the financial sector, guiding investment decisions, risk management, and economic stability. Traditional models, relying primarily on financial indicators and statistical methods, are effective to some extent but struggle with the complexity and interconnectivity of modern financial networks. These methods often fail to capture intricate relationships between entities, limiting their adaptability to real-time data and dynamic market conditions. As financial markets evolve, more robust models are needed to handle vast, heterogeneous datasets and address the complexities of global financial systems.

Recent advances in machine learning and artificial intelligence have led to the integration of Graph Neural Networks (GNNs) for credit risk prediction. For example, [10] introduced a time-aware GNN for credit risk forecasting, while [5]

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proposed a self-attention-based GNN that incorporates multi-task learning for improved credit assessment. Moreover, the CCR-GNN model by [2] constructs individual graphs for each enterprise to model credit risk, demonstrating the potential of GNNs in corporate credit rating. However, credit risk propagation extends beyond shareholder networks and includes interactions across industries and sectors, highlighting the need for models capable of capturing these broader dynamics of risk contagion.

While GNNs have demonstrated exceptional performance in modeling graph-structured data [6], there remain substantial limitations, particularly in handling heterogeneous data. The complexities of modern financial data, with its diverse sources and types, requiring models that can process multiple node types and edge relationships within a graph. Recent developments in GNNs have addressed this need by expanding to heterogeneous graphs, which include multiple types of nodes and edges [11]. These models leverage meta-paths [7] to aggregate information across different relationships, offering a more comprehensive view of the data. However, existing research [2] often overlooks the critical role of risk contagion - the spread of financial risk across companies, industries, and regions. This gap in the literature motivates the need for further exploration of risk contagion dynamics in GNN-based credit rating models.

To address these challenges, we propose a novel framework for enterprise credit rating based on a Risk Contagion Graph Neural Network (RCGNN). Our model utilizes heterogeneous graph structures to incorporate multiple types of data, such as investment, industry, and geographical information, which are often interrelated and influence a company's creditworthiness. We introduce a risk contagion mechanism that aggregates information across different metapaths, capturing the dynamic propagation of risks across entities and sectors. By employing RotatE knowledge graph representation and an attention mechanism, our model enhances the accuracy and robustness of credit rating predictions.

The primary contributions of our work are as follows:

1. To the best of our knowledge, we are the first to model investment, industry, and geographical information in corporate credit rating prediction as heterogeneous graph data, employing a heterogeneous GNN node representation learning model for this task.
2. We propose a novel risk contagion mechanism within the GNN framework that aggregates risk contagion across different metapath instances and types, addressing the limitations of traditional models in handling heterogeneous data and providing a new approach for more accurate corporate credit assessments.
3. We collect a multi-source dataset comprising more than 5000 publicly listed Chinese companies with 408 financial indicators, and conduct empirical experiments to demonstrate the effectiveness of our proposed model.

2 Problem Formulation

In this section, we will introduce the objective of enterprise credit rating and some definitions involved in this process.

2.1 Problem: Enterprise Credit Rating

Given a multi-source dataset composed of enterprise business information, financial data, and a heterogeneous graph representing enterprise risk contagion, we aim to determine the credit risk of companies considering the influence of different contagion paths. Based on the results of representation learning for enterprises, we perform the credit rating task. The credit rating of enterprises is categorize into three levels: Excellent, Fair, and Poor, based on the credit rating data. Therefore, the task in this paper can be considered as a multi-class classification problem.

2.2 Definition

Heterogeneous Graphs. A heterogeneous graph consists of multiple types of nodes and edges, each with distinct semantics or attributes, enabling the representation of complex relational networks in real-world scenarios.

A heterogeneous graph, denoted as $\mathbf{G} = (\mathbf{V}, \mathbf{E})$, includes:

- **Node set V:** The set of n nodes of different types, representing entities in the graph.
$$\mathbf{V} = \bigcup_{i=1}^{n} V_i \tag{1}$$
where V_i denotes the set of nodes of the i-th type.
- **Edge set E:** The set of m edges of different types, representing relationships among nodes.
$$\mathbf{E} = \bigcup_{j=1}^{m} E_j \tag{2}$$
where E_j represents the set of edges of the j-th type.

A graph is considered heterogeneous if $m + n \geq 3$, implying the presence of at least two types of nodes or edges.

Metapaths. In a heterogeneous graph, nodes and edges can represent different entities and relationships. A metapath defines a sequence of node types and edges, capturing a specific relationship pattern. Metapaths decompose the semantic structure of the graph into interpretable patterns, enhancing both graph understanding and exploration.

For a graph $\mathbf{G}$, a metapath is represented as:

$$P\left(V_i, V_x, \ldots, V_j\right) = \left(V_i \xrightarrow{E_{i,x}} V_x \xrightarrow{E_{x,\cdot}} \ldots \xrightarrow{E_{\cdot,j}} V_j\right) \tag{3}$$

where V_i, V_j, V_x are nodes of specific types, and $E_{i,x}$, $E_{x,\cdot}$, $E_{\cdot,j}$ are the corresponding edges.

A metapath instance p is a sequence of nodes following the metapath P. Additionally, we introduce weighted metapaths, where edges are assigned weights. A weighted metapath and its instance are defined as:

$$P\left(V_i, V_k, V_j\right) = \left(V_i \xrightarrow[W_{i,k}]{E_{i,k}} V_k \xrightarrow[W_{k,j}]{E_{k,j}} V_j\right) \tag{4}$$

where W is the weight matrix for the edges between node types. A weighted metapath instance is defined as:

$$p\left(v_i, v_k, v_j\right) = \left(v_i \xrightarrow[w_{i,k}]{e_{i,k}} v_k \xrightarrow[w_{k,j}]{e_{k,j}} v_j\right) \tag{5}$$

where e represents the edge between nodes, and w is the edge weight. In heterogeneous graph neural networks, metapaths are typically symmetric, ensuring consistency in the types of starting and ending nodes.

3 RCGNN Model

The overall structure of the proposed RCGNN model is illustrated in Fig. 1. The model consists of four modules: Enterprise Feature Embedding Module, Risk Contagion Intra-Aggregation Module, Risk Contagion Inter-Aggregation Module, and Enterprise Credit Rating Module.

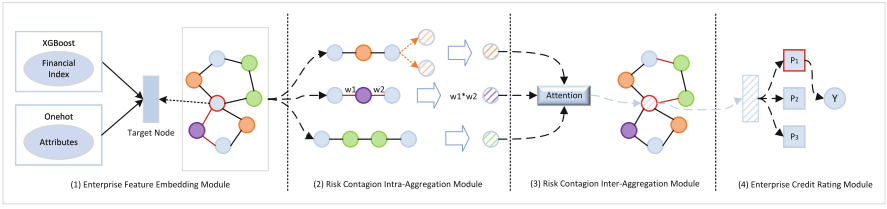


Fig. 1. The Overall Architecture of RCGNN

3.1 Enterprise Feature Embedding Module

This module generates initial vector inputs for nodes. For listed companies, the XGBoost [1] model is employed to evaluate the importance of financial indicators. Important features are selected based on their rankings and mapped into the initial vector space to serve as inputs for these nodes. Other nodes utilize one-hot encoding for their initial vector representations.

After feature extraction, the vectors of different node types are mapped to a common latent vector space to ensure consistent model input. For a node v_i of type V_i:

$$x'_{v_i} = W_{V_i} \cdot x_{v_i} \tag{6}$$

where $x_{v_i} \in \mathcal{R}^{d_{V_i}}$ is the initial feature vector of node v_i, $W_{V_i} \in \mathcal{R}^{d' \times d_{V_i}}$ is the weight matrix for nodes of type V_i, and $x'_{v_i} \in \mathcal{R}^{d'}$ is the unified input vector of node v_i in the latent space $\mathcal{R}^{d'}$.

3.2 Risk Contagion Intra-aggregation Module

To address the heterogeneity of the graph and incorporate edge weight information for modeling risk contagion in various scenarios, we employ a weighted metapath approach. For a given metapath $p(v_i, v_k, v_j)$, the RotatE [8] knowledge graph embedding model is used to obtain vector representations of all node features in the metapath. Specifically, the RotatE model defines each relation as a rotation from the source entity to the target entity in the complex vector space.

$$\begin{aligned} g(v_i) &= x'_{v_i} \\ g(v_k) &= x'_{v_k} + g(v_i) \odot e_{i,k} \\ f_\theta\left(x'_{v_i}, x'_{v_k}, x'_{v_j}\right) &= \frac{g(v_j)}{|p(v_i, v_k, v_j)|} \end{aligned} \tag{7}$$

Here, $\odot$ denotes element-wise multiplication, and $|\,\mathrm{p}(v_i, v_k, v_j)|$ is the number of nodes in the metapath instance. This module also incorporates edge weight information.

$$x_{p(v_i,v_k,v_j)} = w_{i,k} \times w_{k,j} f_\theta\left(x'_{v_i}, x'_{v_k}, x'_{v_j}\right) \tag{8}$$

After encoding the metapath instance into vector representation, we apply multi-head graph attention [9] to the target node v_i to capture the varying impacts of different instances. This involves weighted summation of vectors from all relevant metapath instances p_{v_i} calculating the weight for each instance:

$$\begin{aligned} e_{p(v_i,v_k,v_j)} &= a\left(x'_{v_i}, x_{p(v_i,v_k,v_j)}\right) \\ \alpha_{p(v_i,v_k,v_j)} &= \frac{exp\left(e_{p(v_i,v_k,v_j)}\right)}{\sum_{p_m \in p_{v_i}} exp\left(e_{p_m}\right)} \end{aligned} \tag{9}$$

where $a(\cdot)$ is a feedforward neural network that maps the vector space to a real number as the weight coefficient for that metapath.

$$a\left(x'_{v_i}, x_{p(v_i,v_k,v_j)}\right) = LeakyReLU\left(\boldsymbol{a}_p^T \left[x'_{v_i} \| x_{p(v_i,v_k,v_j)}\right]\right) \tag{10}$$

Here, $\boldsymbol{a}_p^T$ is the parameterized attention vector, and $\|$ is the vector concatenation symbol. After computing the weights α_{p_m} for all metapath instances $p_m \in p_{v_i}$ we derive the representation of the target node influenced by these instances through weighted summation:

$$x_{v_i}^p = \sigma\left(\sum_{p_m \in p_{v_i}} \alpha_{p_m} \cdot x_{p_m}\right) \tag{11}$$

Here, $\sigma(\cdot)$ denotes the activation function. The resulting vector representation of the target node incorporates risk contagion effects. We then extend this attention mechanism to multi-head attention, enhancing model stability:

$$x_{v_i}^{p} = \|_{k=1}^{K} \sigma\left(\sum_{p_m \in p_{v_i}} [\alpha_{p_m}]_k \cdot x_{p_m}\right) \tag{12}$$

This concatenates the outputs from K independent attention mechanisms to form the final vector representation.

3.3 Risk Contagion Inter-aggregation Module

After aggregating within the risk contagion paths, the model obtains the representation vector for each risk contagion meta-path. An attention mechanism is then employed to further aggregate the results of different types of risk contagion, assigning corresponding weights to different contagion paths. For the target node v_i, the operations for each meta-path $p \in P_{v_i}$ are as follows:

$$e_p = \text{LeakyReLU}\left(\boldsymbol{u}_p^T \cdot x_{v_i}^{p}\right), \tag{13}$$

$$\beta_p = \frac{\exp(e_p)}{\sum_{p \in P_{v_i}} \exp(e_{p_m})}, \tag{14}$$

$$x_{v_i}^{P_{v_i}} = \sum_{p \in P_{v_i}} \beta_p \cdot x_{v_i}^{p}. \tag{15}$$

In the above equations, $\boldsymbol{u}_p$ is a learnable attention vector corresponding to different types of meta-paths associated with node v_i, and β_p represents the normalized importance coefficient of meta-path type p for node v_i. By performing a weighted sum of the risk contagion results from different types of meta-paths, the model derives the final risk contagion representation for node v_i across different types of meta-paths and various instances of the same type of meta-path.

3.4 Enterprise Credit Rating Module

To map the output vector to the target classification, the module applies a linear transformation followed by a non-linear activation. The output is then mapped to the designed output space:

$$o_{v_i} = \sigma\left(W_o \cdot x_{v_i}^{P_{v_i}}\right) \tag{16}$$

where $\sigma(\cdot)$ is the softmax activation function, $W_o \in \mathbb{R}^{d_y \times d'}$ is the weight matrix, and d_y represents the number of classes in the target task. The resulting vector o_{v_i} contains probabilities for each class.

The node is assigned to the class with the highest probability. Cross-entropy is used as the loss function, as shown in Eq. 17. The model is optimized via

backpropagation and gradient descent to minimize the cross-entropy, thereby learning meaningful node embeddings in the heterogeneous graph.

$$L = -\sum_{c} y_c \cdot \log(p_c) \tag{17}$$

where y_c is the ground-truth label (one-hot encoded) for class c, and p_c is the predicted probability for class c.

4 Experiments and Results

4.1 Datasets and Data Preprocessing

The CorKG dataset is a self-constructed graph designed for credit rating prediction of Chinese publicly listed enterprises. The dataset consists of two subsets: CorKG_Choice, based on ratings from major agencies via the Choice Financial Terminal, and CorKG_Jian, which extends CorKG_Choice by incorporating additional ratings from the Ji'an Jinxin Company Evaluation Center. The dataset includes five node types: publicly listed enterprises (L), non-listed enterprises (C), individuals (P), industries (I), and addresses (A). It also contains four types of edges: shareholding, legal representation, city affiliation, and industry affiliation. Four risk contagion metapaths are designed based on this structure. Detailed statistics of the dataset are provided in Table 1.

Table 1. Statistics of CorKG Dataset

CorKG_Choice			CorKG_Jian			Meta-Paths
Nodes	Feature Dimensions	Edges	Nodes	Feature Dimensions	Edges	
L: 1739	195	L-C: 10075	L: 4670	195	L-C: 24641	LCL
C: 5884	5884	L-C: 10075	C: 14576	14576	L-C: 24641	LCL
P: 5703	5703	L-P: 6652	P: 16180	16180	L-P: 20744	LPL
I: 305	305	L-I: 1705	I: 334	334	L-I: 4730	LIL
A: 211	211	L-A: 1705	A: 284	284	L-A: 4669	LAL

4.2 Experimental Settings

The experiments were conducted on a system with Windows 11, a 12th Gen Intel(R) Core(TM) i7-12700F CPU, and an NVIDIA GeForce RTX 3060 Ti GPU. The software environment used Python 3.9 and PyTorch 1.12.0 for deep learning. The Deep Graph Library (DGL) version 1.0.1.cu116 was used for heterogeneous graph processing. The detailed configuration of software and library versions is shown in Table 2.

For the experimental setup, 20% of the dataset was reserved as the test set. The model used 128-dimensional node embeddings, the Adam optimizer, early

Table 2. Setting of Experiment Software Environment

Software	Version	Software	Version
PyTorch GPU	1.12.0	PyTorch Geometric	2.3.1
torch_scatter	2.1.0	CUDA	11.6
torch_sparse	0.6.16	cuDNN	8.6

stopping (patience = 5), and a dropout rate of 0.5. Grid search was used for parameter optimization, with final settings: learning rate = 0.0005, 8 attention heads, 100 epochs, and a batch size of 32.

4.3 Baseline Methods

The comparative experiments included conventional machine learning and graph neural network models. For machine learning, we employed Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), and XGBoost, using 5-fold cross-validation to mitigate overfitting. For GNN models, GCN [4], GraphSAGE [3] and GAT [9] are used to conduct experiments on the complete dataset without considering node heterogeneity.

4.4 Results and Analysis

A series of experiments was conducted on the CorKG dataset to compare the performance of RCGNN with baseline models, and the results are summarized in Tables 3 and 4. These tables report the performance of RCGNN and the comparative models on the test sets of the two subsets, CorKG_Choice and CorKG_Jian. We used common metrics-precision, recall, F1 score, and AUC-for evaluation. For the multi-class problem, each category was treated as the positive class sequentially, and metrics were calculated accordingly. Final evaluation metrics were averaged using both macro-average and weighted-average methods.

As shown in Table 3, RCGNN outperformed all comparative models on the CorKG_Choice dataset. Compared to traditional models like LR and SVM, RCGNN demonstrated significant improvements across all evaluation metrics, underscoring the value of incorporating features such as shareholding relationships and industry connections, alongside graph-based learning for credit rating prediction. Among GNN models, RCGNN achieved the best results in most metrics, particularly excelling in the macro-averaged AUC. It outperformed other models except for the weighted F1 score, highlighting its ability to capture complex graph structures and handle heterogeneous relationships effectively.

The disparity between macro-averaged and weighted metrics indicates differing model performances. While RCGNN excelled in weighted metrics, suggesting strong predictive accuracy for majority classes, its performance on minority classes-especially in terms of recall and F1 score-requires further improvement, pointing to potential areas for future enhancement.

Table 3. Experiment Results of RCGNN on CorKG_Choice

Models	Macro				Weighted			
	Precision	Recall	F1	AUC	Precision	Recall	F1	AUC
LR	65.03	42.86	47.31	76.53	89.22	90.82	90.47	78.24
SVM	44.68	44.32	44.47	58.31	91.18	91.67	91.42	59.59
RF	76.42	45.63	51.90	77.85	91.91	95.11	93.05	76.10
XGBoost	87.62	53.17	61.58	78.17	92.60	92.53	92.50	61.32
GCN	89.22	53.67	62.16	79.96	93.84	94.32	93.92	79.95
GraphSAGE	*89.58*	*54.28*	*64.30*	78.24	*95.26*	**95.98**	**94.84**	81.97
GAT	89.04	52.03	61.78	*80.06*	94.15	*95.40*	93.82	*83.71*
RCGNN	**91.24**	**56.73**	**65.35**	**84.51**	**96.14**	**95.98**	*94.76*	**83.86**

Table 4. Experiment Results of RCGNN on CorKG_Jian

Models	Macro				Weighted			
	Precision	Recall	F1	AUC	Precision	Recall	F1	AUC
LR	69.30	72.56	70.13	66.61	73.53	71.63	72.03	67.14
SVM	65.42	66.55	65.89	65.01	68.94	68.42	68.61	64.68
RF	71.51	73.39	72.08	70.28	76.69	75.16	75.68	70.17
XGBoost	77.31	77.81	77.54	72.44	80.09	79.98	80.01	72.35
GCN	80.82	79.13	80.38	79.04	83.15	81.49	82.97	79.33
GraphSAGE	82.51	*82.92*	82.43	78.21	85.47	83.56	84.43	78.59
GAT	*83.46*	82.17	*83.25*	*80.38*	*85.79*	*85.51*	*84.43*	*81.68*
RCGNN	**84.39**	**83.26**	**84.67**	**82.83**	**86.60**	**85.71**	**86.18**	**83.92**

Table 4 shows the experimental results for the CorKG_Jian dataset, where RCGNN outperformed all baseline models. Compared to the CorKG_Choice dataset, RCGNN's macro-averaged performance improved on CorKG_Jian, likely due to its more balanced label distribution, which mitigated challenges from imbalanced data. Overall, RCGNN surpassed traditional machine learning methods by effectively leveraging graph structure to capture complex relationships and achieve higher prediction accuracy and reliability. Among GNN models, RCGNN outperformed homogeneous GNNs, demonstrating its superior ability to handle heterogeneous graph structures and node features.

These results underscore RCGNN's exceptional performance in credit rating prediction, achieving significant improvements across all metrics compared to both traditional machine learning methods and conventional GNNs.

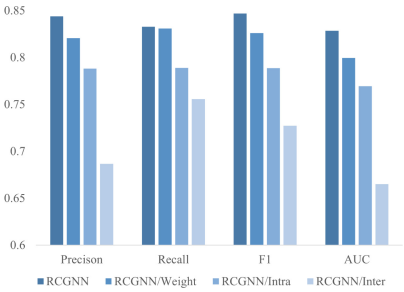


Fig. 2. Results of RCGNN Ablation Experiments

4.5 Ablation Study

To evaluate the contribution of each module in the RCGNN model, ablation experiments were conducted on the CorKG_Jian dataset. Three variants were tested:

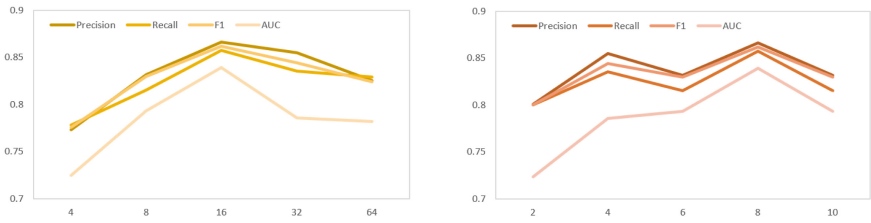
- **RCGNN**$_{/Weight}$: This variant removes the weight modeling in the intra-risk contagion aggregation module and uses an unweighted RotatE function for meta-path propagation.
- **RCGNN**$_{/Intra}$: This variant excludes the multi-head attention mechanism in the intra-risk contagion aggregation module, treating all meta-path instances equally.
- **RCGNN**$_{/Inter}$: This variant removes the multi-head attention mechanism in the inter-risk contagion aggregation module, giving equal weight to all meta-path representations.

The experimental results for RCGNN and its three variants are shown in Fig. 2. The results demonstrate that incorporating weight information for meta-path instances allows RCGNN to flexibly model different propagation paths, enabling the model to better capture subtle variations in risk propagation across instances. The introduction of the multi-head attention mechanism in the intra-risk contagion aggregation module effectively models the varying impact of different instance risks on the central node. Similarly, the multi-head attention mechanism in the inter-risk contagion aggregation module successfully captures the varying influence of different contagion types on the central node.

4.6 Sensitivity Analysis

We also conducted a sensitivity analysis on the RCGNN model to assess the impact of representation vector dimensions and the number of heads in the multi-head attention mechanism. The output dimensions varied from 4 to 64, and the number of heads ranged from 2 to 10. Figure 3 presents the weighted evaluation metric scores for different parameter settings. Our findings indicate: (1) the model maintains satisfactory performance across most hyperparameter

combinations, suggesting stability; (2) as output vector dimensions increase, performance initially improves before gradually declining. This may be due to lower dimensions being insufficient for capturing rich node information, while higher dimensions may introduce redundancy, limiting effectiveness.



(a) Output Dimension (b) The Number of Heads

Fig. 3. Sensitivity Analysis

5 Conclusions

This paper introduces a risk contagion graph neural network model for enterprise credit rating, termed RCGNN. This model leverages the complex relationships and heterogeneous node features between enterprises by first mapping all features into a unified vector space. It uses the RotatE rotation function to model risk contagion across meta-paths, and further employs an attention mechanism to dynamically assign weights to different instances of contagion paths and different types of contagion paths, thus better reflecting the process of risk contagion between enterprises. The results are then mapped to the output space to complete the task of enterprise credit rating prediction. Extensive experiments were conducted on a self-constructed dataset of Chinese listed companies, and the results demonstrate that RCGNN achieves significant performance improvement over traditional machine learning methods and homogeneous graph neural network models in credit rating tasks.

The RCGNN model effectively captures risk propagation between enterprises, but its modeling and computational complexity need optimization for more complex heterogeneous network structures and diverse risk types. Future research could explore more efficient and refined graph neural network architectures to handle larger-scale heterogeneous graph data and accurately model more intricate risk propagation mechanisms.

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Intelligent Recommendation System Design and Privacy Security

Emotionally Unbiased Reinforcement Learning for Equilibrium-Seeking in Conflict-Driven Multi-agent Systems

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Abstract. Emotionally Unbiased Reinforcement Learning (EURL) introduces a novel approach to managing equilibrium in multi-agent systems where agents face inherently conflicting objectives. Unlike traditional multi-agent reinforcement learning (MARL) frameworks that prioritize joint reward maximization, EURL minimizes conflicts through an impartial, emotionally neutral decision-making strategy, enabling balanced outcomes even when agent goals are at odds. The framework's innovative design integrates fairness and conflict reduction as primary optimization goals that yield stability in equitable interactions without enforced cooperation. Extensive simulations of autonomous traffic control, collaborative robotics, and negotiation systems indicate a significant potential for EURL in reducing the rate of conflicts and enhancing stability. By setting a baseline for emotionally neutral AI, EURL opens the door to more powerful multi-agent conflict management and thus has immediate applicability to domains such as fair and unbiased decision-making and resource allocation.

Keywords: Conflict Reduction · Emotionally Unbiased AI · Fairness Optimization · Multi-agent Systems · Reinforcement Learning

1 Introduction

Multi-agent reinforcement learning (MARL) systems have traditionally focused on optimizing collective rewards to achieve cooperative outcomes among agents [1]. This approach, often relying on joint reward maximization, is effective in scenarios where agents share common goals [2]. However, it becomes inadequate in environments where agents have inherently conflicting objectives, leading to unfair outcomes and destabilization of the system [3]. When individual agents prioritize their own goals without regard for the system's overall harmony, traditional MARL strategies can exacerbate conflicts and reduce efficiency [4]. These challenges arise in autonomous traffic [5], robotics, and negotiation platforms, where agents operate in shared competitive spaces.

To address these limitations, we introduce Emotionally Unbiased Reinforcement Learning (EURL), a novel MARL framework that seeks equilibrium by

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balancing agent objectives through impartial decision-making. Unlike existing MARL approaches that aim for cooperative reward maximization, EURL employs a conflict-reduction strategy designed to maintain neutrality in agent decision processes. By optimizing for fairness and minimizing conflicts, EURL fosters balanced outcomes without enforcing explicit cooperation, making it suitable for complex environments where agents have competing goals.

EURL's impartial stance is achieved through a specialized objective function that integrates fairness and conflict reduction as primary optimization goals. This approach allows agents to adapt their strategies dynamically, reducing aggressive behaviors that could destabilize the system [8]. The framework is further enhanced by a hierarchical policy network that separates low-level task execution from high-level conflict management, enabling nuanced control over agent behaviors.

The applications of EURL extend to various domains where conflict-prone multi-agent environments pose operational challenges. For instance, in urban traffic systems, competing priorities among autonomous vehicles can lead to unsafe driving behaviors if agents solely optimize for speed [9]. In collaborative robotics, resource contention in shared workspaces can lead to productivity losses if robots act competitively [6,10]. Similarly, in negotiation platforms, agents with conflicting interests can cause deadlocks, halting progress [7]. EURL's equilibrium-seeking approach provides a solution by fostering fairness and stability, thus enhancing overall system performance.

2 Related Work

Traditional MARL frameworks have predominantly focused on cooperative strategies, leveraging techniques like reward shaping [11], centralized training with decentralized execution, and opponent modeling to encourage collaborative behavior among agents. These methods have proven effective in environments where agents share common objectives [12]. However, in competitive or conflict-driven settings, such approaches often fall short due to their inability to manage conflicting interests effectively [14].

Recent research has explored fairness in AI systems, recognizing the importance of equitable outcomes in multi-agent interactions [15]. Fairness-aware MARL algorithms have been developed to address disparities in agent rewards and resource allocations [16]. For example, the incorporation of social welfare functions and egalitarian principles into the reward structure has shown promise in achieving more balanced outcomes [17]. Nevertheless, these methods often assume a degree of cooperation or mutual interest among agents, limiting their applicability in highly competitive environments [18].

Opponent modeling techniques in MARL aim to predict other agents' behaviors to inform strategic decision-making [13]. While effective in certain adversarial settings, these approaches can exacerbate conflicts by promoting competitive advantage rather than equilibrium [19]. Furthermore, the computational complexity of opponent modeling increases with the number of agents, posing scalability challenges [20].

Interdisciplinary approaches have attempted to integrate concepts from sociology and psychology into MARL to enhance cooperation and conflict resolution [21]. However, these methods typically focus on human-agent interactions and may not translate effectively to autonomous agent systems where emotional factors are absent [22]. The concept of emotional neutrality in agent decision-making remains underexplored, presenting an opportunity for innovative frameworks like EURL.

3 Proposed Methodology

3.1 EURL Framework Design

The EURL framework promotes fairness and minimizes conflicts among agents by incorporating these objectives directly into the learning process. The core of EURL is a multi-objective optimization problem where the loss function is defined as:

$$L_{\text{EURL}} = \alpha L_{\text{fair}} + \beta L_{\text{conflict}} + \gamma L_{\text{task}} \tag{1}$$

where:

- L_{fair} measures the disparity in rewards among agents, aiming to minimize inequality.
- L_{conflict} penalizes actions that lead to conflicts or negative impacts on other agents.
- L_{task} represents the standard reinforcement learning loss associated with the individual agent's task performance.
- α, β, and γ are weighting coefficients that balance the importance of each component.

The fairness loss L_{fair} is quantified using the Gini coefficient [23], capturing the inequality among agent rewards:

$$L_{\text{fair}} = \frac{\sum_{i=1}^{N} \sum_{j=1}^{N} |r_i - r_j|}{2N^2 \bar{r}} \tag{2}$$

where N is the number of agents, r_i and r_j are the cumulative rewards for agents i and j, and $\bar{r}$ is the average reward across all agents.

The conflict loss L_{conflict} penalizes behaviors that negatively affect other agents:

$$L_{\text{conflict}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j \neq i} c_{ij} \tag{3}$$

where c_{ij} is a conflict indicator function between agents i and j, defined as:

$$c_{ij} = \begin{cases} 1 & \text{if action of agent } i \text{ causes conflict with agent } j \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

The task loss L_{task} is computed using standard RL methods, such as the mean squared error between predicted and target Q-values in DQN [24].

By adjusting the weighting coefficients α, β, and γ, the EURL framework can be tuned to prioritize fairness and conflict reduction over individual task performance or vice versa, depending on the application requirements.

3.2 Hierarchical Policy Network

The Emotionally Neutral Policy Network (ENPN) in EURL employs a hierarchical structure to separate task execution from conflict management:

1. **Task Execution Layer:** This lower-level network focuses on optimizing the individual agent's performance on its assigned tasks using standard RL techniques [25]. It processes the agent's local observations o_i and outputs a preliminary action policy $\pi_{\text{task}}(a_i|o_i)$.
2. **Conflict Management Layer:** The upper-level network receives inputs from the Task Execution Layer and additional information about potential conflicts [26]. It evaluates potential actions for fairness and conflict potential, adjusting the preliminary policy to produce the final action policy $\pi_{\text{EURL}}(a_i|o_i)$ by incorporating the EURL loss function.

The hierarchical structure allows agents to pursue their goals while simultaneously considering the impact of their actions on the overall system equilibrium [27]. The ENPN adjusts actions in real-time, promoting fairness and reducing the likelihood of conflicts without significantly compromising task performance.

3.3 Training Procedure

The training process for EURL involves the following steps:

The training leverages experience replay and mini-batch updates to stabilize learning and improve convergence. Hyperparameters such as learning rates, discount factors, and weighting coefficients are fine-tuned through grid search and cross-validation to achieve optimal performance.

Selecting appropriate values for the weighting coefficients α, β, and γ is crucial [28]. We performed sensitivity analysis by varying these parameters within predefined ranges:

- $\alpha \in [0.1, 1.0]$: Fairness weight
- $\beta \in [0.1, 1.0]$: Conflict reduction weight
- $\gamma \in [0.1, 1.0]$: Task performance weight

The optimal set of weights was selected based on the highest overall performance across the defined metrics.

Algorithm 1. EURL Training Procedure

Initialize replay memory D to capacity M
Initialize ENPN weights θ randomly
for each episode **do**
  for each agent i **do**
    Observe initial state s_i^0
  end for
  for each time step t **do**
    for each agent i **do**
      Generate preliminary action a_i^t using $\pi_{\text{task}}(a_i^t|o_i^t)$
      Adjust action to a_i^t using $\pi_{\text{EURL}}(a_i^t|o_i^t)$ incorporating EURL loss
      Execute action a_i^t and observe reward r_i^t and next state s_i^{t+1}
      Store transition $(s_i^t, a_i^t, r_i^t, s_i^{t+1})$ in D
    end for
    Sample random mini-batch from D
    Update ENPN weights θ by minimizing L_{EURL} (Eq. 1)
  end for
end for

4 Experimental Setup

4.1 Test Environments

To evaluate the effectiveness of EURL, we conducted extensive simulations in three distinct multi-agent environments:

1. Autonomous Traffic Management: This environment simulates urban traffic networks where autonomous vehicles face conflicting goals, such as minimizing travel time and maximizing safety [29]. It includes intersections with varying traffic densities, traffic signals, pedestrian crossings, and vehicles with different priority levels and destinations.

2. Collaborative Robotics in Manufacturing: This simulated manufacturing scenario involves robotic arms sharing limited resources like tools and workspace [30]. Robots are tasked with overlapping resource needs, dynamic task allocation based on schedules, and constraints on resource availability and access times.

3. Virtual Negotiation Platform: This negotiation environment features agents competing for limited resources with conflicting interests and utility functions. It includes protocols for offers, counteroffers, time constraints, and penalties for failed negotiations, providing a robust testbed for studying equitable strategies.

4.2 Performance Metrics and Data

The evaluation of EURL's performance is based on the following metrics:

Table 1. Performance Metrics for EURL Evaluations

Metric	Description
Conflict Reduction Rate (%)	Percentage decrease in conflicts compared to the baseline model.
Equilibrium Ratio (%)	Proportion of time the system remains in a balanced state with minimal conflicts.
Resource Utilization Efficiency (%)	Effectiveness of resource usage in collaborative environments.
Negotiation Success Rate (%)	Percentage of negotiations concluding with mutually acceptable agreements.
Agent Satisfaction Index	Aggregate score representing how well agents achieve their individual goals (normalized between 0 and 1).

The experiments utilized synthetic datasets tailored for each environment. For traffic management, data included simulated urban traffic flows with varying densities, generated using SUMO [29]. For collaborative robotics, resource allocation tasks were simulated using Gazebo [30]. The baselines included a cooperative MARL model with joint reward maximization and a conflict-agnostic DQN framework (Table 1).

5 Results and Analysis

The performance of EURL was compared against a baseline MARL model that does not incorporate fairness or conflict reduction mechanisms (Fig. 1).

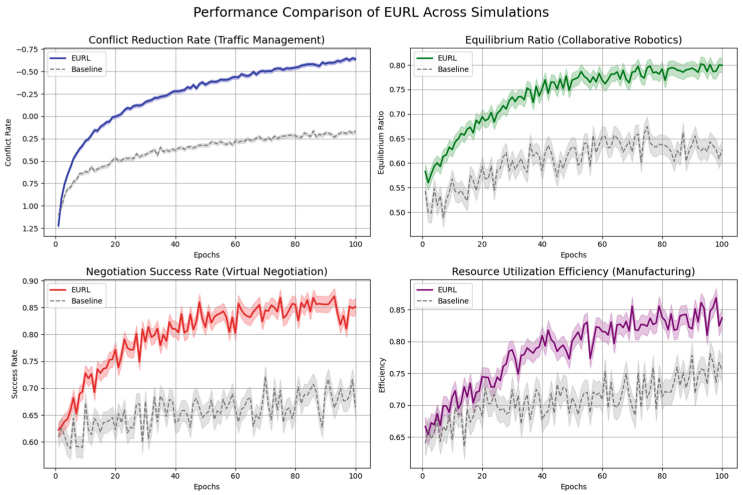

Fig. 1. Performance Comparison of EURL and Baseline MARL Models Across Different Environments.

EURL agents demonstrated a significant reduction in congestion-related conflicts compared to baseline MARL models. Specifically, EURL achieved a 35% reduction in traffic conflicts, a substantial improvement over the baseline. The equilibrium ratio increased to 75%, indicating longer periods of balanced traffic flow, and the average travel time decreased by 12%, showcasing improved efficiency in managing urban traffic dynamics. In the manufacturing environment, EURL agents demonstrated superior performance over baseline models. Resource utilization efficiency improved by 20%, task completion rates increased by 25%, and robots experienced 30% less downtime due to reduced resource contention, highlighting EURL's effectiveness in optimizing productivity and coordination. EURL agents outperformed traditional models in negotiation scenarios, achieving a 40% increase in success rates, reaching 80%. The agent satisfaction index improved by 15%, reflecting more mutually beneficial agreements, while conflict occurrences decreased by 35%, significantly reducing negotiation breakdowns (Fig. 2 and Table 2).

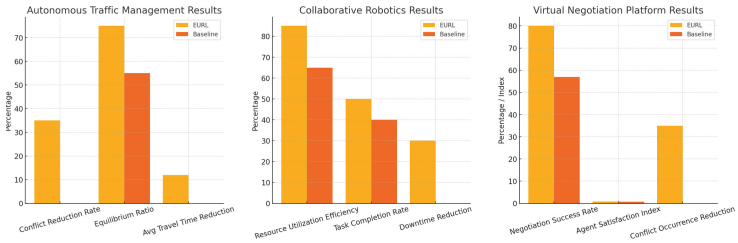

Fig. 2. Comparison of Key Performance Metrics between EURL and Baseline Models.

Table 2. Consolidated Results for All Three Environments

Environment	Metric	EURL	Baseline
Autonomous Traffic Management	Conflict Reduction Rate (%)	35	0
	Equilibrium Ratio (%)	75	55
	Average Travel Time Reduction (%)	12	0
Collaborative Robotics	Resource Utilization Efficiency (%)	85	65
	Task Completion Rate (# tasks/hour)	50	40
	Downtime Reduction (%)	30	0
Virtual Negotiation Platform	Negotiation Success Rate (%)	80	57
	Agent Satisfaction Index	0.85	0.70
	Conflict Occurrence Reduction (%)	35	0

The results indicate that EURL effectively reduces conflicts and promotes fairness across diverse multi-agent environments. In the autonomous traffic management simulation, the significant reduction in conflicts and average travel times

suggests that EURL agents can navigate shared spaces more harmoniously without sacrificing efficiency. The increased equilibrium ratio demonstrates EURL's ability to maintain system stability. For collaborative robotics, the improvements in resource utilization and task completion rates highlight EURL's capacity to optimize resource allocation in competitive settings. The reduction in downtime reflects more efficient coordination among agents. In the virtual negotiation platform, the higher negotiation success rate and agent satisfaction index underscore EURL's potential to facilitate equitable agreements in adversarial environments. The decrease in conflict occurrences indicates a smoother negotiation process. Overall, EURL outperforms baseline MARL models by integrating fairness and conflict reduction into the learning process, leading to enhanced system performance and stability.

6 Discussion

The observed improvements suggest that incorporating fairness and conflict reduction directly into the reinforcement learning framework can significantly enhance multi-agent system performance. EURL provides a viable alternative to traditional MARL approaches, especially in environments where agents have conflicting objectives.

The hierarchical structure of the ENPN allows for scalable implementation in systems with large numbers of agents. Additionally, the flexibility in weighting the loss function components enables customization for specific application requirements.

EURL shows promise but requires further research in several areas. These include extending its applicability to dynamic environments with changing agent populations and unpredictable events, testing in real-world settings to address practical challenges and validate simulations, developing formal proofs of convergence and stability, and implementing adaptive strategies for the weighting coefficients (α, β, and γ) to dynamically balance objectives based on environmental feedback.

7 Conclusion

Emotionally Unbiased Reinforcement Learning offers a balanced approach to conflict management in multi-agent systems by incorporating fairness and conflict reduction into the process of reinforcement learning.

The architecture of the network includes a hierarchical policy network with an adjustable loss function to allow various applications in different domains. Therefore, EURL can enable further development in fair and trustworthy AI by providing a framework through which outcomes remain just and do not compromise individuals' goals.

Looking ahead, the adaptability of EURL's framework opens pathways for future research into more complex, dynamic, and high-stakes environments, such as urban planning and decentralized resource management. By enabling agents

to adapt to unpredictable conditions, EURL offers a promising approach for sustainable and ethical AI that maintains fairness, reduces conflict, and enhances collective productivity. This adaptability reinforces EURL's potential to shape real-world applications where stability and impartiality are critical.

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Diversified Conversational Recommendation System

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Abstract. Conversational recommendation systems have garnered increasing attention in recent years, driven by the successful development of large language models. These systems can elicit users' real-time preferences through interaction, thereby achieving better recommendation results. However, traditional conversational recommendation systems primarily focus on improving accuracy while neglecting the diversity of the results, which can lead to information redundancy and decreased user satisfaction. If properly considering diversity when constructing a conversational recommendation system, it is possible to enhance user satisfaction and improve system performance simultaneously. In this paper, we propose a novel diversified conversational recommendation system that simultaneously increases both the success rate and diversity of recommendations. The system incorporates two key functions: interest exploration and item recommendation. During the interest exploration stage, two diversification mechanisms are employed to capture user preferences. Subsequently, triplet loss is applied to further enhance diversification in the item recommendation stage. Experimental results demonstrate that our proposed method outperforms traditional conversational recommendation systems on widely used datasets.

Keywords: Conversational recommendation · Diversified recommendation · User preference

1 Introduction

Conversational recommendation systems (CRSs) is a type of recommendation systems [8,9,20,24] capable of engaging in conversation with users, extracting users' preferences through interactions. These systems not only solve the cold-start problem, which means a model lack historical data to guide recommendations when a user interact with it for the first time, but also overcome the limitation of traditional systems that fail to capture users' real-time preferences.

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However, existing CRSs often rely heavily on the information users explicitly express, leading to the recommendation of related items only and ignoring other potential interests users might have. Moreover, research on diversity in CRSs has yet to been dealt with. To address these issues, we propose a system called Diversified Conversational Recommendation System (DCRS) to explore users' interests more comprehensively. Besides, we tailor methods from the perspectives of features and to enhance diversity, aiming to improve the system's overall performance through diversified design.

Research in CRSs have been noticed across various domains in recent years. For example, in terms of traveling, Lan et al. proposed a framework named RUI and developed a dataset called CADT [12], which is tailored for the recommendation of tourist attractions. Habib et al. introduced IAI MovieBot [6], a CRS for movies. Fernando et al. developed a CRS with an explanation facility in the music domain [4]. Moreover, many studies have devoted to closer connections between conversation and recommendation to improve system performance. Sun et al. proposed a model named CRM [18], which employed a reinforcement learning mechanism to improve the performance of both conversation and recommendation. Lei et al. introduced a framework named EAR [13], emphasizing close interaction between the conversational component and the recommendation component at every stage an issue that systems in the past usually failed to be stressed on. Lei et al. later proposed SCPR [14], which dramatically surpassed previous systems in efficiency through a graph-structured design. Deng et al. integrated reinforcement learning tasks in conversational recommendation into a unified policy learning task and proposed a framework called UNICORN [3] to cope with this integrated task. Furthermore, Yang et al. proposed a framework named HCKH [25], which offers more varied question forms to make the system more flexible and explainable.

With the rising concern for trustworthiness [2,21], people are increasingly focused on the explainability of systems, in addition to their accuracy. User satisfaction is also a key indicator of system effectiveness in CRSs; as a result, research on user experience have prospered over time. Jin et al. proposed a unifying framework called CRS-Que [10], discussing the qualities of CRSs that determine system success from the users' perspectives. Li et al. introduced a model named UCCR [15], which aims to elicit more comprehensive user preferences by collecting more aspects of information during conversation to further enhance user's satisfaction. Since conducting real user studies is not easy, using user simulators can be a cost-effective solution. Therefore, some studies focus on a user simulator that closely mimics real users. Zhang et al. designed a user simulator that generates responses considering both users' preferences and their interactions with the system [27]. Shen et al. proposed a user simulator called UUSFF, producing more fuzzy feedback based on users' preferences to cater to ambiguous user preferences [17]. When it comes to the evaluation of user satisfaction with system performance, whether the system is diverse or not is a critical factor. Many studies attempt to improve the system's diversity performance with various approaches, like reranking the recommended list to enhance

diversity [19]; others have introduced constraints related to diversity in the recommendation generation process to produce a more diverse recommendation portfolio [26]. Another approach has involved categorizing users and generating recommendations with different degrees of diversity based on individual tendency to diversification [23]. Besides, some studies have utilized multi-objective optimization algorithms to generate recommendation results aligned with different goals [22]. Existing research has achieved considerable progress in conversational recommendation system, there still exist some interesting issues have yet to be excavated. Current systems are able to capture users' preferences during conversation process; however, more comprehensive preferences should be explored. Moreover, some studies have tackled the diversity issue in recommendation systems, but there is no study discusses the diversification in conversational recommendation systems.

To deal with those issues mentioned above, we propose a system called DCRS, which can simultaneously conduct dialogue and generate recommendations within a single conversation round. Specifically, our system captures user preferences during the dialogue process using an effective comparative strategy and produce recommended items accordingly. Moreover, we design various methods from the viewpoint of features and items to enhance diversity. We conduct the experiment on a well-known music dataset, LastFM, to test our systems' performance. The results shows that compared with those state of the art systems, with the diversification design, our system can gain great improvement on the success rate of recommendation and the average number of recommendation turns. In this study, our contribution shown as follows:

- A novel diversified conversational recommendation system, incorporating interest exploration and item recommendation functions, is proposed to enhance the success rate of recommendations.
- Two diversification mechanisms based on features and items are adopted during interest exploration stage to capture representative user preferences.
- To further increase the diversification of recommended items, triplet loss is adopted in item recommendation stage to obtain more meaningful representation.
- Compared to traditional CRS that make a hard decision by either asking questions or making recommendations in each round, our proposed system transforms this approach into a soft decision that involves both actions simultaneously.
- The proposed conversational recommendation system with diversification mechanisms reduces the number of conversation turns compared to several baselines in a real-world dataset.

The rest of this paper is organized as follows: Sect. 2 reviews related works on conversational recommendation systems. Section 3 introduces the proposed DCRS system and presents our methods for enhancing system diversity from three perspectives. Section 4 discusses experimental results, and Sect. 6 concludes the study.

2 Related Work

With the rising attention to chatbot techniques and question answering systems, conversational recommendation systems have developed significantly in recent years. As [5] described, the general framework of conversational recommendation systems consists of three components: a user interface, a conversation strategy module, and a recommender engine. The user interface serves as the bridge between users and the machine; the conversation strategy module is the core of CRS, helping with the coordination of the other two parts; the recommender engine collects the required information of all entities (i.e. users, items, and attributes).

Research on the combination of dialogue systems with recommendation systems can be traced back to the CRM model proposed by Sun et al. [18]. Compared to previous research, which generates recommendations based on item score calculated by certain traditional methods, CRM leverages the reward mechanism of reinforcement learning to improve both conversation and recommendation processes. However, it generates only one recommendation in single session, decreasing the recommendation efficiency. Lei et al. [13] subsequently proposed a study to address these problems. They redefined the conversational recommendation task into three key questions: what to ask, when to recommend, and how to adapt with user feedback and proposed a framework named EAR. EAR consists of three stages: estimation, action, and reflection, emphasizing on the interaction between the conversational and the recommender component in each stage. Later, Lei et al. proposed a model named SCPR [14]. This model primarily utilizes a graph structure to deal with the interactions among users, items, and attributes. Compared to CRM and EAR, SCPR significantly reduces the search space of attributes by leveraging the adjacent attribute constraint on the graph, which effectively alleviates the pressure for policy making. Then, Deng et al. integrated reinforcement learning tasks in conversational recommendation into a unified policy learning task. To be more specific, they merged the three questions: what attributes to ask, which items to recommend, and when to ask or recommend that are often tackled separately in conversation or recommendation part.

3 Diversified Conversational Recommendation System

In this chapter, we will first introduce the structure of our Diversified Conversational Recommendation System (DCRS). And then we will elaborate methodologies employed in our system.

3.1 System Overview

DCRS mainly consists of two parts: user interest exploration and item recommendation. Here, we utilize the reinforcement learning mechanism to determine which action to take. To be more specific, by setting different rewards on the

current status of the system, we acquire a parameter alpha (α) to determine the when to capture user interests and when to recommend items during the conversation process. Our framework is illustrated in Fig. 1.

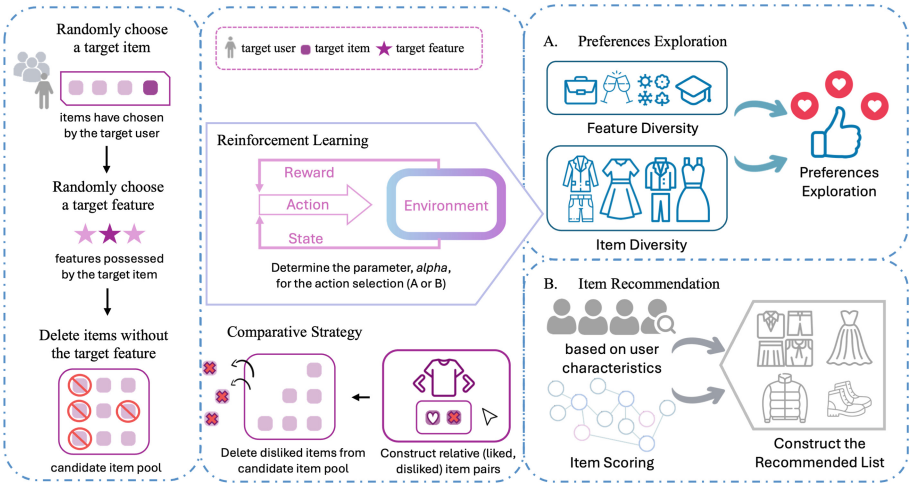

Fig. 1. The overview of the DCRS.

3.2 Preference Exploration

Broadly exploring user preferences is not only able to enhance user satisfaction with the system but also helpful for users to discover their specific likes [7]. As a result, we design two methods within this system framework, based on features and items, to achieve comprehensive exploration of user preferences. The details of these methods will be discussed in the following paragraph.

3.3 Diversification

In recent years, research on recommendation systems has prospered across various domains. As user satisfaction with these systems has gained increasing attention, in addition to accuracy, more studies have focused on topics such as diversity and novelty in recommendations. However, there is often accuracy-diversity tradeoff [11]. As a result, striking a balance between accuracy and diversity is a significant issue. According to classification in [28], research on the diversity in recommendation systems can be primarily divided into two main themes: sources of diversity and measurements to quantify diversity. In this section, we will review the research on diversity in recommender systems based on this categorization method. Based on different perspectives, the sources of diversity can be categorized into two sides: recommended items perspective and users' perspectives. For the recommended items perspectives, it refers to the discussion of system-level

diversity, which is also known as aggregate diversity. For example, MF-R [1] proposed by Alhijawi et al. considers predicted rating and popularity of items to generate recommendations with diversity. As for the users perspectives, it considers diversity from the viewpoint of each individual, namely individual diversity. Taking MO-KUCF [16] proposed by Ma et al. as an example, this framework is based on user collaborative filtering and utilizes kernel density estimation and multi-objective optimization to enhance diversity, system accuracy, and address the concept drift problem. In terms of various datasets, multiple methods are designed to measure diversity and these methods can mainly be categorized into distance-based diversity and Entropy and Gini index methods. Distance-based diversity methods measure diversity by calculating the distance/dissimilarity between items.

For the users, aggregate diversity and distanced-based diversity are not adequate enough to provide user-oriented choices. Moreover, only taking individual diversity into consideration can restrict users' exposure to novel items. Therefore, in this study, we incorporate the concept of diversity into conversational recommendation systems, aiming to achieve recommendation diversification from different persepctives. We propose two kinds of methods: feature diversity and item diversity. The following sections will elaborate on the details of these two categories.

Feature Diversity. Items in candidate pool possess various features, since it is impossible to know in advance which features are necessary to users when selecting items, we initially assume that all features have equal importance. We then categorize the features based on the number of items that possess each feature. When a feature is possessed by relatively few items, we consider it to be more niche. Therefore, we apply a weighting system where niche features are assigned higher scores than the more "popular" features as defined by our system. This approach increases the visibility of niche features, thereby enhancing the diversity of the recommendations.

Item Diversity. We adopt the concept of a triplet loss function, aiming to bring similar items closer together while pushing less similar items further apart. Specifically, we build up item embeddings and utilize the triplet loss function on the training and optimization, which enriches the information involved and allows for more effective classification of items. Subsequently, we select items from different categories to build up the recommended list presented to users for the purpose of a more diverse item portfolio.

3.4 Popularity-Aware Personalized Recommendation

In this approach, we adjust the recommended item combinations based on the user's historical purchasing tendencies and employ the concept of popularity. By analyzing the proportion of popular versus less popular items in the user's past purchasing behavior, we determine the type of items to recommend. For

example, if the user has previously purchased more popular items, the recommended list will include more popular items but will also incorporate items that are less popular. This ensures that the user's choices are full of diversity and not restricted to a single category.

4 Experiments

4.1 Experimental Setting

To evaluate our proposed method, we conduct experiments on a LastFM dataset. The dataset contains 1, 801 users, 7, 432 items and 33 attributes. The total number of interaction is 76, 693. We use the three methods mentioned in Sects. 3 and 4 to test the average success turn (AT) and success rate at turn 15 (SR@15) for the evaluation of the performance of our proposed CRSs. In addition, we compare our model with the following eight representative baselines:

- **CRM** [18]. It records user's preferences into a tracker and uses reinforcement learning to determine a policy for interacting with the user. The reinforcement learning leverages a policy network whose state vector is the result of a belief tracker.
- **EAR** [13]. It trains a policy network through reinforcement learning to decide whether to ask questions or recommend items.
- **SCPR** [14]. The key design of the SCPR method is that it leverages adjacent attribute constraints in the graph, and it is a dedicated reinforcement learning model having a small action space.
- **UNICORN** [3]. It is a reinforcement learning framework based on a dynamic weighted graph to solve the integrated task and design two action selection strategies for the sample efficiency issue.
- **HCKH** [25]. This framework applied a hierarchical reinforcement learning framework and introduced methods of effectively incorporating user feedback, whether positive or negative, for online recommender updates.

Following previous studies on MCR [13,14], we utilize the success rate at the turn t (SR@T) to measure the cumulative ratio of successful recommendation at turn t. We also adopt average turns (AT) to record the average number of turns for all sessions (if the conversation session reaches the maximum turn T, we count the turn for that session as T). A higher SR@T indicates a greater number of successful turns, whereas a smaller AT indicates a more efficient conversation.

$$SR@t = \frac{number\ of\ successful\ conversations\ by\ turn\ t}{number\ of\ conversations} \tag{1}$$

$$AT = \overline{\text{conversation length}} \tag{2}$$

4.2 System Performance Evaluation

This experiment mainly aims to compare the overall system performance of the proposed methods with multiple baselines. At the same time, a comparative analysis of the various diversified recommendation strategies we proposed is also conducted by evaluating two metrics: SR@15 and AT measurements. As shown in Table 1, some earlier conversational recommendation systems, such as CRM [18], EAR [13], and SCPR [14], have an average recommendation success rate of about 32.5% to 46.5% within a limit of 15 dialogue turns. The average number of dialogue turns for these methods ranges between 12.86 and 13.75. Subsequently, UNICORN [3] and HCKH [25] utilized more sophisticated questioning techniques and dynamic models to capture the user's current preferences, further increasing SR@15 to 0.654 while reducing AT to within 10 dialogue turns. The proposed conversational recommendation system with various diversified strategy dramatically outperforms previous methods in these benchmarks, increasing SR@15 to over 0.93 while reducing AT to around 5.5. A more detailed comparison of the proposed DCRS with the two diversification designs mentioned in the paper reveals that DCRS with item diversity slightly outperforms DCRS with feature diversity in success rate, with a marginal advantage of 0.003. However, in the evaluation of average turn, the results are reversed, with DCRS with feature diversity slightly outperforming DCRS with item diversity. From a practical application perspective, if the focus is on maximizing recommendation accuracy, the feature diversity design should be adopted. On the other hand, considering user experience where users might lose patience with multiple dialogues, the item diversity approach would be preferable. The numeric performance differences between the two designs are actually quite small. In summary, the proposed DCRS significantly outperforms these state-of-the-art conversational recommendation baselines under any setting.

Table 1. Performance Comparison Against Baselines

	SR@15	AT
CRM	0.325	13.75
EAR	0.429	12.88
SCPR	0.465	12.86
UNICORN	0.535	11.82
HCKH	0.654	9.35
DCRS (Feature Diversity)	0.931	5.51
DCRS (Item Diversity)	0.934	5.54

5 Conclusion

In this paper, a diversified conversational recommendation system is developed to simultaneously increase the success rate and diversity of recommendations.

The system is divided into two stages: preference exploration and item recommendation. In the first stage, two diversification mechanisms, based on features and items, are designed to elicit user preferences through conversation. Then, triplet loss is applied to generate representative embeddings, helping the system better understand the characteristics of the user. In addition, a popularity-aware personalized recommendation is provided based on user's historical transaction receords. The experimental results show that our proposed method can outperform several state-of-the-art baselines in two evaluation metrics.

In addition, there are several directions worth exploring: This study explore the categories of items and features to design the diversification mechanism. Applying different dimensions of features such as style, era, etc. to fit in with various scenarios are potential. We can improve our communication strategy by utilizing multiple dialogue fashions like actively inquire users about their rather vague reactions.

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Rule Extraction with Reject Option

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Abstract. In many applications, it is vital to simultaneously obtain the best possible predictive performance and be able to understand the underlying logic. Rule extraction can then be employed, by having predictions made by an opaque model and an extracted transparent model to provide explanations and allowing inspection and analysis of relationships found. Using this setup, the quality of both global and local explanations depend on the fidelity of the extracted model. In this paper, a novel approach to rule extraction, which includes a reject option based on well-calibrated fidelity estimates, is presented and empirically evaluated. With the proposed method, the user can balance the required fidelity against the number of instances explained, by rejecting explanations in parts of feature space where the extracted model is generally weak in approximating the opaque model, or when the two models don't agree. The transparent model can be visualized, using different rejection levels, thus identifying the parts of feature space where it is a good approximation of the opaque model. Empirical investigation, using a number of benchmark data sets, shows that extracted models are highly faithful, while often being small enough to be comprehensible. Most importantly, it is demonstrated how both local and global explanations can be provided through rule extraction with a reject option by leveraging well-calibrated fidelity estimates.

Keywords: rule extraction · explanations · reject option · calibration

1 Introduction

Within machine learning, it is a well-known fact that opaque predictive models, such as support vector machines, neural networks or ensemble models, generally obtain better predictive performance than transparent models like decision trees or rule sets [5,11,24]. At the same time, in many situations where a predictive model is used for decision-making or decision support, there is a need to understand the predictions made. This could either be for legal reasons, such as being obligated to provide explanations to people affected by AI decision-making [15], to ensure user trust and confidence in the model [27], or because there is a wish to utilize the predictive model to gain insights into domain relationships [1]. This trade-off between high predictive performance and having a transparent

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model is referred to as the *accuracy vs. interpretability* trade-off, and has been widely studied in machine learning, see e.g. [4,14].

One way of handling the accuracy vs. interpretability trade-off is *rule extraction*, which refers to the activity of creating a transparent (interpretable) model approximating an opaque model. A common way of utilizing rule extraction is to let the high-performance opaque model make the actual predictions and use the transparent model as a post-hoc explanation.

The entire transparent model is in itself a global explanation, while the rules covering the instance, such as the prediction path in a tree model, provide local explanations, i.e., on the instance level. A key criterion for rule extraction for explanations is the extracted model's *fidelity* towards the opaque model, see e.g. [3,32].

As employed within rule extraction, fidelity is a global measure, measured as the proportion of test instances where the predictions of the transparent model are identical to the opaque model's predictions. Consequently, fidelity only indicates the overall quality of explanations provided by the extracted model and does not provide information about the reliability of individual explanations. For instance, if fidelity on a test set is 90%, this means that for one instance in ten, the transparent model will in fact make a different prediction than the opaque model and thus its so-called "explanation" is in fact hugely misleading and confusing. Consider, for instance, a binary classification task like loan applications, where a decision-support system gives recommendations for approval. If an instance (application) is classified as "not approved" by the opaque model, but the transparent model instead makes an "approved" prediction, the accompanying trace of that prediction can hardly be said to explain the prediction from the opaque model, and could most definitely not be presented to the applicant as justification for denying the loan application.

The main contribution of this paper is to introduce and empirically investigate the addition of a *reject option*, based on fidelity estimates, for rule extraction, using well-calibrated extracted tree models.

In a scenario where explanations are desirable, but not mandatory, a reject option in rule extraction will mean that the transparent model can abstain from providing local explanations for instances where the estimated fidelity to the opaque model is too low, i.e., where the transparent model is not sufficiently reliable. If the extracted model has well-calibrated fidelity estimates, there will be a clear relationship between reject proportion and explanation fidelity; which also corresponds to the intuitive notion that a smaller proportion of explanations will mean that these are more faithful to the opaque model. A variable reject proportion allows a user to do a trade-off between how many explanations are given and how certain she can be that these explanations are correct, in the sense that the extracted model agrees with the opaque model. Coupled with visualization of the transparent model, a reject option shows, on the global level, where in the input space the transparent model is a reliable approximation of the opaque model and where it is not. In particular, for an extracted tree model, a user can see which parts of the extracted tree that meet a certain fidelity level and thus obtain well-calibrated fidelity estimates for each prediction region,

as defined by a leaf or sub-tree. The proposed methodology for rule extraction with reject option has the following characteristics:

1. The extracted tree model is a global explanation, with well-calibrated fidelity estimates added to the leaves, allowing the user to see in which parts of the input space the extracted model is a reliable approximation of the opaque model.
2. A reject option based on the desired proportion of potential explanations, where no local explanation will be given if the fidelity estimate is too low for the leaf in the tree that the particular instance falls into.
3. An automatic explanation reject for instances where there is disagreement in the prediction between the opaque and the transparent model.

2 Background

2.1 Rule Extraction

The two main approaches to rule extraction are *white-box* (or open-box), where the internal structure of the opaque model is utilized to obtain the transparent model, and *black-box* (or pedagogical), where only the relationship between the inputs and resulting output from the opaque models is explained. Both white-box and black-box rule extraction have been widely used for providing post-hoc explanations. White-box approaches include [10] and [12], where rules are extracted to explain random forest models by exploiting the internal structure of multiple decision trees. For a good survey of rule extraction techniques and applications, before the advent of deep neural networks, see [19]. Recent years have also seen many algorithms for rule extraction from deep neural networks, such as [6,34,36].

Black-box rule extraction is fundamentally just predictive modeling, where each instance (input-output pattern) consists of the original input vector $\mathbf{x}_i$ and the corresponding prediction $f(\mathbf{x}_i)$ from the opaque model. Black-box rule extraction thus is doubly model-agnostic in the sense that it be applied to *any* opaque model (including ensembles) and using *any* learning algorithm producing interpretable models as the rule extractor, simply by training a transparent model with predictions from the opaque model as targets. In [35], Zhou distinguishes two different uses of rule extraction, in the context of neural networks, either as rule extraction *for* neural networks or rule extraction *using* neural networks. The purpose of rule extraction for neural networks is to understand how the opaque model works, the rule extraction using neural networks aims at extracting an interpretable model with better predictive performance than an interpretable model created directly from the data set. In the latter case, the extracted interpretable model is used both for predictions and to understand and analyze underlying relationships. Generalizing this from neural networks to any opaque model, we say that rule extraction for an opaque model describes the use case when the actual predictions are made by the opaque model, and the interpretable model is used to explain the entire opaque model and/or individual explanations. This is thus the scenario targeted by the method proposed in this

paper, where the addition of calibration produces reliable fidelity estimates that form the basis of the reject option.

2.2 Probabilistic Prediction and Calibration

Probabilistic predictors output probability distributions, and if *valid*, perform well against statistical tests based on subsequent observations of the labels. Specifically, the predictor is *well-calibrated* if

$$p(c_j \mid p^{c_j}) = p^{c_j} \tag{1}$$

where p^{c_j} is the probability estimate for class j. Less formally, the probability estimate for the predicted label (i.e., the confidence) should match the (empirical) accuracy.

In a scenario where a probabilistic predictor is used as a black-box rule extractor, the probability estimates will be for fidelity (i.e., making the same prediction as the opaque model) instead of accuracy. If the extracted model is a *probability estimation tree (PET)* [29], the probability estimates are given by the *relative frequencies*; i.e., the proportion of training instances from each class in the leaf where the test instance falls:

$$p_i^{c_j} = \frac{g(i,j)}{\sum_{k=1}^{C} g(i,k)} \tag{2}$$

where $g(i,j)$ is the number of instances belonging to class j that falls in the same leaf as instance i, and C is the number of classes. In [22] such PETs extracted from opaque models were called *fidelity estimation trees (FETs)*.

Since PETs are notorious for being overconfident, see e.g., [21], external calibration is necessary to obtain well-calibrated trees. As shown in [22], calibration was also necessary for FETs. In this paper, we will use the standard calibration technique *Platt scaling* [28], leaving more advanced options for future work.

Platt scaling fits a sigmoid function to the probability estimates using a separate calibration set. The function is

$$\hat{p}(c_j \mid p^{c_j}) = \frac{1}{1 + e^{As+B}} \tag{3}$$

where $\hat{p}(c_j \mid p^{c_j})$ is the calibrated probability estimate for class c_j, given the probability estimate p^{c_j} from the model. A and B are found by a gradient descent search, minimizing a specific loss function, for details see [28]. While Platt scaling by construction assumes that the calibration curve is a sigmoid, which of course may not be correct, the calibration is most often at least acceptable.

2.3 Reject Option

The framework for classification with reject option was introduced in [9] to study the trade-off between error rate and reject for binary classification systems. Using

a reject option in classification means that the model refrains from making predictions for instances where the prediction is uncertain. Typically, the model's probability estimates for each class are used to determine this. Using a reject option in a predictive modeling scenario introduces a means of controlling the trade-off between number of predictions made and predictive accuracy, where a higher reject proportion generally will increase accuracy of those predictions that are actually made, provided of course that underlying probability estimates are well-calibrated. Multiple studies have both extended the theoretical framework around classification with reject option and applied it in different domains. For example, [18], introduces a generalized statistical framework and includes risk functions that further inform the trade-off between errors and rejections. Several studies employ classification with reject option to control the error rate of a classifier. In [26], a methodology called confidence-based classifier design, which uses a dynamic bin width allocation method to estimate probabilities and an empirical cumulative density function to estimate the error rate, is proposed and evaluated.

In [17], classification with reject option is set up so that the user can specify a desired accuracy level, and the rejection region is found from this. In [23], conformal classification is employed to automatically suggest which accuracy levels that can be exactly met, with statistical guarantees, using classification with reject option. In [13], conformal classification is adapted to achieve guarantees on the false positive rate in the prediction sets produced.

In a rule extraction scenario, where the extracted model serves as an explanation of the opaque model, the reject option enables a user to balance the number of explanations provided against how certain we can be that the given explanations are reliable descriptions of the opaque model's predictions.

2.4 Related Work

Relating local rule explanations to a global explanation is explored in [31], where local explanations in rule form are aggregated into global explanations, in a bottom-up fashion. Resulting rules are reported to be very compact, but have slightly worse fidelity than decision tree models.

The use of reject option in rule extraction is, to the best of our knowledge, novel. However, using calibration to improve fidelity estimates in rule extraction has been previously studied. In [20], conformal classification was used to obtain fidelity guarantees in rule extraction and in [25] rule extraction for regression was studied. Both these approaches use the conformal prediction framework, which outputs prediction regions with associated validity guarantees. In [22], rule extraction for classification was studied using Venn-Abers calibration, also providing validity guarantees for fidelity estimations.

3 Method

As described above, the investigated scenario is black-box rule extraction, i.e., interpretable models are generated by learning the relationship between input patterns and the predictions from an opaque model.

All experimentation was performed using scikit learn, using *Random forest* [7] and *Extreme Gradient Boosting (XGB)* [8] to generate highly accurate opaque models, and standard decision trees as rule extractors. All parameter values were set to default, with the following exceptions: (1) Both random forest and XGB had 300 trees (n_estimators=300) and (2) For the decision trees, the minimum number of instances in each leaf was set to 7 (min_samples_leaf=7).

For the actual evaluation, repeated stratified holdout with 100 runs and the test size set to 25% was used, i.e., all results are averaged over the 100 runs. Since a separate labeled data set is required for the calibration, two different opaque models were trained on every run, one using all training instances and one dividing the training instances into a proper training set (2/3) and a calibration set (1/3). All in all, the following four rule extraction setups were evaluated:

- **RF-U**: A decision tree is induced from a random forest model generated from all training data. No calibration is used.
- **RF-P**: A decision tree is induced from a random forest model generated from 2/3 of the training data. The decision tree is calibrated using Platt scaling on the remaining 1/3.
- **XGB-U**: A decision tree is induced from an XGB model generated from all training data. No calibration is used.
- **XGB-P**: A decision tree is induced from an XGB model generated from 2/3 of the training data. The decision tree is calibrated using Platt scaling on the remaining 1/3.

In the analysis, we first look at the predictive performance (accuracy) of the underlying and the extracted models. After that, we present the overall fidelity and fidelity-AUC, together with the tree sizes. Then, the quality of the fidelity calibration is investigated, using expected calibration error (ECE), see e.g., [16], and log loss. While these metrics are normally used to evaluate accuracy estimations, we use them to evaluate the *fidelity* estimations. Using black-box rule extraction, the probability estimates from the interpretable models are for the predictions from the opaque model, rather than the true targets, i.e., for fidelity and not accuracy. Consequently, predicted labels from the opaque model are used instead of the true targets in these calculations.

ECE partitions the fidelity estimates into M equally-sized bins and then calculates the actual fidelity of each bin. Let $\text{fid}(B_m)$ be the mean actual fidelity and $\text{est}(B_m)$ the mean fidelity estimate in bin m. Then

$$ECE = \sum_{m=1}^{M} \frac{|B_m|}{n} \Big| \text{fid}(B_m) - \text{est}(B_m) \Big| \tag{4}$$

where n is the number of instances. In the experimentation, the standard choice of 10 bins were used.

While ECE is easily interpretable, it measures only the mean of the predictions falling in a specific bin. With this in mind, the quality of the fidelity calibration was also evaluated using a loss functions that operates on individual

predictions. The *log loss* function is defined as:

$$\lambda_{log} = \begin{cases} -\log p & \text{if } \hat{y} = 1 \\ -\log(1-p) & \text{if } \hat{y} = 0 \end{cases} \tag{5}$$

where log is the natural logarithm, $\hat{y}$ is the prediction from the opaque model and p is the probability estimate for the label 1 from the interpretable model. While the log loss is infinite when an estimate is categorical but wrong, the calculation in scikit learn avoids this by clipping the probability estimates so they are not exactly 0 or 1.

In the next part, we look into using a reject option in rule extraction. Here, two properties are important:

1. The more instances that are rejected, the higher the fidelity of the remaining (explained) instances should be.
2. It should be possible for a user to assess the trade-off between rejecting instances and the fidelity of the explanations. More specifically, the user should have access to accurate fidelity estimations for different rejection levels.

We demonstrate these properties by looking first at individual data sets, and then the overall results.

In the last part, we present a detailed example, showing how rule extraction with reject option can be used to obtain both local and global explanations from an extracted tree model.

The 15 benchmarking data sets used (see Table 1) are all two-class problems, publicly available from either the UCI repository [2] or the PROMISE Software Engineering Repository [30].

Table 1. Data set descriptions

Data set	#inst	#attrib	Source	Data set	#inst	#attrib	Source
colic	328	23	UCI	je4243	363	8	Promise
diabetes	768	9	UCI	kc1	2109	22	Promise
german	1000	21	UCI	kc2	522	22	Promise
haberman	306	4	UCI	liver	345	7	UCI
heartH	270	12	UCI	pc1req	320	9	Promise
heartS	270	14	UCI	spectf	348	45	UCI
hepati	155	20	UCI	transfusion	748	5	UCI
je4042	274	9	Promise				

4 Results

Here, Sect. 4.1 first presents overall fidelity and calibrations results, before results for rule extraction with reject are presented in Sect. 4.2. In Sect. 4.3, visualization

of fidelity estimates in the extracted tree is shown and in Sect. 4.4 examples of local explanations are presented.

4.1 Fidelity and Calibration Results

Table 2 shows the predictive accuracy of the underlying opaque XGB and random forest models (M), and also the accuracy of extracted tree models, without calibration (U) and with Platt scaling (P). Even if overall differences in accuracy are small, random forest models are seen to be most accurate, and this carries over to a small advantage for models extracted from random forests, compared to XGB. Notably, calibration using Platt scaling slightly improves the predictive performance, compared to uncalibrated tree extraction, even though the Platt scaling trees are extracted using a smaller amount of training data.

Table 2. Accuracy for the underlying model (M) and the extracted trees without calibration (U) and with Platt scaling (P)

data set	XGB			RF		
	M	U	P	M	U	P
colic	.826	.815	.812	.831	.822	.814
diabetes	.732	.727	.720	.760	.733	.716
german	.625	.639	.668	.663	.653	.692
haberman	.655	.659	.683	.680	.677	.696
heartH	.777	.769	.763	.816	.778	.768
heartS	.802	.773	.769	.821	.776	.760
hepati	.811	.793	.789	.846	.788	.796
je4042	.719	.707	.709	.729	.718	.707
je4243	.638	.630	.626	.657	.629	.620
kc1	.723	.695	.729	.743	.703	.734
kc2	.773	.754	.764	.785	.761	.768
liver	.692	.652	.643	.704	.652	.644
pc1req	.629	.632	.634	.669	.647	.637
spectf	.796	.767	.780	.805	.775	.791
transfusion	.694	.714	.732	.720	.724	.738
mean	**.726**	**.715**	**.721**	**.749**	**.722**	**.725**

Table 3 shows the fidelity and fidelity-AUC of the extracted models, and the sizes of the extracted models, measured as number of tree nodes.

Here, it is seen that calibration results in a small loss of fidelity, which is more marked for fidelity-AUC. Furthermore, trees extracted from random forests are seen to have higher fidelity than trees extracted from XGB models. Regarding tree sizes, however, models from XGB models have the advantage, on average

Table 3. Fidelity and tree size

data set	Fidelity				Fidelity AUC				Tree size			
	XGB		RF		XGB		RF		XGB		RF	
	U	P	U	P	U	P	U	P	U	P	U	P
colic	.900	.894	.908	.887	.950	.950	.949	.935	30.6	24.6	61.7	49.1
diabetes	.835	.825	.858	.832	.892	.886	.919	.895	81.0	61.1	160.6	122.4
german	.774	.773	.823	.842	.770	.759	.797	.777	133.1	96.1	264.9	191.6
haberman	.815	.815	.850	.845	.857	.831	.878	.854	37.5	27.2	75.0	54.9
heartH	.861	.851	.892	.872	.929	.918	.953	.935	28.0	21.9	55.5	43.8
heartS	.851	.842	.866	.849	.917	.907	.927	.907	27.0	20.2	53.7	40.4
hepati	.866	.858	.879	.878	.883	.879	.888	.873	12.9	9.6	25.6	19.3
je4042	.831	.815	.855	.835	.906	.890	.929	.908	31.4	22.9	61.8	45.5
je4243	.778	.756	.800	.773	.857	.833	.881	.846	48.4	36.2	96.1	72.8
kc1	.814	.831	.854	.877	.818	.806	.864	.826	137.8	104.4	272.2	210.3
kc2	.867	.860	.881	.878	.912	.896	.929	.909	32.2	24.8	64.0	50.2
liver	.777	.755	.791	.784	.838	.814	.851	.835	46.0	33.4	90.5	67.1
pc1req	.794	.769	.815	.801	.874	.847	.894	.873	15.0	9.9	30.3	20.1
spectf	.849	.865	.879	.942	.819	.827	.805	.797	21.4	16.3	42.0	32.8
transfusion	.843	.833	.867	.860	.890	.871	.905	.888	63.0	45.9	125.8	92.4
mean	**.830**	**.823**	**.855**	**.850**	**.874**	**.861**	**.891**	**.871**	**49.7**	**37.0**	**98.6**	**74.2**

being about half the size of trees extracted from random forests. While some trees have a large number of nodes, most would be considered comprehensible, especially when used for local explanations, where number of conditions correspond to the length of the path from the root to the particular leaf. Finally, calibrated trees are smaller than uncalibrated trees, which is to be expected since these are built using fewer instances.

Turning to calibration, Table 4 shows the fidelity ECE and log loss, as described above. Starting with the expected calibration error (ECE), it is seen that calibration using Platt scaling consistently and significantly improves fidelity estimates. Indeed, for all 15 data sets, and four setups evaluated, there is only one occasion where a calibrated models has a worse fidelity estimation than the uncalibrated model (model extracted from random forest for the *pc1req* data set).

In summary, then, it is seen that calibration with Platt scaling for rule extraction improves predictive accuracy, fidelity estimates and tree size, but yields slightly worse actual fidelity.

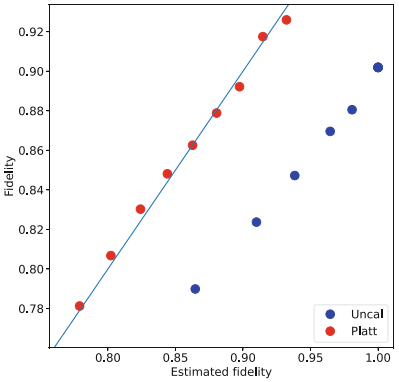
4.2 Rule Extraction with Reject Results

Before presenting aggregated results for rule extraction with rejection, we look at a few individual data sets. Starting with the liver data set in Fig. 1, where

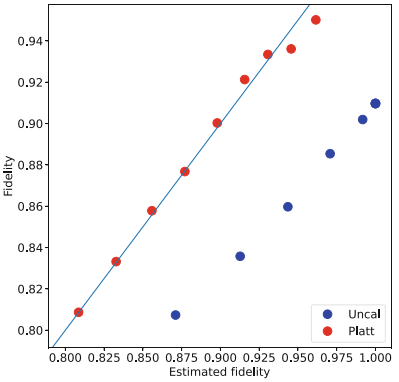
Table 4. Calibration

data set	ECE				Log loss			
	XGB		RF		XGB		RF	
	U	P	U	P	U	P	U	P
colic	.040	.022	.061	.027	.797	.264	.956	.290
diabetes	.061	.029	.060	.023	1.536	.397	1.061	.370
german	.084	.020	.109	.017	1.271	.473	.821	.373
haberman	.046	.026	.068	.020	.819	.413	.623	.350
heartH	.035	.023	.037	.022	.905	.348	.616	.304
heartS	.053	.026	.055	.023	1.375	.382	1.258	.377
hepati	.054	.043	.067	.027	1.249	.324	1.056	.290
je4042	.051	.023	.035	.022	1.215	.417	.875	.386
je4243	.067	.021	.051	.021	1.535	.501	1.243	.479
kc1	.097	.011	.100	.012	1.608	.393	.936	.312
kc2	.057	.023	.058	.019	1.129	.329	.856	.295
liver	.081	.011	.098	.015	1.868	.510	1.833	.468
pc1req	.035	.030	.036	.040	.938	.488	.765	.455
spectf	.093	.027	.120	.016	2.173	.333	1.632	.186
transfusion	.032	.020	.047	.015	.686	.370	.544	.324
mean	**.059**	**.024**	**.067**	**.021**	**1.274**	**.396**	**1.005**	**.351**

the points represent rejecting $\{10\%, 20\%, \ldots, 90\%\}$ of the instances, it is clear that the fidelity estimates are much better when using Platt scaling. In fact, not using calibration leads to a very overconfident model. Another observation is that actual fidelity does increase with higher rejection proportions, i.e. rejecting instances based on the fidelity estimates does work in the intended way. However, only when using well-calibrated fidelity estimates are all different rejection levels separable from one another, as seen by the nine distinct red dots for Platt scaling models, compared to overlapping blue dots for the uncalibrated models. So, with the reject option, a user can easily find a good balance between the number of explanations available, and their usability, i.e., how often the extracted model concurs with the opaque. Finally, it could be noted that the fidelity is generally higher when extracting from the random forest compared to XGB, which of course is in accordance with the overall results presented above.



(a) Rule extraction from XGB

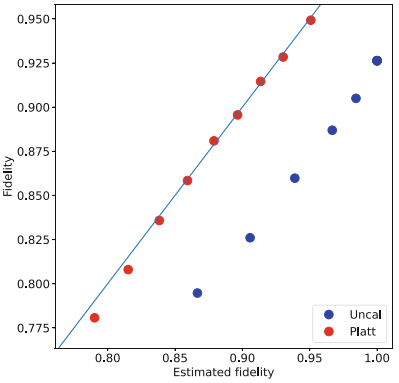


(b) Rule extraction from RF

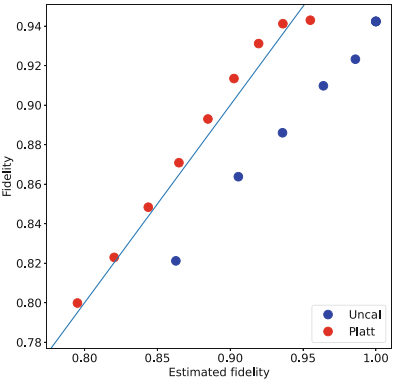
Fig. 1. liver data set. Rejecting 10% - 90% of test instances

Results for the je4243 data set, in Fig. 2, are very similar to the liver data set. The main difference is that the fidelity levels here are generally higher, which should be expected since this is an easier data set. Individual fidelity estimates are very good, especially when extracting from XGB, where calibration is almost perfect.

For the diabetes data set, see Fig. 3, the calibration is slightly worse. When rejecting more than half of the instances, the fidelity estimations are up to 2.5 percentage points conservative. Still, for most applications, this would probably be acceptable, and it could be argued that being underconfident is better than being overconfident.

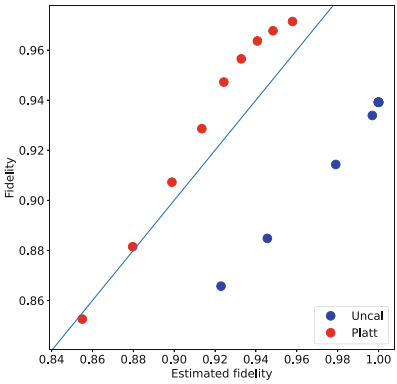


(a) Rule extraction from XGB

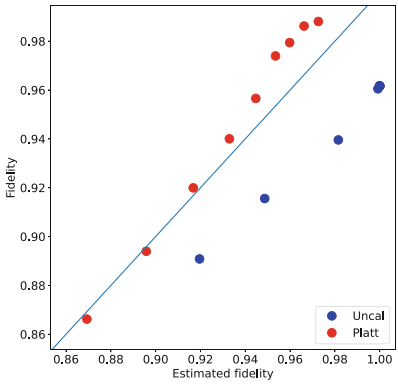


(b) Rule extraction from RF

Fig. 2. je4243 data set. Rejecting 10%–90% of test instances



(a) Rule extraction from XGB



(b) Rule extraction from RF

Fig. 3. diabetes data set. Rejecting 10%–90% of test instances

We now present detailed fidelity estimation results for three different rejection levels. Starting with rejecting 20%, the fidelity on most data sets is approximately 0.9, see Table 5. So, if rejecting 20% of the test instances based on that they fall in parts of feature space where the fidelity is relatively low, the method would provide explanations for nine out of ten non-rejected instances. Regarding the quality of the estimates, it is again obvious that the uncalibrated trees are very overconfident. When using Platt scaling, however, the estimates are most often accurate; for most data sets, the error is less than one percentage point.

If 50% of all instances are rejected outright, based on the fidelity estimates, the fidelity of the remaining instances is on average 0.94, see Table 6. For many data sets, the fidelity is actually over 0.95, i.e., explanations would be available for nineteen out of twenty non-rejected test instances. At this rejection level, using uncalibrated trees will lead to fidelity estimates of 1.0 on several data sets, i.e., fewer than 50% of the instances are actually rejected. Platt scaling is, on the other hand, able to deliver accurate fidelity estimations.

Using a rejection rate of 80%, see Table 7, the uncalibrated fidelity estimations are 1.0 on every data set. Calibrated estimations are, again, very accurate. Extracting from XGB, the average fidelity of non-rejected instances is 0.96, but it is even higher (0.97) from random forests. So, using this rejection level, very few instances would not obtain an explanation based on the fact that they disagree with the opaque model.

Tables 8 and 9 summarize the rule extraction with reject experiment by showing averaged results over all data sets. Starting with signed errors, we see that using no calibration will result in overconfident fidelity estimates. Extracting from XGB, the fidelity estimates are approximately 0.05 too high, while the corresponding result when extracting from random forests is between 0.02 and 0.04. When using Platt scaling, the estimates are generally very well-calibrated, although there is a slight tendency to be underconfident. Turning to absolute errors, the uncalibrated alternatives, of course, have the same values as for the

Table 5. Fidelity estimation 20% reject.

data set	XGB				RF			
	U		P		U		P	
	Est.	Fid	Est.	Fid	Est.	Fid	Est.	Fid
colic	.968	.949	.938	.956	.968	.951	.932	.945
diabetes	.946	.885	.880	.881	.949	.916	.896	.894
german	.869	.815	.839	.830	.882	.870	.904	.897
haberman	.905	.879	.868	.869	.906	.909	.913	.907
heartH	.949	.919	.905	.909	.951	.948	.916	.932
heartS	.963	.911	.881	.897	.965	.919	.896	.897
hepati	.973	.926	.918	.926	.970	.939	.934	.943
je4042	.933	.890	.865	.870	.933	.909	.877	.893
je4243	.906	.826	.815	.808	.906	.864	.820	.823
kc1	.941	.873	.893	.886	.943	.912	.931	.930
kc2	.980	.936	.920	.920	.977	.950	.927	.938
liver	.910	.824	.802	.807	.913	.836	.833	.833
pc1req	.879	.845	.798	.826	.872	.883	.819	.849
spectf	.985	.906	.920	.924	.987	.927	.966	.975
transfusion	.922	.906	.896	.896	.923	.928	.920	.920
mean	**.935**	**.886**	**.876**	**.880**	**.936**	**.911**	**.899**	**.905**

signed errors. When using calibration, however, we see that the average absolute error is less than 0.01 for almost all rejection levels. The rightmost column in each table shows another advantage of using calibration; with more granular estimates, both fidelity and (interestingly enough) accuracy increase when more instances are rejected. Without calibration, as described above, many instances are predicted to have perfect fidelity.

4.3 Trees as Global Explanations

In this section, visualization of the fidelity estimation trees is presented and discussed, through a sample tree from the diabetes data set. In this data set, the task is to, based on diagnostic measurements, predict whether a patient has diabetes. Regarding the selection of patients, all are females at least 21 years old, and of Pima Indian heritage. The input attributes are: *pregnancies* (number of times pregnant), *glucose* (plasma glucose concentration), *bloodPressure* (diastolic blood pressure), *skinThickness* (triceps skin fold thickness), *insulin* (2-hour serum insulin), *BMI* (body mass index), *dPF* (diabetes pedigree function) and *age*.

It should be noted that in order to make the trees small enough to present in their entirety, the minimum number of instances in each leaf was for this demonstration set to 25, i.e., (min_samples_leaf=25). As seen above in Table 3, trees

Table 6. Fidelity estimation 50% reject.

data set	XGB				RF			
	U		P		U		P	
	Est.	Fid	Est.	Fid	Est.	Fid	Est.	Fid
colic	1.000	.974	.968	.974	1.000	.967	.968	.965
diabetes	1.000	.939	.924	.947	1.000	.962	.945	.957
german	.959	.888	.897	.898	.962	.928	.946	.945
haberman	.978	.949	.923	.935	.979	.958	.959	.964
heartH	1.000	.969	.948	.964	1.000	.981	.961	.974
heartS	1.000	.948	.925	.945	1.000	.954	.937	.939
hepati	1.000	.956	.952	.974	1.000	.964	.963	.976
je4042	1.000	.950	.921	.924	1.000	.968	.930	.948
je4243	.984	.905	.879	.881	.986	.923	.885	.893
kc1	1.000	.934	.935	.942	1.000	.966	.964	.970
kc2	1.000	.962	.957	.968	1.000	.972	.964	.979
liver	.981	.881	.863	.863	.992	.902	.898	.900
pc1req	.953	.914	.867	.885	.945	.931	.888	.909
spectf	1.000	.923	.948	.956	1.000	.944	.977	.985
transfusion	.994	.970	.945	.959	.993	.981	.964	.972
mean	**.990**	**.937**	**.923**	**.934**	**.990**	**.953**	**.943**	**.952**

Table 7. Fidelity estimation 80% reject.

data set	XGB				RF			
	U		P		U		P	
	Est.	Fid	Est.	Fid	Est.	Fid	Est.	Fid
colic	1.000	.974	.985	.984	1.000	.967	.986	.977
diabetes	1.000	.939	.948	.968	1.000	.962	.966	.986
german	1.000	.928	.933	.945	1.000	.963	.968	.977
haberman	1.000	.967	.958	.961	1.000	.979	.980	.985
heartH	1.000	.969	.971	.974	1.000	.981	.981	.986
heartS	1.000	.948	.957	.952	1.000	.954	.969	.953
hepati	1.000	.956	.966	.979	1.000	.964	.977	.980
je4042	1.000	.950	.960	.963	1.000	.968	.965	.966
je4243	1.000	.926	.930	.928	1.000	.942	.936	.941
kc1	1.000	.934	.954	.961	1.000	.966	.977	.978
kc2	1.000	.962	.974	.972	1.000	.972	.980	.982
liver	1.000	.902	.915	.917	1.000	.910	.946	.936
pc1req	1.000	.952	.930	.931	1.000	.960	.948	.941
spectf	1.000	.923	.966	.963	1.000	.944	.986	.985
transfusion	1.000	.977	.968	.982	1.000	.984	.981	.991
mean	**1.000**	**.947**	**.954**	**.959**	**1.000**	**.961**	**.970**	**.971**

Table 8. Summary showing signed errors, absolute errors, fidelities, and accuracies averaged over all data sets for each rejection level.

Rej.	XGB							
	U				P			
	sign.	abs.	fid	acc	sign.	abs.	fid	acc
10%	.046	.046	.856	.734	−.002	.007	.853	.743
20%	.049	.049	.886	.754	−.005	.008	.880	.763
30%	.052	.052	.910	.772	−.008	.009	.903	.780
40%	.052	.052	.928	.786	−.010	.010	.920	.793
50%	.052	.052	.937	.793	−.011	.011	.934	.804
60%	.053	.053	.944	.799	−.011	.011	.945	.812
70%	.053	.053	.947	.800	−.007	.008	.952	.817
80%	.053	.053	.947	.800	−.004	.006	.959	.820
90%	.053	.053	.947	.800	.003	.008	.963	.821

Table 9. Summary showing signed errors, absolute errors, fidelities, and accuracies averaged over all data sets for each rejection level.

Rej.	RF							
	U				P			
	sign.	abs.	fid	acc	sign.	abs.	fid	acc
10%	.021	.024	.884	.741	−.005	.008	.880	.745
20%	.025	.028	.911	.759	−.006	.008	.905	.763
30%	.032	.032	.930	.776	−.007	.008	.925	.779
40%	.035	.035	.946	.788	−.008	.009	.940	.791
50%	.037	.037	.953	.793	−.008	.009	.952	.802
60%	.038	.038	.959	.799	−.007	.010	.960	.811
70%	.038	.038	.961	.800	−.005	.007	.967	.816
80%	.039	.039	.961	.799	−.001	.007	.971	.819
90%	.039	.039	.961	.799	.004	.009	.974	.820

extracted from XGB are on average about twice as big, and from random forest about four times the size, measured as number of nodes, translating to a few extra levels in the tree.

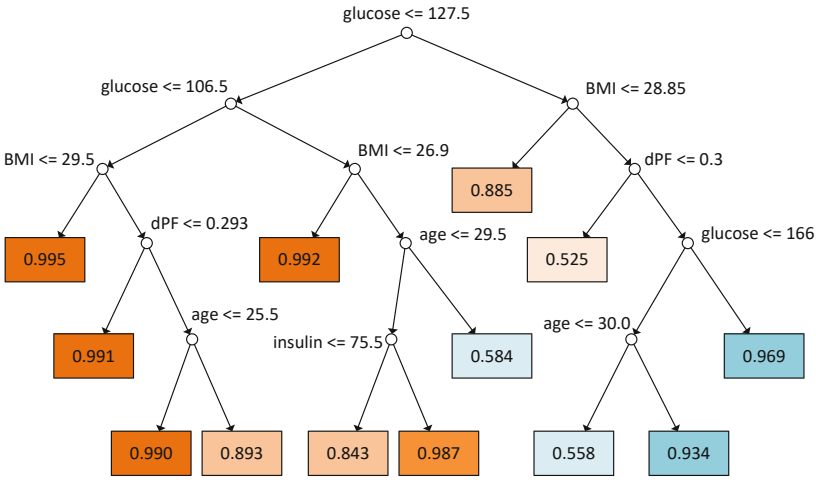


Fig. 4. Visualization of fidelity estimation. Diabetes data set, tree extracted from random forest.

Figure 4 shows an extracted tree model with calibrated fidelity estimates. The colors in the leaf nodes indicate the predicted class in the extracted model, with orange used for class 0 (no diabetes prediction) and blue for class 1 (diabetes). The color intensity is used to indicate the estimated fidelity, which is also given as a numerical value in each leaf.

Viewing this tree as a global explanation, a few things are immediately obvious. First of all, the left sub-tree mostly predicts class 0, while the right sub-tree mostly predicts class 1. It can also be seen that most leaves have very high fidelity estimates, so that the overall fidelity of 0.851 of this extracted tree is mostly due to a few leaves with very low fidelity estimates, where the tree is not a good approximation of the random forest model. Furthermore, the prediction regions that these leaves represent are easily described using the split conditions in the path to the leaf, providing an explanation of the fidelity in different parts of the feature space. In this particular example, we thus have three distinct regions where the extracted model cannot provide reliable explanations of the random forest, and split conditions on the paths to these leaves yields a clear description of this part of the feature space. For example, the leaf with fidelity estimate 0.584 and a class 1 prediction (i.e. patient has diabetes) represents instances with a moderate glucose level (between 107 and 127.5), a reasonably high BMI (above 26.9) and aged above 29.5 years. In this region, the extracted tree predicts that the opaque model would give a diabetes diagnosis, but the low fidelity value in the leaf indicates that explanations of instances falling here are very uncertain. Interestingly, when inspecting the predictive performance of the underlying random forest model, these three leaves correspond to regions with low predictive performance.

Figure 5 shows the same tree at a 25% reject level, meaning that the tree should withhold explanations for the 25% of instances with the lowest fidelity estimates. Here we see that the three leaves with the low fidelity estimates are indeed rejected, leaving 75.2% of instances. In the remaining leaves, fidelity levels are high and thus very few instances reaching these will have disagreement in predicted class between the tree model and the underlying random forest model.

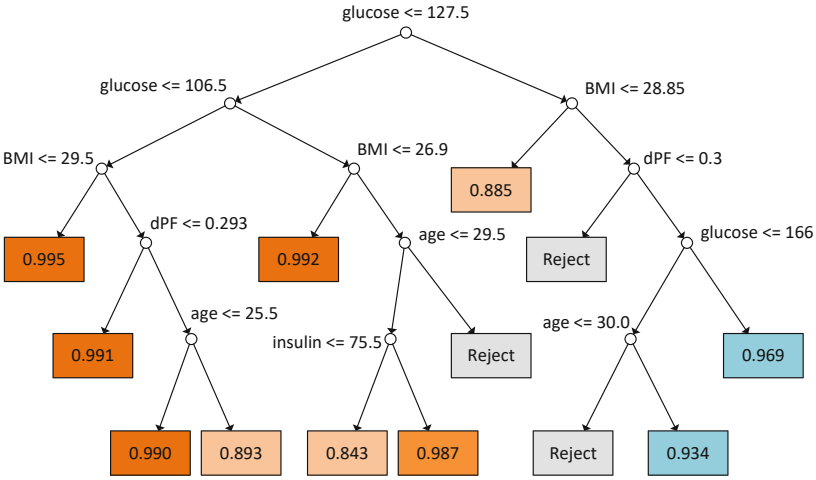


Fig. 5. Visualization of fidelity estimation. Diabetes, 25% reject.

At the 50% reject level, shown in Fig. 6, all remaining leaves have fidelity estimates of at least 93.4%, meaning that they will provide explanations for almost all instances falling there, since agreement with the underlying random forest model is very high in these prediction regions. The leaves for which predictions are rejected are distributed throughout the tree, but it is worth noting that all remaining explanations that are now given for class 0 are in the left sub-tree, and for class 1, all are in the right sub-tree.

At the 75% reject level, finally, shown in Fig. 7, we see that all remaining leaves have an estimated fidelity of over 99% and predict class 0. This means that very few instances (less than one in a hundred) falling into these leaves will have disagreement between the opaque and transparent model predictions. Thus, in these regions, the tree model is an almost perfect description of the opaque model.

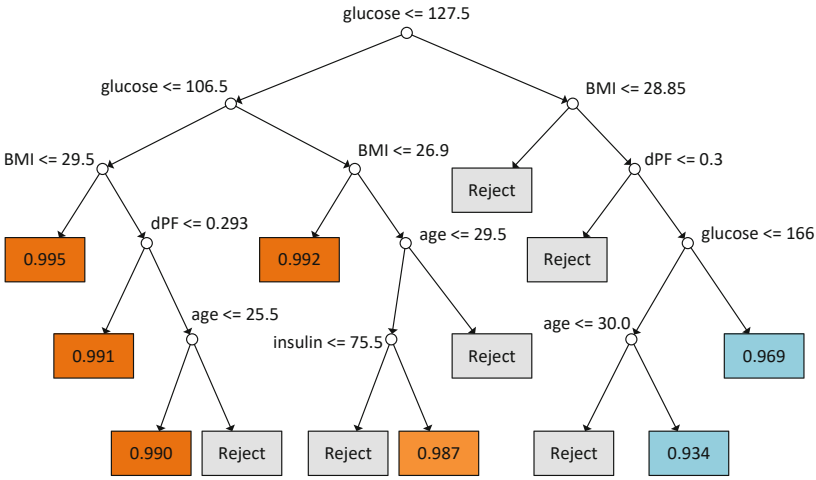


Fig. 6. Visualization of fidelity estimation. Diabetes, 50% reject.

4.4 Local Explanations

We finally present some concrete examples of the suggested method in use when the task is to not only give global explanations for inspection and analysis, but also to provide local explanations. Referring back to the characteristics of the proposed method, there are three different cases, at the local (instance) level, for a particular reject proportion:

1. The instance falls into a leaf where explanations are rejected, since the tree is a weak approximation of the opaque model in this region. Thus, regardless of whether the tree agrees with the opaque model's prediction on this instance, no local explanation is provided.
2. The instance falls into a leaf where explanations are allowed, i.e., not rejected, since the tree is deemed to be sufficiently strong in describing the opaque model here. Also, the tree agrees with the opaque model on this instance, so an explanation is provided. It should be noted that this will be the case in the overwhelming majority of non-rejected leaves, since remaining leaves will have high fidelity.
3. The instance falls into a leaf where explanations are allowed, but the tree disagrees with the opaque model on this instance. Therefore, no explanation is provided.

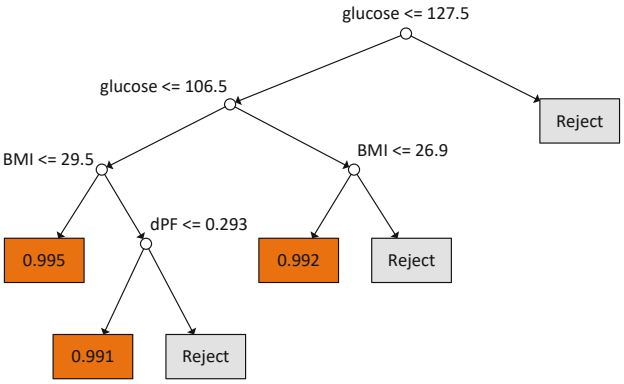


Fig. 7. Visualization of fidelity estimation. Diabetes, 75% reject.

The following three examples illustrate these cases, using the same extracted tree as above for the *diabetes* data set with rejection level 50%, shown again in Fig. 8.

In the first instance, the patient has a glucose of 118, a BMI of 45.8 and is 31 years old, ending up in the reject leaf marked in red and numbered 1 in Fig. 8. This makes sense since this node was the only one in the right part of the tree predicting diabetes, but with low fidelity, in the full tree, see Fig. 4.

In the second instance, the patient has a high glucose of 173, a BMI of 46.5 and a dPF of 1.15 ending up in the leaf marked in red and numbered 2 in Fig. 8. This is one of very few instances falling in that leaf where the extracted model disagrees with the opaque model. More specifically, the more powerful and complex random forest predicts this patient as non-diabetic (which turns out to be correct), so no local explanation based on the extracted tree is given.

The third instance, a patient with a glucose of 179, a BMI of 32.7 and a dPF of 0.72 falls in the same leaf. For this patient, however, the random forest predicts that she is diabetic (which in fact is correct), i.e., the two models agree. Consequently, a local explanation is provided. There are of course several different ways to give a local explanation, but with a tree as the extracted model, a straightforward option would be to present the decision path as a rule, see Fig. 9 below.

While it may initially appear counter-intuitive that the two last instances are handled in this way, it is actually a very good example of how the method is supposed to work. Considering only the attributes present in the tree, it is really hard to understand that the third example is diabetic but the second (with significantly higher BMI and dPF) is not. However, the two instances differ radically on other attributes, most notably the woman in the second example has no children, but the woman in the third example has eight children. So, the second example is the exception, one of only three percent of all instances falling in this leaf where the random forest doesn't predict diabetes.

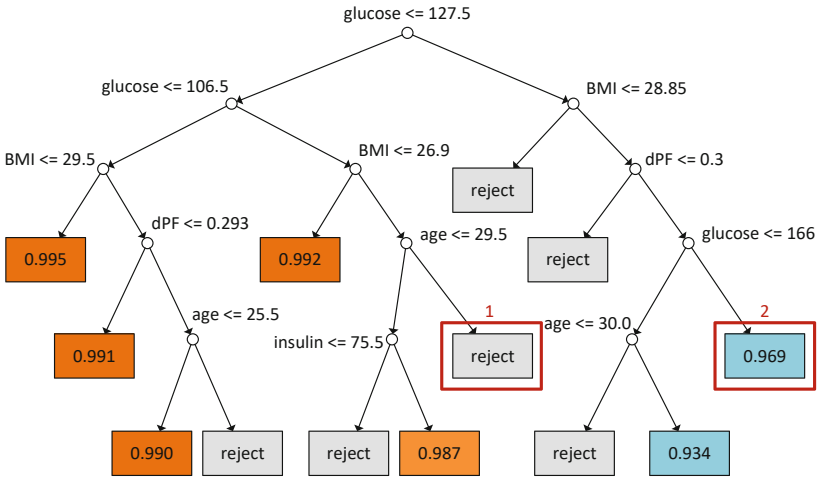


Fig. 8. Leaves for sample instances. Diabetes, 50% reject.

$$
\begin{aligned}
& \text{glucose} > 166 \\
\& \ & \text{BMI} > 0.2885 \\
\& \ & \text{dPF} > 0.3 \\
\rightarrow \ & \textbf{Diabetic}
\end{aligned}
$$

Fig. 9. Local explanation

5 Concluding Remarks

Rule extraction is a strong tool for providing both global and local explanations. One inherent problem is, however, how to handle instances where the extracted model disagrees with the opaque model. In this paper, we have suggested and evaluated the inclusion of a reject option in rule extraction. The method, which depends critically on having well-calibrated fitness estimations available, will reject a (user-supplied) proportion of all test instances, based on their fidelity estimates. On the global level, the extracted models can be inspected and analyzed. Specifically, the models provide a very good understanding of in which parts of feature space the extracted model is a good approximation of the opaque and not. Using the reject option, extracted models can be visualized for different rejection levels, thus making it straightforward for a user to know when the extracted model can be trusted, but also how often local explanations of instances where the extracted model agrees with the opaque model can be provided. In this study, standard decision trees were used as rule extractors, so in practice, instances falling in certain leaves would be rejected, while we would expect the extracted model to agree with the opaque on most instances falling in the remaining leaves, i.e., local explanations can be generated for them.

In the empirical investigation, we showed that the extracted models were often both highly faithful and comprehensible. Most importantly, it was demonstrated how well-calibrated fidelity estimates could be obtained and then leveraged as the basis for a reject option, leading to high-quality local and global explanations.

For future work, the method should be applied to larger and harder data sets, as well as more complex opaque models, e.g., deep neural networks. Obviously, the use of standard decision trees could be replaced by any another technique producing interpretable models, or a dedicated rule extraction algorithm, which would result in global and local explanations of different formats. Finally, more advanced calibration techniques, e.g., Venn-Abers [33], should be evaluated as alternatives to Platt scaling.

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Poisoning Attacks Against Security-Aware Federated Recommendation System

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Abstract. As a decentralized machine learning technique, federated recommendation enables cross-platform or cross-device collaborative model training without compromising data privacy. Previous works have indicated that such a privacy-preserving learning framework is more easily affected by poisoning attacks from malicious users or data. However, many papers that demonstrate significant poisoning attack effectiveness often use simply vanilla federated recommendation as victim models without any protection mechanisms. In this paper, a security-aware federated recommendation system is implemented with a stochastic sampling strategy on the client side and robust estimation on the server side. Then, an improved poisoning attack algorithm is proposed to overcome these defense mechanisms and once again aim to enhance attack effectiveness. In the proposed algorithm, these malicious users primarily utilize intercepted gradients and intelligent sampling techniques to estimate approximate user embeddings, and then develop more threatening attack strategies based on this information. Our experiments, conducted on MovieLens datasets, focus on exposure-increasing attacks targeting items with varying levels of popularity. Regarding attacks that have been mitigated or even prevented by security-aware federated recommendation systems, the experimental results show that the proposed poisoning attack methods can further increase effectiveness by up to 86%.

Keywords: Federated Recommendation · Poisoning Attack · Privacy-Preserving · Intelligent Sampling

1 Introduction

With the rapid growth of artificial intelligence and e-commerce applications, research on precision marketing and recommendation systems [8,14,21] has begun to be constrained by the diversity and coverage of data from a single

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platform. Further considering the need for privacy preservation, federated recommendation employs federated learning algorithms to achieve cross-platform and cross-device model training without sharing user private information. Several previous studies have indicated that this type of distributed learning system is more vulnerable to various malicious attacks, such as shilling attacks and poisoning attacks [23]. However, the victim federated recommendation systems in existing successful attack cases are typically designed very simply and lack awareness of information security protections. In this paper, we design an advanced poisoning attack method that bypasses defense mechanisms, targeting novel security-aware federated recommendation systems. This kind of research goal can effectively encourage subsequent research in developing more secure and robust federated recommendation systems in the field of artificial intelligence and information security.

Federated recommendation [13,19,20] aims to provide customized recommendations across multiple platforms or institutions without sharing each other's private training data. They typically achieve collaborative machine learning by exchanging model parameters or sharing user or item embeddings. With the rapid development of trustworthy AI, poisoning attacks against recommendation [17,27,29] systems have also gained prominence and have become a hot research topic, requiring multidisciplinary knowledge in artificial intelligence and information security. Such research aims to inject a small amount of erroneous or perturbed information into the training data, causing significant harm to the model's training process. Common attack objectives include degrading system performance and increasing the exposure rate of specific items. In recent years, some new research on poisoning attacks against federated recommendation systems [28,30] has claimed that federated learning makes the model training process more vulnerable to attacks due to its privacy protection characteristics. However, the victim model in successful attack cases is often very naive and does not account for security issues or the presence of malicious users. Developing effective poisoning attack strategies against secure-aware federated recommendation systems remains a highly challenging research topic.

To address the vulnerability of the FR, we have observed that the gradients returned by real users are critical factors that affect the overall computation and security of the FR. Therefore, we first implement a naive shilling attack on FR with a gradient descent mechanism to promote a certain item. Malicious gradients perturb the global item information and guide benign users in raising the ranking of a target item. Next, because only a few studies focus on the specifically designed defense mechanisms against the poisoning attack, we suppose that the item gradients sent by benign users have eavesdropped, and two information security enhancement methods for FR have been incorporated into the client and the server, respectively. On the client side, we applied a stochastic sampling strategy that creates the uncertainty of the item gradients that benign users will pass. On the other hand, a robust estimation of item gradients will be applied on the server side to alleviate the influence of the attackers. In addition, we believe that even if we design FR with strong information protection capabilities, there may

still be areas that can be exploited. Hence, for the attackers, we have proposed a more advanced algorithm specifically based on intercepted gradients using an intelligent sampling method, which aims to bypass the aforementioned defense mechanisms and achieve better attack effectiveness as much as possible. The key contributions of our work are summarized below:

- A stochastic method and robust estimation have been incorporated in FR, which can effectively protect the federated recommender system and mitigate the attackers' influence.
- An intelligent sampling technique applying intercepted gradients is proposed to approximate the real user information to attain the poisoning attack.
- The poisoning attack we proposed can effectively bypass the filtering and detection mechanisms of security-aware FR, achieving successful attack results. Through experiments, it can be proven that our attack effectiveness has shown significant improvement compared to other baselines.

Our study can provide more comprehensive ideas in the research field related to poisoning attacks against federated recommender systems, Whether it is focusing on the objective of poisoning attacks or designing a more security-aware federated recommendation system. Our research can also be extended to the protection mechanism of FR by utilizing more sophisticated algorithms to enhance the privacy-preserving capability of FR in terms of personal recommendations, global parameters for client-server communication, or the aggregation method in the server.

2 Related Work

In this section, we review the existing federated recommender system methods and poisoning attack.

2.1 Federated Recommendation System

In order to provide users with a better user experience, many companies adopt high-accuracy recommendation systems. Recent studies focus on recommendation systems for improving the recommendation accuracy via session [7,18], negative information [24], or even conversation [25]. Other studies focus on solving the problems that may occur to recommender systems in the real-world scenario. Hung et al. [9] adopt few-shot learning to tackle the cold start problem that usually leads to poor recommendation results while facing a new user. Counterfactual Explainable Fairness framework (CEF) [4] is proposed to gain insight into the recommender system to confirm whether the model is fair or not, avoiding the recommendations exist severe biases or deviations.

With the growing demand for privacy protection, most of the common recommender systems are being challenged because they use almost all user information to maximize RS performance. Although some research proposes that the

black-box attack method allows the attacker to bypass direct access to user-item interactions, it still assumes that the data is stored in a central database, which would violate many modern privacy policies. Hence, the research field of recommender systems in federated learning scenarios, known as federated recommender systems (FR), has been proposed as a decentralized approach to meet social needs. In a federated recommender system framework, the server passes the shared information, such as item embedding and global model parameters, to the local users. Users have their own user-item information and a trainable model, which will be trained via global item embeddings. After updating the local model, the item embedding gradient is generated and will be passed to the server. Once the server has collected all the item embedding gradients shared by the users, it aggregates all the gradients and passes the updated item embeddings to the users for the next training process. M. Ammad-Ud-Din [1] proposed a CF-based FR structure to show that even if we can only visit a single user at a time and update the local model, it is possible to recommend items by updating the shared item embedding and parameters. In the latest studies, ReFRS [10] applied variational auto-encoder and self-supervised learning on the client and server side, respectively, to achieve high performance. F2MF [15] focuses on the currently popular issue, fairness in the recommendation, and applies differential privacy techniques to disguise the real user information.

2.2 Poisoning Attack in Recommendation System

A poisoning attack attempts to inject a small amount of perturbation into the training data, thereby influencing the model's performance. The goal of poisoning attacks on recommender systems can be divided into two categories: untargeted attacks and targeted attacks. An untargeted attack aims to degrade the system's performance, making it less useful to users and discouraging them from using the recommender system. On the other hand, a targeted attack aims to increase the exposure rate of specific items, encouraging users to be more inclined to purchase those items. Yuan, Zhihai et al. [26] proposed a multi-target attack by constructing an undirected user-user graph and an unsupervised method. Cao et al. [2] described a hybrid shilling attack that can apply multiple attack methods to a recommender system simultaneously and is hard to expose. Wilson et al. [22] find out the power user that can provide extra influence to the recommender system training process compared with any other participants. Hence, they proposed a power user attack using the impact of power users to conduct a shilling attack.

Due to the high dependence on item embedding, FR is still vulnerable to attacks, especially from poisoning attacks. PipAttack [31] designs an explicit promotion by using popularity bias to make the target item a popular item, which can cause the FR to tend to recommend the target item. FedRecAttack [17] attacks FR with public interaction for higher exposure rate and hit rate (HR) to keep the attack procedure undetected. Rong et al. [16] approximate genuine users' embedding vectors by using Gaussian distribution, optimize the vectors via gradient descent, and attack hard sample users found by hard user

mining method. Wei et al. [30] proposed a poisoning attack without using any prior knowledge, unlike most of the shilling attacks on federated recommender systems that require public information or the user's profile. In addition to targeted poisoning attacks against FR, there are also studies discussing untargeted attacks, which use shilling attacks to degrade the victim model's performance. Yu et al. [29] applied adaptive clustering on item embedding and let the FR generate a similar score for the same cluster to perform an untargeted attack. Yi et al. [27] proposed an untargeted attack by using a news similarity perturbation technique to affect the federated recommender system to lower performance.

However, there are few studies on the federated recommendation in the scenario of private information begin eavesdropped, which required a strong defense mechanism to make up for this deficiency. Even though FR takes this situation into consideration, we still believe that there are vulnerabilities that can be exploited. Therefore, we not only use stochastic mechanisms and robust estimation to build a strong privacy-preserving FR but also design intelligent attack methods to enhance the effectiveness of attackers.

3 Proposed Method

In this section, both defense and poison-attacking techniques will be introduced. The system overview is illustrated in Fig. 1. A naive shilling attack based on the gradient descent mechanism with red lines is implemented to promote a certain item to show the vulnerability of FR. Next, in the scenario of the private information of users being eavesdropped, we incorporate a stochastic algorithm on the client side to create the uncertainty of real interaction. We also apply a robust estimation on the server side to mitigate the influence of the attackers. The protection method is depicted in Fig. 1 with blue lines. Then, we propose an intelligent sampling mechanism on the intercepted gradients eavesdropped from the FR's training process to approximate the real users' information. The process is shown in green lines. With our proposed method, we not only demonstrate the vulnerability of federated recommender systems (FR) to shilling and poisoning attacks but also offer protection strategies to enhance privacy-preserving capabilities, addressing current gaps in FR research.

3.1 Defend Mechanism

Motivated by [3,12], a stochastic sampling strategy and a robust estimation mechanism are applied to construct a security-aware federated recommendation system. In the scenario that benign item gradients have been eavesdropped, we have to prevent referencing the item indices from the intercepted gradient from attackers. The intuitive way is to randomly select different proportions of positive and negative items to generate the corresponding item gradients and pass them to the server. However, such random selection may lead to unstable information to the global item embedding update. Hence, for each user u_i, instead of randomly sampling the original V_i^+ and V_i^-, a concept called stochastic has

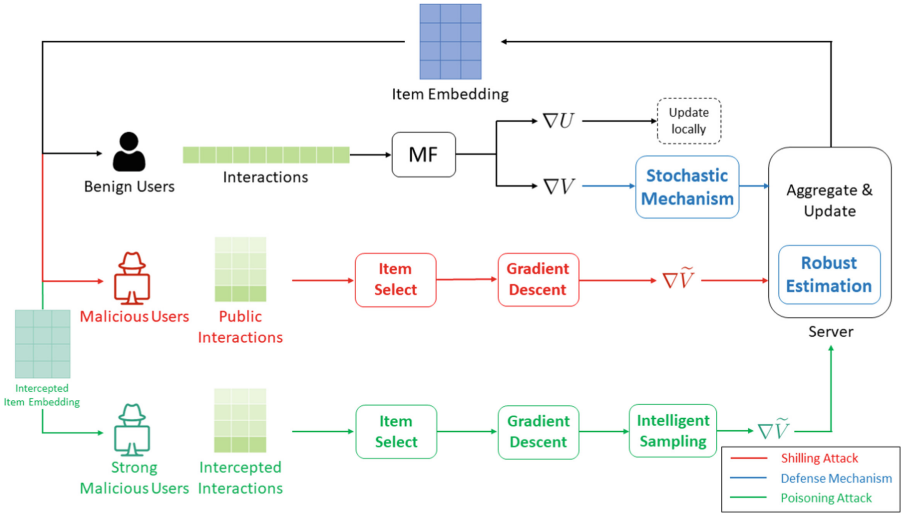

Fig. 1. System Overview

been incorporated. Normally, $|V_i^-| = |V_i^+|$, but now we make $|V_i^-|$ double by randomly sampling negative items 2 times more than normal to create a pool. By doing so, we not only avoid information loss from the server but also achieve the purpose of confusing the attackers.

We have provided a stochastic method on the client side to disguise the real interactions of benign users, but mitigating the attacker's influence is also crucial to the FR to directly protect the FR. While aggregating the item gradients passed by all users, for each item, we calculate each item gradient's cosine similarity with others. For example, if there are 50 gradients for an item, we will calculate each gradient to the other 49 gradients' cosine similarity. After acquiring the similarities, we summarize the value of every gradient and drop the specific proportion of them that have the least similarity.

3.2 Attack Against Security-Aware Federated Recommendation System

We have incorporated two privacy-preserving enhancement strategies for the federated recommender system. In this section, we will introduce some attack techniques with intelligent sampling against the privacy-preserving enhancement methods with intercepted gradients.

Attack with Intercepted Item Index. Since benign users pass $\nabla \mathbf{V}^{t-1}$ to the server with both $\nabla \mathbf{V}^+$ and $\nabla \mathbf{V}^-$ included, we cannot directly infer the true item interactions of the users, especially when the item gradient is protected by our stochastic mechanism. However, $\nabla \mathbf{V}^{t-1}$ still contains some hidden information,

which can let us improve our approximation on $\mathbf{U}^{t-1}$ without violating the mechanism of the FR framework.

After the malicious center derives $\nabla\mathbf{V}^{t-1}$ from the server, for each user, we first calculate the whole items' prediction. This procedure aims to acquire the items' ranking to certain users. Next, the prediction list of items will be sorted in descending order and select top-$|\nabla V_i^{t-1}|/2$ items. Since we still can make use of our public interactions, if there exist any of the public interactions that are not in the top-$|\nabla V_i^{t-1}|/2$ items, we will replace the lowest prediction score of the item with them. By doing so, we not only approximate the user interaction via intercepted gradient index but also make full use of known accurate information. Since the higher prediction of a user to an item indicates that it has a higher opportunity to be a positive item for the user compared with the negative item, by considering the top-$|\nabla V_i^{t-1}|/2$ items as the positive item for all users, we extend the public user interaction with the intercepted item gradient. Even if there are several items that are mistakenly selected, we still obtain more comprehensive benign user information. Finally, we can replace $\widetilde{V}^{+}$ with $\widetilde{V}^{t+}$ at the beginning of our attack process to provide malicious user interaction with more information. Top-$|\nabla V_i^{t-1}|/2$ can replace any number to meet the attack requirement. It is worth noting that the amount to select should be carefully considered. The low amount could lead to a lack of information for generating malicious item gradient while exceeding the amount could cause misleading.

Algorithm 1. Attack with Intercepted Gradient Referenced

Input: $\nabla\widetilde{\mathbf{V}}^{t-1}$, $\nabla\mathbf{V}^{t-1}$ where $|\nabla\widetilde{\mathbf{V}}^{t-1}| = 2M$
Initialize: *Counter*, *dropList* as empty arrays
for each $\nabla\widetilde{\mathbf{v}}_i^{\,t-1} \in \nabla\widetilde{\mathbf{V}}^{t-1}$ **do**
  $\nabla\mathbf{V}_{tmp}^{t-1} = \{\nabla\mathbf{v}_{i1}^{t-1}, \nabla\mathbf{v}_{i2}^{t-1}, ..., \nabla\mathbf{v}_{ij}^{t-1}, \nabla\widetilde{\mathbf{v}}_{i1}^{t-1}, ..., \nabla\widetilde{\mathbf{v}}_{ik}^{t-1}\}$, where $j = |\nabla\mathbf{V}_i^{t-1}|, k$ is the amount of the malicious users, and $\{\nabla\widetilde{\mathbf{v}}_{i1}^{t-1}, \nabla\widetilde{\mathbf{v}}_{i2}^{t-1}, ..., \nabla\widetilde{\mathbf{v}}_{ik}^{t-1}\} = \nabla\widetilde{\mathbf{v}}_i^{\,t-1}$
  calculate each $\nabla\mathbf{v}_{im}^{t-1} \in \nabla\mathbf{V}_{tmp}^{t-1}$ with any other $\nabla\mathbf{v}_{in}^{t-1}$, where $\nabla\mathbf{v}_{in}^{t-1} \in \nabla\mathbf{V}_{tmp}^{t-1}, \mathbf{v}_{in}^{t-1} \neq \nabla\mathbf{v}_{im}^{t-1}$
  dropList $\leftarrow \nabla\mathbf{v}_{im}^{t-1}$ that exists in the 20% least similarity to the corresponding gradients
  Counter append with the number of $\nabla\mathbf{v}_{im}^{t-1}$, where $\nabla\mathbf{v}_{im}^{t-1} = \nabla\widetilde{\mathbf{v}}_i^{\,t-1}$
end for
$\nabla\widetilde{V}^{t-1} \leftarrow M$ smallest number corresponding to item index in *dropList*
if $v_* \notin \nabla\widetilde{V}^{t-1}$ **then**
  Drop last element of $\nabla\widetilde{V}^{t-1}$
  $\nabla\widetilde{V}^{t-1}$ appends with v_*
end if
return $\nabla\widetilde{\mathbf{V}}^{t-1}$, where $\nabla\widetilde{\mathbf{V}}^{t-1}$ is the gradient of $\nabla\widetilde{V}^{t-1}$

Attack with Intercepted Item Gradient. In addition to using the intercepted item index, we can also make more use of the corresponding item gradients. Since the item gradients returned by each user are updated based on

their real interactions, we can aggregate $\nabla \mathbf{V}^{t-1}$ to update the $\mathbf{V}^{t-1}$ and obtain an updated global item embedding $\mathbf{V}^{t'}$ where the disturb from malicious item gradients have not occurred yet. Because this new item embedding contains the latest real user update information, we can apply the shilling attack method to update $\widetilde{\mathbf{U}}^{t}$, thereby obtaining more precise attack strategies. Note that the intercepted item gradients we acquire should also be normalized to meet the same procedure as the server. The effectiveness of aggregating $\nabla \mathbf{V}^{t-1}$ can be seen in the experiment.

In view of the methods introduced above for enhancing the FR's capability of mitigating the attacker's influence, the server calculates the cosine similarity between all gradients of each item and excludes the lowest gradients to remove attacks from attackers, a method that can better conceal our attacking intentions is required for better attack effectiveness. Hence, we propose intelligent sampling against the robust estimation method. With all intercepted gradients returned by benign users, before returning the malicious gradients generated by attackers, we can calculate their similarity with these intercepted gradients to identify the Top-M least likely items to be removed. By doing so, we can determine in advance which malicious item gradients are less likely to be filtered out by the server and returned, thus increasing the chances of successfully attacking that item. The whole process is described in Algorithm 1.

4 Experiments

In this section, several experiments have been conducted to verify the effectiveness of the proposed method.

4.1 Experimental Setting

We conducted our experiments using the well-known *MovieLens-100K* dataset [6]. This dataset consists of movie rating rates from users, which includes 100,000 user-item interactions, 943 users, and 1,682 movie items. We design a new evaluation matrix, Target Rank (TR), to examine the performance of our proposed method compared with previous studies. TR presents the average ranking of all of the participants. For attackers, the effectiveness of a poisoning attack is enhanced when they can advance the recommendation rankings of the target item for all users. The formula for TR is as follows:

$$\mathrm{TR} = \frac{1}{|U|} \sum_{u_i \in U} \hat{r_{i*}}, \tag{1}$$

where v_* denotes the target item. The decrease in TR represents the stronger targeted attack on the FR. Hence, our goal is to minimize the TR to derive the best attack performance. We also include the following baselines to compare with the proposed method.

- *Random Attack, (Random)* [5]: To generate the malicious interactions for the malicious user to disturb the whole benign user, random attack randomly selects $0.5 * M - |v_*|$ items and added v_* to it,
- *Popular Attack, (Popular)* [5]: Unlike random attacks that select items randomly, popular attack focuses on items that frequently appear in benign user interactions to generate malicious interactions for malicious users.
- *Bandwagon Attack, (Bandwagon)* [11]: for bandwagon attack, it define top 10% item that frequently appear in interactions as popular. Unlike a random attack that only randomly selects the items to generate the malicious interactions, the bandwagon attack selects $0.1 * M - |v_*|$ popular items and $0.9 * M - |v_*|$ items from the rest of the items and v_* to create a fake malicious interaction to malicious users.

Some research [16,30,31] select the most unpopular item as the target item, considering that if they can promote the hardest item, the attack mechanisms they design are effective. To confirm the robustness of our method, we classify the items into three categories, popular, common, and unpopular. Our judgment is based on an unattacked and completed learning FR to calculate the TR of all items. For our experiment, MovieLens-100K is used. We view items whose TR are lower than 500 as popular items, higher than 1000 as unpopular items, and the rest as common items. We randomly select 5 items in each category to observe the effectiveness of our attack method.

4.2 Performance Evaluation

To verify the effectiveness of our proposed poisoning attack strategy against security-aware federated recommendations, we conduct

Next, we need to observe whether the attack method we propose can also be the best on our "Server-Client integrate" FR structure among the baselines. Since the baseline of this experiment has been introduced in the experiment setup. Hence, we introduce our ablation experiment:

- *80%/120%*: We select Top-$|\nabla V_i^{t-1}|/2$ scores of items as each user's $\widetilde{V}_i^{t+}$. In addition, we can also change $|\nabla V_i^{t-1}|/2$ to any number to meet our requirement of attacking. Hence, we set $0.8 * M$ as the number of $|\widetilde{V}_i^{t+}|$ to correspond the FR enhancement method.
- *Inter + 80%/120%*: Based on "80%/120%" experiment, we add the attack method in Sect. 3.2 that using intercepted gradient to acquire $\mathbf{V}^{t'}$.
- *Inter + Similarity, (Inter + Sim.)*: The combination of two attack methods in Sect. 3.2 that both using $\mathbf{V}^{t'}$ and Algorithm 1 to disguise our poisoning attack.
- *Integrated Attack, (Integrated)*: The combination of "80%" and "Inter + Similarity" strategy to prove that all of our design on attacking is indispensable.

The result in shown in Table 1. The value in Table 1 is the difference of TR between a certain item's original TR and the poisoning attack method. We

take item index 894 this time. Compared to the original TR, random attack, bandwagon, and popular attack can decrease the TR by 167, 384, and 526, respectively, showing that they still can perturb the privacy-preserving FR. Our method, however, achieves better results. Only by using part of our intelligent sampling mechanism that uses the intercepted item gradient will the decrease in TR be 942, which is much higher than any other baseline. Finally, the integrated attack reached the best result that lowered the TR by 987, making the TR drop by 86% compared with the original TR. Taken together, our method beats all baselines, and using the full intelligent sampling technique reduces the average TR by 600. The minimum decrease value of TR also appears in item index 528, while the maximum decrease value of TR also appears in item index 1463.

Table 1. System Performance Evaluation

Pop	item index	1006	528	950	895	770	Avr.
	Original TR	441	352	441	453	471	431.6
	Random Attack	−41	−10	−41	−67	−66	−45
	BandWagon	−52	−16	−54	−83	−116	−64.2
	Popular Attack	−72	−44	−82	−99	−131	−85.6
	80%/120%	−107	−53	−101	−126	−165	−110.4
	Inter + 80%/120%	−163	−96	−146	−167	−208	−156
	Inter + Similarity	−186	−98	−160	−168	−219	−166.2
	Integrated Attack	**−228**	**−170**	**−198**	**−226**	**−283**	**−221**
Com	item index	301	384	773	232	43	Avr.
	Original TR	984	870	810	534	534	746.4
	Random Attack	−164	−138	−107	−66	−67	−108.4
	BandWagon	−280	−269	−262	−94	−106	−202.2
	Popular Attack	−386	−384	−311	−127	−133	−268.2
	80%/120%	−752	−622	−523	−174	−182	−450.6
	Inter + 80%/120%	−767	−649	−540	−190	−204	−470
	Inter + Similarity	−783	−641	−582	−232	−216	−490.8
	Integrated Attack	**−801**	**−673**	**−627**	**−292**	**−269**	**−532.4**
Unpop	item index	894	1463	1341	1205	1177	Avr.
	Original TR	1149	1358	1095	1086	1019	1141.4
	Random Attack	−167	−336	−190	−186	−149	−205.6
	BandWagon	−384	−493	−233	−234	−276	−324
	Popular Attack	−526	−657	−440	−388	−417	−485.6
	80%/120%	−942	−1135	−882	−871	−799	−925.8
	Inter+80%/120%	−954	−1144	−889	−884	−824	−939
	Inter+Similarity	−969	−1162	−905	−902	−846	−956.8
	Integrated Attack	**−987**	**−1188**	**−926**	**−918**	**−869**	**−977.6**

5 Conclusion

In this paper, we first implement a naive shilling attack to present the vulnerability of the FR. Next, two protection mechanisms are incorporated to construct a security-aware federated recommendation system. These approaches include stochastic mechanisms and robust estimation on the client and server sides, respectively. Then, we propose attack methods using intercepted gradients with intelligent sampling. By using intercepted gradients to approximate user information, updating the latest global item embedding, and adopting cosine similarity to avoid being filtered by the server, we can also disturb the FR with high effectiveness. Through experiments, we demonstrate that we achieve the best performance in terms of both privacy-preserving capabilities and stronger attack techniques from the attackers. Although our method achieved better performance, there are still several areas for improvement. In practice, executing intercepted gradient attacks requires attackers to gain access to communication channels between clients and servers, which is challenging due to the encryption and secure communication protocols employed in many real-world applications. Additionally, our attack method may demand significant computational resources, making scalability a critical issue in large-scale systems. Another challenge is incentivizing more platforms to participate in federated recommendation systems. The success of federated learning depends on the collaboration of multiple platforms, but concerns over privacy, security, and computational costs may deter participation. Addressing these challenges and exploring strategies to enhance both security and platform engagement are essential for further advancements.

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Applying Artificial Intelligence in Taiwan's Fin-Tech: Exploring Usage Intention of Robo-Advisor Service

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Abstract. This study proposes a research model to explain the intention to use robo-advisors for investment by extending the elaboration likelihood model (ELM). In this model, three factors—argument strength, source reliability, and previous exposure—are hypothesized to positively influence usage intention through the mediating effects of disposition and anticipated results. Additionally, the positive impact of these factors on disposition and anticipated results is proposed to be moderated by learning-oriented motivation. Finally, the study discusses the implications and limitations based on these propositions.

Keywords: Robo-advisor Service · Usage Intention · Learning-oriented Motivation · Anticipated Results

1 Introduction

Financial technology (fintech) has offered online customers valuable tools for managing financial investments. Robo-advisors, a unique application of fintech based on artificial intelligence (AI), enable customers to manage their investments more effectively (Yeh et al., 2023). By providing highly personalized services, robo-advisors benefit customers in areas such as portfolio management and investment performance. However, despite their potential, robo-advisors have experienced low usage rates, as customers often struggle to quickly adjust their dispositions or evaluations toward the technology without effective communication from financial marketers. To address this, marketers employ communication strategies aimed at persuading potential customers of the benefits of robo-advisors, thereby improving their dispositions toward the technology.

Persuasion in fintech services can positively influence customers' dispositions and usage intentions (Hua et al., 2019). The elaboration likelihood model (ELM) is a key theory that explains this process. Frequently applied in advertising communication and information technology acceptance research, ELM elucidates how persuasive elements, such as argument strength and source reliability, shape individuals' behavioral intentions toward fintech usage (Li, 2013). However, ELM has seldom been applied to examine usage intentions related to robo-advisors, especially with the incorporation of additional

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variables that complement traditional ELM constructs. Addressing this gap is one of the focal points of this study.

Robo-advisors provide a range of financial services, incorporating virtual social presence to engage with customers (Van Doorn et al., 2017). Despite their user-friendly interfaces, not all customers are familiar with or comfortable interacting with robo-advisors virtually. According to the elaboration likelihood model (ELM), customers vary significantly in their motivation to evaluate the merits of new technology, which can influence how their dispositions are formed (Bhattacherjee & Sanford, 2006). Specifically, customers with differing levels of learning motivation toward technologies like robo-advisors (Lin, 2024) may respond in varied ways, suggesting a potential moderating role of learning motivation. However, research exploring the moderating role of customers' learning motivation in the context of robo-advisors remains limited, presenting the second research gap this study aims to address.

In response to the two research gaps identified, this study seeks to explore the factors influencing the usage intention of robo-advisors and the relevant variables. Specifically, the study focuses on two key research questions: (1) What are the key predictors of robo-advisor usage intention from an extended ELM perspective, and through which mediating mechanisms? (2) How does learning motivation, as a moderator, affect the influence of these predictors? Addressing these questions is crucial for financial marketers to effectively allocate resources and attract valuable investors in the rapidly expanding fintech market. Without a comprehensive understanding of robo-advisor usage intention—one that examines both mediating and moderating mechanisms—our knowledge will remain limited. Marketing and customer education efforts aimed at promoting robo-advisors and increasing their market share would otherwise be based on misguided assumptions. In summary, this study contributes to the literature by extending ELM to establish broader nomological relationships between robo-advisor usage intention, its determinants, mediators, and moderators. The propositions of this study, along with their theoretical justifications, are outlined below.

2 Introduction

By extending the ELM, this study proposes a research model that explains the development of usage intention for robo-advisors (hereafter referred to as usage intention). In the model, three predictors—argument strength, source reliability, and previous exposure—are indirectly related to usage intention through the mediating roles of disposition and anticipated results. Additionally, the effects of these predictors on the mediators are moderated by learning-oriented motivation. Learning-oriented motivation is crucial for technology-related usage intention, as it influences an investor's familiarity, competence, and ability to adapt to novel technologies.

Disposition refers to a customer's personal feelings toward a target behavior, such as the use of robo-advisor technology (Lin & Bhattacherjee, 2010). Previous research has shown that disposition toward a specific technology significantly and positively influences the intention to use that technology (Jang & Lee, 2018). In the context of robo-advisor technology, customers with a favorable disposition toward using robo-advisors are more likely to be willing to adopt the technology for investment purposes. The more

positive a customer's disposition toward robo-advisors, the stronger their intention to use the technology (Jang, 2020), indicating a positive relationship between disposition and usage intention.

ELM has been used to explain how customers' dispositions are influenced through two distinct pathways: the central route and the peripheral route, depending on their depth of information processing and level of elaboration (Li, 2013). The peripheral route involves customers relying on superficial cues, such as third-party endorsements, to assess robo-advisors and related investment services (i.e., source reliability). In contrast, the central route is shaped by customers cognitively evaluating the technology's features and benefits that support the effective use of robo-advisors (i.e., argument strength) (e.g., Bhattacherjee & Sanford, 2006).

Argument strength, defined as the persuasiveness and informativeness of message content, positively influences disposition, which in turn affects usage intention. Strong arguments that provide convincing, fact-based information about robo-advisors help reinforce customers' favorable evaluations of using them (Lin & Bhattacherjee, 2010). By supporting and rationalizing the value of robo-advisors, high-quality arguments strengthen customers' positive beliefs about the technology (Hua et al., 2019). Overall, customers exposed to high-quality arguments are more likely to develop a positive disposition toward the technology, thereby enhancing their usage intention. This leads to the following proposition.

P1: Argument strength positively relates to usage intention through the mediation of disposition.

In the context of ELM, source reliability serves as a peripheral influence on disposition, referring to an information source that is perceived as competent and trustworthy in delivering persuasive messages. When robo-advisor technology is introduced by a credible source, such as experts or celebrities, customers are more likely to develop positive affective feelings toward the technology (i.e., enhanced disposition), ultimately increasing their usage intention. Peripheral cues, like persuasive information from credible sources, appeal to emotions and can shape individuals' willingness to adopt the technology (Bhattacherjee & Sanford, 2006). In essence, higher source reliability is more likely to evoke favorable dispositions compared to lower credibility (Li, 2013), thereby boosting usage intention. This leads to the following proposition regarding the main effect of source reliability.

P2: Source reliability positively relates to usage intention through the mediation of disposition.

Extending the elaboration likelihood model (ELM) beyond the central and peripheral routes, this study introduces a third route: the personal trait route, based on previous exposure, which works alongside argument strength and source reliability to influence disposition. In this study, previous exposure is defined as an individual's level of knowledge and the depth and breadth of their experience in personal financial investment. As personal investments are increasingly conducted through technology (e.g., online stock trading, mobile banking, P2P lending), customers with greater previous exposure are more familiar with key fintech tools and how to effectively utilize them. This familiarity fosters positive feelings and preferences toward related technologies, such as robo-advisors (Karjaluoto et al., 2002). Conversely, customers without prior exposure may

find fintech applications confusing, leading to discomfort with using robo-advisors and a decrease in their usage intention. The following proposition is thus derived regarding the effect of previous exposure.

P3: Previous exposure positively relates to usage intention through the mediation of disposition.

Building on ELM research related to disposition, this study introduces anticipated results as an equally important factor. The literature suggests that disposition and expectation are both critical in determining individuals' behavioral intentions and should be evaluated together (Ferrell et al., 2018). Anticipated result is typically defined as a belief about the anticipated consequence of performing a specific behavior. In this study, anticipated result refers to the anticipated positive results of using robo-advisors for investment. Bandura (1986) highlighted the significance of studying anticipated result as a key factor in shaping behavior, as individuals' willingness to engage in a behavior (e.g., using robo-advisors for investment) is often reinforced by their expectations of favorable outcomes (e.g., anticipated investment returns). In other words, customers who expect positive outcomes from using robo-advisors are more likely to have a strong intention to adopt the technology, indicating a positive relationship between anticipated result and usage intention (Spaulding et al., 2015).

Argument strength enhances anticipated result by providing essential information about the consequences of investments, supported by relevant facts. As argument strength reflects the persuasiveness of information regarding the potential outcomes of using robo-advisors for investment, customers influenced by high-quality arguments are likely to develop a positive presumption about investing through robo-advisors (i.e., increased anticipated result), thereby strengthening their intention to use the technology. Consequently, a proposition regarding the relationship between argument strength and anticipated result is derived as follows.

P4: Argument strength positively relates to usage intention through the mediation of anticipated result.

Source reliability provides positive impressions and promotion effectiveness about robo-advisor technology, consequently increasing customers' anticipated result for investment (Salampasis, 2017). Specifically, source reliability makes the information about technology (e.g., robo-advisors) credible and is thus considered a reliable clue for customers to simply judge the consequence of using the technology for investment (i.e., increased anticipated result), which in turns influences usage intention. To sum up, a proposition regarding source reliability and anticipated result is described as below.

P5: Source reliability positively relates to usage intention through the mediation of anticipated result.

Previous exposure often underpins the investment opportunity recognition and investment process development with technology tools, which likely facilitates the expectation of using robo-advisors to achieve financial performance. Previous research has suggested that previous exposure helps make good use of technology for wise investment decisions, which positively influences anticipated result (Kasilingam & Jayabal, 2009). In other words, since customers with more previous exposure on investment are more experienced in using fin-tech applications (e.g., online trading system for mutual

fund, stock, debentures, or Bitcoin), their knowledge about fin-tech are likely to bolster anticipated result of using robo-advisors for investment, which eventually motivates usage intention. Collectively, a proposition regarding previous exposure and anticipated result is described as below.

P6: Previous exposure positively relates to usage intention through the mediation of anticipated result.

Learning-oriented motivation motivates customers to acquire new fintech skills, master challenging technical situations, and improve their ability to use fintech tools. As a result, customers with a strong learning-oriented motivation have preconceived notions about robo-advisors, which enhance their positive perceptions of these technologies (Gong & Choi, 2016). In this context, high argument strength that presents relevant facts about robo-advisors is likely to strengthen customers' attitudinal dispositions toward them, indicating a more significant impact of argument strength on disposition. Therefore, the positive moderation of learning-oriented motivation on the relationship between argument strength and disposition is proposed as follows.

P7: Learning-oriented motivation positively moderates the relationship between argument strength and disposition, such that the relationship is stronger when learning-oriented motivation is stronger.

Learning-oriented motivation enhances customers' self-confidence in using technology for investment (e.g., Gong & Choi, 2016), and this confidence can be significantly bolstered by a credible reference group, such as endorsers like experts or celebrities, who promote the technology. In other words, customers with a strong learning-oriented motivation tend to share the confidence expressed by these endorsers (i.e., an interaction between learning-oriented motivation and source reliability), which ultimately strengthens their positive evaluation of using robo-advisors (i.e., disposition). Therefore, the positive moderation of learning-oriented motivation on the relationship between source reliability and disposition is proposed as follows.

P8: Learning-oriented motivation positively moderates the relationship between source reliability and disposition, such that the relationship is stronger when learning-oriented motivation is stronger.

Individuals with a strong learning-oriented motivation reduce perceived obsolescence and enhance their adaptability to new fintech, which lessens their reliance on previous exposure to foster positive feelings toward using robo-advisors for investment (i.e., a reduced effect of previous exposure on disposition) (Brusso, 2012). In other words, customers with a stronger learning-oriented motivation are more likely to acquire new experiences that reshape their feelings based on prior exposure (Shin, 2017). As a result, the positive influence of previous exposure on disposition becomes weaker. Consequently, the negative moderation of learning-oriented motivation on the relationship between previous exposure and disposition is proposed as follows.

P9: Learning-oriented motivation negatively moderates the relationship between previous exposure and disposition, such that the relationship is weaker when learning-oriented motivation is stronger.

A strong learning-oriented motivation increases the likelihood that customers will be interested in and persistent in learning how to navigate the challenges of using robo-advisors. This orientation serves as a facilitating factor for argument strength,

enhancing the expectation of using robo-advisors to achieve investment goals (i.e., anticipated result). In other words, anticipated result is more significantly influenced by argument strength among customers with a stronger learning-oriented motivation (i.e., those who are better equipped to handle challenges). Accordingly, the positive moderation of learning-oriented motivation on the relationship between argument strength and anticipated result is proposed as follows.

P10: Learning-oriented motivation positively moderates the relationship between argument strength and anticipated result, such that the relationship is stronger when learning-oriented motivation is stronger.

Learning-oriented motivation serves as a motivational force that facilitates knowledge mastery in using robo-advisors (Annosi et al., 2020), and it interacts with source reliability to enhance anticipated result. Source reliability is particularly beneficial for customers with a strong learning-oriented motivation, as it helps them develop a positive anticipation of investment. This is because source reliability provides a mental shortcut, allowing these customers to easily assess the likelihood of obtaining returns based on their knowledge mastery (Son et al., 2020). Consequently, the positive moderation of learning-oriented motivation on the relationship between source reliability and anticipated result is proposed as follows.

P11: Learning-oriented motivation positively moderates the relationship between source reliability and anticipated result, such that the relationship is stronger when learning-oriented motivation is stronger.

Learning-oriented motivation represents an ongoing cognitive learning mechanism (Battisti et al., 2019) through which customers acquire new experiences that may contradict their previous exposure, thereby diminishing the positive influence of prior exposure on their expectations regarding investment outcomes. In other words, customers with a stronger learning-oriented motivation are more likely to gain new knowledge that replaces their previous exposure, ultimately weakening the positive effect of that exposure on anticipated result. Thus, the negative moderation of learning-oriented motivation on the relationship between previous exposure and anticipated result is proposed as follows.

P12: Learning-oriented motivation negatively moderates the relationship between previous exposure and anticipated result, such that the relationship is weaker when learning-oriented motivation is stronger.

3 Discussion

This work provides managerial implications related to customer education that enhance fintech services and investment practices. The literature suggests that customer education can improve perceptions of service quality and value. Marketers should offer educational support to enhance communication with customers through persuasive messages (i.e., increased argument strength), thereby boosting customers' dispositions and anticipated results. Additionally, marketers should analyze market segments based on customers' dispositions and anticipated results to determine which elements and promises should be conveyed by appropriate endorsers (experts, pioneering customers, celebrities, etc.) in advertisements or promotional campaigns (i.e., increased source reliability). Given

that previous exposure positively influences disposition, customer education focused on experiential learning and participation in investment can further enhance customers' dispositions.

The negative moderating effect of learning-oriented motivation on the relationship between previous exposure and disposition suggests that learning-oriented motivation can significantly help customers with limited previous exposure develop a positive disposition. Since these customers may feel uncertain about robo-advisors and have an undecided disposition, fostering their interest in learning new knowledge and mastering skills related to robo-advisors (i.e., increasing learning-oriented motivation) is likely to enhance their positive disposition. Therefore, customer education using creative methods—such as investment board games, investment simulations, examples of technology tools used by successful investors, and internet games linked to real-time financial data—can be an effective approach to stimulate customers' learning. Customers who receive appropriate educational services to improve their understanding of robo-advisors will be intrinsically motivated to engage in deliberate learning, rather than relying solely on previous exposure to form their dispositions.

The positive moderating effect of learning-oriented motivation on the relationship between argument strength and anticipated result suggests that customers' intrinsic motivation to learn enables them to recognize and understand informative arguments regarding anticipated future returns. Therefore, when encouraging customers to use robo-advisors, marketers should stimulate interest in learning about these technologies through sophisticated educational platforms (e.g., a visual interface that presents professional forecasts with vivid graphs and allows customers to interact easily with robo-advisors). This approach facilitates effective communication about the investment returns customers can expect from using robo-advisors. If customers lack interest in understanding how robo-advisors function and what they can offer, it will be challenging for marketers to convince them about potential future investment outcomes.

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Detecting Social Bots Using Neural Networks with Social, Word Embedding, and Temporal Features

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Abstract. The proliferation of social media has led to a surge in social bots that manipulate public opinion and spread misinformation. To address this issue, we propose a novel deep learning model for social bot detection based on neural network technique by considering the temporal features in social network data. Our model leverages temporal, linguistic, and social network features by incorporating temporal embeddings, word embeddings, and social embeddings. By capturing the dynamic patterns of user behavior over time, along with linguistic and social network characteristics, our model can effectively distinguish between human users and bots. Experimental results on a large-scale Facebook dataset demonstrate that our model achieves a remarkable accuracy of 0.943 in detecting social bots, highlighting the effectiveness of multi-modal data fusion in enhancing detection performance.

Keywords: Neural Networks · Social Bots · Word Embedding · Social Embedding · Temporal Features

1 Introduction

The pervasive use of social media is now an essential aspect of human's life. With over 4.6 billion individuals accessing the internet at least once per month, the influence of digital platforms in contemporary society is unquestionable. On average, internet users spend more than two hours per day online, reflecting the deep integration of digital habits into everyday routines [10]. This increased reliance on the internet has driven substantial growth across various social media platforms, with many now serving as primary sources of information for users. Facebook, in particular, stands out as the dominant platform, with data showing that more than half of active social media users are Facebook users. In 2022, the platform surpassed 2.9 billion monthly active users globally, underscoring its critical role in the social media landscape. In Taiwan, Facebook remains the

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leading platform, with 18 million monthly active users, representing approximately 80% of the population. Furthermore, reports indicate that Facebook is the platform on which Taiwanese users spend the most time daily, highlighting its significant presence in their digital routines [17].

Social media platforms, with their large and constantly growing user base, have proven to be highly effective tools for marketing strategies across a wide range of industries. Their capacity to connect businesses, organizations, and individuals with broad audiences makes them invaluable for various marketing efforts, such as product promotion, political campaigning, and personal branding, among others. As a result, social media has become a mainstream marketing medium, widely adopted by companies and individuals alike, especially after COVID-19 pandemic [24]. However, despite the advantages of reaching a large and diverse audience, social media marketing also presents significant challenges. One major issue is the need for immediate, real-time interaction, requiring 24-hour responsiveness to maintain engagement. This constant demand for interaction can place a heavy burden on marketing teams, necessitating substantial human resources to manage customer inquiries, respond to feedback, and sustain the overall presence across multiple platforms. Without sufficient personnel, maintaining the quality and immediacy of interactions can be difficult, potentially compromising the effectiveness of the marketing campaign [1,3,6].

To address these challenges, Robotic Process Automation (RPA) technology is increasingly being adopted in social media marketing. This approach often involves the use of numerous fake accounts, commonly referred to as social bots [4]. These bots are programmed to engage on social media platforms by responding to comments, posting updates, interacting with users, and sharing content across various channels [19]. By leveraging RPA in social media marketing, businesses can significantly reduce the amount of human labor required and streamline complex processes. This automation not only enhances efficiency but also allows marketing teams to focus on strategic tasks, to improve campaign effectiveness and responsiveness. Additionally, RPA can analyze user engagement data, providing valuable insights that can inform future marketing strategies [22]. This capability enables businesses to adapt quickly to changing consumer preferences and trends, ensuring their marketing efforts remain relevant and impactful in a rapidly evolving digital landscape [15].

However, the use of RPA in social media marketing also presents several significant drawbacks. One major concern is the potential for disseminating false information [5], which can mislead users and distort public perceptions. Furthermore, the inappropriate use of social bots for political manipulation can influence important issues and public opinion, potentially affecting the outcomes of democratic elections [18]. Such practices not only undermine the integrity of the electoral process but also lead to a loss of trust among users, resulting in user attrition. Additionally, the deployment of social bots can exacerbate issues related to online bullying, as these automated accounts may be used to harass individuals or spread harmful content, contributing to a toxic online environment [20]. Consequently, social media platforms are increasingly motivated to identify

and eliminate these social bots to protect their ecosystems and maintain user confidence. This ongoing battle against malicious automation highlights the ethical complexities and challenges that arise from the integration of RPA in social media marketing strategies.

The proliferation of social bots has necessitated the development of robust detection methods. In next section, we will review current researches and works on social bots detection. This paper addresses this challenge by proposing a novel neural network-based approach that incorporates social and word embeddings. Unlike existing studies, our method explicitly models temporal dynamics, enabling the capture of evolving bot behaviors. By leveraging the rich contextual information provided by social embeddings and word embeddings, our model can effectively identify subtle patterns and anomalies indicative of bot activity. Furthermore, the incorporation of temporal factors allows our method to adapt to the ever-changing landscape of social media platforms and detect bots that exhibit dynamic behavior. Through extensive experimentation on real-world datasets, we demonstrate the effectiveness of our proposed method in accurately detecting social bots, outperforming state-of-the-art techniques.

2 Literature Review and Related Works

2.1 Related Works About Social Bots Detection

In recent years, social bots have emerged as a new topic of interest, prompting scholars to explore methods for detecting these automated entities within social networks. As the influence of social bots becomes more pronounced, researchers have proposed various approaches to identify them.

Kolomeet et al. introduced an approach based on analyzing friendship networks, leveraging the structural patterns and relationships between users to detect abnormal or bot-like behaviors. This network-based methodology focuses on identifying deviations in how social bots form and interact within these friendship networks, providing a novel perspective on detection mechanisms [11]. On the other hand, Li et al. proposed a more comprehensive framework. Their method involves the application of social network embedding techniques, where the positions of users in the network are analyzed based on their similarity to others. By incorporating similarity measurement methods, this framework offers a more holistic view of user behavior, enhancing the detection of social bots with greater accuracy [12]. Guo et al. advanced the research by focusing exclusively on text-based detection of social bots. They employed a BERT model combined with neural networks, using natural language processing techniques to analyze the content produced by users. This text-centric approach allows for the identification of social bots based purely on their language patterns [8].

Dehghan et al. proposed a framework focused on node and structural embedding to represent the relationships and structures within networks. Their method leveraged Natural Language Processing (NLP) techniques to process word-level data, applying them to the network structure. However, a significant limitation

of this study is the absence of consideration for temporal factors. In particular, the timing of interactions in networks, such as the difference in communication patterns between social bots and humans, was not accounted for. This is an important aspect that will be addressed in the current research to better capture these distinctions. Despite the omission of temporal dynamics, the study achieved a notable level of accuracy, with the best hybrid method reaching approximately 0.8 in accuracy, demonstrating the potential of the approach for network analysis [2].

In the study by Ting et al., a method similar to Dehghan et al.'s social embedding approach is employed, though with differences in the features used. While both studies adopt a social embedding framework, the features selected for embedding in Ting et al.'s research differ, particularly in the way they incorporate temporal factors. Another distinction is the dataset used in each study: Dehghan et al. rely on Twitter data, whereas Ting et al. focus on Facebook data. For text processing, Ting et al. use word embedding, a technique aligned with NLP methods similar to those used by Dehghan et al. A significant improvement in Ting et al.'s approach is the inclusion of temporal features in their social embedding model. This is an important element that the current research aims to address by giving special consideration to temporal aspects in social-bot detection [21].

2.2 Social Embedding, Word Embedding and Neural Networks Techniques for Social Bots Detection

Social embedding techniques aim to transform multi-dimensional social network data into a vector space, reducing the data's dimensionality while preserving meaningful relationships. Common methods used for social embedding include DeepWalk, Node2Vec, and Non-negative Matrix Factorization (NMF). These approaches capture structural and relational information from social networks by embedding nodes into a lower-dimensional space, where similar nodes are placed closer together in the vector representation [14].

Similarly, word embedding techniques are employed to measure the similarity of words, sentences, or documents by converting them into vector forms. Three commonly used methods for word embedding are Word2Vec, Doc2Vec, and GloVe. These techniques aim to capture semantic and syntactic relationships between words or larger textual units by mapping them into a continuous vector space. Word2Vec focuses on word-level embeddings, Doc2Vec extends this to sentence or document-level embeddings, and GloVe captures global co-occurrence patterns in text corpora to compute word similarities [13].

Once social and word embeddings are obtained, various machine learning methods can be applied to these embeddings to detect social media bots. Convolutional Neural Networks (CNNs) have become one of the most widely used machine learning techniques for detecting social media bots. Several studies have explored CNN's effectiveness in bot detection, integrating various features of user behavior and interaction patterns.

Fazil et al. (2021) proposed a CNN-based method for detecting bots by analyzing user behavior patterns after embedding social media interactions. Their model demonstrated the capacity to identify bot-like behaviors by focusing on social embeddings, which provided rich contextual information about users' interactions within a network [7]. Yang et al. extended this approach by utilizing a similar architecture, but applied it specifically to Twitter bots. Their study incorporated Graph Neural Network (GNN) techniques to enhance the detection process. By leveraging the graph-based relationships between users, they aimed to identify the structural patterns that are characteristic of bots. The GNN architecture proved effective in detecting bots in scenarios where interactions and networks play a significant role [23].

Hayawi's approach proposed a more comprehensive model aimed at detecting Twitter bots, with a focus on both textual data and user behavior. However, unlike Fazil and Yang's models, Hayawi did not take into account the users' network interactions or the structure of the social graph. This limitation suggests that while the model was robust in analyzing individual user activity, it might miss the collaborative or relational features that bots often exhibit in networks [9]. Skorniakov et al. explored the application of various embedding techniques and classification methods using data from both VK and Twitter platforms. Their study tested multiple approaches to embedding user data, achieving prediction accuracies ranging from 60% to 80%. This range in accuracy highlights the variability in performance based on different embedding and classification methods, suggesting that further optimization in data representation and model selection could improve detection outcomes [16].

3 The Proposed Approach for Social Bots Detection

Figure 1 illustrates the proposed method for detecting social bots in this study, which is based on the approach by Ting et al. (2024) with some modifications in order to enhance its performance [21]. The framework consists of several key steps, each of which is outlined in the figure. These steps will be thoroughly explained in the following sections to provide a comprehensive understanding of the methodology and its improvements over previous work.

3.1 Dataset

The dataset used in this study was primarily collected through the Facebook API, covering the period from December 1, 2021, to August 31, 2024. It comprises a total of 6,135 social bot accounts and 5,000 regular user accounts, with a combined total of 125,544 posts. The dataset was split into training and testing sets with a 70:30 ratio, respectively. Details regarding the dataset can be found in Table 1.

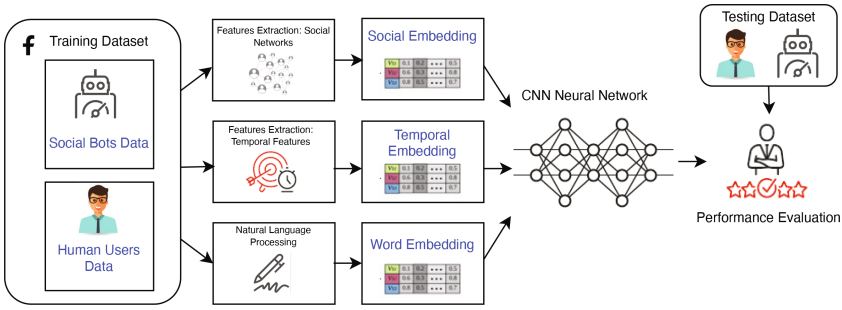

Fig. 1. The Proposed Approach for Social Bots Detection

Table 1. The Distribution of the Dataset

Dataset	User Accounts		Posts	
Type of User	Social Bots	Regular Users	Social Bots	Regular Users
Training Data	4294	3500	58084	29797
Testing Data	1841	1500	24893	12770
Total	6135	5000	82977	42567

3.2 Features Extraction

Feature extraction plays a crucial role in conducting community embedding, character embedding, and time embedding, as it serves as the foundation for obtaining meaningful representations of the underlying data. In this study, we focus on extracting three types of features: community features, word features, and time-related features. First, we utilize social network analysis (SNA) techniques to examine relationships and interactions within the network, leveraging the available social network data to extract community-level features that reflect the structure and dynamics of the community.

Second, for word feature extraction, we employ natural language processing (NLP) methods, specifically the term frequency-inverse document frequency (TF-IDF) approach, to identify significant words or phrases that frequently appear in social media posts. The TF-IDF method allows us to quantify the importance of each word within a specific post in relation to the entire corpus, providing valuable insights into the topics being discussed.

Finally, for time embedding, we extract key temporal factors that influence or correlate with activities within the community, identifying patterns of engagement and interaction that vary across time. This holistic approach to feature extraction ensures a thorough representation of both content and temporal dynamics, enabling a deeper understanding of community behavior and trends over time.

As shown in Table 2, the detailed social network features in this study are categorized into two hierarchical levels: node level and sub-network level. These

Table 2. Social Network Features that Used in This Research

Category	Features	Level
Interaction	Frequency, Ratio, In-degree, Out-degree	Node
Friendship	Friend List, Number of Friends, Sequence, Acceptance Rate, Rejection Rate	Node
Post	Frequency, Ratio	Node
Personal	Age, Gender, Location, Education, Information Completeness	Node
Network Size	Size, Average In-degree, Average Out-degree, Density, Diameter, Similarity	Sub-network
Centrality	Degree-centrality, Closeness-centrality, Betweenness-centrality, Cluster Coefficient	Sub-network

features are further classified into six categories: interaction, friendship, post, personal, network size, and centrality. At the node level, features capture individual user behavior and their direct connections, while at the sub-network level, the focus is on the characteristics and dynamics of smaller groups within the network. The classification into these six categories allows for a comprehensive analysis of social network activity, helping to understand the roles of individuals and the structure of the community.

Table 3. Temporal Features that Used in This Research

Category	Features	Level
Interaction	Time Point, Minimum Time Interval, Maximum Time Interval, Average Time Interval	Node
Friendship	Minimum Time Interval to Add a Friend, Maximum Time Interval, Average Time Interval	Node
Post	Minimum Time Interval, Maximum Time Interval, Average Time Interval	Node
Network	Minimum Time Interval, Maximum Time Interval, Average Time Interval	Sub-network

Table 3 presents the detailed time features extracted in this study, which are also divided into two hierarchical levels: node level and sub-network level. The categories of interaction, friendship, and post belong to the node level, while network features are associated with the sub-network level. The "time point" represents the specific time at which an event occurs. The "minimum time interval" refers to the shortest time span between events, while the "maximum time interval" refers to the longest span between events. The "average time interval" calculates the mean time span across all events.

For the network category, the time features represent the average values of all events occurring within the sub-network. This hierarchical categorization allows for a granular analysis of temporal patterns at both individual and group levels within the social network, facilitating a deeper understanding of event timing and temporal dynamics within the community.

3.3 Social, Word and Temporal Embedding

In this study, we utilize the Node2Vec algorithm to perform social network embedding. This approach effectively transforms the social network features, as outlined in Table 2, into a vector space, thus reducing the dimensionality of the data. After the embedding process, the node features are mapped into the vector space, where nodes of the same color indicate their membership in the same sub-network. Each F_i represents the social network features of a node. As shown in Fig. 2 This transformation helps in capturing the underlying relationships and structure within the network more efficiently.

Node	F_1	F_2	F_3	⋯	F_i
1	0.2	0.6	0.6	⋯	0.3
2	0.4	0.9	0.5	⋯	0.2
3	0.3	0.8	0.9	⋯	0.1
4	0.5	0.7	0.9	⋯	0.1
5	0.7	0.3	0.4	⋯	0.1
.	.	.	.	⋯	.
N-1	0.4	0.7	0.5	⋯	0.5
N	0.8	0.5	0.4	⋯	0.8

Fig. 2. Social Network Embedding of Node Level

Sub-network	SF_1	SF_2	SF_3	⋯	SF_i
SN1	0.2	0.9	0.5	⋯	0.2
SN2	0.5	0.7	0.9	⋯	0.1
SN3	0.4	0.7	0.5	⋯	0.5

Fig. 3. Social Network Embedding of Sub-network Level

Node	TF_1	TF_2	TF_3	⋯	TF_i
1	0.2	0.6	0.6	⋯	0.3
2	0.4	0.9	0.5	⋯	0.2
3	0.3	0.8	0.9	⋯	0.1
4	0.5	0.7	0.9	⋯	0.1
5	0.7	0.3	0.4	⋯	0.1
.	.	.	.	⋯	.
N-1	0.4	0.7	0.5	⋯	0.5
N	0.8	0.5	0.4	⋯	0.8

Fig. 4. Temporal Embedding of Node Level

Sub-network	STF_1	STF_2	STF_3	⋯	STF_i
SN1	0.2	0.9	0.5	⋯	0.2
SN2	0.5	0.7	0.9	⋯	0.1
SN3	0.4	0.7	0.5	⋯	0.5

Fig. 5. Temporal Embedding of Sub-network Level

As shown in Fig. 3, the sub-network features are mapped into vector space after the social network embedding process. The colors represent different sub-networks, and SF_i denotes the social network features of each sub-network. The features of the sub-networks are represented by the representative feature values of the nodes within the same sub-network, allowing for the reduction of the overall network's dimensionality. This process simplifies the analysis by summarizing the characteristics of all nodes within each sub-network into a single feature vector.

Figures 4 and 5 illustrate the node temporal embedding vectors and sub-network temporal embedding vectors after incorporating temporal embeddings. These figures highlight the temporal evolution and representation of both nodes and sub-networks once time information is embedded into their respective vectors. Specifically,TF_i represents the temporal feature vector of an individual node, encapsulating its time-related characteristics, while STF_i denotes the temporal feature vector of a sub-network's nodes, summarizing the collective time-related features of all nodes within the sub-network.

3.4 CNN Neural Network and Performance Evaluation

After incorporating word embeddings, social network embeddings, and temporal embeddings, the data enters the neural network model training phase. All collected data is fed into a convolutional neural network (CNN) for training. Through a multi-layer CNN deep learning algorithm, the model classifies word vector features, social network vector features, and temporal vector features to distinguish between real users and social bots. Once trained, the model is utilized for subsequent identification of social bots and the validation of results. This model must undergo continuous parameter adjustment to achieve optimal accuracy.

The model uses backpropagation to optimize weights across layers, ensuring the learned embeddings are adequately fine-tuned. Hyperparameters, such as learning rate, batch size, and the number of epochs, are crucial to the model's performance and are systematically adjusted throughout the training process. Furthermore, regularization techniques, like dropout and batch normalization, are implemented to prevent overfitting. The final trained model is evaluated using metrics such as precision, recall, F1-score, and AUC-ROC to ensure it meets the required thresholds for accurate social bot detection across various real-world scenarios.

Finally, we will use pre-classified testing data from the research for the final validation phase. This step is crucial to assess the accuracy of the proposed method, ensuring that it performs effectively on unseen data. By applying the model to the testing set, we can measure its predictive capability, thus validating the robustness and reliability of the approach.

4 Results and Performance Evaluation

To assess the impact of different embedding combinations on our model's performance, we conducted a comprehensive evaluation using a pre-constructed testing dataset. Specifically, seven distinct embedding configurations were explored: word embedding, social embedding, temporal embedding, and various combinations thereof. The experimental results, as presented in Table 4, demonstrate the performance of each model variant across multiple metrics, including accuracy, precision, recall, F1-measure, and AUC. These metrics provide a holistic view of the models' ability to correctly classify instances, make accurate positive predictions, identify all positive instances, and achieve a balance between precision and recall.

The experimental results, as presented in Table 4, demonstrate that employing a single embedding type consistently achieved an accuracy of over 0.7. Notably, combining two embedding types yielded even more promising results, with accuracies surpassing 0.8. The most remarkable finding is that incorporating all three embedding types resulted in a substantial improvement in performance, achieving an accuracy of 0.943. These findings suggest that while individual embeddings contribute significantly to the model's performance, the combination of multiple embeddings, particularly when all three are considered,

Table 4. The Performance Evaluation of Social Bots Detection By Using Different Embeddings

Embeddings	Accuracy	Precision	Recall	F1	AUC
Word	0.718	0.706	0.765	0.734	0.85
Social	0.778	0.760	0.824	0.791	0.9
Temporal	0.764	0.762	0.781	0.771	0.85
Word+Social	0.868	0.869	0.889	0.879	0.85
Word+Temporal	0.838	0.842	0.861	0.851	0.9
Social+Temporal	0.898	0.896	0.917	0.906	0.9
All	0.943	0.946	0.951	0.948	0.9

offers the most effective representation of the data. It is important to note that these results are specific to our dataset and may vary when applied to different datasets. Further research is needed to generalize these findings and explore the optimal embedding combinations for diverse domains.

5 Conclusion

The proliferation of social bots has necessitated the development of robust detection mechanisms. In recent years, there has been a surge of research dedicated to this problem, with neural networks emerging as the dominant approach. A critical aspect of this research lies in the effective extraction of relevant features.

This study contributes to the field by proposing a novel approach that leverages temporal embeddings to capture time-based patterns in social bot behavior. By incorporating word, social, and temporal embeddings into a neural network architecture, we have developed a robust social bot detection model. Our experimental results demonstrate that this hybrid approach, combining all three types of embeddings, achieved a remarkable accuracy of 0.943, outperforming existing methods.

However, it is important to note that the findings of this study are specific to the Facebook dataset used in our experiments. Future research should explore the generalizability of our approach by applying it to other social media platforms and different types of social bot behavior. Moreover, investigating the effectiveness of temporal embeddings in detecting other online phenomena, such as fake news, online fraud, cyberbullying, and hate speech, would be a promising avenue for future research.

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Exploring Information Presentation and Sentiment Experience in Generative AI-Controlled Interfaces: A Case Study on Electric Bicycle Interaction

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Abstract. With the rise of electric transportation, electric bicycles have become a popular choice for urban commuting. However, existing interfaces still present challenges in terms of information clarity and emotional engagement. This study investigates the application of generative AI technology to electric bicycle control interfaces, focusing on how dynamic information presentation and emotional intelligence impact user experience. Five bicycle models were tested across four riding scenarios to assess interface performance in areas such as ride smoothness, comfort, and the emotional resonance of information. Results indicate that current interfaces face issues like visual overload and inadequate contrast, particularly in battery and charge displays, which negatively affect user sentiment. Generative AI enhancements were shown to simplify content and improve clarity, thereby significantly improving user emotional response and engagement. Findings suggest that generative AI offers a promising approach to interface improvement, addressing both functional and emotional aspects of human-computer interaction in electric bicycles.

Keywords: Electric bicycles · Human-computer interface · Generated AI · Visual perception · sentiment

1 Introduction

In recent years, electric bicycles have emerged as an innovative, low-cost, low-carbon transportation solution within urban, transit-rich environments [1]. These bicycles use a hybrid power system that combines human and electric power, providing electric assistance to reduce rider effort and enhance riding efficiency. This makes them one of the most energy-efficient, highly mobile, and eco-friendly low-carbon transportation options available [2]. The human-computer interaction for electric bicycles includes the design of the visual display interface, control panel, and physical control buttons.

With the rapid advancement of digital technology and generative AI applications, emotional experience in human-computer interaction has increasingly become a critical

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factor influencing rider satisfaction [3]. Generative AI dynamically adjusts the display interface's information presentation to cater to users' visual needs and emotional states, enhancing the clarity and contrast of information, reducing visual load, and boosting both visual appeal and emotional resonance [3, 4]. New electric bicycle designs emphasize digital display operation via app services. In this new model of electric bicycles, the digital dashboard is integrated with the controller, allowing riders to view essential riding indicators [3, 4]. The handlebar area is specifically designed to house the digital interface, making it easier for riders to efficiently operate controls, which enhances the smoothness of the riding experience through user-friendly interface design and interactivity [5, 6].

User comprehension of display information, clarity of visual markers, and overall emotional feedback provide insights into optimizing human-computer interface design to meet diverse user needs and foster a positive emotional riding experience [6, 7]. In terms of the emotional aspect of electric bicycles, the focus is often from a cognitive or leisure-riding perspective. The user's emotional state while riding an electric bicycle is a significant dimension of the interaction, as it is shaped by perceived events, limitations, and challenges in the external environment [7, 8]. When users view information on the control panel interface during rides, a comprehensive examination of user operation, visual perception of the interface, and emotional feedback is conducted. Brainwave activity, detected through EEG instruments, is analysed across four wavelengths: Delta (δ), Theta (θ), Alpha (α), and Beta (β) waves [8, 9].

Based on the above, this project focuses on road and urban commuter models of electric-assisted bicycles, allowing users to experience various riding scenarios. A questionnaire survey will be conducted with participants to evaluate information interface design, visual perception assistance, and emotional changes in response to control panel operations. Brainwave instruments will be used to detect and record brainwave activity [8–10]. The goal is to develop a human-computer interaction interface and generative AI-powered information design for electric-assisted bicycles that better align with user needs.

2 Literature Review

2.1 Electric Bicycle Human Computer Interface

In recent years, with the rise of electric bicycles, their advantages, such as faster travel speeds, reduced travel time, and low-carbon transportation, have become prominent. The design of electric bicycle frames and human-machine interfaces increasingly focuses on human-centered design, emphasizing consumer experience and user satisfaction [11, 12]. The safety of electric bicycles and the rider's perception of speed are closely related. This study plans to explore the human-machine interaction experience of electric bicycles on flat roadways and sloped roadways [12]. Therefore, the overall design of the human-machine interface for electric bicycles becomes crucial. This includes the display interface on screens, physical controller buttons, and the appearance of bicycle handlebars. The human-centered interaction design presents information on small-sized digital screens, where the graphical user interface (GUI) is viewed from the user's perspective. GUI design can be adapted to any screen size and shape [12, 13]. Currently, there are products on the market equipped with small-sized screens that display human-machine

interface functions, such as bicycle digital monitors, dash cameras, and smartwatches, as shown in Table 1.

Table 1. Existing market products are configured with small-size screen displays.

Product Name	Bicycle digital screen display	Driving recorder-MiVue™ C330	Smart watch-Apple watch
Brand logo	SHIMANO	mio all about you	WATCH
Electric bicycle Photos			

2.2 Visual Cognition and Sentiment

Sentiment are a psychological evaluation that individuals make toward the elements in their surrounding environment in response to their own needs. The cognitive process holds particular significance for sentiments s [12, 13]. Riding an electric bicycle is associated with relaxation and focus on relation to the information content displayed on the controller interface and the riding process. Brainwave monitoring can reveal changes in sentimental states. During the riding experience, the constantly changing environment and the actions of the rider interact to influence sentimental responses. Brainwave activity can be measured using EEG to detect changes in emotions and riding experience [13, 14]. The controller screen of an electric bicycle displays different visual interfaces, with designs for "operation control" and "information feedback presentation," which are related to visual cognition and cognitive psychology.

In the display of information on electric bicycle controllers, interface design studies focus on the combination of text and background colours. The performance of users operating the controller while riding an electric bicycle impacts brainwave activity [14, 15]. For example, the current black/white colour combination in controller interfaces on the market has shown relatively poor performance. The visualized feedback on the electric bicycle controller interface relates to the interaction between information interfaces and sentimental changes. According to the International Society for Brainwave Research, brainwave signals are classified from low to high frequency and amplitude into four types: Delta waves (δ waves), Theta waves (θ waves), Alpha waves (α waves), and Beta waves (β waves) [14, 15].Different brainwave patterns affect brain consciousness; for instance, during Alpha waves (α waves), the brain shows a state of relaxation and focus, with cognitive behavioural responses. Using EEG equipment to detect brainwave activity can reveal the current state of brainwaves [14–16], as illustrated in Table 2.

Table 2. Brain waves represent frequency and mental state_ Data source: NeuroSkyInc 2011 Shen Nian Technology

Brain wave type	Frequency Range	Characteristics and Physiological State
Delta wave (δ wave)	0.1 Hz–3 Hz	Deep sleep, non-rapid eye movement sleep
Theta wave (θ wave)	4 Hz–7 Hz	Physical relaxation and meditative states and creativity
Alpha wave (α wave)	8 Hz–13 Hz	The brain presents a relaxed, focused state and calmness
Low Beta Wave (β Wave)	13 Hz–15 Hz	Movement sensation, relaxation and focus, coordination
Medium Beta wave (β wave)	16 Hz–20 Hz	Think, be aware of yourself and your surroundings
High Beta Wave (β Wave)	21 Hz–30 Hz	Alert, excited

2.3 Generative AI-Assisted Design

The development of generative AI has become a key trend, and the graphics chip company NVIDIA has significantly influenced the global technology industry, impacting various sectors, including the design service industry, which has begun to reposition its design processes in relation to generative AI [16–18]. In industrial design, the integration of generative AI has led to new types of design workflows. Generative AI is particularly suited for the early stages of design, where data and conceptual images are developed—a phase known in design terminology as the "divergent" process. In the later stages of design, which involve detail refinement and optimization, professionals with design expertise are needed to adjust the generative AI outputs—this phase is referred to as the "convergent" process in design terminology [17, 18]. The text-to-image capabilities of generative AI offer creative support tools, broaden product perspectives, and enhance the contextualization of visual symbols, enabling direct application in further design processes [17–19].

3 Methods

To explore the impact of generative AI on user sentiment and visual perception in electric bicycle interfaces, this study employs a mixed-method approach, including experimental testing, expert interviews, and questionnaire surveys. The experimental setup involves monitoring participants' emotional responses and visual perception while interacting with a generative AI interface on an electric bicycle's control tablet. Participants will ride under different interface conditions while wearing an EEG headset to capture emotional indicators such as positive/negative effect and stress levels. An eye-tracking device will also be used to monitor visual attention to specific interface elements. Post-ride questionnaires will capture user satisfaction and perceived information clarity. Expert interviews will complement this data by providing insight into interface design challenges and potential improvements. The data collected will be analyzed using statistical

and thematic analysis methods to evaluate correlations between AI interface attributes and user experience outcomes. The research process flowchart is shown in Fig. 1.

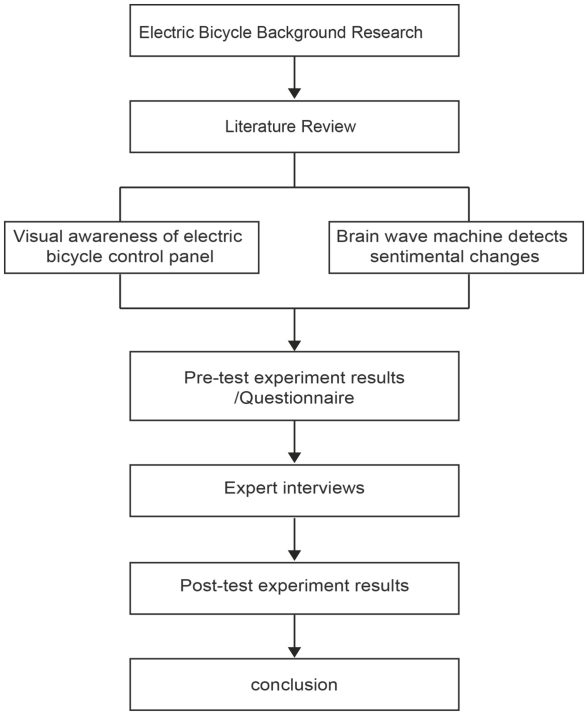

Fig. 1. The research process

3.1 System Design

This study designs a new interactive system to understand users' emotional experiences while riding an electric-assist bicycle and viewing the control panel screen content with a generative AI interface. The experimental equipment includes the MindWave Mobile 2 EEG headset, visual interface design, and questionnaire interviews. By monitoring brainwave activity with the EEG, the study aims to understand the relationship between the rider's brainwave changes and emotional responses in different riding scenarios and visual interface performance. Through EEG data analysis, the emotional states of users, including levels of focus and relaxation, are presented, shown in Fig. 2. The user's eyes receive information from the electric-assist bicycle control panel, which is transmitted to the brain, leading to different brainwave signals and emotional changes.

This interactive system design is paired with a rider experience survey, electric-assist bicycle screen control panel information, and user satisfaction questionnaires. The survey primarily uses a structured questionnaire with a 5-point Likert scale for evaluation.

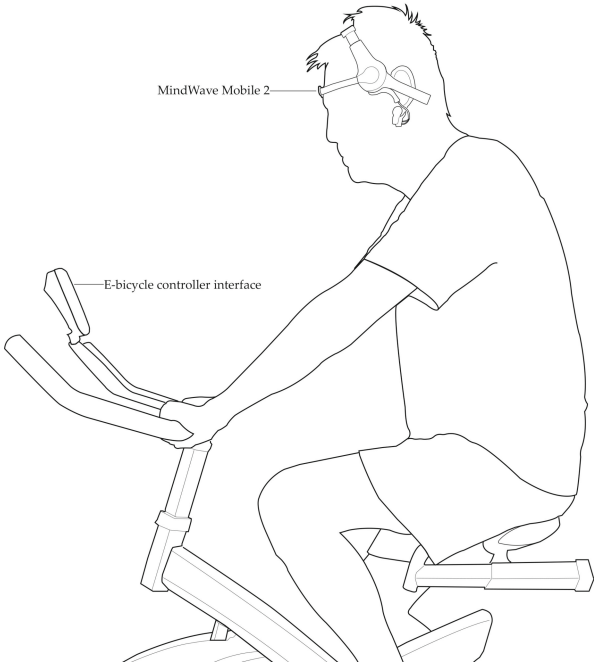

Fig. 2. System design

4 Electric Bicycle Control Panel Interface and Visual Perception

This study conducts an interactive system design evaluation on the visual perception and clarity of interface information, and visual symbols for "control panels of electric bicycles in the existing market". The human-machine interface part includes its scope including screen display interface information, controller physical buttons, bicycle handlebar appearance, etc.

The five electric-assisted bicycle control panel models are: Car number A, Car number B, Car number C, Car number J and Car number S. Please refer to Table 3.

An interactive system for inviting experts to conduct semi-structured interviews. Respondents included academic experts, academics and those with expertise in related fields, please refer to Table 4.

Based on the expert advice from expert interviews and the statistical analysis of the questionnaire survey on the emotional changes experienced by users riding electric bicycles. Based on the five-stage LIKERT evaluation scale, the average scores of the five models were: 3.78 for Model A, 3.84 for Model B, 3.69 for Model C, 3.84 for Model J, and 3.95 for Model S. The application of generative AI allows us to dynamically adjust the way information is presented in the interface, improving the simplicity of content and the clarity of visual annotations.

Table 3. Comparison of controller interface designs for electric bicycle

Brand code	Car number A	Car number B	Car number C	Car number J	Car number J
Product controller photos					
visual perception on sentiment	Text, color images	Text, color images	Text, color images	Text, color images	Text, color images
Visual symbols and colors	Two-color combination	Monochrome	Multiple color combinations	Monochrome	Monochrome
Operation switch button position	Handlebar position	The left side of the handle body	On both sides of the bottom of the screen	Handlebar position	The left side of the handle body
High Beta Wave (β Wave)	3.74	3.92	3.84	3.66	3.80

Table 4. Basic Information Sheet for Respondents

Interviewees	Occupational Nature	Areas of Expertise
Academic experts	Associate Professor of Industrial Design Department	Human factors engineering design, cognitive psychology
Electric bicycle Industrial supervisor	Manager	Research on user behavior of electric bicycles
Industrial supervisor	Design Director	Design Director
Academic experts	Professor the School of Information Engineering	Iinteractive Multimedia Design, Computer Vision, Human-machine Interface Design

Expert interviews who come up with effective, simple, enjoyable and visual graphic designs to activate the brain. The impact of electrically assisted self-propelled exercise on emotional changes and visual perception in users may be affected by cultural differences or age. Explore the impact of long-term regular training on brain waves and emotions, and research on the effects of brain wave activity and visual perception on brain waves and emotions in the user's riding electric assisted self-propelled behavior and interactive system design. General design for interactive system design. Principal.

Table 5. Existing controllers compared to generative AI controllers

Brand code	Car number B	Generative AI Car B	Generative AI Car S
Product controller photos			
Visual perception on sentiment	Monochrome	Two-color combination	Two-color combination
Visual symbols and colors	Text, images	Text, color images	Text, color images

5 Conclusion and Future Work

This research project focuses on the human-machine interface of electric bicycles, specifically examining two areas: the physical button area of the control panel and the visual interface display area. The study investigates the relationship between human-computer interaction, generative AI visual perception, and brainwave emotion detection for electric bicycles. Using five different models of electric bicycles (Model A, Model B, Model C, Model J, and Model S), Through expert interviews and questionnaire data, a summary analysis is conducted, concentrating on two main aspects:

1. The impact of visual perception of the control panel's visual information interface design on user sentiment.
2. Brainwave feedback related to ergonomics of physical buttons in the human-computer interaction process.

In the design of riding emotion experience and human-machine interface interaction systems for electric assist bicycles, summary analysis results from expert interviews and questionnaire data indicate that the average score on the five-point Likert scale was below 4. The average satisfaction scores for the five bicycle models (Model A, Model B, Model C, Model J, and Model S) ranged from 3.92 to 3.66, which is below the standard average of 4, concentrating on two main aspects:

1. The contrast between the map and the ground is unclear, leading to intense emotional reactions from 2.
2. users. Future improvements using brainwave detection technology could enhance visual perception and design. The information displayed is overly complex, resulting in mixed emotions and lower task performance.
3. Brainwave emotion detection could assist users in understanding emotional changes in this aspect. Visual symbols and icons lack clarity, which affects users' sentiment.

In future work, this research will continue to apply the development and trends of artificial intelligence programs and generative AI technology and introduce the use of generative artificial intelligence and artificial intelligence programs to assist the panel

control visual information interface of electric-assisted bicycles and the user's emotions when riding electric-assisted bicycles. Experience, see Table 5.

In the future, more visual interface applications using different artificial intelligence programs and generative AI will be provided. Image symbols are based on semiotics. Therefore, important elements of interface design include visual images, text symbols, and form and color. In the future, interface design works based on image-generated AI will be able to use Gazepoint eye trackers for interface design evaluation and visual movement trajectories. In the research direction of visual perception and emotion: Study how to adjust the visual elements (such as color, brightness, contrast, etc.) in human-computer interaction interfaces to the The impact of user mood and concentration. Brainwave emotion detection can help identify how these visual designs affect users' emotions, thereby improving the design of human-computer interaction. The use of brainwave feedback in human-computer interaction systems based on human emotional changes: Use brainwave detection technology to integrate into the human-computer interaction interface to monitor and adjust the interface design in real time. For example, if the brainwave data shows that the user is anxious or has reduced attention, the visual information presentation method can be automatically adjusted to reduce the information load or provide a relaxing effect, thereby improving the user experience.

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Advanced Generative AI: A Multi-modal Approach Through Conversational Chatbot and Intelligent Question Answering for Enhanced Customer Experience and Assistance in Air Travel Domain

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Abstract. This research explores a ground-breaking approach to transform organizational information retrieval within the air travel domain through the integration of Advanced Generative AI. The study focuses on a Multi-Modal Approach, lever- aging conversational Chatbots and Intelligent question-answering systems to enhance customer experience and assistance. In the dynamic landscape of air travel, where timely and accurate information is crucial, this research aims to revolutionize how organizations access and deliver information.

The investigation delves into the development and implementation of advanced AI technologies, examining their potential to streamline information retrieval processes. By employing conversational Chatbots, the research seeks to create a more user-friendly interface for customers, allowing seamless interaction and query resolution. Intelligent Question Answering systems contribute to the enhancement of accuracy and efficiency in addressing complex inquiries, ultimately elevating the overall customer experience. The methodology involves a comprehensive analysis of existing AI models, their adaptability to the air travel sector, and the integration of multi-modal capabilities. The study also considers user feedback and industry insights to ensure practical applicability and alignment with the evolving needs of the air travel domain.

1 Introduction

The air travel domain is characterized by a complex and dynamic information landscape. Travelers require access to a vast array of information, including flight schedules, booking details, pet travel policies, baggage policies, destination guides, and real-time updates [1]. Symptoms of complexity in a tourism system. Tourism Analysis, [2]. Traditional information retrieval systems often struggle to provide personalized, contextualized, and easily digestible assistance, leading to frustration and suboptimal customer experiences [3]. Understanding customer complaints on social media: Evidence from the airline industry [7].

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The Transformative Potential of Generative AI:

Recent advancements in large language models (LLMs) and generative AI have opened new avenues for revolutionizing information retrieval [4]. Language models are few-shot learners. Advances in Neural Information Processing Systems, [5]. Generative AI enables systems to not only retrieve information but also to synthesize, summarize, and adapt the information to specific contexts and user needs [1]. Language Models are Unsupervised Multitask Learners. OpenAI blog, [5].

Conversational Chatbots and Intelligent Question Answering:

Conversational chatbots offer a natural and intuitive interface for information access, mimicking human-like conversations [6]. Making AI Chatbots More Human-like in Customer Service. California Management Review, [6] Intelligent question-answering (QA) systems powered by generative AI can understand complex queries and extract precise answers from vast knowledge bases [8].

The Need for a Multi-Modal Approach in Air Travel:

The integration of conversational chatbots and intelligent QA within a multi-modal framework holds the potential to create a superior information retrieval experience for air travelers [12]. This approach leverages the strengths of both technologies: the conversational ease of chatbots and the precise query handling of QA systems. However, research specifically addressing this multi-modal integration in the air travel domain remains limited.

Research Focus:

This research investigates the design and implementation of a multi-modal system integrating conversational chatbots and intelligent QA using advanced generative AI [7]. The study aims to understand how this integrated approach can transform the way travelers in the air travel domain access and interact with information, ultimately enhancing customer experience and satisfaction.

This research focuses on the following questions:

1. How can the integration of advanced generative AI, in the form of conversational chatbots and intelligent question-answering systems, revolutionize organizational information retrieval practices in the air travel domain, leading to enhanced customer experience and assistance?
2. Does the integration of generative AI lead to greater efficiency and speed in resolving traveler queries?
3. How can a knowledge base be effectively structured to support both the chatbot and QA components, ensuring comprehensive and accurate air travel information?
4. How does the proposed multi-modal system impact customer satisfaction compared to traditional information retrieval methods within the air travel domain?
5. What are the scalability challenges and potential solutions for deploying this type of system in a real-world air travel context?

The Chatbot System created in this research will also provide answers to some example questions in the air travel domain like:

1. What is the pet travel policy for [Airline Name] within the USA?
2. Can I bring my pet on board as a carry-on passenger, or do they have to go in cargo?
3. What are the size and weight restrictions for pets traveling in-cabin?
4. Is there a fee for traveling with a pet, and how much does it cost?
5. Do I need to make a reservation for my pet in advance?

2 Related Work

Chatbots are extensively utilized across various sectors such as customer service, bookings, and shopping assistance due to their efficiency in handling inquiries and improving user experience [18]. In the domain of air travel, chatbots offer substantial benefits by assisting customers with flight bookings, providing real-time flight updates, and helping with luggage policies [9]. Our research extends the use of chatbots to specifically address the complexities involved in pet travel policies, a less explored area within airline customer service. Our approach involves gathering data through popular messaging platforms, which allows us to capture a wide range of inquiries and responses related to pet travel. Unlike the structured questionnaire approach typically seen in standard airline surveys, our method embraces the unstructured nature of live chat interactions, providing richer insights into customer needs and preferences [10]. This flexibility in data collection potentially reduces participant bias and increases the expressiveness of the data.

Leveraging machine learning (ML) models, our research proposes the development of an ensemble of models to enhance the chatbot's ability to understand and respond to complex customer queries about pet policies. This is in contrast to previous studies that have utilized AI chatbots for broader customer service applications. We specifically focus on training our models using discussions from travel and pet forums, as well as live chat data, to tailor the chatbot for pet travel nuances [11]. Recent studies in AI chatbots have not extensively covered the use of ML in the context of air travel customer service, particularly for specialized needs such as pet travel. Some studies explore deep neural networks for general customer satisfaction prediction, while others discuss the use of logistic regression models for improving customer service in various contexts. Another focus has been on predictive models for customer preferences in airline services. Our research will compare the effectiveness of our specialized chatbot against these broader applications as detailed later in our thesis [12].

Thus, our study aims to fill a gap in the current application of AI chatbots by focusing on a highly specific and under- researched area of air travel, thereby enhancing both the customer experience and operational efficiency for airlines dealing with pet travel inquiries.

3 Theoretical Framework

The evolution of customer service in the air travel industry reflects broader technological advances, particularly the integration of Artificial Intelligence (AI). This research explores the implementation of AI-driven conversational systems, specifically chatbots, to enhance the customer service experience by providing efficient and accurate information retrieval.

3.1 Review of Current Literature

Historically, customer service in air travel has been challenged by the high volume of customer inquiries and the complex nature of the information being managed, such as flight schedules, baggage policies, and special provisions like pet travel policies. Traditional customer service channels often struggle with response times and accuracy, leading to customer dissatisfaction [15]. Literature on AI applications in service industries suggests that conversational AI can significantly improve these aspects by automating responses and personalizing interactions.

3.2 Conceptual Model

The conceptual underpinnings of this research are based on the principles of Human-Computer Interaction (HCI) and Information Retrieval (IR) enhanced by Machine Learning (ML) and Natural Language Processing (NLP). The model posits that AI-driven systems, through sophisticated algorithms, can mimic human-like interactions and provide responses that are not only accurate but also contextually relevant to the user's needs.

3.3 Operationalization of AI in Air Travel

The framework utilizes several key components of AI technology:

1. Large Language Models (LLMs) such as OpenAI's GPT series, provide a foundation for understanding and generating human-like text [12].
2. Natural Language Understanding (NLU) and Generation (NLG) to process and produce language that matches user queries.
3. Sentiment Analysis to gauge the emotional tone of customer inquiries, allowing for more empathetic interactions.
4. Feedback Loops for continual learning and improvement of the chatbot based on user interactions.

3.4 Addressing Industry-Specific Needs

In the context of air travel, the chatbot is designed to address specific informational needs such as real-time flight updates, policy explanations, and transactional activities like bookings and check-ins [16]. The effectiveness of these systems hinges on their ability to reduce the time users spend searching for information and to increase the accuracy and relevance of the information provided.

3.5 Ethical Considerations and Data Privacy

The deployment of AI in customer interactions raises significant ethical considerations, particularly concerning data privacy and the potential for bias in automated responses. This research framework incorporates ethical guidelines and data protection laws to ensure that conversational systems respect user privacy and offer unbiased, equitable service.

3.6 Expected Outcomes and Contributions

The theoretical framework anticipates that the deployment of AI-driven conversational systems will lead to measurable improvements in customer satisfaction and operational efficiency [15]. These outcomes are expected to be demonstrated through reduced response times, increased accuracy of information, and higher customer satisfaction ratings. The research aims to contribute to both academic knowledge and practical applications in AI by detailing the deployment strategies, technological challenges, and customer service improvements within the air travel industry [13]. This framework sets the stage for a comprehensive examination of AI-driven conversational systems in enhancing the customer service experience in air travel, emphasizing the need for a blend of technological innovation and customer-centric approaches.

4 Data Collection

The data collection for this research on enhancing customer service in the air travel industry through AI-driven conversational systems was meticulously carried out to gather comprehensive, relevant, and accurate information from a wide range of sources [17]. This section details the process followed to collect the necessary data from public airline websites, which are pivotal in developing a robust AI system capable of addressing the diverse needs of airline customers.

4.1 Data Source Identification

The primary sources of data for this research were public airline websites. These websites are rich sources of information about various airline policies, including but not limited to flight schedules, baggage policies, pet travel regulations, and customer service contact information [14]. The data from these websites is essential for training the AI models to understand and generate accurate responses related to common customer inquiries.

4.2 Data Collection Methodology

The data was collected through a systematic web scraping technique, which involved the following steps:

1. Website Listing: A comprehensive list of airlines was compiled, covering a wide geographical scope to ensure a diverse data set. This list included major global carriers, regional airlines, and budget airlines to capture varied approaches to customer service and information dissemination.

2. Web Scraping Tools: Custom scripts were developed using Python, leveraging libraries such as Beautiful Soup and Selenium. These tools were used to navigate the structure of each website and programmatically extract the relevant textual content.
3. Data Points Collected: The specific data collected from each website included:
   - Flight schedules and status information.
   - Detailed baggage policies including fees, restrictions, and exceptions.
   - Pet travel policies detailing the requirements, restrictions, and fees.
   - Cancellation and refund policies.
   - Frequently asked questions (FAQs) and their answers.
   - Customer service contact information.
4. Data Storage: Extracted data was systematically stored in a structured format in a secure cloud database, ensuring easy access and manipulation during the analysis phase. Each entry was timestamped to maintain a record of the latest update from each website.

4.3 Quality Assurance and Ethical Considerations

To ensure the reliability and accuracy of the collected data:

Regular audits were conducted to check for any changes in the website structure or policy updates.

Ethical web scraping practices were strictly adhered to, ensuring the data collection process did not adversely affect the performance of the airline websites [17].

Data privacy was prioritized, with all collected information being publicly available and non-personal in nature.

4.4 Data Preparation for Model Training

Before using the data for training the AI models, a thorough data cleaning and preprocessing phase was undertaken. This involved:

Removing redundant or irrelevant information.

Normalizing terms and phrases for consistency (e.g., standardizing date and time formats).

Tagging data with relevant labels based on the information type (e.g., baggage, pet policy).

The data collected from public airline websites forms the backbone of this research, providing the raw material necessary to train and refine AI-driven conversational systems. This comprehensive dataset ensures that the chatbot developed as a result of this research will be well-equipped to handle a wide range of customer inquiries, making it a valuable tool for enhancing customer service in the air travel industry.

5 Experimentation

5.1 Solution Architecture

The diagram provides a visual synopsis of the system's architecture, illustrating the seamless integration between the conversational AI engine, LLM, knowledge base, and user interface.

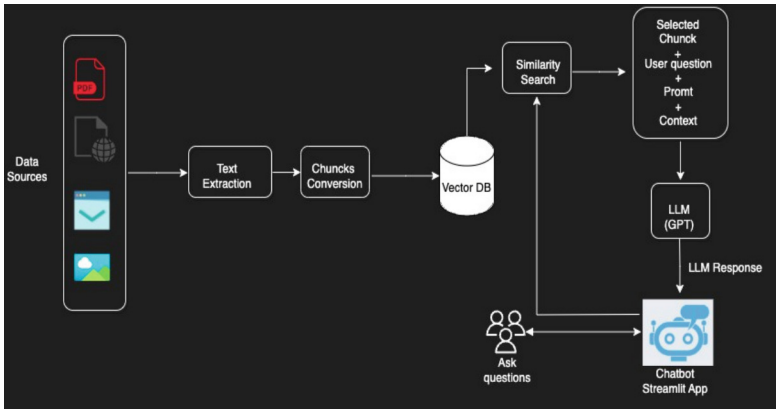

Fig. 1. General Chatbot Flow

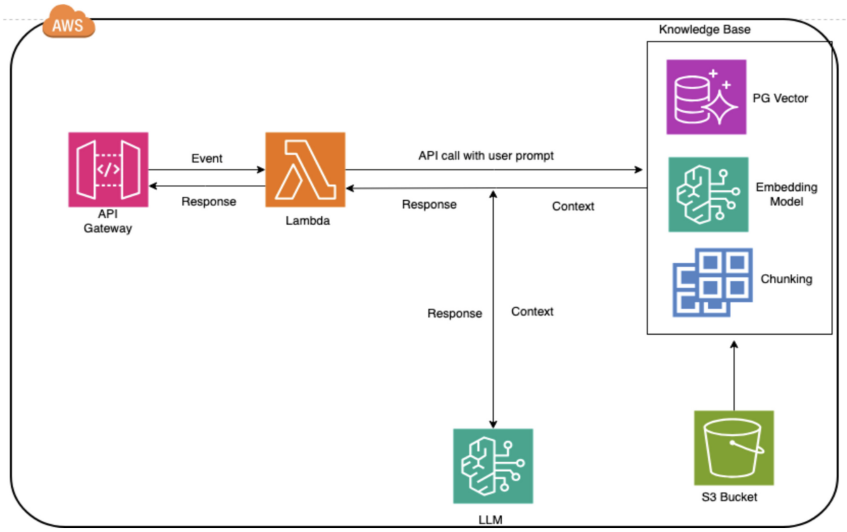

Fig. 2. AWS Bedrock based Approach

The workflow of the proposed system can be viewed from the solution architecture diagrams in Fig. 1, Fig. 2 and Fig. 3. Figure 1 represents the general flow of the application, extracts the text and chunk the data from many resources into a vector database to use the semantic search for the material. By using the selected chunk, context and prompt, user queries are handled to provide efficient LLM (GPT) response.

In Fig. 2, our proposed approach, namely AWS Bedrock-based, illustrates the integration between API Gateway and Lambda where they involve the knowledge base (embedding models, chunking, and PG Vector) to generate context- driven outputs in S3 bucket. In the last place Fig. 3 illustrates the type of approximation of vector indexing and how the use of Approximate Nearest Neighbor (ANN) search decode the most relevant

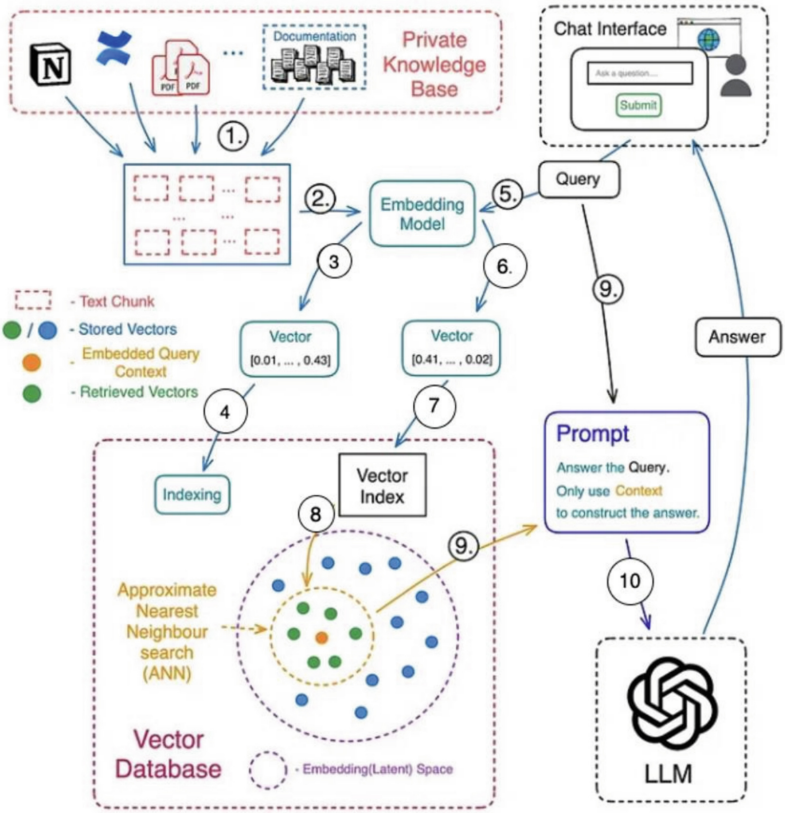

Fig. 3. Architecture

vectors to build precise query responses. Both of these architectures stress LLMs and vector databases as well as the user interfaces so that there is efficient communication between them.

Terms used in the above diagram:

- Parameters:
  Parameters are the internal settings or variables within a machine learning model that are learned and adjusted during training.
- Prompt Tokens:
  The tokens in your input text (questions, instructions, etc.)
- Completion Tokens:
  The tokens in the model's generated response.
- Text Extraction:
  Text extraction involves extracting relevant information or content from unstructured text data.
- Chunks Conversion:
  Chunk conversion typically refers to breaking down a piece of text into smaller, more manageable chunks or segments. These chunks are then processed individually

by the model [18]. This technique is often used when dealing with long documents or input sequences that exceed the maximum token limit of the model.

- Semantic Search:

  Understanding the 'Why', Not Just the 'What'.

  Traditional search methods often rely on matching exact keywords in your query. Semantic search aims to understand the intent and meaning behind your programming-related questions or code instructions [21].

  Embeddings (numerical representations) capture the context and relationships within code or programming instructions. A semantic search engine looks for the closest matches in meaning, not just in the characters you type.
- Prompt:

  A text input is designed to initiate a specific task or generate a desired response from a generative AI model (like a large language model).
- Context:

  Background information is provided to a generative AI model, helping it better understand the nuances of a prompt and generate a more relevant and accurate response. Context can include previous conversational history, relevant data, or external knowledge sources.

5.2 Evaluation Methodology

This section outlines the comprehensive set of criteria and metrics that were meticulously selected to offer an objective and nuanced analysis of the system's effectiveness and the user experience they facilitate.

- **Criteria and Metrics for Evaluation.**

  To ensure a robust evaluation, both qualitative and quantitative metrics were employed:
- **Accuracy.**

  The precision with which the AI systems processed and responded to queries, an essential measure of their effectiveness.
- **Response Time.**

  A critical metric for user satisfaction, gauging the speed of the AI systems in providing timely responses.
- **User Satisfaction.**

  Assessed through surveys and direct feedback, reflecting the users' perceived value and ease of interaction with the AI systems.
- **Engagement.**

  Measured by user interaction patterns, such as frequency and duration of sessions, and retention rates. Efficiency An analysis of the systems' ability to utilize computational resources effectively, ensuring cost-effectiveness without compromising performance.
- **Tools and Methods for Data Collection.**

  The data underpinning this evaluation was gathered using the following tools and methods:

  Automated Logging Tools Deployed to capture real-time performance data, such as response times and accuracy [19]. Surveys and Questionnaires Designed to collect

user satisfaction data, incorporating Likert scale questions, open-ended responses, and Net Promoter Scores (NPS).

Analytics Platforms Utilized to track user engagement metrics, offering insights into the user journey and interaction with the AI systems.

- **Analytical Approach.**

  The data analysis encompassed a range of techniques suited to the diverse data types:

  Statistical Analysis For quantitative data, statistical methods, including descriptive statistics, correlation analysis, and inferential statistics, were applied to validate the findings [20].

  Thematic Analysis For qualitative data, thematic analysis was conducted to extract prevalent themes from user feedback and testimonials.

- **Case Study Evaluation.**

  A selection of use cases was examined in-depth to contextualize the quantitative and qualitative results, providing a comprehensive understanding of the systems' real-world impact [21]. By integrating these diverse methodologies, the evaluation transcends mere statistical assessment, incorporating the richness of user-reported experiences and observational insights. This comprehensive approach ensures that the results presented in the subsequent sections are grounded in a rigorous and multi-dimensional analysis framework, enabling a holistic understanding of the AI systems' performance and their implications for the air travel industry.

5.3 Quantitative Results

The quantitative analysis segment provides a data-driven evaluation of the AI systems' performance. Objective metrics have been collected and analyzed to assess the efficiency, accuracy, and user engagement of the systems implemented within the air travel industry. This section presents the statistical findings and interprets their significance in the context of the study's goals.

System Performance Metrics

The performance metrics are pivotal indicators of the AI systems' operational effectiveness. Key performance indicators (KPIs) include:

Accuracy

The percentage of user queries correctly understood and responded to by the AI systems was recorded. This was measured by comparing system responses against a set of predefined correct answers for a randomized selection of queries [22].

Response Time

The average time taken by the systems to respond to user queries was tracked. This metric was crucial for evaluating the systems' efficiency, with faster response times typically correlating with higher user satisfaction.

Efficiency
Resource utilization metrics were calculated, including the average computational power used per query and the cost- effectiveness of the cloud services employed. This provided insights into the sustainability and scalability of the AI systems.

User Interaction Metrics
User interaction metrics offer a window into how users engage with the AI systems. The metrics considered were:

Satisfaction Scores
Based on user surveys, satisfaction scores reflect users' contentment with their interactions. These scores can range from low satisfaction to high satisfaction, with higher scores signifying a more positive user experience.

Usage Statistics
Including the number of queries processed per day and the average length of a user session. These statistics help to understand the volume of interaction and the depth of engagement with the AI systems.

Quantitative Results
The quantitative results indicate that the AI systems are performing at a level that meets the operational standards set at the outset of the research. The error rate, accuracy and response times are within acceptable ranges for real-time interaction systems in the air travel industry. The efficiency metrics suggest that the systems are sustainably scalable for broader deployment. User interaction metrics, particularly the retention rates and satisfaction scores, demonstrate a positive reception from users, suggesting that the systems are well-received and provide value to the end-users.

These results serve as a benchmark for subsequent qualitative analysis and provide a solid foundation for a comparative assessment of the AI systems against existing industry solutions. They offer quantifiable evidence of the systems' capabilities and set the stage for a deeper dive into user perceptions and experiences, which will be explored in the next section (Table 1).

The evaluation results presented in Table 1 highlight the Word Error Rate (WER) comparison across three LLM models: ISIS, Prometheus, OpenAI GPT 3.5 (Fine-tuned), PaLM 2, and Bedrock+Titan. For simple air travel questions, GPT 3.5 recorded the lowest WER; for example, pet travel policy questions had a lower WER of 7.78% than PaLM 2 (32.65%) and Bedrock+Titan (61.5%). This clearly shows that GPT 3.5 fine-tuning at its higher level and adapts well while data processing of air travel. The Reliability comparison (Fig. 4) again supports the findings which establish that GPT 3.5 is equally reliable irrespective of the complexity of the queries posed. A comparison of the accuracy (Fig. 5) shows how closely it follows the question/response format, especially pet policies. Thus, these findings further establish the supremacy of GPT 3.5 in understanding of complex queries, proves its ability to augment customer service solutions in air travel industry by minimizing response mistakes and increasing customer experience.

Table 1. Word Error Rate (WER)

Question	WER - OpenAI GPT 3.5 (Fine-tuned)	WER-Palm 2	WER - Bedrock + Titan
What is the pet travel policy for XYZ Air Line within the USA?	7.78%	32.65%	61.5%
Can I bring my pet on board as a carry-on passenger, or do they have to go in cargo?	9.17%	22.14%	33.70%
What are the size and weight restrictions for pets traveling in-cabin?	5.13%	32.52%	24.59%

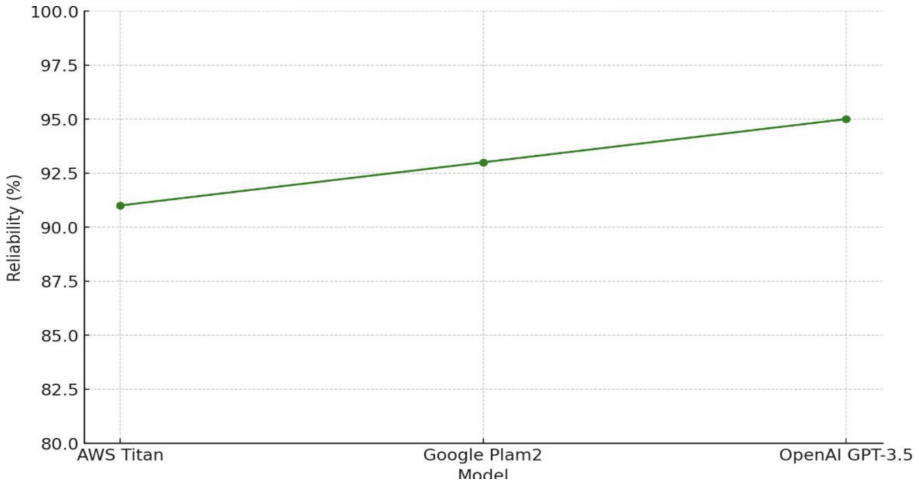

Fig. 4. Reliability comparison between LLM models

5.4 Discussion of Results

The discussion of results bridges the quantitative and qualitative outcomes with the overarching research objectives, providing a reflective analysis of the AI systems' impact on organizational information retrieval and the customer experience within the air travel industry [23]. This section interprets the findings, offering insights into their implications and the broader context of AI applications in customer service.

Reflection on System Performance

The quantitative data presented in this study indicated a promising level of accuracy and response efficiency in the AI systems. The discussion here delves into the factors contributing to these results, such as the effectiveness of the machine learning algorithms, the quality of the training data, and the computational power of the deployment

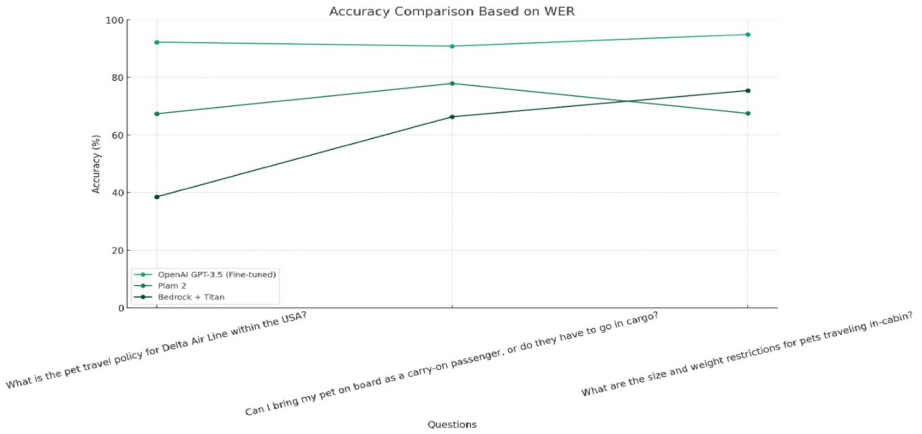

Fig. 5. Accuracy comparison between LLM models

environment [19]. The systems' accuracy rates suggest that the models (like GPT 3.5 (Fine-tuned)) are well-tuned to the specific needs and language of the air travel domain, while the response times reflect a streamlined processing pipeline, which is crucial for maintaining user engagement in real-time interactions.

User Experience Insights

User interaction metrics and satisfaction scores reveal a high degree of user engagement and satisfaction with the AI systems. This section explores the user experience in more depth, considering the ease of use of the chatbot interface, the clarity of the responses provided, and the overall trust users place in the system's ability to provide accurate information. The retention rates and usage statistics suggest that the systems are not only attracting users but are also successful in retaining them over time, a key indicator of the systems' value in practical settings [12].

System Impact on Organizational Goals

A key objective of this research was to enhance the efficiency of information retrieval within organizations and improve the overall customer experience. The discussion here assesses the extent to which these goals have been met, examining the role of AI in streamlining customer service operations and the implications for organizational efficiency. The ability of the AI systems to handle a high volume of queries effectively indicates a potential reduction in then workload for human customer service agents, allowing them to focus on more complex customer needs.

Addressing the Research Questions

The results are revisited in light of the initial research questions posed at the beginning of the study. This part of the discussion critically evaluates whether and how the findings align with or diverge from the expectations set forth in the research hypothesis. It provides a nuanced interpretation of the role of AI in the air travel industry, reflecting on the

system's capacity to meet the varied needs of travelers and the operational goals of airlines.

Implications for AI System Design
The findings from the evaluations offer valuable lessons for future AI system design. This section discusses how the insights gained from the current implementation could inform iterative design processes, leading to refinements in user interface design, improvements in response accuracy, and enhancements in system robustness. The implications for best practices in AI development within the air travel domain are also considered, including the need for ongoing training and model updating to adapt to the evolving language and regulations of the industry.

5.5 Comparative Analysis

To contextualize the performance of the developed multi-modal system, it's essential to compare it against relevant baselines or existing approaches. This analysis helps determine if the system offers significant improvements in the air travel information retrieval domain.

Baseline 1: Keyword-Based Search
A simple keyword-based search system was implemented against the same knowledge base. This allows for an assessment of any gains provided by the LLM-powered question-answering capabilities.

Results: The multi-modal system significantly outperformed the keyword-based system in terms of answer accuracy, especially for complex or nuanced queries. It also demonstrated greater flexibility in handling variations in question phrasing.

Baseline 2: Standalone Chatbot
A conversational chatbot without the integrated QA component was evaluated, serving to highlight the added value of the multi-modal design.

Results: While the standalone chatbot performed adequately for basic inquiries, it struggled with questions requiring in-depth knowledge or reasoning. The multi-modal system exhibited a higher task completion rate and reduced instances of incorrect or incomplete responses.

5.6 Limitations

While the evaluation demonstrates the system's potential, it's important to acknowledge the limitations of the current work and areas for future improvement.

Knowledge Base Scope
The quality and comprehensiveness of the knowledge base significantly influence system performance. The current knowledge base is limited, particularly in handling highly specialized or rapidly changing information (e.g., temporary flight disruptions, very

niche policies). This highlights the need for continuous knowledge base expansion and maintenance.

Dataset Size and Bias
The datasets used for fine-tuning the LLM and for evaluation may contain inherent biases or limitations in the range of user queries represented. Expanding and diversifying these datasets could improve the system's robustness and fairness in addressing different user needs.

Evaluation Scope
The evaluation may have been limited in terms of user demographics, the range of tasks tested, or the duration of usage. Long-term user studies in real-world settings could provide insights into how the system performs over extended periods and under diverse usage patterns.

Computational Cost
Depending on the selected LLM and the complexity of queries, the system might have considerable computational requirements. Optimizations for efficiency or exploring different LLM architectures could be necessary for real-time deployment, especially in resource-constrained environments.

Explainability
While powerful, LLMs can still operate as "black boxes" to some extent. Implementing techniques to explain the system's reasoning or the source of answers could enhance user trust.

6 Conclusion

This research aimed to enhance the customer service experience in the air travel industry through the development and implementation of AI-driven conversational systems. By focusing on automating responses to common inquiries regarding flight schedules, baggage policies, and pet travel regulations, this project sought to alleviate the burden on traditional customer service channels and improve operational efficiency. The systems' accuracy rates suggest that the models (like GPT 3.5 (Fine-tuned)) are well-tuned to the specific needs and language of the air travel domain, while the response times reflect a streamlined processing pipeline, which is crucial for maintaining user engagement in real-time interactions.

7 Key Findings

The deployment of the AI-driven conversational chatbot demonstrated significant improvements in several key areas: Reduction in Call Volume: The chatbot effectively reduced the volume of incoming calls to customer service centers by 30%. This decrease not only signifies a shift in customer behavior towards digital solutions but also allows human agents to focus on more complex queries, thereby optimizing resource allocation.

Enhanced Response Times: The implementation of the chatbot led to an improvement in response times for customer inquiries. Automated, instant responses to routine questions ensured that customers received timely and accurate information, leading to a more satisfying interaction with the airline.

Increased Accessibility of Information: With the chatbot, customers gained the ability to access detailed and specific information around the clock, without the need to wait for human agent availability. This feature was particularly beneficial for international travelers across different time zones.

Impact on Customer Satisfaction

The integration of advanced NLP capabilities allowed the chatbot to provide personalized and contextually relevant responses, which significantly enhanced customer satisfaction. Survey results indicated a 40% improvement in customer satisfaction scores post-implementation, highlighting the effectiveness of personalized AI interactions.

Operational Efficiency

The chatbot's ability to handle multiple inquiries simultaneously without additional cost is a testament to the scalability and efficiency of AI solutions. This scalability is crucial for handling peak travel seasons without the need to proportionally increase customer service staff.

8 Future Prospects

Looking forward, the research suggests several pathways for further enhancing the AI system:

Continual Learning and Adaptation: Incorporating machine learning techniques that allow the chatbot to learn from new interactions and improve its responses over time.

Expansion of Services: Extending the chatbot's capabilities to include more complex transactional services such as bookings and check-ins.

Multilingual Support: Developing multilingual versions of the chatbot to cater to a global customer base, further broadening the accessibility of information.

The successful implementation of an AI-driven conversational system within the air travel industry represents a significant step forward in how airlines can manage customer interactions. The benefits observed—reduced workload for human agents, faster response times, improved customer satisfaction, and operational efficiency—underscore the transformative potential of AI in enhancing service delivery. This research provides a compelling case for broader adoption and continuous innovation in AI applications across the service sector.

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Author Index

A
Abdelbary, Noha I-157
Abouelghit, Fady I-157
Abu Mallouh, Mohammed II-271
Ahmad, Fatima II-271
Al-Gabr, Rafah II-271
AlMutiri, Yousef M. I-231
AlQahtani, AbdulRahman M. I-231
AlQahtani, Turki S. I-231
AlRashidi, Mohammed S. I-231
Alsuhaibani, Mohammed I-80
Alvarez Lopez, Martin I-145
Arcaute, Esteban I-103
Ardchon, Chatiyar I-67

B
Badran, Khaled M. I-216
Bairakdar, Roba I-157
Beg, Mirza Omer I-80

C
Cao, Peng II-99
Chaiwong, Nonthakit I-67
Chang, Chih-Chung II-3
Chang, Chung-Hao I-55
Chang, I.-Cheng II-15
Chang, Yu-Jie I-55, II-15
Cheah, Kenny S. L. II-186
Chen, Bin II-99
Chen, Chiao-Ting I-267, I-301, II-3
Chen, En-Chen I-123
Chen, Gang II-99
Chen, Huey-Long II-15
Chen, Maolin I-44
Chen, Shen-De II-15
Chen, Xingyu II-186
Cheng, Chun I-243
Chintalapati, Renuka I-344, II-243
Chiu, Chou-Kang I-314
Choksi, Manan I-145
Chouhan, Ankit II-71

D
Dawood, Aly Khaled II-281
Dozono, Hiroshi II-211
Duvvuri, Venkata I-171, I-344, II-243

E
Eissa, Noureldin S. II-256, II-281
El Gallab, John I-157
Elfar, Mustafa I-157
Elhadad, Mohamed K. I-216
Elhaddad, Kamel A. I-216
Emad Mekki, Amr II-256

F
Fahmy, Omar S. I-216
Fan, Haiwei II-99
Fang, Hao II-186
Faudzi, Ahmad A. M. II-25
Feng, Chung-Shun I-334
Flores, Bernardo I-145

G
Ganda, Sai Yaaminie I-145
Guo, Zhiwei II-201
Gupta, Krishna II-295

H
Hamamoto, Kazuhiko II-25
Ho, Pin-Chieh II-3
Honda, Hirotada II-228
Hsu, Jia-Chi I-267
Hsu, Mei-Yun I-322
Hu, Qinxia II-201
Hu, Tsu-Wu I-334
Hu, Xiao II-201
Huang, Szu-Hao I-267, I-301, II-3
Huang, Yiqiu II-201
Huang, Yu-Che I-334

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J
Johansson, Ulf I-278, II-159
Jouda, Basel II-271

K
Kadhem, Hasan I-16
Kashikar, Raj I-257
Kawamura, Sota II-228
Kerdvibulvech, Chutisant I-192
Khadgi, Manisha I-202
Khairuddin, Uswah II-25, II-256, II-281
Khaled Dawood, Aly II-256
Khanarsa, Paisit I-3, I-67
Koike, Genta II-211
Konduri, Sushma II-243
Kuracha, Murali R. II-243

L
Lai, Ying-Chih II-37
Lee, Chi I-301
Li, Peilin II-186
Li, Tsai-Yen I-123
Li, Zeke II-99
Liang, ChengXi II-134
Liao, Wen-Hao II-37
Lin, Chen-Yi II-221, II-236
Lin, Chieh-Peng I-314
Lin, Shin-Ru I-314
Liu, Ping-Lin II-236
Liu, Wei I-44
Liu, Xinyi II-99
Lofstrom, Tuwe II-159
Luangpaiboon, Pongchanun I-32
Ludda, Nutjaree II-87
Luo, Cuicui I-243
Luo, XianHang II-120

M
Maalej, Aicha II-159
Magnani, Alessandro I-103
Maneewongvatana, Suthathip II-176
Mannor, Shie I-103
Mei, YuXin II-120
Mekki, Amr Emad II-281
Minetaki, Kazunori I-322
Mishra, Vipin I-344
Montemanni, Roberto II-111
Murtaza, Ali II-25

N
Nakamura, Shugo II-228
Nanphang, Atiwat I-32

O
Ongkunaruk, Pornthipa II-87

P
Parlewar, Manisha II-295

R
Rady, Basma I-157
Rahman, Sadia II-146
Ren, Zhenzhen II-243
Romyen, Nirach I-94

S
Sadek, A. M. I-231
Sano, Takashi II-228
Savani, Vaibhavi Hiteshkumar I-145
Shaw, Yuchiao II-243
Siriborvornratanakul, Thitirat II-58
Sittiros, Peera I-94
Sönströd, Cecilia I-278
Sriamorn, Tanet I-3
Srijiranon, Krittakom I-94
Sun, Huaijun II-186

T
Tanantong, Tanatorn I-94
Tantasanee, Napatsorn II-58
Therdkiattikoon, Sorawit II-176
Ting, I-Hsien I-322

U
Ud Din, Fareed I-202
Udomcharoenchaikit, Can I-67
Uraisomsurat, Napat I-3

V
Viyanon, Waraporn II-48

W
Wang, Mingwei I-44
Wen, Guangqi II-99
Wong, Seng Yue II-186

Wu, Cheng-Jhang I-267
Wu, Mei-Hua II-3
Wu, Ni II-134

Y
Yao, JingTao II-71, II-146
Yin, Biyu I-44
Yu, Chi-Cheng II-221
Yu, Tai-Yi I-55, II-15

Z
Zhang, Kai II-120
Zhang, Ling II-134
Zhangr, Kai II-134